National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Prevention and Treatment of Cancer-Related Infections

Version 3.2024 — September 23, 2024

NCCN.org

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NCCN 肿瘤临床实践指南（NCCN 指南®）

癌症相关感染的预防和治疗

版本 3.2024—September 23，2024

NCCN.org

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NCCN Guidelines Version 3.2024
Prevention and Treatment of Cancer-Related Infections

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重试 错误原因

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NCCN 指南第 3.2024 版癌症相关感染的预防和治疗

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NCCN 指南面板披露

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NCCN Guidelines Version 3.2024
Prevention and Treatment of Cancer-Related Infections

NCCN Prevention and Treatment of Cancer-Related Infections Panel Members
Summary of the Guidelines Updates

Prevention/Prophylaxis
- Antimicrobial Prophylaxis Based on Overall Infection Risk in Patients with Cancer (INF-1)
- Prevention of Fungal Infections (INF-2)
- Prevention of Herpes Simplex Virus (HSV) and Varicella Zoster Virus (VZV) Reactivation or Disease (INF-3)
- Prevention of Cytomegalovirus (CMV) Reactivation or Disease (INF-4)
- Management of Hepatitis B Virus (HBV), Hepatitis C Virus (HCV), and Human Immunodeficiency Virus (HIV) Reactivation or Disease (INF-5)
- Prevention of *Pneumocystis Jirovecii (Pneumocystis Carinii)* Infection (INF-6)
- General Recommendations for Vaccination in Patients with Cancer (INF-7)
- Recommended Vaccination Schedule After Autologous or Allogeneic HCT (INF-8)
- Immune and Targeted Treatments (INF-A)

Evaluation and Treatment
- Initial Evaluation of Fever and Neutropenia (FEV-1)
- Initial Risk Assessment for Patients with Febrile Neutropenia (FEV-2)
- Outpatient Therapy for Patients at Low Risk (FEV-3)
- Initial Inpatient Empiric Therapy for Uncomplicated Fever and Neutropenia (FEV-5)
- Site-Specific Evaluation and Therapy:
  - Mouth/Mucosal Membrane, Esophagus, and Sinus/Nasal (FEV-6)
  - Abdominal Pain, Perirectal Pain, Diarrhea, and Urinary Tract Symptoms (FEV-7)
- Cellulitis/Skin and Soft Tissue Infections, Vesicular Lesions, Disseminated Papules or Other Lesions, Vascular Access Devices, and Central Nervous System (CNS) Symptoms (FEV-8)
  - Lung Infiltrates (FEV-9)
- Treatment of *Clostridioides Difficile* Infections (CDI) in Patients with Cancer (FEV-10)
- Results of Daily Monitoring, Follow-Up Therapy (FEV-11)
- Follow-Up Therapy for Responding Disease (FEV-12)

COVID-19
- Management of Concurrent COVID-19 and Cancer in Patients (COV-1)

Clinical Trials: NCCN believes that the best management for any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions.

NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

Antibacterial Agents Tables (FEV-A)
Antifungal Agents Tables (FEV-B)
Antiviral Agents Tables (FEV-C)
Risk Assessment Resources (FEV-D)

Abbreviations (ABBR-1)

The NCCN Guidelines® are a statement of evidence and consensus of the authors regarding their views of currently accepted approaches to treatment. Any clinician seeking to apply or consult the NCCN Guidelines is expected to use independent medical judgment in the context of individual clinical circumstances to determine any patient's care or treatment. The National Comprehensive Cancer Network® (NCCN®) makes no representations or warranties of any kind regarding their content, use or application and disclaims any responsibility for their application or use in any way. The NCCN Guidelines are copyrighted by National Comprehensive Cancer Network®. All rights reserved. The NCCN Guidelines and the illustrations herein may not be reproduced in any form without the express written permission of NCCN. ©2024.

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NCCN 预防和治疗癌症相关感染专家组成员指南更新摘要

预防/预防
·基于癌症患者总体感染风险的抗菌预防（INF–1）
·预防真菌感染（INF–2）
·预防单纯疱疹病毒（HSV）和水痘带状疱疹病毒（VZV）再激活或疾病（INF–3）
·预防巨细胞病毒（CMV）再活化或疾病（INF–4）
• B 型肝炎病毒（HBV）、丙型肝炎病毒（HCV）和人类免疫缺陷病毒（HIV）的管理再激活或疾病（INF–5）
·预防卡氏肺孢子虫（Pneumocystis Carinii）感染（INF–6）
·癌症患者接种疫苗的一般建议（INF–7）
·自体或同种异体 HCT 后推荐的疫苗接种时间表（INF–8）
免疫和靶向治疗（INF–A）

评估和治疗
·发热和中性粒细胞减少症（FEV–1）的初步评估
·发热性中性粒细胞增多症（FEV–2）患者的初始风险评估
·低风险患者的门诊治疗（FEV–3）
·单纯性发热和中性粒细胞减少症（FEV–5）的初始住院经验性治疗
• 研究中心特定评价和治疗：
口腔/粘膜、食管和鼻窦/鼻腔（FEV–6） 腹痛、直肠周围疼痛、腹泻和尿路症状（FEV–7）·蜂窝织炎/皮肤和软组织感染、水疱性病变、播散性丘疹或其他病变、血管通路装置和中枢神经系统（CNS）症状（FEV–8）

肺抑制（FEV–9）
·癌症患者艰难梭菌感染（CDI）的治疗（FEV–10）·每日监测、随访治疗的结果（FEV–11）·疾病应答的随访治疗（FEV–12）

COVID–19
·并发 COVID–19 和癌症患者的管理（COV–1）

临床试验：NCCN 认为，对任何癌症患者的最佳管理是临床试验。特别鼓励参加临床试验。

查找 NCCN 成员机构：
https://www.nccn.org/home/member–机构职能体系

NCCN 证据类别和
协商一致意见：除非另有说明，所有建议均为 2A 类。

参见 NCCN 证据分类和共识。

抗菌药物表（FEV–A）抗真菌药物表（FEV–B）抗病毒药物表（FEV–C）风险评估资源（FEV–D）

缩略语（ABBR–1）

NCCN 指南是作者关于他们对目前接受的治疗方法的观点的证据和共识声明。任何寻求应用或咨询 NCCN 指南的临床医生都应在个体临床情况下使用独立的医学判断来确定任何患者的护理或治疗。美国国家综合癌症网络（NCCN）对它们的内容，使用或应用不作任何形式的陈述或保证，并以任何方式对其应用或使用不承担任何责任。NCCN 指南的版权归国家综合癌症网络所有。All rights reserved.未经 NCCN 明确书面许可，不得以任何形式复制 NCCN 指南和此处的插图。©2024 年。

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NCCN Guidelines Version 3.2024
Prevention and Treatment of Cancer-Related Infections

Terminologies in all NCCN Guidelines are being actively modified to advance the goals of equity, inclusion, and representation.

Updates in Version 3.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 2.2024 include:

MS-1

• Discussion section updated to reflect changes in the algorithm

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Updates in Version 2.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 1.2024 include:

INF-A

• Section extensively modified

Updates in Version 1.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 2.2023 include:

INF-1

• Header line added: See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions
• Antimicrobial Prophylaxis, High, bullet 4 added: Length of prophylaxis depends on immune reconstitution.
• Footnotes modified:
  ▸ a: Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, ~~recipient or donor CMV status~~ *CMV serostatus*, and intensity of immunosuppressive therapy (IST). For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.
  ▸ d: For patients who are intolerant to fluoroquinolone, consider TMP/SMX or an oral third-generation cephalosporin (category 2B). ~~The emergence of multidrug-resistant organisms (MDROs), disruption of the microbiome, and antibiotic toxicities must be taken into consideration when choosing an antimicrobial prophylactic agent.~~
• Footnote removed: Pneumocystis prophylaxis (INF-6). For dosing, spectrum, and specific comments/cautions, see Antibacterial Agents (FEV-A), Antifungal Agents (FEV-B), and Antiviral Agents (FEV-C) as indicated.
• Asterisk added under table: Neutropenia: ≤500 neutrophils/mcL or ≤1000 neutrophils/mcL and a predicted decline to ≤500/ mcL over the next 48 hours

INF-2

• Row added: Immune and targeted treatments

INF-4

• Footnote m modified: Some centers consider the use of letermovir through day 100 post-HCT and continue CMV surveillance for patients at high risk for CMV reactivation. See Antiviral Agents (FEV-C 2 of 4). Letermovir lacks HSV and VZV coverage and HSV/VZV prophylaxis should be continued. *In certain circumstances, up to day 200 can be considered.*

INF-5

• Footnote removed: Diagnostic monitoring and treatment for HBV, HCV, and HIV are evolving fields; consultation with an ID expert or hepatologist should be sought in the treatment of all patients with reactivation or disease.
• Footnotes modified:
  ▸ y: ~~If viral load is consistently undetectable, treatment is considered prophylactic.~~ If viral load fails to drop or previously undetectable PCR becomes positive, consult hepatologist and discontinue anti-CD20 antibody therapy.
  ▸ x: ~~Lamivudine may be considered in certain circumstances with expert consultation~~ *Lamivudine is inferior to entecavir and tenofovir, but may be considered when other agents are unavailable*.

INF-7

• Statement added: For prevention of infections in cancer survivors, including vaccination recommendations, see the NCCN Guidelines for Survivorship.

Continued

UPDATES

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所有 NCCN 指南中的术语都在积极修改，以推进公平、包容和代表性的目标。
NCCN 癌症相关感染预防和治疗指南第 3.2024 版（第 2.2024 版）的更新包括：
MS–1

- 更新讨论章节，以反映算法的变更

NCCN 癌症相关感染预防和治疗指南第 2.2024 版的更新包括：

INF–A
- 广泛修改的章节

NCCN 癌症相关感染预防和治疗指南（2.2023 版）1.2024 版的更新包括：

INF–1
- 增加标题行：剂量、光谱和具体备注/注意事项见抗菌剂（FEV–A）
- 添加了高水平的抗菌预防措施，项目符号 4：预防措施的持续时间取决于免疫重建。
- 修改了脚注： a：风险类别基于几个因素，包括潜在的恶性肿瘤，疾病是否缓解，中性粒细胞减少持续时间，既往

化疗暴露、受体或供体 CMV 状态、CMV 血清状态和免疫抑制治疗（IST）强度。有关感染问题和免疫靶向药物的推荐预防，请参阅 INF–A。

d：对于氟喹诺酮类药物不耐受的患者，考虑 TMP/SMX 或口服第三代头孢菌素（2B 类）。的出现

在选择抗微生物预防剂时，必须考虑多重耐药微生物（MDRO）、微生物组的破坏和抗生素毒性。

- 删除脚注：肺孢子虫预防（INF–6）。关于剂量、光谱和具体注释/注意事项，请参见所示的抗菌药物（FEV–A）、抗真菌药物（FEV–B）和抗病毒药物（FEV–C）。

·在表下添加星号：中性粒细胞：≤500 个中性粒细胞/mcL 或≤1000 个中性粒细胞/mcL，并预测在接下来的 48 小时内下降至≤500 个/ mcL
INF–2
·新增一行：免疫和靶向治疗
INF–4 ·修改脚注：一些中心考虑使用莱特莫韦直至 HCT 后 100 天，并继续对 CMV 再激活高风险患者进行 CMV 监测。抗病毒药物（FEV–C 2/4）莱特莫韦缺乏 HSV 和 VZV 覆盖率，应继续进行 HSV/VZV 预防。在某些情况下，可以考虑到 200 天。INF–5

- 删除脚注：HBV、HCV 和 HIV 的诊断监测和治疗是不断发展的领域;在治疗所有复发或疾病患者时，应寻求 ID 专家或肝病学家的咨询。

- 修改了脚注：
y：如果病毒载量始终检测不到，则认为治疗是预防性的。如果病毒载量未能下降或之前无法检测到的 PCR 变为阳性，请咨询肝病专家并停止抗 CD 20 抗体治疗。
X：拉米夫定可能会考虑在某些情况下与专家咨询拉米夫定是劣于恩替卡韦和替诺福韦，但可能会考虑当其他代理商不可用。INF–7

- 声明补充说：为了预防癌症幸存者的感染，包括疫苗接种建议，请参阅 NCCN 生存指南。

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更新

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Updates in Version 1.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 2.2023 include:

For prevention of infections in pediatric patients, refer to guidance from the CDC.

- General recommendations bullet removed: The safety of vaccines in patients receiving immunostimulatory drugs is unclear. Some emerging data suggest vaccines (eg, influenza) can be given safely.
- Pneumococcal recommendation modified: The pneumococcal conjugate vaccine (PCV20 ~~or PCV15~~) should be administered to adults who are newly diagnosed with cancer who are pneumococcal vaccine-naïve. ~~If PCV15 is used it should be followed by the polysaccharide pneumococcal vaccine (PPSV23) at least 8 weeks later. Additional PPSV23 is not needed for those receiving PCV20. For patients who have previously received PPSV23, the PCV20 or PCV15 dose should be given at least 1 year after the last PPSV23 dose. Regardless of if PCV15 or PCV20 is given, an additional dose of PPSV23 is not recommended. For those who have received PCV13 with or without PPSV23, give PPSV23 as previously recommendedkk. See Pneumococcal Vaccine Timing for Adults for specific guidance. The incremental public health benefits of providing PCV15 or PCV20 to adults who have received PCV13 only or both PCV13 and PPSV23 have not been evaluated.~~ *According to the CDC, additional pneumococcal polysaccharide vaccine (PPSV23) is not needed for those receiving PCV20. Alternatively, PCV15 can be given, followed by PPSV23 at least 8 weeks later. For patients who have previously received PPSV23, PCV20 (preferred) or PCV15 can be given. For patients who have previously received PCV13 only, they can receive PCV20 at least 1 year later, rather than PPSV23. For patients who have previously received PCV13 and 1 or 2 doses of PPSV23, they can receive PCV20 at least 5 years later. See CDC recommendations for pneumococcal vaccination.*
- Meningococcal recommendation modified: ... MenACWY vaccine is given in 2 doses ≥8 weeks apart; serogroup B vaccine is available in a 2- or 3-dose series, depending on the vaccine formulation used. *Patients with ongoing risk for meningococcal disease should receive a booster dose 5 years after completion of the primary series and every 5 years thereafter.*
- Respiratory Syncytial Virus (RSV), recommendation modified: The RSV vaccine is approved by the FDA and available for those ≥60 y. Its effectiveness in patients with cancer is unknown. The *long-acting* RSV monoclonal antibody (mAb) (nirsevimab) is approved for infants <24 months of age to prevent RSV infection.
- Vaccination modified: Travel *and Other Vaccines*
- Travel and Other Vaccines, Recommendation, bullet added: For recommendations on travel vaccines, refer to the CDC Yellow Book.

INF-7A

- Footnotes removed:
  - For prevention of infection in cancer survivors, including vaccination recommendations, see the NCCN Guidelines for Survivorship.
  - See Centers for Disease Control and Prevention (CDC) website for more updated recommendations.
  - For adults who have received PCV13 but have not completed their recommended pneumococcal vaccine series with PPSV23, one dose of PCV20 may be used if PPSV23 is not available. If PCV20 is used, their pneumococcal vaccinations are complete.
- Footnote ff modified: ~~Pneumococcal antibody responses to some serotypes in PCV13 were decreased following co-administration of the meningococcal conjugate vaccine, MenACWY-D, and PCV13. Therefore, PCV13 should not be given with MenACWY-D but can be given with MenACWY-CRM. Similar precautions should be used for PCV15.~~ *PCV20 or PCV15 should not be given with meningococcal conjugate vaccine, quadrivalent (MenACWY-D) but can be given with MenACWY-CRM.*

INF-8

- Inactivated, Subunit, or Toxoid Vaccines, row 1 modified: ~~DTaP (Diphtheria/Tetanus/Acellular Pertussis)~~ *Tdap*.
- Pneumococcal vaccination
  - Recommended timing after HCT modified: ~~6–12 months ≥12 months~~ *3-6 months*
  - Number of doses modified: ~~3,1~~ *3-4*

INF-8A

- Footnotes modified:
  - gg: Emerging therapies such as CAR T-cell therapy appear to behave similar to ~~patients who have undergone~~ allogenic transplant in terms of vaccine

Continued

UPDATES

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NCCN 癌症相关感染预防和治疗指南第 1.2024 版（第 2.2023 版）的更新包括：关于儿科患者感染的预防，请参阅 CDC 的指南。

- 删除一般建议：接受免疫刺激药物的患者接种疫苗的安全性尚不清楚。一些新的数据表明疫苗（如流感）可以安全地接种。

- 肺炎球菌建议修改：肺炎球菌结合疫苗（PCV 20 或 PCV 15）应给予成年人谁是新
diagnosed with cancer who are pneumococcal vaccine–naïve. If PCV15 is used it should be followed by the polysaccharide pneumococcal vaccine (PPSV23) at least 8 weeks later. Additional PPSV23 is not needed for those receiving PCV20. For patients who have previously received PPSV23, the PCV20 or PCV15 dose should be given at least 1 year after the last PPSV23 dose. Regardless of if PCV15 or PCV20 is given, an additional dose of PPSV23 is not recommended. For those who have received PCV13 with or without PPSV23, give PPSV23 as previously recommendedkk. See
Pneumococcal Vaccine Timing for Adults for specific guidance. The incremental public health benefits of providing PCV15 or PCV20 to adults who have received PCV13 only or both PCV13 and PPSV23 have not been evaluated. According to the CDC, additional pneumococcal polysaccharide vaccine
对于接受 PCV 20 的患者，不需要 PPSV 23。或者，可以给予 PCV 15，然后在至少 8 周后给予 PSV 23。对于既往接受过 PPSV 23、PCV 20（首选）或 PCV 15 的患者。对于既往仅接受过 PCV 13 的患者，他们可以在至少 1 年后接受 PCV 20，而不是 PSV 23。对于既往接受过 PCV 13 和 1 剂或 2 剂 PPSV 23 的患者，他们可以在至少 5 年后接受 PCV 20。参见 CDC 关于肺炎球菌疫苗接种的建议。

- 修订脑膜炎球菌建议：. MenACWY 疫苗分 2 剂接种，间隔≥8 周;血清群 B 疫苗分 2 剂或 3 剂接种
根据所使用的疫苗配方。持续存在脑膜炎球菌病风险的患者应在完成主要系列研究后 5 年接受加强剂量，此后每 5 年接受一次。

- 呼吸道合胞病毒（RSV），建议修改：RSV 疫苗已获得 FDA 批准，可用于≥60 岁的人群。它对癌症患者的有效性尚不清楚。长效 RSV 单克隆抗体（mAb）（nirsevimab）已获批用于<24 个月的婴儿，以预防 RSV 感染。

- 疫苗接种：旅行和其他疫苗
·旅行和其他疫苗，建议，补充项目：有关旅行疫苗的建议，请参阅 CDC 黄皮书。
INF–7A
- Footnotes removed:
For prevention of infection in cancer survivors, including vaccination recommendations, see the NCCN Guidelines for Survivorship.
See Centers for Disease Control and Prevention (CDC) website for more updated recommendations.
For adults who have received PCV13 but have not completed their recommended pneumococcal vaccine series with PPSV23, one dose of PCV20 may be used if PPSV23 is not available. If PCV20 is used, their pneumococcal vaccinations are complete.
- Footnote ff modified: Pneumococcal antibody responses to some serotypes in PCV13 were decreased following co–administration of the meningococcal conjugate vaccine, MenACWY–D, and PCV13. Therefore, PCV13 should not be given with MenACWY–D but can be given with MenACWY–CRM. Similar precautions should be used for PCV15. PCV20 or PCV15 should not be given with meningococcal conjugate vaccine, quadrivalent (MenACWY–D) but can be given with MenACWY–CRM.
INF–8
灭活、亚单位或类毒素疫苗，第 1 行修改：DTaP（白喉/破伤风/无细胞百日咳）Tdap。
·肺炎球菌疫苗接种 HCT 改良后的推荐时间：6–12 个月≥12 个月 3–6 个月 修改的剂量数：3，1 3–4 INF–8A ·修改的脚注： gg：新兴疗法，如 CAR T 细胞疗法，在疫苗方面似乎与接受同种异体移植的患者表现相似

继续
更新

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Updates in Version 1.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 2.2023 include:

boosting recommendations.
- ii: DTaP (diphtheria, tetanus, and acellular pertussis) is not approved for use in ages >7 (FDA/ACIP-approved 3-dose series for ≥7 year olds is Tdap/Td/Td (tetanus-diphtheria) ~~vs. considerations of Tdap/Tdap/Tdap~~). ~~Other than 3 doses of DTaP as stated, 3 doses of Tdap, or 1 dose of Tdap followed by 2 doses of Td are also acceptable options.~~
- jj ~~For patients with GVHD, PCV15 or PCV20 may be considered instead of PPSV23 as a fourth dose. The CDC is currently evaluating the use of a primary series of PCV20 and PCV15 post-HCT.~~ If PCV20 is used, 4 doses should be administered. First 3 doses are generally 1-2 months apart, with the fourth dose 6 months after the third dose. There is no need to give PPSV23. If PCV15 is used, 3 doses should be administered, followed by PPSV23 6-12 months post primary series. Following the primary series of 3 PCV doses, a dose of the PPSV23 to broaden the immune response might be given. For patients with chronic GVHD who are likely to respond poorly to PPSV23, a fourth dose of PCV20 or PCV15 should be considered instead of PPSV23.
- ll: Meningococcal B vaccine should be considered for patients at high risk, such as patients with *chronic GVHD,* asplenia, or complement deficiency or patients receiving a complement C5 inhibitor (eg, eculizumab, ravulizumab).

• Footnote qq added: Pergam SA, et al. Biol Blood Marrow Transplant 2019;25: e321-e330.

INF-A 1 of 13
• Column 1 modified: *Drug Class*/Mechanism of Action (Also for INF-A 2 of 12 through 10 of 12)
• Major Uses column removed from Table (Also for INF-A 2 of 12 through 10 of 12)
• Phosphatidylinositol-3-kinase (PI3K) inhibitors, Recommendations and Comments, bullet 1 modified: ~~Consider CMV surveillance in CMV seropositive patients~~ *Monitor for CMV reactivation in patients at high risk.*

INF-A 6 of 13
• CD30 target, Recommendations and Comments, bullet 1 modified: ~~Consider CMV monitoring in CMV-seropositive patients~~ *Severe or fatal CMV infections reported. Monitor for CMV reactivation in patients at high risk.* (Also for CCR4 target on INF-A 7 of 12)

INF-A 9 of 13
• Recommendations and Comments, bullet 5 modified: *Consider* PJP prophylaxis if high-dose steroid use (≥20 mg per day of prednisone x4 weeks).

INF-A 10 of 13
• Recommendations and Comments, bullets added:
- Consider immunoglobulin replacement therapy in the setting of low IgG and recurrent or severe sinopulmonary infections.
- For vaccine recommendations, see INF-7 and INF-8.

FEV-6
• Mouth/mucosal membrane, necrotizing ulceration, evaluation, sub-bullet removed: Consider leukemic infiltrate
• Esophagus, evaluation
- Bullet removed: Consider CMV esophagitis in patients at high risk for CMV disease
- Bullet added: Histopathologic examination for viral and fungal pathogens

• Footnote n modified: Posaconazole or isavuconazonium sulfate can be considered for patients who have invasive, refractory infections or who have intolerance to amphotericin B formulations. ~~Posaconazole is not approved by the FDA as primary therapy or secondary therapy for refractory invasive fungal infections.~~

FEV-7
• Diarrhea, evaluation, bullet 4 modified: Depending on clinical circumstances, consider *diagnostic* testing for viral, *bacterial, and/or parasitic* pathogens ~~(eg, adenovirus, rotavirus, norovirus, CMV), bacterial cultures, and/or parasite exam~~

Continued
UPDATES

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NCCN 癌症相关感染预防和治疗指南第 2.2023 版第 1.2024 版的更新包括：加强建议。

ii：DTaP（白喉、破伤风和无细胞百日咳）未被批准用于>7 岁的儿童（FDA/ACIP 批准的≥7 岁奥尔兹的 3 剂系列为 Tdap/Td/Td（破伤风–白喉）与 Tdap/Tdap/Tdap 的考虑）。除了如上所述的 3 剂 DTaP 外，3 剂 Tdap 或 1 剂 Tdap 随后 2 剂 Td 也是可接受的选择。

jj 对于患有 GVHD 的患者，可以考虑将 PCV 15 或 PCV 20 代替 PPSV 23 作为第四剂。CDC 目前正在评估 HCT 后 PCV 20 和 PCV 15 的主要系列的使用。如果使用 PCV 20，应给予 4 剂。前 3 剂通常间隔 1–2 个月，第 4 剂在第 3 剂后 6 个月。没有必要给 PPSV 23。如果使用 PCV 15，应给予 3 剂，然后在主要系列后 6–12 个月给予 PSV 23。在 3 次 PCV 给药的主要系列后，可给予 PPSV 23 剂量以扩大免疫应答。对于可能对 PPSV 23 反应较差的慢性 GVHD 患者，应考虑使用第四剂 PCV 20 或 PCV 15 代替 PPSV 23。

ll：脑膜炎球菌 B 疫苗应考虑用于高风险患者，如慢性 GVHD、无脾或补体缺陷患者或接受补体 C5 抑制剂（如依库珠单抗、ravulizumab）的患者。

·添加脚注 qq：Pergam SA，et al. Biol Blood Marrow Transplant 2019;25：e321–e330.
INF–A 1/13
·第 1 列修改：药物类别/作用机制（也适用于 INF–A 2/12 至 10/12）·从表中删除主要用途列（也适用于 INF–A 2/12 至 10/12）
磷脂酰肌醇–3–激酶（PI 3 K）抑制剂，建议和评论，项目符号 1 修改：考虑对 CMV 血清阳性患者进行 CMV 监测，监测高危患者的 CMV 再激活。INF–A 6/13

• CD 30 靶点，建议和评论，项目符号 1 修改：考虑在 CMV 血清阳性患者中进行 CMV 监测重度或致死性 CMV 感染报告。监测高危患者的 CMV 再激活。(Also INF–A 7/12）INF–A 9/13 的 CCR 4 目标·建议和评论，项目符号 5 修改：如果使用高剂量类固醇（≥20 mg/天泼尼松 x4 周），则考虑 PJP 预防。

INF–A 10/13
• 建议和评论，添加了项目符号：
在低 IgG 和复发或严重鼻窦炎感染的情况下考虑免疫球蛋白替代疗法。
For vaccine recommendations, see INF–7 and INF–8. FEV–6 • Mouth/mucosal membrane, necrotizing ulceration, evaluation, sub–bullet removed: Consider leukemic infiltrate • Esophagus, evaluation Bullet removed: Consider CMV esophagitis in patients at high risk for CMV disease Bullet added: Histopathologic examination for viral and fungal pathogens
重试 错误原因

Footnote n modified: Posaconazole or isavuconazonium sulfate can be considered for patients who have invasive, refractory infections or who have intolerance to amphotericin B formulations. Posaconazole is not approved by the FDA as primary therapy or secondary therapy for refractory invasive fungal infections. FEV–7
重试 错误原因

• Diarrhea, evaluation, bullet 4 modified: Depending on clinical circumstances, consider diagnostic testing for viral, bacterial, and/or parasitic pathogens (eg 腺病毒、轮状病毒、诺如病毒、CMV）、细菌培养物和/或寄生虫检查
重试 错误原因

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更新

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Updates in Version 1.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 2.2023 include:

FEV-8
- Vesicular lesions, evaluation modified: Aspiration or scraping for PCR/DFA, ~~and/or herpes virus cultures if PCR unavailable~~ *and/or HSV cultures if PCR unavailable.*
- CNS symptoms
  - Evaluation, bullet 3 added: Infectious disease (ID) consult
  - Treatment Modifications, bullet 1 modified: Initial empiric therapy ~~pending ID consult~~

FEV-9
- Title added: Evaluation of Patients with Possible Respiratory Infection
- Evaluation column extensively modified

FEV-12
- Follow-up therapy, bullet 4, sub-bullet added: Catheter removal highly recommended if persistent positivity
- Suggested minimum duration...,bullet 2, sub-bullet 3 modified: S. aureus: typically requires 4 weeks *(some institutions may use a shorter duration based on ID consultation)* after first negative blood culture; ID consult strongly recommended *(ID consult is associated with decreased mortality)*

FEV-A 1 of 3
- Daptomycin, Comments/Precautions, bullet removed: ID consult strongly recommended
- Tedizolid row added

FEV-B 1 of 5
- Itraconazole, Comments/Precautions, bullet 2 modified: H2 blockers and proton pump inhibitors (PPIs) may inhibit absorption of capsule formulation. Oral liquid *formulation* is preferred for improved absorption.
- Footnote b modified: TDM is routinely used in managing *itraconazole, posaconazole, and voriconazole*. *TDM is not routinely used for isavuconazole.*

FEV-B 2 of 5
- Posaconazole, Comments/Cautions
  - Bullet 1 modified: ~~Evaluated as treatment of refractory infection (but not FDA-approved) in several invasive fungal diseases~~ *Used for treatment of refractory infection (but not FDA-approved) in several invasive fungal diseases.*
  - Bullet 3 modified: *IV posaconazole or* alternative antifungal therapy should be considered for patients who cannot eat a full meal or tolerate an oral nutritional supplement.
  - Bullet added: FDA-approved for invasive aspergillosis
- Voriconazole, Comments/Cautions, bullet 2 modified: Long-term complications ~~resulting from metabolic irregularities~~ may include increased risk for squamous cell carcinoma and hyperphosphatemia

FEV-B 3 of 5
- Footnote d added: AmB-D is not preferred whenever L-AMB or ABLC is available.

FEV-C 1 of 4
- Valacyclovir: Comments/Cautions, statement removed: CMV in allogeneic HCT recipients (2 gm PO QID).
- Valganciclovir, Typical Dosing Based on Indication, bullet modified: Preemptive therapy *and treatment* for CMV...

FEV-C 4 of 4
- Footnote removed: In general, the strategy of CMV surveillance testing by PCR followed by preemptive anti-CMV therapy for a positive result is favored over universal long-term prophylaxis in patients receiving allogeneic HCT.

COV-5
- Bullet added: For unresolved COVID-19, ID consult is recommended.

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UPDATES

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NCCN 癌症相关感染预防和治疗指南（2.2023 版）1.2024 版的更新包括：

FEV–8
• 水泡性病变，修改评价：抽吸或刮擦用于 PCR/DFA，和/或疱疹病毒培养（如果 PCR 不可用）和/或 HSV 培养（如果 PCR 不可用）。

• CNS symptoms Evaluation, bullet 3 added: Infectious disease (ID) consult Treatment Modifications, bullet 1 modified: Initial empiric therapy pending ID consult FEV–9

重试 错误原因

• 增加标题：评估可能发生呼吸道感染的患者
·评估栏大幅修改
FEV–12
• 随访治疗，项目符号 4，添加子项目符号：如果持续阳性，强烈建议拔除导管
• 建议的最短持续时间.，项目符号 2，子项目符号 3 修改：S。aureus：通常需要 4 周（某些机构可能使用较短的持续时间）基于 ID 咨询）首次血培养阴性后;强烈建议 ID 咨询（ID 咨询与死亡率降低相关）FEV–A 1/3

• 达托霉素，备注/注意事项，删除项目符号：强烈建议咨询 ID
·添加了 Tedizolid row
FEV–B 1/5
• 伊曲康唑，备注/注意事项，项目符号 2 修改：H2 阻滞剂和质子泵抑制剂（PPI）可能抑制胶囊制剂的吸收。
口服液体制剂优选用于改善吸收。
·修改脚注 B：TDM 常规用于管理伊曲康唑、泊沙康唑和伏立康唑。艾沙康唑未常规使用 TDM。
FEV–B 2/5
• 泊沙康唑，备注/注意事项
修改项目符号 1：评价为治疗几种侵袭性真菌疾病的难治性感染（但未获得 FDA 批准）用于治疗几种侵袭性真菌疾病的难治性感染（但未获得 FDA 批准）。

修改项目符号 3：静脉泊沙康唑或替代抗真菌治疗应考虑患者谁不能吃一顿饱饭或耐受口服营养补充剂。

添加项目符号：FDA 批准用于侵袭性曲霉病
• 伏立康唑，注释/注意事项，项目符号 2 修改：代谢异常导致的长期并发症可能包括鳞状细胞癌和高磷血症风险增加 FEV–B 3/5 ·添加脚注 d：当 L–AMB 或 ABLC 可用时，AmB–D 不是首选。

FEV–C 1/4
• 伐昔洛韦：备注/注意事项，删除声明：同种异体 HCT 接受者中的 CMV（2 gm PO QID）。
·缬更昔洛韦，基于适应症的典型剂量，修改的子弹：CMV 的抢先治疗和治疗...
FEV–C 4/4
删除脚注：一般来说，在接受同种异体 HCT 的患者中，通过 PCR 进行 CMV 监测检测，然后对阳性结果进行先发制人的抗 CMV 治疗的策略优于普遍的长期预防。
COV–5

• Bullet 补充道：对于未解决的 COVID–19，建议进行身份咨询。

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Updates in Version 1.2024 of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections from Version 2.2023 include:

COV-6
- Footnote g modified: COVID-19 convalescent plasma obtained from those who have recovered from the ~~Omicron variant~~ *recent circulating variants* and have been previously vaccinated is preferred. COVID-19 convalescent plasma can be acquired via the Blood Centers of America.

COV-7
- Patient hospitalized for acute symptomatic COVID-19, Comments, bullet 2, sub-sub bullet modified: Although investigational (not standard of care), consider extending remdesivir duration to 10 days *for patients hospitalized for COVID-19* if PCR Ct is still low after 5 days and the patient remains symptomatic or is not improving.

COV-8
- Persistent symptomatic COVID-19 infection; particularly B-cell impairment, Comments
  - Bullet 2 modified: To determine potential benefit of COVID-19 convalescent plasma ~~or monoclonal antibody therapies~~, some providers (via clinical investigational approach) will first check:
  - Bullet 2, sub-bullet 2 modified: ~~SARS-CoV-2 PCR Ct for determination of viral load/burden (higher viral load corresponding to lower PCR Ct).~~ *Consult ID to assess viral load burden by PCR.*
- Persistent asymptomatic SARS-CoV-2–positive testing, Comments, bullet 3, sub-bullet removed: Low SARS-CoV-2 PCR Ct correlates with high viral load and may further suggest potential for subclinical active disease, although is currently not validated for clinical use and is considered investigational.

COV-9
- Remdesivir, Dosing/Duration, bullet removed: Caution with moderate-severe renal dysfunction (eg, CrCl <30 mL/min)

Removed from Guidelines
- COV-10 through COV-15, and COV-A 4 of 4

COV-A 3 of 3
- References 16–19 removed

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NCCN 癌症相关感染预防和治疗指南（2.2023 版）1.2024 版的更新包括：

COV-6
修改脚注 g：最好从从 Omicron 变种最近传播的变种中康复并之前接种过疫苗的人中获得 COVID-19 恢复期血浆。COVID-19 恢复期血浆可以通过美国血液中心获得。
COV-7

• 患者因急性症状性 COVID-19 住院，评论，项目符号 2，修改子项目符号：尽管是研究性（非标准治疗），
对于因 COVID-19 住院的患者，如果 5 天后 PCR Ct 仍然较低，且患者仍有症状或没有改善，则考虑将 Remdesivir 的持续时间延长至 10 天。COV-8

•持续症状性 COVID-19 感染;特别是 B 细胞损伤，评论 修改项目 2：为了确定 COVID-19 恢复期血浆或单克隆抗体治疗的潜在益处，一些提供者（通过临床

调查方法）将首先检查：
Bullet 2, sub-bullet 2 modified: SARS-CoV-2 PCR Ct for determination of viral load/burden (higher viral load corresponding to lower PCR Ct). Consult ID to assess viral load burden by PCR.
• Persistent asymptomatic SARS-CoV-2—positive testing, Comments, bullet 3, sub-bullet removed: Low SARS-CoV-2 PCR Ct correlates with high viral load and may further suggest potential for subclinical active disease, although is currently not validated for clinical use and is considered investigational. COV-9
• Remdesivir, Dosing/Duration, bullet removed: Caution with moderate-severe renal dysfunction (eg, CrCl <30 mL/min) Removed from Guidelines • COV-10 through COV-15, and COV-A 4 of 4 COV-A 3 of 3 • References 16—19 removed

重试 错误原因

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ANTIMICROBIAL PROPHYLAXIS BASED ON OVERALL INFECTION RISK IN PATIENTS WITH CANCER

See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions

Overall Infection Risk in Patients with Cancer[a]	Disease/Therapy Examples	Antimicrobial Prophylaxis
Low	• Standard chemotherapy regimens for most solid tumors • Anticipated neutropenia* <7 days	• Bacterial - None • Fungal - None • Viral - None unless prior HSV episode
Intermediate	• Autologous HCT • Lymphoma[c] • Multiple myeloma[c] • CLL[c] • Purine analog therapy (ie, fludarabine, clofarabine, nelarabine, cladribine) • Anticipated neutropenia* 7–10 days • CAR T-cell therapy	• Bacterial - Consider fluoroquinolone prophylaxis during neutropenia[d] • Fungal - Consider prophylaxis during neutropenia and for anticipated mucositis (INF-2); consider PJP prophylaxis (INF-6) • Viral - During neutropenia and longer depending on risk (INF-3, INF-4, INF-5) • See Immune and Targeted Treatments (INF-A 11 of 13)
High[b]	• Allogeneic HCT including cord blood • Acute leukemia ▸ Induction ▸ Consolidation/maintenance • Alemtuzumab therapy • Moderate to severe GVHD • Anticipated neutropenia* >10 days	• Bacterial - Consider fluoroquinolone prophylaxis during neutropenia[d] • Fungal - Consider prophylaxis during neutropenia (INF-2); consider PJP prophylaxis (INF-6) • Viral - During neutropenia and longer depending on risk (INF-3, INF-4, INF-5) • Length of prophylaxis depends on immune reconstitution.

*Neutropenia: ≤500 neutrophils/mcL or ≤1000 neutrophils/mcL and a predicted decline to ≤500/ mcL over the next 48 hours.

[a] Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, cytomegalovirus (CMV) serostatus, and intensity of immunosuppressive therapy (IST). For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.

[b] In patients at high risk, additional prophylaxis may be necessary; for example, consider penicillin and trimethoprim/sulfamethoxazole (TMP/SMX) for allogeneic hematopoietic cell transplant (HCT) recipients with chronic graft-versus-host disease (GVHD). In those with an allergy history, a careful reassessment of the allergy is recommended.

[c] This is a heterogenous disease. Therefore, treatment modalities and the type of malignancy affect risk level.

[d] For patients who are intolerant to fluoroquinolone, consider TMP/SMX or an oral third-generation cephalosporin (category 2B).

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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ANTIMICROBIAL PROPHYLAXIS BASED ON OVERALL INFECTION RISK IN PATIENTS WITH CANCER

有关剂量、光谱和具体备注/注意事项，请参见抗菌剂（FEV-A）

重试 错误原因

总感染患者的风险与癌症	疾病/治疗实例	抗微生物预防
Low	·大多数实体瘤的标准化疗方案 ·预期中性粒细胞减少症 * <7 天	·细菌–无 ·真菌–无 ·病毒–无，除非先前 HSV 发作
中间	·自动 HCT ·淋巴瘤 ·多发性骨髓瘤 · CLL • 嘌呤类似物治疗（即，氟达拉滨，氯法拉滨，奈拉滨、克拉屈滨）·预期中性粒细胞减少症 * 7–10 天· CAR T 细胞疗法	• Bacterial – Consider fluoroquinolone prophylaxis during neutropenia • Fungal – Consider prophylaxis during neutropenia and for anticipated mucositis (INF–2); consider PJP prophylaxis (NF6) • Viral – During neutropenia and longer depending on risk (INF3, INF–4, INF–5) • See Immune and Targeted Treatments (INF–A 11 of 13) 重试 错误原因
High	• Allogeneic HCT including cord blood • Acute leukemia Induction Consolidation/maintenance • Alemtuzumab therapy • Moderate to severe GVHD • Anticipated neutropenia* >10 days 重试 错误原因	• Bacterial – Consider fluoroquinolone prophylaxis during neutropenia • Fungal – Consider prophylaxis during neutropenia (INF–2); consider PJP prophylaxis (INF–6) • Viral – During neutropenia and longer depending on risk (INF–3, INF–4, INF–5) • Length of prophylaxis depends on immune reconstitution. 重试 错误原因

*Neutropenia: ⩽500 neutrophils/mcL or ⩽1000 neutrophils/mcL and a predicted decline to ⩽500/ mcL over the next 48 hours.

重试 错误原因

a 风险类别基于几个因素，包括潜在的恶性肿瘤、疾病是否处于缓解期、中性粒细胞减少持续时间、既往化疗暴露、巨细胞病毒（CMV）血清状态和免疫抑制治疗（IST）强度。有关感染问题和免疫靶向药物的推荐预防，请参阅 INF–A。B 在高危患者中，可能需要额外的预防;例如，对于患有慢性移植物抗宿主病（GVHD）的异基因造血细胞移植（HCT）受者，考虑使用青霉素和甲氧苄啶/磺胺甲恶唑（TMP/SMX）。对于那些有过敏史的人，建议对过敏进行仔细的重新评估。

c 这是一种异质性疾病。因此，治疗方式和恶性肿瘤的类型影响风险水平。
d 对于氟喹诺酮类药物不耐受的患者，考虑 TMP/SMX 或口服第三代头孢菌素（2B 类）。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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PREVENTION OF FUNGAL INFECTIONS

See Antifungal Agents (FEV-B) for dosing, spectrum, and specific comments/cautions

Overall Infection Risk in Patients with Cancer[a]	Disease/Therapy Examples	Consider Antifungal Prophylaxis Based on Patient- and Center-Specific Risk Factors See Antipneumocystis Prophylaxis (INF-6)	Duration
Intermediate to high	ALL	• Fluconazole[f] or an echinocandin[g] • Amphotericin B products[h] (category 2B)	Typically until resolution of neutropenia
	MDS (neutropenic)	• Posaconazole[f] (category 1) • Voriconazole,[f] isavuconazole,[f] an echinocandin,[g] amphotericin B products[i], or fluconazole (if mold activity not needed)[f] (all category 2B)	
	AML (neutropenic)		
	Autologous HCT with mucositis[e]	• Fluconazole[f] or an echinocandin[g] (both category 1)	
	Autologous HCT without mucositis	No prophylaxis (category 2B)	N/A
	Allogeneic HCT (neutropenic)[a]	• Fluconazole[f] or an echinocandin[g] (both category 1) • Voriconazole,[f] posaconazole,[f] isavuconazole,[f] or amphotericin B products[i] (all category 2B)	Continue during neutropenia[h]
	Immune and targeted treatments	See Immune and Targeted Treatments (INF-A)	Depends on nature and duration of treatment
	Significant acute GVHD (especially grade 3/4) receiving IST	• Posaconazole[f] (category 1) • Voriconazole,[f] echinocandin,[g] amphotericin B products,[i] or isavuconazole,[f] (all category 2B)	Until resolution of significant GVHD

[a] Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, CMV serostatus, and intensity of IST. For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.
[e] Mucositis is a risk factor for candidemia in patients with hematologic malignancies and HCT recipients not receiving antifungal prophylaxis.
[f] Itraconazole, voriconazole, and posaconazole are more potent inhibitors of hepatic cytochrome P450 3A4 isoenzymes than fluconazole and may significantly decrease the clearance of several agents used to treat cancer (eg, vincristine). In select circumstances when standard therapy is contraindicated, due to drug interactions or the risk of QTc prolongation, some centers consider using echinocandins, amphotericin B at prophylactic doses, or isavuconazole.
[g] All three agents in the echinocandin class (micafungin, caspofungin, and anidulafungin) are considered by many to be interchangeable. Echinocandins are active against *Candida* and *Aspergillus*.
[h] Some studies/centers continue prophylaxis for up to day 75. Prophylaxis may be extended based on individual risk.
[i] A lipid formulation of amphotericin is generally preferred based on less toxicity.

Note: All recommendations are category 2A unless otherwise indicated.
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预防真菌感染
有关剂量、谱和具体备注/注意事项，请参见抗真菌药（FEV–B）

总感染患者的风险与癌症	疾病/治疗实例	根据患者情况考虑抗真菌预防，中心特定风险因素见抗肺孢子虫预防（INF–6）	持续时间
中间 to high	ALL	两性霉素 B 产品（类别 2 B）	通常直至中性粒细胞减少症消退（青霉
	MDS（血小板减少症）	伏立康唑、艾沙康唑、棘白菌素、阿替霉素 B 产品或氟康唑（如果不需要霉菌活性）（均为 2 类 B）	
	AML（血小板减少）		
	有粘膜炎的自动 HCT·氟康唑或棘白菌素（均为 1 类）无粘膜炎的自动 HCT 无预防（2B 类）N/A 同种异体 HCT 素）·氟康唑或棘白菌素（均为 1 类）		
		·伏立康唑、泊沙康唑、艾沙康唑或阿替霉素 B 产品（所有类别 2 B）	内继续进行细胞减少
	免疫和靶向治疗	免疫和靶向治疗（INF–A）	取决于治疗的性质和持续时间
	严重急性 GVHD（尤其是 Level 3/4（receiving IST）（IST）	·波沙康唑（类别 1）·Voriconazole，echinocandin，amphotericin B products，or isavuconazole，（all category 2B）	直到解决显著 GVHD

[a] 风险分类基于几个因素，包括潜在的恶性肿瘤，疾病是否缓解，中性粒细胞减少持续时间，既往化疗暴露，CMV 血清状态和 IST 强度。有关感染问题和免疫靶向药物的推荐预防，请参阅 INF–A。
e 粘膜炎是血液系统恶性肿瘤患者和未接受抗真菌预防的 HCT 接受者中念珠菌血症的危险因素。
[f] 伊曲康唑、伏立康唑和泊沙康唑是比氟康唑更强效的肝细胞色素 P450 3A 4 同工酶抑制剂，几种用于治疗癌症的药物（如长春新碱）的清除率。在某些情况下，由于药物相互作用或 QTc 间期延长的风险，标准治疗是禁忌的，一些中心考虑使用棘白菌素、预防性剂量的阿替霉素 B 或艾沙康唑。
[g] 棘白菌素类的所有三种药物（米卡芬净、卡泊芬净和阿尼芬净）被许多人认为是可互换的。棘白菌素对念珠菌属和曲霉属有活性。

h 一些研究/中心继续预防直至第 75 天。预防措施可根据个体风险延长。
[i] 基于较低的毒性，通常优选阿替霉素的脂质制剂。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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PREVENTION OF HERPES SIMPLEX VIRUS (HSV) AND VARICELLA ZOSTER VIRUS (VZV) REACTIVATION OR DISEASE

See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions

For CMV prophylaxis, see INF-4. For HBV, HCV, and HIV prophylaxis, see INF-5. For general vaccine recommendations, see INF-7.

Overall Infection Risk in Patients with Cancer[a]	Disease/Therapy Examples	Minimum Duration of Antiviral Prophylaxis
Low	• Standard chemotherapy regimens for solid tumors	No prophylaxis unless prior HSV episode; if needed, treat during active therapy including periods of neutropenia
Intermediate	• Autologous HCT • Lymphoma[b] • Multiple myeloma[b] • CLL[b] • Purine analog therapy (eg, fludarabine)	HSV prophylaxis[j] • Consider during active therapy and possibly longer depending on degree of immunosuppression VZV prophylaxis[j] • Consider for at least 6–12 months after autologous HCT
High	• Acute leukemia	HSV prophylaxis during active therapy including periods of neutropenia[j]
	• Proteasome inhibitors	VZV prophylaxis during active therapy including periods of neutropenia[j]
	• Alemtuzumab therapy • Allogeneic HCT • GVHD requiring significant escalation of immunosuppression	HSV prophylaxis[j] • Minimum of 2 months after alemtuzumab and until CD4 ≥200 cells/mcL VZV prophylaxis[j] • Prophylaxis should be considered for at least 1 year after allogeneic HCT

[a] Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, CMV serostatus, and intensity of IST. For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.

[b] This is a heterogenous disease. Therefore, treatment modalities and the type of malignancy affect risk level.

[j] In pediatrics, HSV prophylaxis is indicated in children who are seropositive and prophylaxis for VZV is not routinely given unless there is a history of recurrent zoster infections or after first zoster episode while on myelosuppressive therapy, even if they are seropositive or vaccinated.

Note: All recommendations are category 2A unless otherwise indicated.
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预防单纯疱疹病毒（HSV）和带状疱疹病毒（VZV）再活化或疾病

有关剂量、频谱和具体备注/注意事项，请参见抗病毒药物（FEV-C）

关于 CMV 预防，参见 INF-4。关于 HBV、HCV 和 HIV 预防，请参见 INF-5。关于一般疫苗建议，见 INF-7。

癌症患者的总体感染风险	疾病/治疗实例	抗病毒预防治疗的最短持续时间
Low	·实体瘤的标准化疗方案	除非既往 HSV 发作，否则不进行预防;如需要，在活性治疗期间进行治疗，包括中性粒细胞减少期
中间	·自动 HCT ·淋巴瘤 ·多发性骨髓瘤 · CLL • 嘌呤类似物治疗（例如，氟达拉滨）	HSV prophylaxis • Consider during active therapy and possibly longer depending on degree of immunosuppression VZV prophylaxis • Consider for at least 6—12 months after autologous HCT 重试 错误原因
High	·急性白血病	活性治疗期间的 HSV 预防，包括中性粒细胞减少期
	·蛋白酶体抑制剂	活性治疗期间的 VZV 预防，包括中性粒细胞减少期
	·阿仑单抗疗法 ·同种异体 HCT • 需要显著升级的 GVHD 免疫抑制	HSV 预防性 ·阿仑单抗给药后至少 2 个月，直至 CD 4 ⩾200 个细胞/mcL VZV 预防性治疗·应在同种异体 HCT 后至少 1 年内考虑预防性治疗

[a] 风险分类基于几个因素，包括潜在的恶性肿瘤，疾病是否缓解，中性粒细胞减少持续时间，既往化疗暴露，CMV 血清状态和 IST 强度。有关感染问题和免疫靶向药物的推荐预防，请参阅 INF-A。

[b] 这是一种异质性疾病。因此，治疗方式和恶性肿瘤的类型影响风险水平。

[j] 在儿科，HSV 预防适用于血清阳性的儿童，除非有复发性带状疱疹感染史或在骨髓抑制治疗期间首次带状疱疹发作后，否则不常规给予 VZV 预防，即使他们是血清阳性或接种疫苗。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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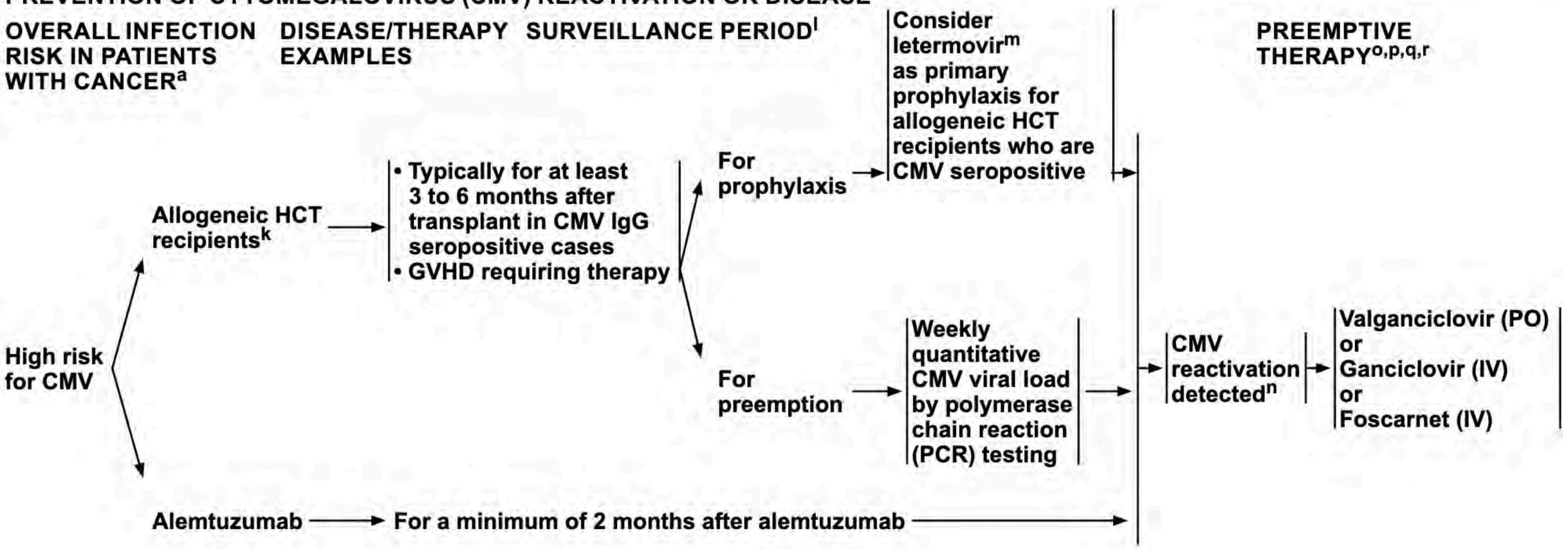
PREVENTION OF CYTOMEGALOVIRUS (CMV) REACTIVATION OR DISEASE



[a] Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, CMV serostatus, and intensity of IST. For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.
[k] Higher risk transplant subgroups may exist and require different management strategies.
[l] CMV surveillance consists of weekly monitoring by PCR (thresholds for treatment vary at individual sites).
[m] Some centers consider the use of letermovir through day 100 post-HCT and continue CMV surveillance for patients at high risk for CMV reactivation. See Antiviral Agents (FEV-C 2 of 4). Letermovir lacks HSV and VZV coverage and HSV/VZV prophylaxis should be continued. In certain circumstances, up to day 200 can be considered.
[n] Consider testing for drug resistance if clinically significant breakthrough infection is detected.
[o] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions.
[p] Preemptive therapy is defined as administration of antiviral agents to patients who are asymptomatic with laboratory markers of viremia in order to prevent CMV disease in patients who are at high risk. Duration of antiviral therapy is for at least 2 weeks and until CMV is no longer detected.
[q] Typically therapy is initiated with oral valganciclovir unless there are absorption or toxicity issues. However, some centers prefer ganciclovir over valganciclovir. Choice of agent may depend on institutional preference and/or concern for myelosuppression and nephrotoxicity.
[r] For refractory or resistant infections, an infectious disease (ID) consultation is recommended.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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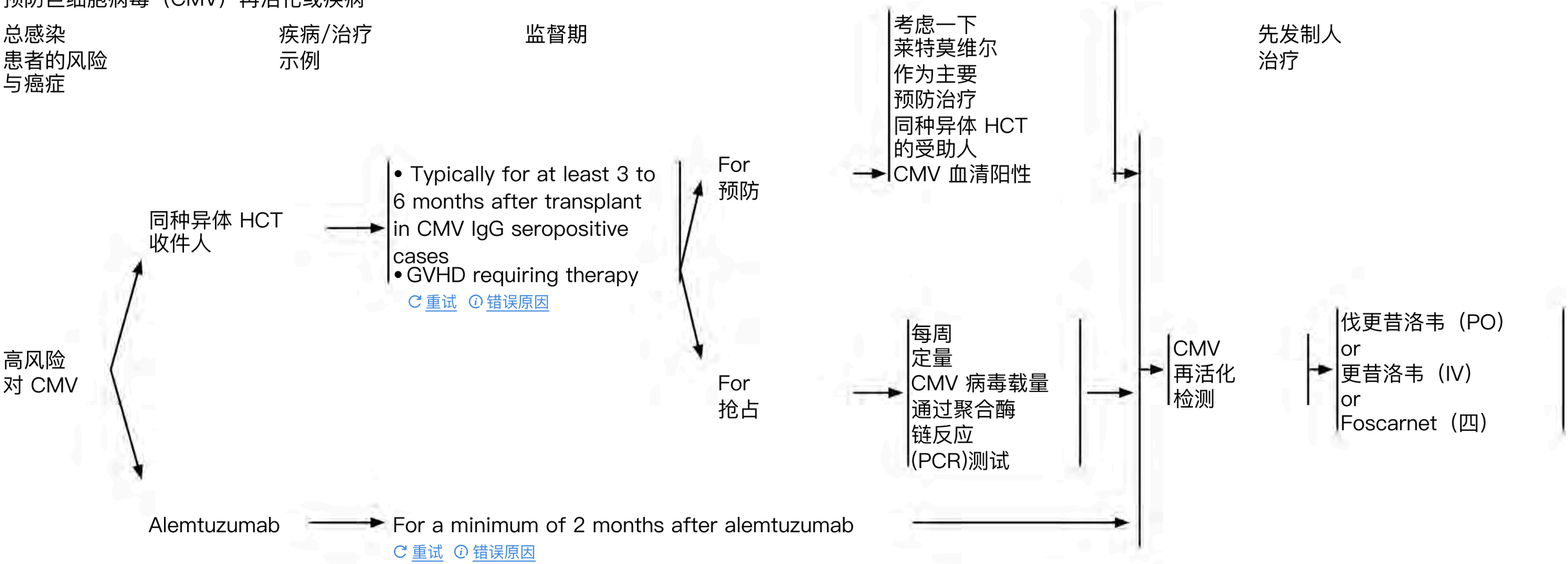
预防巨细胞病毒（CMV）再活化或疾病



a Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, CMV serostatus, and intensity of IST. For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.

k Higher risk transplant subgroups may exist and require different management strategies.

l CMV surveillance consists of weekly monitoring by PCR (thresholds for treatment vary at individual sites).

m Some centers consider the use of letermovir through day 100 post-HCT and continue CMV surveillance for patients at high risk for CMV reactivation. See Antiviral Agents (FEV-C 2 of 4). Letermovir lacks HSV and VZV coverage and HSV/VZV prophylaxis should be continued. In certain circumstances, up to day 200 can be considered.

如果检测到具有临床意义的突破性感染，则考虑进行耐药性检测。

o 有关剂量、频谱和具体注释/注意事项，请参见抗病毒药物（FEV-C）。

p 抢先治疗定义为对无症状且实验室病毒血症标志物的患者给予抗病毒药物，以预防高危患者的 CMV 疾病。抗病毒治疗的持续时间至少为 2 周，直到不再检测到 CMV。

q 通常治疗开始于口服缬更昔洛韦，除非有吸收或毒性问题。然而，一些中心更喜欢更昔洛韦而不是缬更昔洛韦。药物的选择可能取决于机构偏好和/或对骨髓抑制和肾毒性的关注。

r 对于难治性或耐药性感染，建议进行传染病（ID）咨询。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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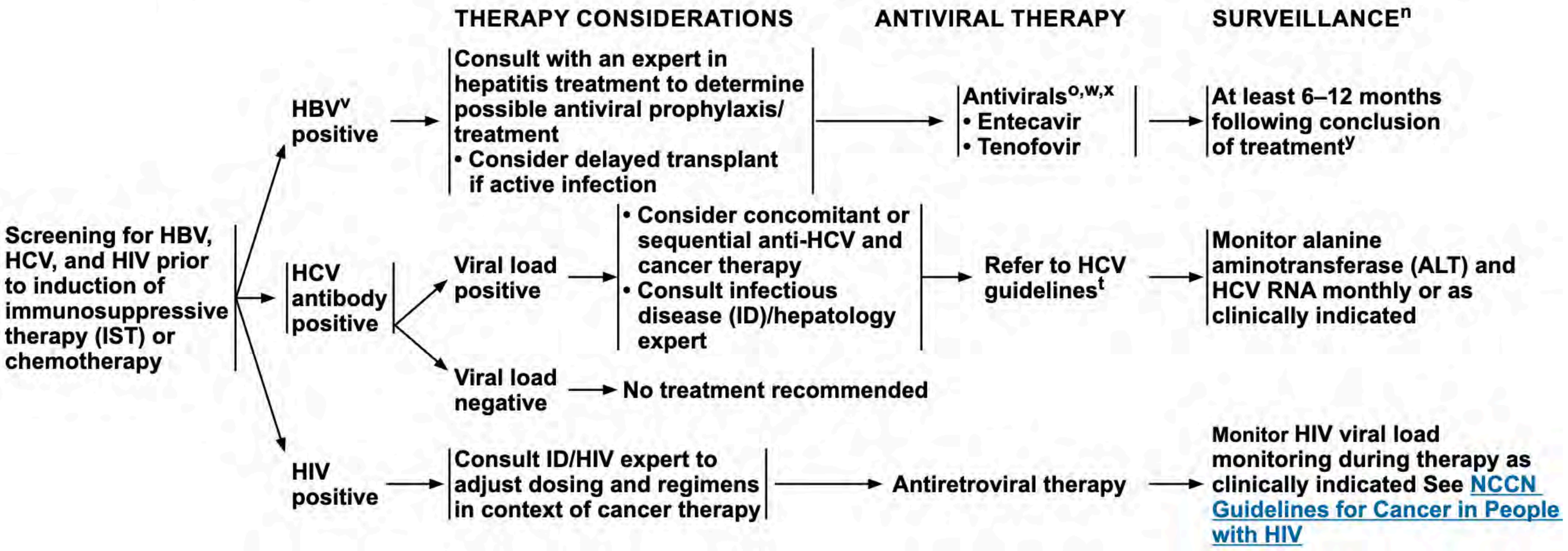
MANAGEMENT OF HEPATITIS B VIRUS (HBV),[s] HEPATITIS C VIRUS (HCV),[t] AND HUMAN IMMUNODEFICIENCY VIRUS (HIV)[u] REACTIVATION OR DISEASE



[n] Consider testing for drug resistance if clinically significant breakthrough infection is detected.
[o] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/ cautions.
[s] Hepatitis B Virus Screening and Management for Patients With Cancer Prior to Therapy: ASCO Provisional Clinical Opinion Update | Journal of Clinical Oncology (ascopubs.org).
[t] Therapy should be given by a provider experienced in treating hepatitis C. See American Association for the Study of Liver Diseases/Infectious Diseases Society of America HCV Guidelines.
[u] See current HIV management Guidelines: https://www.iasusa.org/resources/guidelines and https://clinicalinfo.hiv.gov/en/guidelines.
[v] High risk of HBV is defined as patients with HBsAg+ serology or HBcAb+ serology or with increasing HBV viral load in patients planned for allogeneic HCT or B-cell–depleting therapy.
[w] Duration of therapy may depend on various factors and typically needs to be continued beyond the completion of immunosuppression.
[x] Lamivudine is inferior to entecavir and tenofovir, but may be considered when other agents are unavailable.
[y] If viral load fails to drop or previously undetectable PCR becomes positive, consult hepatologist and discontinue anti-CD20 antibody therapy.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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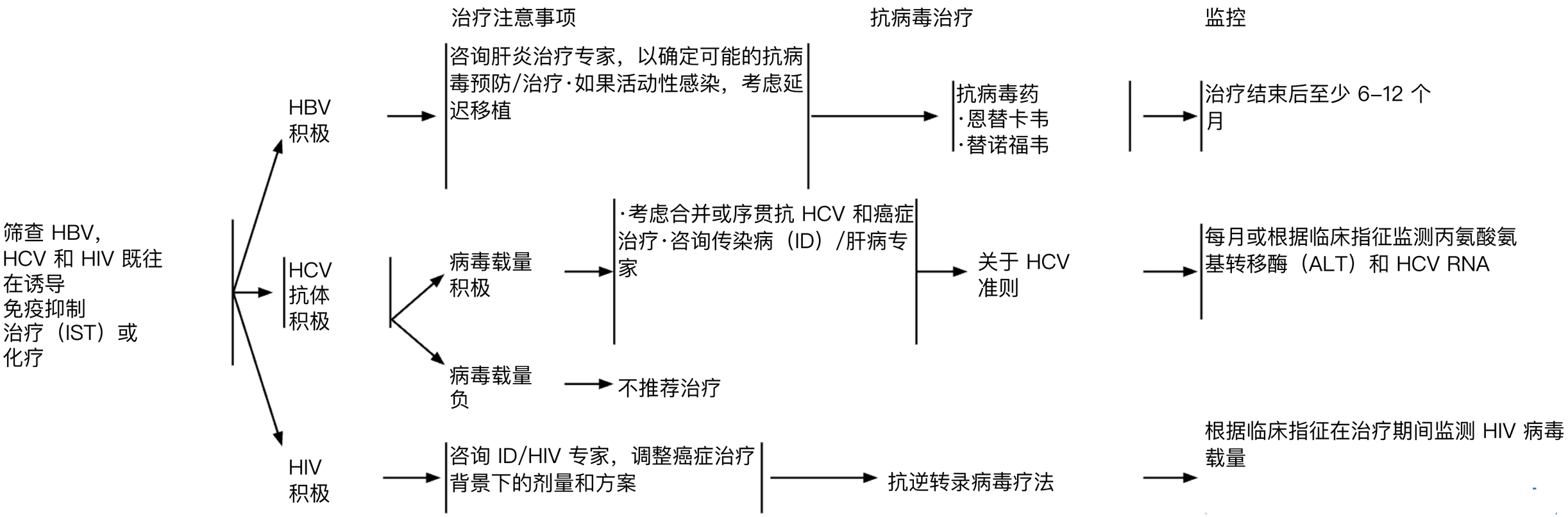
B 型肝炎病毒（HBV）、丙型肝炎病毒（HCV）和人免疫缺陷病毒（HIV）再活化或疾病的管理



如果检测到具有临床意义的突破性感染，则考虑进行耐药性检测。

[o] 有关剂量、频谱和具体注释/注意事项，请参见抗病毒药物（FEV–C）。

癌症患者在接受化疗前的 B 型肝炎病毒筛查和管理
治疗：ASCO 临时临床意见更新|Journal of Clinical Oncology（ascopubs.org）.

治疗应该由有治疗丙型肝炎经验的提供者提供。参见美国肝病研究协会/美国传染病学会 HCV 指南。

[u] 见现行艾滋病毒管理准则：https://www.iasusa.org/resources/准则和 https://clinicalinfo.hiv.gov/en/guidelines。

[v] HBV 高风险定义为 HBsAg+血清学或 HBcAb+血清学的患者，或计划接受同种异体 HCT 或 B 细胞耗竭治疗的患者中 HBV 病毒载量增加的患者。

治疗的持续时间可能取决于各种因素，通常需要在免疫抑制完成后继续治疗。

[x] 拉米夫定是劣于恩替卡韦和替诺福韦，但可以考虑当其他代理商不可用。

[y] 如果病毒载量未能下降或之前无法检测到的 PCR 变为阳性，请咨询肝病专家并停止抗 CD20 抗体治疗。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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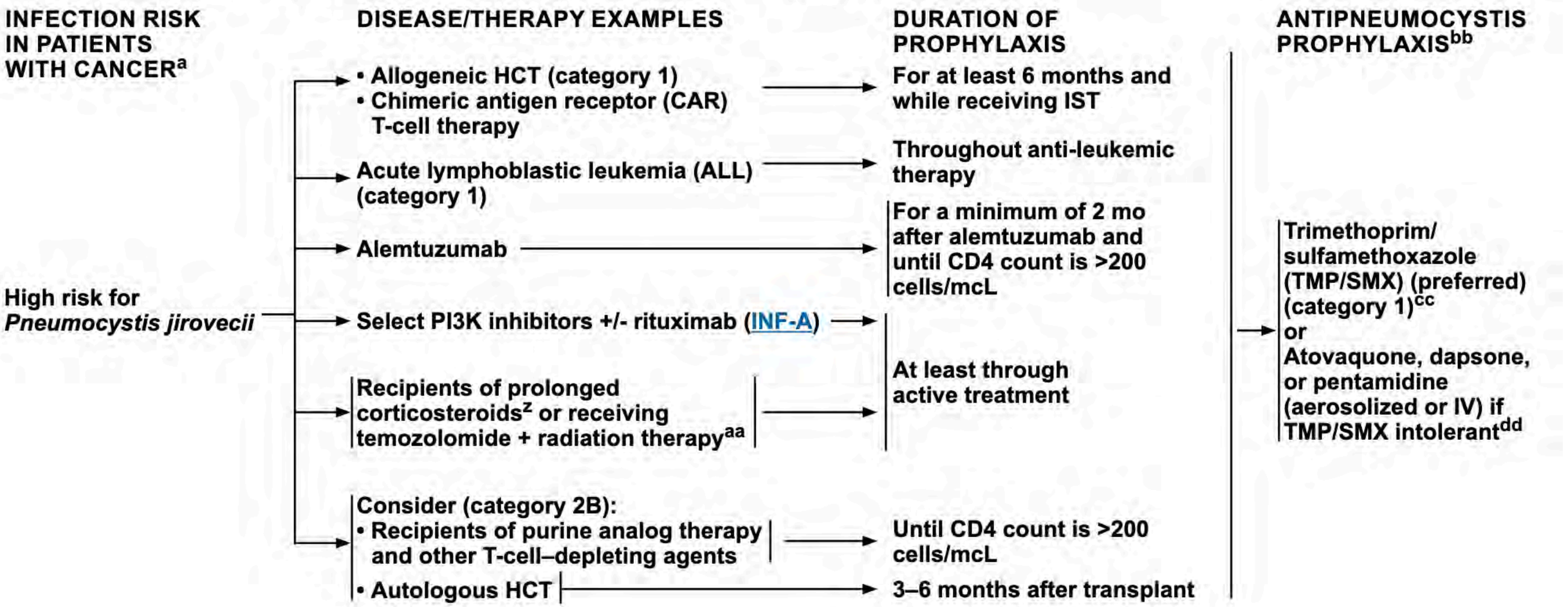
PREVENTION OF *PNEUMOCYSTIS JIROVECII (PNEUMOCYSTIS CARINII)* INFECTION



[a] Categories of risk are based on several factors, including underlying malignancy, whether disease is in remission, duration of neutropenia, prior exposure to chemotherapy, CMV serostatus, and intensity of IST. For infection concerns and recommended prophylaxis for immune-targeted agents, see INF-A.

[z] Risk of *Pneumocystis jirovecii* pneumonia (PJP) is related to the daily dose and duration of corticosteroid therapy. Prophylaxis against PJP can be considered in patients receiving the prednisone equivalent of 20 mg or more daily for 4 or more weeks.

[aa] PJP prophylaxis should be continued until recovery from lymphocytopenia.

[bb] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.

[cc] TMP/SMX, when appropriately dosed, may have activity against other pathogens including *Nocardia, Toxoplasma*, and *Listeria.* Atovaquone may have activity against *Toxoplasma*.

[dd] The list of agents is alphabetical and does not reflect preference. Consider TMP/SMX desensitization or atovaquone, dapsone, or pentamidine (aerosolized or IV) when PJP prophylaxis is required in patients who are TMP/SMX-intolerant. For patients receiving dapsone, assessing G6PD levels prior to initiating therapy is recommended.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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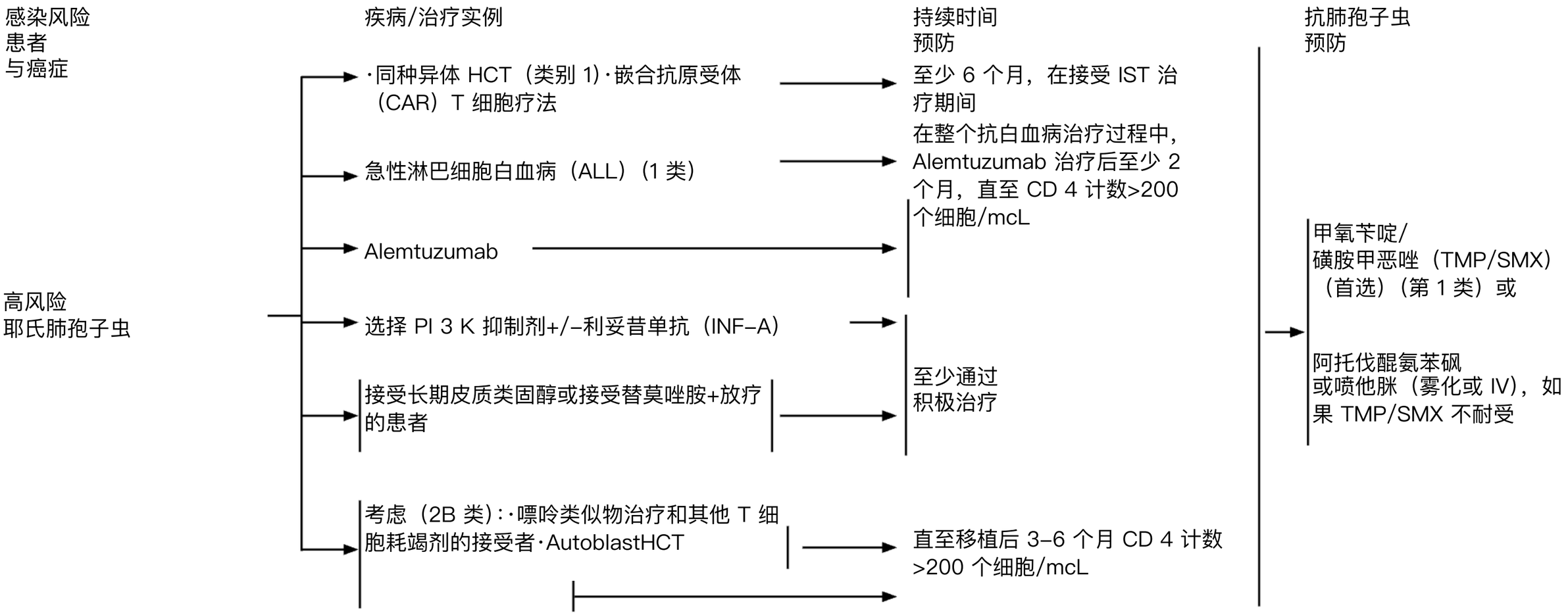
预防卡氏肺孢子虫感染



a 风险类别基于几个因素，包括潜在的恶性肿瘤、疾病是否处于缓解期、中性粒细胞减少持续时间、既往化疗暴露、CMV 血清状态和 IST 强度。有关感染问题和免疫靶向药物的推荐预防，请参阅 INF−A。z 耶氏肺孢子虫肺炎（PJP）的风险与皮质类固醇治疗的每日剂量和持续时间有关。对于每天接受相当于 20 mg 或更高剂量泼尼松治疗 4 周或更长时间的患者，可以考虑预防 PJP。aa PJP 预防应持续至淋巴细胞减少症恢复。

bb 有关剂量、光谱和具体备注/注意事项，请参见抗菌剂（FEV−A）。
cc TMP/SMX 在适当给药时，可具有抗其他病原体的活性，包括诺卡氏菌、弓形虫和李斯特菌。阿托伐醌可能具有抗弓形虫活性。
dd 代理人名单按字母顺序排列，不反映优先顺序。当 TMP/SMX 不耐受患者需要 PJP 预防时，考虑 TMP/SMX 脱敏或阿托伐醌、氨苯砜或喷他脒（雾化或 IV）。对于接受氨苯砜治疗的患者，建议在开始治疗前评估 G6 PD 水平。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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GENERAL RECOMMENDATIONS FOR VACCINATION IN PATIENTS WITH CANCER[ee]

For prevention of infections in cancer survivors, including vaccination recommendations, see the NCCN Guidelines for Survivorship. For prevention of infections in pediatric patients, refer to guidance from the CDC.

VACCINATION	RECOMMENDATION
General	• Live vaccines should NOT be administered during chemotherapy or periods of significant immunosuppression, such as treatment of GVHD. • All household members should be up-to-date with vaccines.
Influenza	• Patients with hematologic or solid tumor malignancies should receive inactivated or recombinant influenza vaccine annually. See Key Facts about Seasonal Flu Vaccine for specific guidance.
COVID-19	• All persons with cancer, or who have been previously treated for cancer, should receive COVID-19 vaccination. See The CDC's Stay Up to Date with COVID-19 Vaccines for specific guidance. For additional information on COVID-19 vaccinations and their use, also see the CDC for Use of COVID-19 Vaccines in the United States.
Pneumococcal[ff]	• The pneumococcal conjugate vaccine (PCV20) should be administered to adults who are newly diagnosed with cancer who are pneumococcal vaccine-naïve. According to the CDC, additional pneumococcal polysaccharide vaccine (PPSV23) is not needed for those receiving PCV20. Alternatively, PCV15 can be given, followed by PPSV23 at least 8 weeks later. For patients who have previously received PPSV23, PCV20 (preferred) or PCV15 can be given. For patients who have previously received PCV13 only, they can receive PCV20 at least 1 year later, rather than PPSV23. For patients who have previously received PCV13 and 1 or 2 doses of PPSV23, they can receive PCV20 at least 5 years later. See CDC recommendations for pneumococcal vaccination.
Meningococcal[ff]	• The addition of serogroup B meningococcal vaccination has been recommended for patients at increased risk for meningococcal disease. Patients at increased risk for meningococcal disease should receive quadrivalent MenACWY vaccine series and monovalent meningococcal serogroup B vaccine series. Patients at risk include those with persistent complement component deficiencies, those taking a complement C5 inhibitor (eg, eculizumab, ravulizumab), or those with anatomic or functional asplenia. MenACWY vaccine is given in 2 doses ≥8 weeks apart; serogroup B vaccine is available in a 2- or 3-dose series, depending on the vaccine formulation used. Patients with ongoing risk for meningococcal disease should receive a booster dose 5 years after completion of the primary series and every 5 years thereafter.
Human papillomavirus (HPV) vaccination	• The recombinant 3-dose HPV vaccine should be offered to patients of all sexes up to 26 years of age and may be considered in patients up to 45 years of age.
Recombinant zoster vaccine	• The administration of recombinant zoster vaccine (RZV) is recommended for adult patients aged ≥50 years and those ≥18 years who are at increased risk for herpes zoster (HZ). The RZV vaccine is given in 2 doses ≥2–6 months apart. For adults who are at risk and ≥18 years of age, a second dose can be given 1–2 months after the first dose if they will benefit from a shorter vaccination schedule. For patients who have previously received the live-attenuated zoster vaccine live (ZVL), RZV should be given at least 2 months after the last ZVL dose.
Tdap (tetanus/ diphtheria/pertussis)	• Given every 10 years. See CDC recommendations for Tdap vaccinations.
Respiratory syncytial virus (RSV)	• The RSV vaccine is approved by the FDA and is available for those ≥60 years. Its effectiveness in patients with cancer is unknown. The long-acting RSV monoclonal antibody (mAb) (nirsevimab) is approved for infants <24 months of age to prevent RSV infection.
Travel and other vaccines	• ID consult for travel vaccines is recommended. • For recommendations on travel vaccines, refer to the CDC Yellow Book.

Footnotes on INF-7A

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癌症患者接种疫苗的一般建议

关于癌症幸存者感染的预防，包括疫苗接种建议，请参阅 NCCN 生存指南。对于预防儿科患者感染，请参阅 CDC 的指导。

一般建议	
	·在化疗或显著免疫抑制期间，如 GVHD 治疗期间，不应接种活疫苗。 ·所有家庭成员都应接种最新疫苗。
流感	·血液病或实体瘤恶性肿瘤患者应每年接种灭活或重组流感疫苗。有关季节性流感疫苗的具体指导，请参阅关键事实。
COVID–19	·所有癌症患者或曾接受过癌症治疗的人都应接种 COVID–19 疫苗。请参阅 CDC 的 Stay Up to Date with COVID–19 Vaccines 以获取具体指导。有关 COVID–19 疫苗接种及其使用的更多信息，请参阅 CDC 关于在美国使用 COVID–19 疫苗的信息。
肺炎球菌	·肺炎球菌结合疫苗（PCV 20）应给予新诊断患有癌症的未接种过肺炎球菌疫苗的成年人。根据 CDC 的说法，接受 PCV 20 的人不需要额外的肺炎球菌多糖疫苗（PPSV23）。或者，可以给予 PCV 15，然后在至少 8 周后给予 PSV 23。对于既往接受过 PPSV23、PCV 20（首选）或 PCV 15 的患者。对于既往仅接受过 PCV 13 的患者，他们可以在至少 1 年后接受 PCV 20，而不是 PSV 23。对于既往接受过 PCV 13 和 1 剂或 2 剂 PPSV23 的患者，他们可以在至少 5 年后接受 PCV 20。参见 CDC 关于肺炎球菌疫苗接种的建议。
脑膜炎	·建议脑膜炎球菌病风险增加的患者增加血清群 B 脑膜炎球菌疫苗接种。脑膜炎球菌病风险增加的患者应接种四价 MenACWY 疫苗系列和单价脑膜炎球菌血清群 B 疫苗系列。处于风险中的患者包括具有持续补体成分的患者 缺乏者，服用补体 C5 抑制剂（例如，依库珠单抗，ravulizumab）的患者，或解剖学或功能性无脾的患者。MenACWY 疫苗分 2 剂接种，间隔≥8 周;血清群 B 疫苗有 2 剂或 3 剂系列，具体取决于 使用的疫苗配方。持续存在脑膜炎球菌病风险的患者应在完成主要系列研究后 5 年接受加强剂量，此后每 5 年接受一次。
人乳头瘤病毒（HPV）疫苗接种	• 3 剂重组 HPV 疫苗应提供给 26 岁以下的所有性别患者，并可考虑在 45 岁以下的患者中使用。
重组带状疱疹疫苗	• 对于年龄≥50 岁的成年患者和带状疱疹（HZ）风险增加的≥18 岁患者，建议接种重组带状疱疹疫苗（RZV）。RZV 疫苗分 2 剂接种，间隔≥2–6 个月。 对于有风险且年龄≥18 岁的成人，如果他们将从较短的剂量中获益，则可以在第一剂后 1–2 个月给予第二剂。 疫苗接种时间表对于既往接受过带状疱疹减毒活疫苗（ZVL）的患者，应在最后一剂 ZVL 后至少 2 个月给予 RZV。
破伤风/白喉/百日咳	·每 10 年一次。参见 CDC 关于 Tdap 疫苗接种的建议。
呼吸道合胞病毒（RSV）	·RSV 疫苗已获得 FDA 批准，适用于 60 岁以上的人群。它对癌症患者的有效性尚不清楚。长效 RSV 单克隆抗体（mAb）（nirsevimab）已获批用于<24 个月的婴儿，以预防 RSV 感染。
旅行和其他疫苗	·建议咨询旅行疫苗的身份证。 有关旅行疫苗的建议，请参阅 CDC 黄皮书。

INF–7A 的脚注

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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GENERAL RECOMMENDATIONS FOR VACCINATION IN PATIENTS WITH CANCER
FOOTNOTES

[ee] Appropriate timing of vaccination should be assessed in patients whose disease is unlikely to respond (eg, patients who received anti–B-cell antibodies within 6 months, induction and consolidation chemotherapy for acute leukemia).

[ff] PCV20 or PCV15 should not be given with meningococcal conjugate vaccine, quadrivalent (MenACWY-D) but can be given with MenACWY-CRM.

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癌症足型患者接种疫苗的一般建议

ee 对于疾病不太可能应答的患者（例如，在 6 个月内接受抗 B 细胞抗体、急性白血病诱导和巩固化疗的患者），应评估接种疫苗的适当时机。

ff PCV 20 或 PCV 15 不应与四价脑膜炎球菌结合疫苗（MenACWY-D）一起接种，但可与 MenACWY-CRM 一起接种。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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RECOMMENDED VACCINATION SCHEDULE AFTER AUTOLOGOUS OR ALLOGENEIC HCT[gg]

Inactivated, Subunit, or Toxoid Vaccines[hh]	Recommended Timing After HCT	Number of Doses[nn]
Tdap[ii]	6–12 months	3
Haemophilus influenzae type b (Hib)	6–12 months	3
Pneumococcal vaccination[jj]	3–6 months	3–4
Hepatitis A[kk] (Hep A)	6–12 months	2
Hepatitis B[kk] (Hep B)	6–12 months	2–3
Meningococcal conjugate vaccine[ll]	6–12 months	2–3
Influenza (injectable)	6 months	1, annually[oo]
Inactivated polio vaccine	6–12 months	3
Recombinant zoster vaccine[mm]	50–70 days after autologous HCT May be considered after allogeneic HCT	2
HPV vaccine	>6–12 months For patients ≤26 years, consider up to age 45	3
COVID-19	6 months	1 or more doses per CDC recommendations[pp]

Live Vaccines[mm]	Recommended Timing After HCT	Number of Doses[jj]
Measles/mumps/rubella (MMR)	≥24 months (may vaccinate earlier when risk:benefit ratio suggests)[qq]	1–2
Varicella vaccine	≥24 months (if no GVHD or ongoing immunosuppression and patient was seronegative for varicella pretransplant)	2

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Footnotes on INF-8A

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自体或异体 HCT 后的推荐接种计划

灭活、亚单位或类毒素疫苗	HCT 后的推荐时间	剂量数
Tdap	6–12 个月	3
流感嗜血杆菌 B 型（HiB）	6–12 个月	3
肺炎球菌疫苗	3–6 个月	3–4
甲型肝炎（Hep A）	6–12 个月	2
B 型肝炎（Hep B）	6–12 个月	2–3
脑膜炎球菌结合疫苗	6–12 个月	2–3
流感（注射）	6 个月	1、每年
灭活脊髓灰质炎疫苗	6–12 个月	3
重组带状疱疹疫苗	自体 HCT 后 50–70 天可考虑在同种异体 HCT 后	2
HPV 疫苗	> 6 – 12 个月 对于≤26 岁的患者，考虑年龄不超过 45 岁	3
COVID–19	6 个月	1 或更大剂量/CDC 建议

活疫苗	HCT 后的推荐时间	剂量数
麻疹/腮腺炎/风疹（MMR）	≥24 个月（当风险：获益比提示时，可提前接种疫苗）	1–2
水痘疫苗	≥24 个月 （if 无 GVHD 或持续免疫抑制，患者移植前水痘血清反应阴性）	2

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

INF–8A 的脚注

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RECOMMENDED VACCINATION SCHEDULE AFTER AUTOLOGOUS OR ALLOGENEIC HCT
FOOTNOTES

[gg] Emerging therapies such as CAR T-cell therapy appear to behave similar to allogenic transplant in terms of vaccine boosting recommendations.
[hh] Inactivated, subunit, or toxoid vaccines may be given as a combined vaccine. Vaccination may be postponed for patients receiving >20 mg of prednisone.
[ii] DTaP (diphtheria, tetanus, and acellular pertussis) is not approved for use in ages >7 (FDA/ACIP-approved 3-dose series for ≥7 year olds is Tdap/Td (tetanus-diphtheria)/Td).
[jj] If PCV20 is used, 4 doses should be administered. First 3 doses are generally 1-2 months apart, with the fourth dose 6 months after the third dose. There is no need to give PPSV23. If PCV15 is used, 3 doses should be administered, followed by PPSV23 6-12 months post primary series. Following the primary series of 3 PCV doses, a dose of the PPSV23 to broaden the immune response might be given. For patients with chronic GVHD who are likely to respond poorly to PPSV23, a fourth dose of PCV20 or PCV15 should be considered instead of PPSV23.
[kk] Strongly consider if clinically indicated. May consider Hep A and B combined vaccine if immunization for both is needed.
[ll] Meningococcal B vaccine should be considered for patients at high risk, such as patients with chronic GVHD, asplenia, or complement deficiency or patients receiving a complement C5 inhibitor (eg, eculizumab, ravulizumab).
[mm] Efficacy in allogeneic HCT, in the presence of GVHD, or ongoing immunosuppression has not been established.
[nn] Number of doses depends on which vaccine formulation is used.
[oo] Refer to 2-1-8 rule as proposed by Carpenter and Englund. Carpenter PA, et al. Blood 2016;127:2824-2832.
[pp] May consider early vaccination at 3 months during community outbreaks and high disease activity. See INF-7 for additional information.
[qq] Pergam SA, et al. Biol Blood Marrow Transplant 2019;25: e321-e330.

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自体或异体 HCT 足迹后的推荐接种计划

gg 新兴疗法如 CAR T 细胞疗法在疫苗加强建议方面似乎与同种异体移植表现相似。
[hh] 灭活疫苗、亚单位疫苗或类毒素疫苗可以作为联合疫苗给予。对于接受>20 mg 泼尼松的患者，疫苗接种可推迟。
[ii] DTaP（白喉、破伤风和无细胞百日咳）未被批准用于>7 岁的儿童（FDA/ACIP 批准的≥7 岁奥尔兹的 3 剂系列为 Tdap/Td（破伤风白喉）/Td）。

jj 如果使用 PCV 20，应给予 4 剂。前 3 剂通常间隔 1–2 个月，第 4 剂在第 3 剂后 6 个月。没有必要给 PPSV 23。如果使用 PCV 15，应给予 3 剂，然后在主要系列后 6–12 个月给予 PSV 23。在 3 次 PCV 给药的主要系列后，可给予 PPSV 23 剂量以扩大免疫应答。对于可能对 PPSV 23 反应较差的慢性 GVHD 患者，应考虑使用第四剂 PCV 20 或 PCV 15 代替 PPSV 23。

[kk] 如果有临床指征，应慎重考虑。如果需要接种甲型和 B 型肝炎联合疫苗，可考虑接种。
[ll] 脑膜炎球菌 B 疫苗应考虑用于高风险患者，如慢性 GVHD、无脾或补体缺陷患者或接受补体 C5 抑制剂（如依库珠单抗、ravulizumab）的患者。

在存在 GVHD 或持续免疫抑制的情况下，在同种异体 HCT 中的功效尚未确定。
[nn] 接种次数取决于使用的疫苗制剂。
[oo] 参考 Carpenter 和 Englund 提出的 2–1–8 规则。Carpenter PA，et al. Blood 2016;127：2824–2832.
pp 在社区爆发和疾病活动性高的情况下，可考虑在 3 个月时提前接种疫苗。有关更多信息，请参见 INF–7。
[qq] Pergam SA，et al. Biol Blood Marrow Transplant 2019;25：e321–e330.

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IMMUNE AND TARGETED TREATMENTS[a,b]

Table 1. Targeted Therapies[i]

Drug Class/Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
Ubiquitin-proteasome pathway inhibitors[1]	Bortezomib Carfilzomib Ixazomib	• Respiratory tract infection • VZV • HBV • PML	• Recommend VZV prophylaxis • VZV vaccination in patients seronegative for VZV at least 1 month prior to initiation • Consider HZ vaccination in patients seropositive for VZV • Drug-induced neutropenia and pneumonitis
Bruton tyrosine kinase (BTK) inhibitors[2]	Acalabrutinib Ibrutinib Pirtobrutinib Zanubrutinib	• VZV • HBV • PJP	• Consider HSV/VZV prophylaxis • Consider prophylaxis against PJP and opportunistic fungal infections in patients with additional risk factors • Drug-induced neutropenia
BCR::ABL tyrosine kinase inhibitors[2,3,5]	Asciminib Bosutinib Nilotinib Imatinib Dasatinib Ponatinib	• CMV (dasatinib) • VZV •HBV	• Second-generation agents are associated with greater risk of drug-induced pancreatitis and hepatotoxicity • Drug-induced neutropenia • Drug-induced pleural effusion (most frequently dasatinib)
Phosphatidylinositol-3-kinase (PI3K) inhibitors[3]	Copanlisib Idelalisib Duvelisib Alpelisib	• CMV • VZV • PML • Opportunistic fungal infections • PJP	• Consider CMV surveillance in patients seropositive for CMV • Consider PJP prophylaxis • Drug-induced neutropenia • Drug-induced pneumonitis, colitis, and hepatitis • Alpelisib has not been associated with PJP risks.

Footnotes on INF-A 12 of 13
References

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免疫和靶向免疫

表 1.靶向治疗

药物类别/作用机制	剂	感染问题	建议和意见
泛素 途径抑制剂	Bortezomib Carfilzomib Ixazomib	·呼吸道感染 · VZV · HBV · PML	·推荐 VZV 预防 ·在开始前至少 1 个月对 VZV 血清阴性的患者中接种 VZV 疫苗
布鲁顿酪氨酸激酶（BTK） 抑制剂	Acalabrutinib 伊曲替尼	· VZV · HBV · PJP	·考虑 HSV/VZV 预防 ·在有其他风险因素的患者中考虑预防 PJP 和机会性真菌感染
BCR:: ABL 酪氨酸 激酶抑制剂	阿西替尼 Bosutinib 重试 错误原因 Nilotinib Imatinib 重试 错误原因 Dasatinib Ponatinib 重试 错误原因	· CMV（达沙替尼） · VZV •HBV	·第二代药物与药物性胰腺炎和肝毒性的风险增加相关 • 药物性胸腔积液（最常见达沙替尼）
Phosphatidylinositol-3kinase (PI3K) inhibitors 重试 错误原因	Copanlisib 重试 错误原因 Idelalisib Duvelisib Alpelisib 重试 错误原因	• CMV 重试 错误原因 • VZV 重试 错误原因 • PML 重试 错误原因 ·真菌感染 · PJP	·考虑对 CMV 血清阳性患者进行 CMV 监测 ·考虑 PJP 预防 ·药物引起的中性粒细胞减少症 ·药物性肺炎、结肠炎和肝炎 Alpelisib 与 PJP 风险无关。

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临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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IMMUNE AND TARGETED TREATMENTS[a,b]

Table 1. Targeted Therapies (continued)[i]

Drug Class/Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
KRAS G12C inhibitors	Adagrasib Sotorasib	• No significantly increased risks for infection	• Drug-induced gastrointestinal (GI) side effects (nausea, vomiting, diarrhea, hepatoxicity) • Drug-induced pneumonitis and interstitial lung disease
mTOR inhibitors[2]	Everolimus Temsirolimus Sirolimus	• VZV • HBV • HCV • PML • PJP • TB	• Screen for latent TB, treat as indicated • Consider PJP prophylaxis in patients with additional risk factors • Drug-induced pneumonitis and stomatitis • Associated with impaired wound healing
Histone deacetylase inhibitors	Vorinostat Romidepsin Belinostat	• HBV • HIV	• May reverse HIV and HBV latency
Janus kinase (JAK) inhibitors[2,3,5]	Momelotinib Fedratinib Pacritinib Ruxolitinib	• CMV • HBV • HSV • Opportunistic fungal infections • PJP • PML • TB • VZV	• Screen for latent TB and HBV, treat as indicated • Consider PJP prophylaxis (depending on additional risk factors) and HSV/VZV prophylaxis • Monitor for drug withdrawal syndrome with taper or discontinuation when used for PV or myelofibrosis. • Fedratinib can be associated with serious and sometimes fatal Wernicke-like encephalopathy • Drug-induced neutropenia
Isocitrate dehydrogenase 1 (IDH1) and isocitrate dehydrogenase 2 (IDH2) inhibitors	Enasidenib Ivosidenib Olutasidenib	No significantly increased risks for infections	• Monitor for differentiation syndrome[f] when used for AML • Drug-induced hepatotoxicity
BRAF kinase inhibitors[2]	Dabrafenib Encorafenib Vemurafenib	No significantly increased risks for infections	• Drug-induced rash (including serious hypersensitivity reactions), fever, arthralgias, neutropenia, and lymphopenia • Drug-induced pneumonitis and interstitial lung disease reported with single and combination therapies (eg, BRAF kinase + MEK kinase inhibitors) • Drug-induced hepatotoxicity, especially with vemurafenib • Adverse effect profile impacted by combination MEK kinase inhibitor therapy

Footnotes on INF-A 12 of 13
References

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表 1.靶向治疗（续）

药物类别/作用机制	剂	感染关切	建议和意见
KRAS G12C 抑制剂	阿达格拉西布 索托拉西布	·感染风险没有显著增加	·药物引起的胃肠道（GI）副作用（恶心、呕吐、腹泻、肝毒性） ·药物性肺炎和间质性肺病
mTOR 抑制剂	维莫司 西罗莫司	· VZV · HBV · HCV · PML · PJP • TB	·筛查潜伏性结核病，按指示治疗 ·在有其他风险因素的患者中考虑 PJP 预防 ·药物性肺炎和口腔炎 ·与伤口愈合受损有关
组蛋白脱乙酰酶抑制剂	Vorinostat 罗米地辛贝利司他	· HBV ·艾滋病毒	·可能逆转 HIV 和 HBV 潜伏期
Janus 激酶（JAK）抑制剂	Momelotinib 非拉替尼 Pacritinib Ruxolitinib	· CMV · HBV · HSV 机会主义真菌感染 · PJP · PML • TB · VZV	·筛查潜伏性 TB 和 HBV，按指示治疗 ·考虑 PJP 预防（取决于其他风险因素）和 HSV/VZV 预防 ·当用于 PV 或骨髓纤维化时，监测药物戒断综合征，逐渐减少或停止。 ·非拉替尼可能与严重的、有时是致命的韦尼克样脑病有关
异柠檬脱氢酶 1（IDH1）和异柠檬酸脱氢酶 2（IDH2）抑制剂	依那地尼 Ivosidenib 重试 错误原因 Olutasidenib 重试 错误原因	无显著风险增加的感染	·用于 AML 时监测分化综合征 • Drug-induced hepatotoxicity 重试 错误原因
BRAF kinase inhibitors 重试 错误原因	Dabrafenib 恩可非尼 维穆拉非尼	无显著风险增加的感染	·药物诱导的皮疹（包括严重的超敏反应）、发热、关节痛、中性粒细胞减少症和淋巴细胞减少症·单一和联合治疗（例如，BRAF 激酶+ MEK 激酶抑制剂）报告的药物诱导的肺炎和间质性肺病·药物诱导的肝毒性，尤其是维罗非尼 • MEK 激酶抑制剂联合治疗影响的不良反应特征

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IMMUNE AND TARGETED TREATMENTS[a,b]

Table 1. Targeted Therapies (continued)[i]

Drug Class/ Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
MEK kinase inhibitors[2]	Binimetinib Cobimetinib Trametinib Selumetinib	No significantly increased risks for infections	• Drug-induced rash (including serious hypersensitivity reactions) and fever • Drug-induced hepatotoxicity, neutropenia, and lymphopenia • Drug-induced pneumonitis and interstitial lung disease reported with single and combination therapies (eg, BRAF kinase + MEK kinase inhibitors) • Adverse effect profile impacted by combination BRAF kinase inhibitor therapy
Bcl-2 (B-cell lymphoma 2) inhibitors[2]	Venetoclax	No significantly increased risks for infections	• Drug-induced neutropenia and lymphopenia • Dose reduction is required when used with P-gp inhibitors or strong/moderate CYP3A inhibitors (eg, posaconazole and other azoles) • Consider monitoring for fungal infections depending on additional risk factors
FLT3 (FMS-like tyrosine kinase 3) inhibitors	Gilteritinib Midostaurin Quizartinib	No significantly increased risks for infections	• Monitor for differentiation syndrome with gilteritinib[f] • Drug-induced neutropenia • Drug-induced pneumonitis
Nuclear export inhibitor	Selinexor	No significantly increased risks for infections	• Drug-induced side effects (nausea, vomiting, and diarrhea) and neutropenia
Multi-target protein kinase inhibitors	Pralsetinib Entrectinib Repotrectinib	No significantly increased risks for infections	• Toxicities vary with agent but include drug-induced neutropenia, lymphopenia, skin rash, hepatotoxicity, CNS effects, pneumonitis, interstitial lung disease, and GI effects including perforation • Pralsetinib is associated with impaired wound healing
Rho-associated coiled-coil–containing protein kinase 2 (ROCK2) inhibitor	Belumosudil	No significantly increased risks for infections	• Drug-induced neutropenia and lymphopenia • Associated with impaired wound healing

Footnotes on INF-A 12 of 13
References

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表 1.靶向治疗（续）

药物类别/机制行动	剂	感染关切	建议和意见
MEK 激酶抑制剂	Binimetinib Cobimetib Trametib Selumetib	无显著风险增加的感染	·药物诱导的皮疹（包括严重超敏反应）和发热·药物诱导的肝毒性、中性粒细胞减少症和淋巴细胞减少症·单药和联合治疗（例如，BRAF 激酶+ MEK 激酶抑制剂）报告的药物诱导的肺炎和间质性肺病 • 受 BRAF 激酶抑制剂联合治疗影响的不良反应特征
Bcl-2（B 细胞淋巴瘤 2）抑制剂	维奈托克	无显著风险增加的感染	·药物引起的中性粒细胞减少症和淋巴细胞减少症 • 当与 P-gp 抑制剂或强/中效药物联合使用时，需要降低剂量 CYP3A 抑制剂（例如泊沙康唑和其他唑类）·根据其他风险因素考虑监测真菌感染
FLT3（FMS 样酪氨酸激酶 3）抑制剂	Gilteritinib Midostaurin Quizartinib	无显著风险增加的感染	·使用 gilteritinib 监测分化综合征 ·药物性中性粒细胞减少症
核出口抑制剂	Selinexor	无显著风险增加的感染	• 药物引起的副作用（恶心、呕吐和腹泻）和中性粒细胞减少症
多靶点蛋白激酶抑制剂	普拉塞替尼 恩曲替尼 瑞泊替尼	无显著风险增加的感染	·毒性因药物而异，但包括药物诱导的中性粒细胞减少症、淋巴细胞减少症、皮疹、肝毒性、CNS 效应、肺炎、间质性肺病和 GI 效应（包括穿孔）
rho 相关盘绕的含有蛋白质激酶 2（ROCK2）抑制剂	Belumosudil	无显著风险增加的感染	·药物诱导的中性粒细胞减少症和淋巴细胞减少症

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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IMMUNE AND TARGETED TREATMENTS[a,b]

Table 1. Targeted Therapies (continued)[i]

Drug Class/Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
Vascular endothelial growth factor receptor (VEGFR) inhibitors	Axitinib Fruquintinib Sorafenib Sunitinib Pazopanib Vandetanib Regorafenib Lenvatinib Tivozanib Cabozantinib	• Increased risk of infections, including fatal infections, reported with some agents	• May impair wound healing • GI perforation and fistula formation may rarely occur
ALK inhibitors[3]	Alectinib Brigatinib Ceritinib Crizotinib Lorlatinib	No significantly increased risks for infections	• Drug-induced pneumonitis and hepatotoxicity • Development of renal cysts with potential secondary infection seen with crizotinib
CDK4/6 inhibitors	Abemaciclib Palbociclib Ribociclib Trilaciclib	No significantly increased risks for infections	• Drug-induced neutropenia, hepatotoxicity, and rash
Fibroblast growth factor receptor (FGFR) kinase inhibitor	Futibatinib Pemigatinib Erdafitinib	No significantly increased risks for infections	• Monitor for hyperphosphatemia and ocular toxicity including retinal pigment epithelial detachment

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References

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表 1.靶向治疗（续）

药物类别/作用机制	剂	感染问题	建议和意见
血管内皮生长因子受体（VEGFR）抑制剂	Axitinib 果昔替尼 Frucininib Sorafenib 的 Sunitin Pazopanib 的 Vandetanib 的 Regorafenib 的 Lenvatinib 的 tivozanib 的 Cabozantinib	·增加感染的风险，包括致命感染，报告与一些代理	·可能损害伤口愈合 ·胃肠道穿孔和瘘管形成可能很少发生
ALK INHIBITOR 创始人	Alectinib Brigatinib Ceritinib Crizotinib Lorlatinib	无显著增加的风险的感染	·克唑替尼治疗后出现肾囊肿并可能继发感染
cdk 4/6 抑制剂	Abemaciclib Palbociclib Ribociclib Trilaciclib	无显著增加的风险的感染	·药物引起的中性粒细胞减少症、肝毒性和皮疹
成纤维细胞生长因子受体（FGFR）激酶抑制剂	呋替巴替尼 Pemigatinib 的 关于 Erdafitinib	无显著增加的风险的感染	·监测高磷血症和眼毒性，包括视网膜色素上皮脱落

关于 INF-A 的脚注 12/13
引用

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Table 2. Monoclonal Antibodies and Fusion Proteins[i] IMMUNE AND TARGETED TREATMENTS[a,b]

Drug Class/ Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
Bispecific CD19-directed CD3 T-cell engager (BiTE)[8]	Blinatumomab	• Bacterial infection • CMV • HSV/VZV • HBV • PML • Opportunistic fungal infections • PJP	• Consider PJP and HSV/VZV prophylaxis • Monitor for cytokine release syndrome • Drug-induced neurotoxicity, leukoencephalopathy, pancreatitis, hepatoxicity, neutropenia, and hypogammaglobulinemia
	Tafasitamab-cxix	• Bacterial infections • HBV	• Drug-induced neutropenia
Bispecific BCMA-directed CD3 T-cell engager (BiTE)	Teclistamab-cgyv Elranatamab-bcmm	• Bacterial infection • HSV/VZV • Adenovirus • CMV • PML • HBV • PJP • Opportunistic fungal infection	• Recommend PJP and HSV/VZV prophylaxis • Consider CMV screening based on epidemiologic risks • Monitor for CRS, drug-induced neutropenia, neurotoxicity, and hepatotoxicity
Bispecific C20-directed CD3 T-cell engager (BiTE)	Epcoritamab-bysp Glofitamab-gxbm Mosunetuzumab-axgb	• Bacterial infections • HBV (high risk)	• Recommend PJP and HSV/VZV prophylaxis • Consider CMV screening based on epidemiologic risks • Monitor for CRS, drug-induced neutropenia and neurotoxicity
Bispecific G protein coupled receptor class C group 5 member D (GPRC5d) directed CD3 T-cell engager (BiTE)	Talquetamab-tgvs	• Bacterial infections • HSV/VZV • CMV • Adenovirus • HBV • Candidiasis	• Recommend PJP and HSV/VZV prophylaxis • Consider CMV screening based on epidemiologic risks • Skin and nail toxicities • Monitor for CRS, drug-induced neutropenia, neurotoxicity, and hepatotoxicity
CD19 target	Loncastuximab tesirine-lpyl	• Limited data, but severe or fatal infections reported • Likely HBV risk based on effects on B-cells	• Drug-induced pleural effusion, pericardial effusion, ascites, and myelosuppression (ie, neutropenia, lymphocytopenia)

Note: All recommendations are category 2A unless otherwise indicated.
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References

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表 2.单克隆抗体和融合蛋白

药物类别/机制行动	剂	感染问题	建议和意见
双特异性 CD 19–引导 CD 3 T 细胞增殖剂（BiTE）	Blinatumomab	·细菌感染 · CMV · HSV/VZV · HBV · PML ·真菌感染 · PJP	·考虑 PJP 和 HSV/VZV 预防 ·监测细胞因子释放综合征 ·药物诱导的神经毒性、白质脑病、胰腺炎、肝毒性、中性粒细胞减少症和低丙种球蛋白血症
	他法他单抗–cxix	·细菌感染· HBV	·药物引起的中性粒细胞减少症
双特异性 BCMA–定向 CD3 T 细胞 BiTE（BiTE）	Teclistamab–cgyv Elranatamab–bcmm	·细菌感染 · HSV/VZV ·腺病毒 · CMV · PML · HBV · PJP ·真菌感染	·建议 PJP 和 HSV/VZV 预防·考虑根据流行病学风险进行 CMV 筛查·监测 CRS、药物诱导的中性粒细胞减少症、神经毒性和肝毒性
双特异性 C20–定向 CD3 T 细胞 BiTE（BiTE）	Epcoritamab–bysp 格洛非他单抗 莫苏奈	·细菌感染 · HBV（高风险）	·推荐 PJP 和 HSV/VZV 预防 ·根据流行病学风险考虑 CMV 筛查 ·监测 CRS、药物诱导的中性粒细胞减少症和神经毒性
双特异性 G 蛋白偶联 receptor class C group 5 member D (GPRC5d) directed CD3 T 细胞增殖剂（BiTE）	Talquetamab–tgvs	·细菌感染 · HSV/VZV · CMV ·腺病毒 · HBV ·疟疾	·推荐 PJP 和 HSV/VZV 预防 ·根据流行病学风险考虑 CMV 筛查 ·皮肤和指甲毒性 ·监测 CRS、药物诱导的中性粒细胞减少症、神经毒性和肝毒性
CD 19 靶标	隆卡妥昔单抗 替西林–丙酯	·数据有限，但报告了严重或致命的感染	• 药物性胸腔积液心包积液腹水和骨髓抑制（即中性粒细胞减少症、淋巴细胞减少症）

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Table 2. Monoclonal Antibodies and Fusion Proteins[i] IMMUNE AND TARGETED TREATMENTS[a,b]

Drug Class/ Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
CD20 target[8]	Obinutuzumab Ofatumumab Rituximab	• HBV (high risk) • HCV • HSV/VZV • PML	• Screen for HBV[c], treat as indicated • Consider prophylaxis for VZV/HSV • Consider prophylaxis for PJP, especially if concomitant therapy further increases PJP risk • Drug-induced neutropenia, lymphocytopenia, and hypogammaglobulinemia
CD22 target[9]	Inotuzumab ozogamicin Moxetumomab pasudotox	Limited data on specific infections	• Risk for capillary leak syndrome (moxetumomab) and VOD/ hepatotoxicity (inotuzumab)
CD30 target[9]	Brentuximab vedotin	• PML • CMV • PJP • HSV/VZV	• Consider CMV monitoring in patients seropositive for CMV • Consider PJP and HSV/VZV prophylaxis • Drug-induced neutropenia and lymphocytopenia
CD33 target[9]	Gemtuzumab ozogamicin	• Bacterial infections • Opportunistic fungal infections • PJP	• Drug-induced VOD/hepatotoxicity, neutropenic colitis, and interstitial pneumonitis
CD38 target[9]	Daratumumab Isatuximab	• *Listeria* • HBV • HSV/VZV • CMV • PJP • *Cryptococcus*	• Recommend HSV/VZV prophylaxis • Consider PJP prophylaxis • Drug-induced neutropenia
CD52 target[8]	Alemtuzumab	• *Nocardia* • TB • *Listeria* • HSV/VZV • CMV • ADV • BKV • PML • Opportunistic fungal infections	• Consider CMV monitoring in patients seropositive for CMV • Recommend PJP prophylaxis if CD4 <200 • Recommend VZV/HSV prophylaxis • Risk for prolonged lymphocytopenia

Footnotes on INF-A 12 of 13
References

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免疫和靶向免疫

表 2.单克隆抗体和融合蛋白

药物类别/机制行动	剂	感染问题	建议和意见
CD 20 目标	Obinutumab Ofatumumab 美罗华	HBV（High Risk）高风险 · HCV ● HSV/VZV： · PML	·筛查 HBV，按指示治疗 ·考虑预防 VZV/HSV ·考虑预防 PJP，特别是如果伴随治疗进一步增加 PJP 风险
CD22 靶标	Inotuzumab ozogamicin Moxetumomab pasudotox	具体数据有限 感染	·毛细血管渗漏综合征（moxetumomab）和 VOD/肝毒性（inotuzumab）的风险
CD 30 靶标	维布妥昔单抗	· PML · CMV · PJP · HSV/VZV	·考虑对 CMV 血清阳性患者进行 CMV 监测 ·考虑 PJP 和 HSV/VZV 预防 ·药物诱导的中性粒细胞减少症和淋巴细胞减少症
CD33 靶标	吉妥 ozogamicin	·细菌感染 ·真菌感染 · PJP	·药物诱导的 VOD/肝毒性、血小板减少性结肠炎和间质性肺炎
CD 38 靶标	Daratumumab isatuximab 相关	·李斯特菌 · HBV · HSV/VZV · CMV · PJP • 隐球菌	·建议 HSV/VZV 预防·考虑 PJP 预防·药物诱导的中性粒细胞减少症
CD 52 目标	Alemtuzumab	·Nocardia 的 • TB ·Listeria 列表 ● HSV/VZV： · CMV ·ADV ·BKV · PML ·真菌感染	·考虑对 CMV 血清阳性患者进行 CMV 监测 ·如果 CD 4 <200，建议 PJP 预防 ·推荐 VZV/HSV 预防 ·长期淋巴细胞减少症的风险

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Table 2. Monoclonal Antibodies and Fusion Proteins[i] IMMUNE AND TARGETED TREATMENTS[a,b]

Drug Class/ Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
CD319 (SLAMF-7) target[9]	Elotuzumab	• VZV	• Recommend HSV/VZV prophylaxis • CCR4 target[9]; drug-induced interstitial pneumonitis
CCR4 target[9]	Mogamulizumab	• *Mycobacterium spp.* • CMV • HSV/VZV • HBV • *Candida* • PJP	• Consider CMV monitoring in patients seropositivefor CMV • Recommend PJP and HSV/VZV prophylaxis • Drug-induced dermatologic toxicity
Complement C1s target	Sutimlimab	• Encapsulated bacteria	• Vaccinate against encapsulated bacteria including *Streptococcus pneumoniae, Neisseria meningitidis* (serogroups A, C, W, Y and B) and *Hemophilus influenzae* at least 2 weeks prior to starting treatment.
Complement C5 inhibitor[10]	Eculizumab Ravulizumab	• Encapsultated bacteria, but in particular, *Neisseria spp* (eg, *N. meningitidis, N. gonorrhoeae*) • Opportunistic fungal infections in patients with neutropenia	• Screen for gonorrhea in patients who are at high risk for STIs • Consider prophylaxis with PCN (ciprofloxacin or azithromycin if allergic to PCN) in addition to vaccination. Duration of prophylaxis is to be guided by drug half-life, sC5b-C9/sMAC levels, sC5a, and CH50 complement activity recovery[11] • Vaccinate with both MenACWY and MenB vaccines at least 2 weeks prior to starting treatment (if possible) • Risk for other encapsulated bacterial infections (*Streptococcus pneumoniae* and *Haemophilus influenzae*) is lower. Patients who are not vaccinated should be immunized according to ACIP recommendations. • Non-groupable *Neisseria meningitidis* infection can occur despite vaccination

Footnotes on INF-A 12 of 13
References

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表 2.单克隆抗体和融合蛋白

药物类别/机制 行动	剂	感染问题	建议和意见
CD 319 （SLAMF 7） Target 目的	Elotizumab 其它	·vzv	·CCR 4 目标;药物诱导的间质性肺炎
CCR4 靶标	·分枝杆菌属	· CMV · HSV/VZV · HBV ·念珠菌 · PJP	·考虑对 CMV 血清阳性患者进行 CMV 监测 ·推荐 PJP 和 HSV/VZV 预防 ·药物引起的皮肤毒性
补充 C1s 目标	舒替利尤单抗	·包囊细菌	·在开始治疗前至少 2 周针对包封细菌接种疫苗，包封细菌包括肺炎链球菌、脑膜炎奈瑟菌（血清群 A、C、W、Y 和 B）和流感嗜血杆菌。
补充 C5 抑制剂	Eculizumab * 拉乌利亚沙 *	·包封的细菌，但特别是奈瑟氏菌属（Neisseria spp）（例如，奈瑟氏球菌（N. meningitidis、脑膜炎奈瑟氏球菌（N.淋病） ·中性粒细胞减少症患者的真菌感染	·对性病高危患者进行淋病筛查 • 如果出现以下情况，考虑使用 PCN（环丙沙星或阿奇霉素）进行预防 对 PCN 过敏）。预防的持续时间由药物半衰期、sC 5 b–C9/sMAC 水平、sC 5a 和 CH 50 补体活性恢复指导·在开始治疗前至少 2 周接种 MenACWY 和 MenB 疫苗（如果可能）·其他包囊细菌感染（肺炎链球菌和流感嗜血杆菌）的风险较低。未接种疫苗的患者应根据 ACIP 建议进行免疫接种。·尽管接种了疫苗，仍可能发生非分组性脑膜炎奈瑟菌感染

关于 INF–A 的脚注 12/13

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Table 2. Monoclonal Antibodies and Fusion Proteins[i] IMMUNE AND TARGETED TREATMENTS[a,b]

Drug Class/ Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
IL-6 inhibitor[10]	Tocilizumab Siltuximab	• Bacterial infections • Mycobacteria (TB, non- TB) • VZV • HBV • Opportunistic fungal infections • PJP	• Screen for latent TB when combined with other immunosuppressive agents in patients at high risk and if epidemiologically indicated • Monitor closely for signs of infection as fever and CRP can be blunted • Drug-induced hepatotoxicity
VEGF inhibitor[6]	Bevacizumab Aflibercept	No significantly increased risks for infection	• Drug-induced neutropenia, bowel perforation, and GI hemorrhage • Associated with impaired wound healing
VEGFR inhibitor[6]	Ramucirumab		
Bispecific EGFR and MET receptor-directed antibody (with EGFR exon 20 insertion mutation)	Amivantamab-vmjw		• Drug-induced skin rash including acneiform dermatitis and interstitial pneumonitis
Epidermal growth factor receptor (EGFR/HER1) inhibitor[6]	Cetuximab Panitumumab		• Avoid sun exposure; use sunscreen • Dermatology consultation for severe rash • Drug-induced neutropenia, severe rash, and acneiform eruptions
HER2 inhibitor[6]	Pertuzumab	Bacterial infections	• Risk for skin and nail infections • Drug-induced rash including acneiform dermatitis

Footnotes on INF-A 12 of 13
References

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免疫和靶向免疫

表 2.单克隆抗体和融合蛋白

药物类别/机制行动	剂	感染问题	建议和意见
IL–6 抑制剂	Tocilizumab 西妥昔单抗	·细菌感染 ·分枝杆菌（TB，非 TB） · VZV · HBV ·真菌感染 · PJP	·在高危患者中与其他免疫抑制剂联合使用时筛查潜伏性结核病，如果有流行病学指征·密切监测感染体征，因为发热和 CRP 可能减弱·药物诱导的肝毒性
VEGF 抑制剂	Bevacizumab Aflibercept	无显著增加感染风险	·药物诱导的中性粒细胞减少症、肠穿孔和胃肠道出血·与伤口愈合受损相关
VEGFR 抑制剂	Ramucirumab		
双特异性 EGFR 和 MET 受体–引导抗体（与 EGFR 外显子 20 插入突变）	阿米万塔马布–vmjw		·药物引起的皮疹，包括痤疮样皮炎和间质性肺炎
表皮生长因子受体 (EGFR/HER1) 抑制剂	西妥 Panitumumab 重试 错误原因		·避免阳光照射;使用防晒霜·严重皮疹的皮肤科咨询·药物引起的中性粒细胞减少症，严重皮疹和痤疮样皮疹
HER2 inhibitorPertuzumab 重试 错误原因		细菌感染	·皮肤和指甲感染的风险·药物引起的皮疹，包括痤疮样皮炎

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Table 2. Monoclonal Antibodies and Fusion Proteins[i] IMMUNE AND TARGETED TREATMENTS[a,b]

Drug Class/Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
Monomethylauristatin E linked to IgG1 antibody targeting CD79b	Polatuzumab vedotin-piiq	• PJP • HSV/VZV • CMV • PML • Fungal infections • Hepatitis B reactivation	• Drug-induced myelosuppression • Drug-induced hepatotoxicity • Consider PJP and HSV/VZV prophylaxis, based on concomitant immunosuppression
Microtubule inhibitor linked to IgG1 folate receptor alpha (FRα) targeting antibody	Mirvetuximab soravtansine- gnyx	• No significantly increased risks for infection	• Monitor for drug-induced ocular disorders, interstitial lung disease/pneumonitis, and peripheral neuropathy

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免疫和靶向免疫

表 2.单克隆抗体和融合蛋白

药物类别/作用机制	剂	感染问题	建议和意见
与靶向 CD79b 的 IgG1 抗体连接的单甲基奥瑞他汀 E	Polatuzumab 韦多廷皮克	· PJP · HSV/VZV · CMV · PML ·真菌感染 ·B 型肝炎复发	·药物诱导的骨髓抑制 ·药物性肝毒性 ·基于伴随的免疫抑制，考虑 PJP 和 HSV/VZV 预防
与 IgG 1 叶酸连接的微管抑制剂 受体α（FRα） 靶向抗体	Mirvetuximab 索拉夫坦辛–格尼克斯	• 无显著增加的风险 感染	·监测药物引起的眼部疾病、间质性肺病/肺炎和周围神经病变

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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IMMUNE AND TARGETED TREATMENTS[a,b]

Table 3. Checkpoint Inhibitors (Monoclonal Antibodies)[1,i]

Drug Class/ Mechanism of Action	Agents	Infection Concerns[12]	Recommendations and Comments[c,d,e,g,h]
Cytotoxic T-lymphocyte–associated antigen 4 (CTLA-4) inhibitor	Ipilimumab Tremelimumab-actl	Increased infection risks from checkpoint inhibitors are thought to be mostly due to immunosuppressive treatment of irAEs (eg, with corticosteroids and/or TNF-alpha antagonists, alemtuzumab, abatacept), but emerging data suggest that dysregulated immunity pathways critical for infection surveillance from checkpoint inhibitors can directly increase infection risks. Infections which simulate irAEs can occur without additional immunosuppression	• Examples of irAEs: colitis, hepatitis, pneumonitis, thyroiditis, myositis, myasthenia gravis, rash, and many others. See NCCN Guidelines for Management of Immunotherapy-Related Toxicities. • Reactivation of latent TB and HBV, and invasive fungal infections have been reported with or without additional immunosuppression for treatment of irAEs ▸ Consider screening for CMV infection in patients with colitis who are not responding to corticosteroids ▸ Corticosteroids risk factor for bacterial infection: most commonly pulmonary, genitourinary, and intraabdominal ▸ Unusual CNS infections, *Listeria*, CMV, VZV have been reported • Screen for HBV and latent TB, treat as indicated • Based on epidemiologic factors, screening for *Coccidioides* and *Strongyloides* may be indicated • Consider PJP prophylaxis if high-dose steroid use (e.g., ≥ 20mg per day of prednisone for 4 weeks), or lower steroid doses but with other concomitant immunosuppression or risk factors
Programmed death-1 (PD-1) inhibitors	Nivolumab Pembrolizumab Cemiplimab-rwlc Dostarlimab-gxly Retifanlimab-dlwr Toripalimab-tpi		
PD ligand-1 (PD-L1) inhibitor	Atezolizumab Durvalumab Avelumab		

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References

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免疫和靶向免疫

表 3.检查点抑制剂（单克隆抗体）

药物类别/机制行动	剂	感染问题	建议和意见
细胞毒性 T 淋巴细胞相关抗原 4（CTLA–4）抑制剂	Ipilimumab 曲美木单抗–actl	检查点抑制剂导致的感染风险增加被认为主要是由于 irAE 的免疫抑制治疗（例如，皮质类固醇和/或 TNF–α拮抗剂、阿仑单抗、阿巴西普），但新出现的数据表明，对检查点抑制剂的感染监测至关重要的免疫途径失调可直接增加感染风险。模拟 irAE 的感染可在无额外免疫抑制的情况下发生	·irAE 的实例：结肠炎、肝炎、肺炎、甲状腺炎、肌炎、重症肌无力、皮疹和许多其他。参见 NCCN 免疫治疗相关毒性管理指南。·在有或无额外免疫抑制治疗 irAE 的情况下，已报告潜伏性 TB 和 HBV 以及侵袭性真菌感染的再活化 考虑在对皮质类固醇无反应的结肠炎患者中筛查 CMV 感染 细菌感染的皮质类固醇风险因素：最常见于肺部、泌尿生殖道和腹腔内 已报告异常 CNS 感染、李斯特菌、CMV、VZV·筛查 HBV 和潜伏性 TB，按指示治疗·根据流行病学因素，可能需要筛查球孢子菌和类圆线虫
程序性死亡–1（PD–1）抑制剂	Nivolumab Pembrolizumab 塞普利单抗–rwlc Dostarlimab–gxly 瑞替尼单抗–dlwr Toripalimab–tpi		
PD 配体–1（PD–L1）抑制剂	Atezolizumab Durvalumab Avelumab		• 如果使用高剂量类固醇（例如，≥ 20 mg 每天泼尼松治疗 4 周），或降低类固醇剂量，但伴有其他免疫抑制或风险因素

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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IMMUNE AND TARGETED TREATMENTS[a,b]

Table 4. Chimeric Antigen Receptor-Engineered T-Cell (CAR T-Cell) Therapy[13-15,i]

Drug Class/ Mechanism of Action	Agents	Infection Concerns	Recommendations and Comments[c,d,e,g,h]
CD19-directed	Axicabtagene ciloleucel Brexucabtagene autoleucel Tisagenlecleucel Lisocabtagene maraleucel	Risk factors for infection: • Pre-infusion: underlying malignancy, prior chemotherapy +/- HCT, antecedent infection, neutropenia • Post-infusion: CRS/ICANS and associated treatment (e.g. high-dose steroids, IL-6 inhibitors), neutropenia, lymphopenia, and hypogammaglobulinemia Infection Timeline Within 30 days: • Bacterial infections predominate • Highest risk period Beyond 30 days: • Respiratory tract viral infections common Fungal infection risk low but varies based upon prior therapies and degree of immunosuppression as well as other relevant risks	• Relevant serologic screening includes HIV, HBV and HCV. Consider CMV and additional screening based on epidemiologic risks. • Consider antibacterial and antifungal prophylaxis while neutropenic. • Consider mold-active antifungal prophylaxis if additional risks such as prolonged neutropenia, previous allogeneic HCT or augmented IST for CRS/ICANS. • PJP and HSV/VZV prophylaxis are recommended. • Monitor for CRS, which may mimic sepsis. See NCCN Guidelines for Management of Immunotherapy-Related Toxicities.
B-cell maturation antigen (BCMA)-directed	Idecabtagene vicleucel Ciltacabtagene autoleucel		

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References

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免疫和靶向药物嵌合抗原受体工程化 T 细胞（CAR T 细胞）
疗法

药物类别/机制行动	剂	感染问题	建议和意见
CD19 导向	Axicabtagene ciloleucel	·输注前：潜在恶性肿瘤，既往化疗+/– HCT、既往感染、中性粒细胞减少症·输注后：CRS/ICANS 和相关治疗（例如，高剂量类固醇、IL–6 抑制剂）、中性粒细胞减少症、淋巴细胞减少症和低丙种球蛋白血症 感染时间轴 30 天内：·细菌感染占主导地位·最高风险期 超过 30 天：·呼吸道病毒感染常见 真菌感染风险较低，但根据既往治疗和免疫抑制程度以及其他相关风险而有所不同	·相关血清学筛查包括 HIV、HBV 和 HCV。考虑 CMV 和基于流行病学风险的额外筛查。·考虑抗菌和抗真菌预防，而血小板减少。·如果存在其他风险，例如长期中性粒细胞减少症、既往同种异体 HCT 或 CRS/ICANS 的强化 IST，请考虑使用霉菌活性抗真菌预防药物。 ·推荐 PJP 和 HSV/VZV 预防。 ·监测 CRS，其可能模拟败血症。参见 NCCN 免疫治疗相关毒性管理指南。
	Brexucabtagene 的 Autoleucel 汽车		
	Tisagenlecleucel 的		
	Lisocabtagene。Maraleucel 公司		
b 细胞成熟抗原（BCMA）–引导	艾代卡巴他烯维克卢塞尔		
	西塔卡巴他金自体白细胞		

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IMMUNE AND TARGETED TREATMENTS FOOTNOTES

[a] The information in this table is continuously evolving and is not an exhaustive list. Refer to the FDA-approved labeling for these agents for further information on the appropriate use and further details on potential toxicities and drug interactions. The infection risk of these agents should be weighed according to the cancer being treated, the patient's relative medical comorbidities, and other antineoplastic therapies used during treatment.

[b] Additional agents in these and other categories have been FDA-approved, but their infection risk profile has not been fully established.

[c] All patients anticipating systemic anticancer therapy should be tested for HBV prior to the start of therapy. Risk assessment, including the need for HBV-directed treatment and prophylaxis, should be undertaken in patients with findings of chronic or past HBV infection (INF-5).[7]

[d] Tuberculosis (TB) screening should at minimum be performed in those with risk factors (eg, individuals [or caregivers and household members] from high-incidence TB countries, recent exposure, health care workers, residents and employees of homeless shelters/correctional facilities) and with planned use of agents associated with an increased risk for TB infection.

[e] Vaccination history should be assessed and updated (when relevant) in all patients (INF-7 and INF-8).

[f] Clinical features of differentiation syndrome can include fever, shortness of breath, rapid weight gain, pleuro-pericardial effusions, lung infiltrates, hypoxia, and hypotension.

[g] Consider monitoring and/or prophylaxis for opportunistic fungal infections.

[h] Many of these agents can cause QTc prolongation.

[i] Agents listed in this table may be used either as monotherapy or in combination regimens. Please refer to the full prescribing information and/or other cancer-specific NCCN Guidelines for the appropriate use of these agents.

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免疫和靶向治疗脚注

[a] 本表中的信息不断变化，并非详尽无遗的清单。请参阅 FDA 批准的这些药物的标签，以了解有关适当使用的更多信息以及有关潜在毒性和药物相互作用的更多详细信息。这些药物的感染风险应根据所治疗的癌症，患者的相关医学合并症和治疗期间使用的其他辅助治疗进行权衡。

B 这些和其他类别的其他药物已获得 FDA 批准，但其感染风险特征尚未完全确定。

[c] 所有预期接受全身抗癌治疗的患者在开始治疗前都应进行 HBV 检测。对于发现慢性或既往 HBV 感染的患者，应进行风险评估，包括 HBV 导向治疗和预防的必要性（INF-5）。

[d] 至少应在具有风险因素的人群（例如，来自结核病高发国家的个人[或护理人员和家庭成员]、近期接触、卫生保健工作者、无家可归者收容所/教养机构的居民和雇员）以及计划使用与结核病感染风险增加相关的药物的人群中进行结核病（TB）筛查。

e 应对所有患者的疫苗接种史进行评估和更新（如相关）（INF-7 和 INF-8）。

f 辨证综合征的临床特征可包括发热、呼吸急促、体重快速增加、胸膜-心包积液、肺浸润、缺氧和低血压。

[g] 考虑监测和/或预防机会性真菌感染。

[h] 这些药物中的许多可引起 QTc 延长。

i 本表所列药物可作为单一疗法或联合疗法使用。请参阅完整的处方信息和/或其他癌症特异性 NCCN 指南，以了解这些药物的适当使用。

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注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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CLINICAL PRESENTATION

Fever:
- Single temperature equivalent to ≥38.3°C orally

or
- Equivalent to ≥38.0°C orally over 1-hour period

Neutropenia:
- ≤500 neutrophils/mcL

or
- ≤1000 neutrophils/mcL and a predicted decline to ≤500/mcL over the next 48 hours

→

INITIAL EVALUATION OF FEVER AND NEUTROPENIA

- Complete history and physical (H&P) including supplemental history:
  - Major comorbid illness
  - Type and time since last chemotherapy
  - History of prior significant infections
  - Recent antibiotic therapy/prophylaxis
  - Medications
  - Use of devices
- Epidemiologically relevant exposures

- Laboratory/radiology assessment:
  - Complete blood count (CBC) with differential, comprehensive metabolic panel
  - Consider chest x-ray and urinalysis

→

MICROBIOLOGIC EVALUATION

- Blood culture x at least 2 sets (one set = 2 bottles)
  - One peripheral + one catheter (preferred but not required)[a]
- Urine culture (only if patient has symptoms or abnormal urinalysis; exercise caution in interpreting results if urinary catheter is present)
- Site-specific diagnostics:
  - Diarrhea (*Clostridioides difficile* [*C. difficile*] assay, enteric pathogen screen)
  - Skin (aspirate/biopsy of skin lesions or drainage)
- Viral diagnostics:
  - PCR- and/or direct fluorescence antibody (DFA)-based tests for vesicular/ulcerated lesions on skin or mucosa
  - Throat or nasopharynx for respiratory virus symptoms, especially during outbreaks

→ Initial Risk Assessment (FEV-2)

[a] Preferred for distinguishing catheter-related infections from secondary sources.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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临床表现

发热：·单一温度相当于

口服≥38.3°C
or
• 相当于≥38.0°C
口服 1 小时以上
期间
中性化：
• ≤500 个中性粒细胞/mcL
or
• ≤1000 个中性粒细胞/mcL
以及预计的
至≤500/mcL
未来 48 小时

→

发热和中性粒细胞减少症的初步评价

·完整的历史和物理（H&P），包括补充历史： 重大共病类型和自末次化疗后的时间

既往重大感染史
近期抗生素治疗/预防 药物 器械使用·流行病学相关暴露

·实验室/放射学评估： 全血细胞计数（CBC）和分类计数，
综合代谢小组 考虑胸部 X 光和尿液分析

→

微生物评价

·血液培养 x 至少 2 套（1 套= 2 瓶） 一个外周+一个导管

（首选但非必需）
·尿培养（仅当患者有症状或尿液分析异常时;如果存在导尿管，则在解释结果时应谨慎）

• 站点特定诊断：
腹泻（艰难梭菌[C. 艰难梭菌试验，肠道病原体筛查） 皮肤（皮肤病变或引流的抽吸/活检）·病毒诊断： 基于 PCR 和/或直接荧光抗体（DFA）的皮肤或粘膜水疱/溃疡病变检测 咽喉或鼻咽检查呼吸道病毒症状，尤其是在爆发期间

→

初始风险评估
（FEV–2）

[a] 首选用于区分导管相关感染和继发感染。

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临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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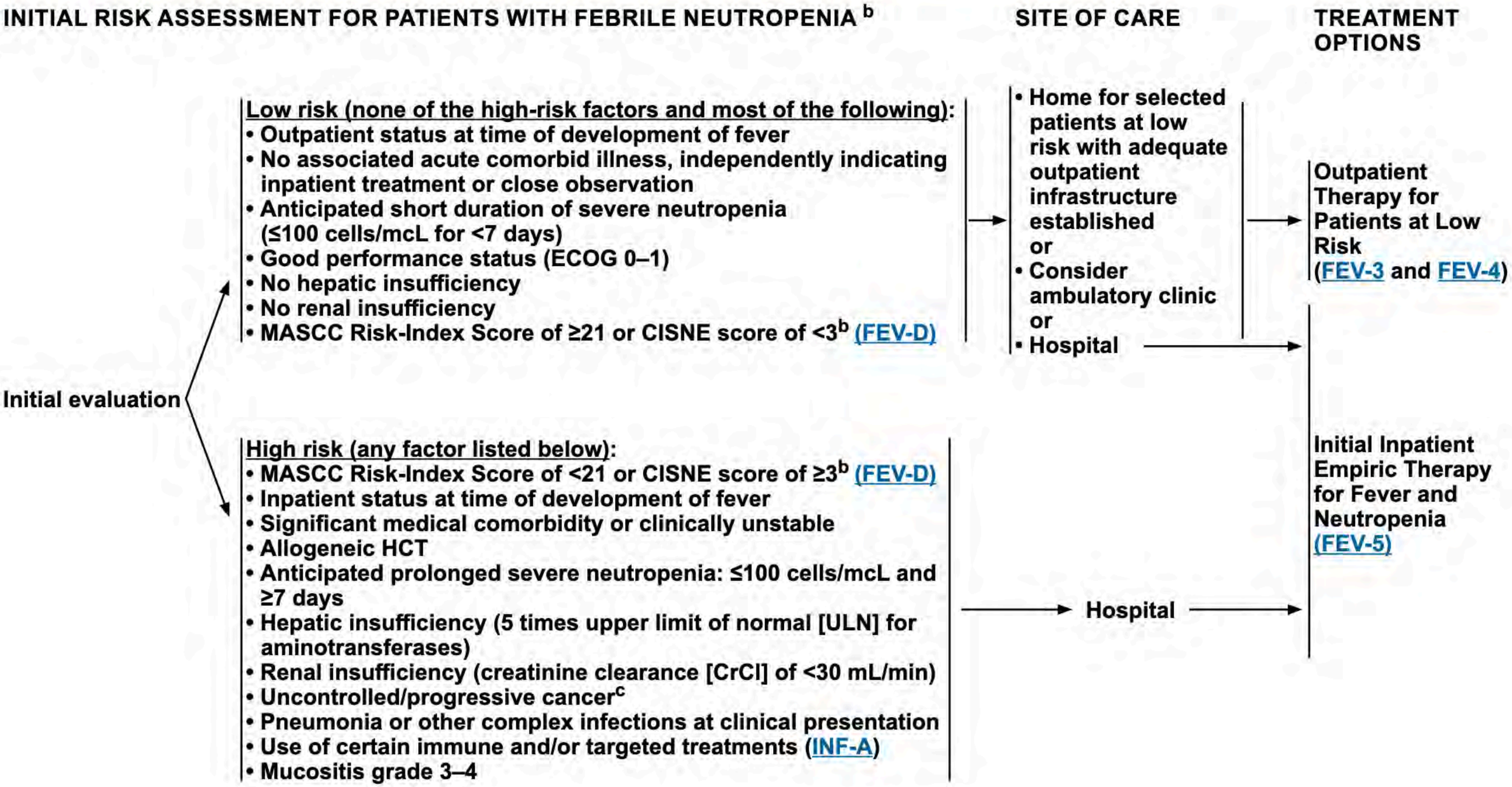

[b] Risk categorization refers to risk of serious complications, including mortality, in patients with neutropenic fever. Risk stratification is validated in adults; no generalizable, cross-validated, risk-stratified management exists for pediatric patients with febrile neutropenia. See Risk Assessment Resources (FEV-D).

[c] Uncontrolled/progressive cancer is defined as any patients with leukemia not in complete remission, or patients with other cancers and evidence of disease progression after more than 2 courses of chemotherapy.

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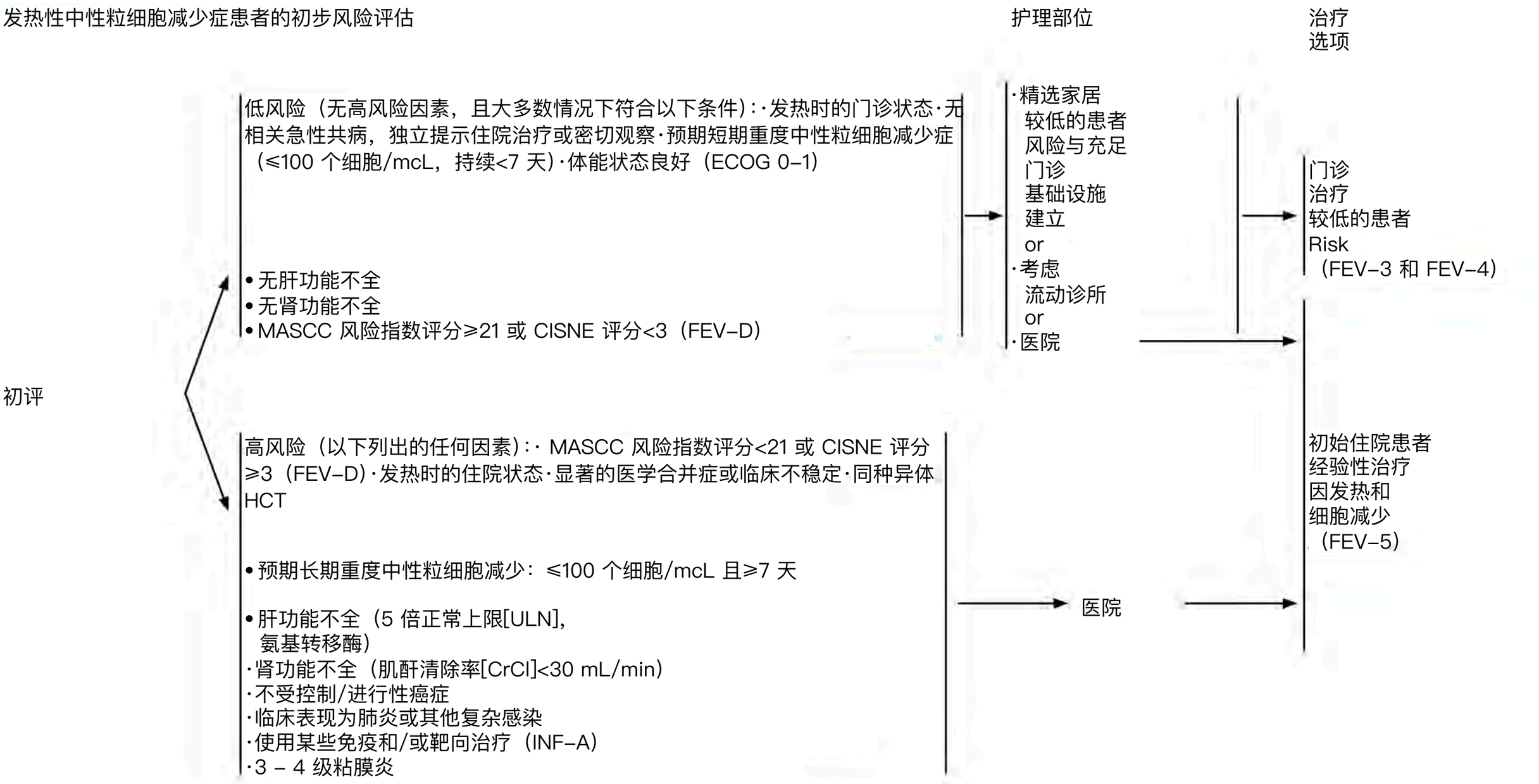
发热性中性粒细胞减少症患者的初步风险评估



[b] 风险分类是指贫血热患者发生严重并发症的风险，包括死亡率。风险分层在成人中得到验证;对于发热性中性粒细胞减少症的儿科患者，尚无可推广的、交叉验证的风险分层管理。参见风险评估资源（FEV–D）。
[c] 未控制/进展性癌症定义为任何未完全缓解的白血病患者，或患有其他癌症并在超过 2 个疗程的化疗后有疾病进展证据的患者。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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OUTPATIENT THERAPY FOR PATIENTS AT LOW RISK

INDICATION	ASSESSMENT	MANAGEMENT
Patient at low risk with fever and neutropenia[b]	• Careful examination • Review lab results: no critical values • Review social criteria for home therapy ▸ Patient consents to home care ▸ 24-hour home caregiver available ▸ Telephone ▸ Access to emergency facilities ▸ Adequate home environment ▸ Distance within approximately one hour of a medical center or treating physician's office • Assess for oral antibiotic therapy ▸ No nausea and vomiting ▸ Able to tolerate oral medications ▸ Not on prior fluoroquinolone prophylaxis	See Treatment and Follow-up (FEV-4) OR Consider observation period (2–12 hours) (category 2B) in order to: • Confirm low-risk status and ensure stability of patient • Observe and administer first dose of antibiotics and monitor for reaction • Organize discharge plans to home and follow-up • Provide patient education • Follow up by telephone within 12–24 hours

[b] Risk categorization refers to risk of serious complications, including mortality, in patients with neutropenic fever. Risk stratification is validated in adults; no generalizable, cross-validated, risk-stratified management exists for pediatric patients with febrile neutropenia. See Risk Assessment Resources (FEV-D).

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OUTPATIENT THERAPY FOR PATIENTS AT LOW RISK

重试 错误原因

指示	评估	管理
发热和中性粒细胞减少的低风险患者	·仔细检查·审查实验室结果：没有关键值·审查家庭治疗的社会标准 患者同意接受家庭护理 24–提供小时家庭护理 电话 使用应急设施 适当的家庭环境 距离医疗中心或治疗中心约一小时 医生办公室 ·评估口服抗生素治疗 无恶心呕吐 能够耐受口服药物 既往未使用氟喹诺酮类药物 预防	参见治疗和随访（FEV–4） OR 考虑观察期（2–12 小时）（2B 类），以便： • 确认低风险状态，确保患者稳定 • 观察和给予第一剂抗生素并监测 ·组织出院回家计划和随访·提供患者教育·12–24 小时内电话随访

[b] 风险分类是指贫血热患者发生严重并发症的风险，包括死亡率。风险分层在成人中得到验证;对于发热性中性粒细胞减少症的儿科患者，尚无可推广的、交叉验证的风险分层管理。参见风险评估资源（FEV–D）。

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临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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OUTPATIENT THERAPY FOR PATIENTS AT LOW RISK

TREATMENT OPTIONS

- Intravenous (IV) antibiotics at home
- Daily long-acting IV agent ± oral therapy
  - Home or office
- Oral therapy only[d]:
  - Ciprofloxacin plus amoxicillin/clavulanate[e] (category 1)
  - Levofloxacin
  - Moxifloxacin[f] (category 1)

→

FOLLOW-UP

- Patient should be monitored daily
- Daily assessment for the first 72 hours to assess response, toxicity, and adherence; if responding, then telephone follow-up daily thereafter.
- Specific reasons to return to clinic:
  - Any positive culture from blood or other sterile source
  - New signs/symptoms reported by the patient
  - Persistent or recurrent fever at 3–5 days
  - Inability to continue prescribed antibiotic regimen (ie, oral intolerance)
  - Office visit for infusion of IV antibiotics

[d] Criteria for oral antibiotics: no nausea or vomiting, patient able to tolerate oral medications, and patient not on prior fluoroquinolone prophylaxis.
[e] Use clindamycin in place of amoxicillin-clavulanate for patients who are allergic to penicillin.
[f] Insufficient activity against *Pseudomonas aeruginosa*. Recommended for patients at low risk who may not require *Pseudomonas aeruginosa* coverage.

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低风险患者的门诊治疗

治疗选择	后续行动
·在家中静脉注射（IV）抗生素·每日长效 IV 剂±口服治疗　家庭或办公室·仅口服治疗：　环丙沙星+阿莫西林/克拉维酸（类别 1） 左氧　莫西沙星（1 类）	·患者应每天进行监测 • 前 72 小时的每日评估，以评估反应、毒性，如有回应，则每日电话跟进。 • 返回诊所的具体原因： 任何来自血液或其他无菌来源的阳性培养物　患者报告的新体征/症状　持续或反复发热 3–5 天　无法继续处方抗生素治疗方案（即，口服不耐受）　因静脉输注抗生素而就诊

d 口服抗生素的标准：无恶心或呕吐，患者能够耐受口服药物，患者既往未接受氟喹诺酮预防。
e 对青霉素过敏的病人用克林霉素代替阿莫西林–克拉维酸。
f 对铜绿假单胞菌的活性不足。推荐用于可能不需要铜绿假单胞菌覆盖的低风险患者。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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INITIAL INPATIENT EMPIRIC THERAPY FOR UNCOMPLICATED FEVER AND NEUTROPENIA[g]

Initial antibiotic therapy should be based on:
- Infection risk assessment (FEV-2)
- Broad-spectrum coverage including antipseudomonal activity
- Colonization with or prior infection with multidrug-resistant organisms (MDROs)[h]
- Site of infection
- Local antibiotic susceptibility patterns
- Organ dysfunction/drug allergy
- Previous antibiotic therapy

IV antibiotic therapy[i]
- Typically monotherapy
  - Cefepime (category 1)
  - Imipenem/cilastatin (category 1)
  - Meropenem (category 1)
  - Piperacillin/tazobactam (category 1)
  - Ceftazidime[j] (category 2B)
- IV combination therapy could be considered where antimicrobial resistance is suspected
- Patients with high risk of anaphylaxis, consider ID/allergy consultation[k]

Consider oral antibiotic therapy for select patients at low risk:
- Ciprofloxacin + amoxicillin/clavulanate (category 1)[e]
- Moxifloxacin[f] (category 1)
- Levofloxacin
- Oral antibiotic regimen not recommended if patient received prior quinolone prophylaxis

Site-specific evaluation and therapy:

Mouth, esophagus, and sinus/nasal (FEV-6)

Abdominal pain, perirectal pain, diarrhea, and urinary tract symptoms (FEV-7)

Lung infiltrates (FEV-9)

Cellulitis, vascular access devices (VADs), vesicular lesions, disseminated papules or other lesions, and central nervous system [CNS] symptoms (FEV-8)

OR

Follow-up (FEV-11)

Follow-up (FEV-4)

[e] Use clindamycin in place of amoxicillin-clavulanate for patients who are allergic to penicillin.
[f] Insufficient activity against *Pseudomonas aeruginosa*. Recommended for patients at low risk who may not require *Pseudomonas aeruginosa* coverage.
[g] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.
[h] The CDC defines MDROs as microorganisms that are resistant to one or more classes of antimicrobial agents. Methicillin-resistant *Staphylococcus aureus* (MRSA) is an example of an MDRO.
[i] Choice of antibiotic may depend on local antibiotic susceptibility patterns and individual patient syndromes.
[j] Weak Gram-positive coverage and increased breakthrough infections limit utility.
[k] For severe beta-lactam allergy, consider vancomycin and aztreonam while further evaluation is carried out with ID/allergy consultation.

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无并发症发热和中性粒细胞减少症的初始住院经验性治疗

初始抗生素治疗应基于：·感染风险评估（FEV–2）·广谱覆盖率，包括抗假单胞菌活性·多重耐药微生物（MDRO）的定植或既往感染

·感染部位
·当地抗生素敏感性模式
·器官功能障碍/药物过敏
·既往抗生素治疗

IV 抗生素治疗·通常为单药治疗 头孢吡肟（第 1 类） 亚胺培南/西司他丁（1 类） 美罗培南（1 类） 瑞拉西林/他唑巴坦（1 类） 头孢他啶（2B 类）·如果怀疑存在抗生素耐药性，可考虑静脉联合治疗·过敏反应风险高的患者，考虑 ID/过敏咨询

·考虑对低风险的选定患者进行口服抗生素治疗：

- 环丙沙星+阿莫西林/克拉维酸（第 1 类）
- 莫西沙星（1 类）

·左氧氟沙星
·如果患者既往接受过喹诺酮类预防治疗，则不推荐口服抗生素方案

现场具体评价和治疗：口腔、食管和鼻窦/鼻腔（FEV–6）

腹痛、直肠周围疼痛、腹泻和尿路症状（FEV–7）

肺浸润（FEV–9）

蜂窝织炎、血管通路装置（VAD）、水疱性病变、播散性丘疹或其他病变和中枢神经系统[CNS]症状（FEV–8）

OR

随访（FEV–11）

随访（FEV–4）

e 对青霉素过敏的患者使用克林霉素代替阿莫西林–克拉维酸。
f 对铜绿假单胞菌的活性不足。推荐用于可能不需要铜绿假单胞菌覆盖的低风险患者。
g 有关剂量、光谱和具体注释/注意事项，请参见抗菌剂（FEV–A）。
h CDC 将 MDRO 定义为对一种或多种抗菌剂具有耐药性的微生物。耐甲氧西林金黄色葡萄球菌（MRSA）是 MDRO 的一个例子。

i 抗生素的选择可能取决于当地抗生素敏感性模式和个体患者的综合征。
j 弱革兰氏阳性覆盖率和增加的突破性感染限制了效用。
k 对于重度β–内酰胺过敏，考虑万古霉素和氨曲南，同时通过 ID/过敏咨询进行进一步评估。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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INITIAL CLINICAL FINDING PRESENTATION (DAY 0)		EVALUATION	TREATMENT MODIFICATIONS[g,l,m]
		All patients with febrile neutropenia should receive broad-spectrum antibiotics (FEV-5)	
Mouth/ mucosal membrane	Necrotizing ulceration	• HSV, VZV diagnostics • Culture and Gram stains ▸ Fungal • Biopsy suspicious lesions	• Ensure adequate anaerobic activity • Consider anti-HSV/VZV therapy • Consider systemic antifungal therapy
	Thrush		• Antifungal therapy ▸ Fluconazole first-line therapy ▸ Voriconazole, posaconazole, or echinocandin if refractory to fluconazole
	Vesicular lesions	HSV, VZV diagnostics	Anti-HSV therapy (category 1)
Esophagus	• Retrosternal burning • Dysphagia/odynophagia	• HSV diagnostics • Fungal culture of suspicious oral lesions • Consider endoscopy, if no response to therapy • Histopathologic examination for viral and fungal pathogens	• Initial therapy guided by clinical findings (eg, thrush or perioral HSV) • Antifungal therapy for thrush • Consider HSV treatment
Sinus/ nasal	• Sinus tenderness • Periorbital cellulitis • Nasal ulceration • Unilateral eye tearing	• High-resolution sinus CT/orbit MRI • ENT/ophthalmologic urgent evaluation • Culture and stains/biopsy	• Add vancomycin if periorbital cellulitis noted • Add lipid amphotericin B preparation to cover possible aspergillosis and mucormycosis in patients at high risk with suspicious CT/MRI findings[n]

[g] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.
[l] See Antifungal Agents (FEV-B) for dosing, spectrum, and specific comments/cautions.
[m] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions. The antivirals are not equal in terms of efficacy, side effects, and resistance.
[n] Posaconazole or isavuconazonium sulfate can be considered for patients who have invasive, refractory infections or who have intolerance to amphotericin B formulations.

Follow-up (FEV-11)

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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初始临床呈现 (DAY 0 个)	找到	评价	治疗修改
所有发热性中性粒细胞减少症患者应接受广谱抗生素（FEV–5）			
嘴/粘膜膜	坏死性溃疡	· HSV、VZV 诊断·培养和革兰氏染色 真菌·活检可疑病变	·确保足够的厌氧活动 ·考虑抗 HSV/VZV 治疗·考虑全身抗真菌治疗
	画眉		·抗真菌治疗 氟康唑一线治疗 Voriconazole，posaconazole，or echinocandin if（如果）氟康唑难治性
	水泡性病变	HSV、VZV 诊断	抗 HSV 治疗（1 类）
食管	·胸骨后烧灼 ·吞咽困难/吞咽痛	· HSV 诊断 ·可疑口腔病变的真菌培养·如果治疗无效，考虑内镜检查·病毒和真菌病原体的组织学检查	• 根据临床结果（如鹅口疮或口周 HSV）·鹅口疮的抗真菌治疗·考虑 HSV 治疗
Sinus/nasal	• Sinus tenderness • Periorbital cellulitis • Nasal ulceration • Unilateral eye tearing	·高分辨率鼻窦 CT/眼眶 MRI·耳鼻喉科/眼科紧急评估·培养和染色/活检	·如果发现眶周蜂窝织炎，则添加万古霉素·添加脂质阿替霉素 B 制剂，以覆盖高风险患者中可能的曲霉菌病和毛霉菌病 可疑的 CT/MRI 结果

重试 错误原因

[g] 有关剂量、光谱和具体注释/注意事项，请参见抗菌剂（FEV–A）。
[l] 有关剂量、谱和具体注释/注意事项，请参见抗真菌药（FEV–B）。
[m] 有关剂量、谱和具体注释/注意事项，请参见抗病毒药物（FEV–C）。抗病毒药物在疗效、副作用和耐药性方面是不平等的。
[n] 泊沙康唑或硫酸艾沙康唑可用于侵袭性、难治性感染或对阿替霉素 B 不耐受的患者制剂。

随访（FEV–11）

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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INITIAL CLINICAL PRESENTATION (DAY 0)	EVALUATION	TREATMENT MODIFICATIONS[g,l,m]
		All patients with febrile neutropenia should receive broad-spectrum antibiotics (FEV-5)
Abdominal pain	• Abdominal exam • Abdominal CT (preferred) or ultrasound • Alkaline phosphatase, transaminases, bilirubin, amylase, lipase	• Consider adding anaerobic therapy • Consider early surgical and other subspecialty (eg, gastroenterology, interventional radiology) consultations as clinically indicated
Perirectal pain	• Perirectal inspection • Consider imaging such as abdominal/pelvic CT or MRI	• Ensure adequate anaerobic therapy • Consider local care
Diarrhea[o]	• *C. difficile* testing (if not on laxatives/stool softeners/tube feeds, which may cause diarrhea) • Consider other medical causes of diarrhea: neutropenic enterocolitis, GI GVHD, mucositis • Colonoscopy may be required to evaluate for colitis and determine etiology of diarrhea • Depending on clinical circumstances, consider diagnostic testing for viral, bacterial, and/or parasitic pathogens.	*C. difficile* infection (CDI) confirmed → FEV-10 CDI negative/other cause identified → Treat as appropriate for clinical diagnosis
Urinary tract symptoms	Urine culture	No additional therapy until specific pathogen identified

[g] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.
[l] See Antifungal Agents (FEV-B) for dosing, spectrum, and specific comments/cautions.
[m] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions. The antivirals are not equal in terms of efficacy, side effects, and resistance.
[o] Diarrhea from chemotherapy or antibiotic-associated diarrhea can be confused with true CDI.

Follow-up (FEV-11)

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初始临床呈现 (DAY 0 个)	评价	治疗修改	
		所有发热性中性粒细胞减少症患者应接受广谱抗生素（FEV–5）	
腹部 pain	·腹部 CT（首选）或超声检查·碱性磷酸酶、转氨酶、胆红素、淀粉酶、脂肪酶	·考虑增加厌氧治疗·根据临床指征考虑早期外科和其他专科（例如，胃肠病学，介入放射学）咨询	
直肠周 pain	·考虑腹部/盆腔 CT 或 MRI 等成像	·确保充分的厌氧治疗 ·考虑当地护理	
腹泻	€.艰难梭菌测试（如果不是泻药/粪便软化剂/管饲，可能会导致腹泻）·考虑腹泻的其他医学原因：贫血性小肠结肠炎，GI GVHD，粘膜炎 • 可能需要结肠镜检查来评估 ·根据临床情况，考虑对病毒、细菌和/或寄生虫病原体进行诊断检测。	C.确诊艰难梭菌感染（CDI）	FEV10
		CDI 阴性/其他原因识别	根据临床诊断进行适当治疗
尿路症状	尿培养	在确定特定病原体之前，无需额外治疗	

g 有关抗菌剂的剂量、谱和具体注释/注意事项，请参见抗菌剂（FEV–A）。
l 参见抗真菌药（FEV–B）的剂量、谱和具体注释/注意事项。
m 有关剂量、频谱和具体注释/注意事项，请参见抗病毒药物（FEV–C）。抗病毒药物在疗效、副作用和耐药性方面是不平等的。
o 化疗引起的腹泻或腹泻相关性腹泻可能与真正的 CDI 混淆。

随访（FEV–11）

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INITIAL CLINICAL PRESENTATION (DAY 0) | **EVALUATION** | **TREATMENT MODIFICATIONS[g,l,m]**

All patients with febrile neutropenia should receive broad-spectrum antibiotics (FEV-5)

Cellulitis/skin and soft tissue infections → **Consider aspirate or biopsy for culture** → **Add Gram-positive active agents (consider for methicillin-resistant *Staphylococcus aureus* [MRSA])**

Vesicular lesions → **Aspiration or scraping for PCR/DFA, and/or HSV cultures if PCR unavailable.** → **Consider acyclovir, famciclovir, or valacyclovir**

Disseminated papules or other lesions → **• Aspiration or biopsy for bacterial, fungal, and mycobacterial cultures and histopathology**
• Consider evaluation for VZV → **• Consider vancomycin or other Gram-positive coverage (FEV-A)**
• Consider mold-active antifungal therapy in patients who are at high risk

Vascular access devices (VADs)

Entry or site inflammation → **• Swab entry site drainage (if present) for culture**
• Blood culture from each port of VAD and a peripheral culture → **Vancomycin initially or add if site not responding after 48 hours of empiric therapy**

Tunnel infection/ port pocket infection, septic phlebitis → **Blood culture from each port of VAD and a peripheral culture** → **• Remove catheter and culture surgical wound**
• Add vancomycin

CNS symptoms → **• MRI preferred or CT scan**
• Lumbar puncture (if possible)
• Infectious disease (ID) consult
• Neurology consult → **• Initial empiric therapy**
• Empiric therapy of meningitis should include coverage of *Pseudomonas aeruginosa* and *Listeria*
• For encephalitis, add high-dose acyclovir with hydration and monitor renal function

[g] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.
[l] See Antifungal Agents (FEV-B) for dosing, spectrum, and specific comments/cautions.
[m] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions. The antivirals are not equal in terms of efficacy, side effects, and resistance.

Follow-up (FEV-11)

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初始临床呈现 (DAY 0 个)	评价	治疗修改
		所有发热性中性粒细胞减少症患者应接受广谱抗生素（FEV–5）
蜂窝织炎/皮肤和软组织感染	考虑穿刺或活检进行培养	添加革兰氏阳性活性剂（考虑耐甲氧西林金黄色葡萄球菌 [MRSA]）
水泡性病变	如果 PCR 不可用，则抽吸或刮取 PCR/DFA 和/或 HSV 培养物。	考虑阿昔洛韦，泛昔洛韦，或伐昔洛韦
传播丘疹或其他病变	·细菌、真菌和分枝杆菌培养物和组织病理学的抽吸或活检	·考虑万古霉素或其他革兰氏阳性覆盖率（FEV–A）·在高危患者中考虑霉菌活性抗真菌治疗
Vencular 的 Access 的设备（VAD）	入口或站点炎症 ·从 VAD 的每个端口进行血液培养和外周培养	最初使用万古霉素，或如果经验性治疗 48 小时后部位无反应，则添加万古霉素
	隧道感染/输液港囊袋感染、脓毒性静脉炎 每个 VAD 端口的血培养和外周血培养	·移除导管并培养手术伤口
CNS 症状	·首选 MRI 或 CT 扫描·腰椎穿刺（如果可能）·传染病（ID）咨询·神经科咨询	·初始经验性治疗·脑膜炎的经验性治疗应包括铜绿假单胞菌和李斯特菌的覆盖范围·对于脑炎，添加大剂量阿昔洛韦与水合作用并监测肾功能

g 有关抗菌剂的剂量、谱和具体注释/注意事项，请参见抗菌剂（FEV–A）。
l 参见抗真菌药（FEV–B）的剂量、谱和具体注释/注意事项。
m 有关剂量、频谱和具体注释/注意事项，请参见抗病毒药物（FEV–C）。抗病毒药物在疗效、副作用和耐药性方面是不平等的。

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随访（FEV–11）

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EVALUATION OF PATIENTS WITH POSSIBLE RESPIRATORY INFECTION

INITIAL CLINICAL PRESENTATION (DAY 0)

EVALUATION[p,q]

TREATMENT MODIFICATIONS[g,l,m]

All patients with febrile neutropenia should receive broad-spectrum antibiotics (FEV-5)

Lung infiltrates →

The potential diagnoses and useful tests are extensive. CT scan to define nature and extent of infiltrates and culture of blood and sputum (if available) should be performed. Consider bronchoalveolar lavage (BAL), especially if no response to initial therapy. Based on the nature of the radiologic appearance, epidemiology, risk factors, patient history, and clinical presentation, consider:

Diagnostic tests[r]:

- **Blood**
  - ***Aspergillus* galactomannan**
  - ***Cryptococcal* antigen**
  - ***Coccidioides* serology**
  - ***Blastomyces* and/or *Histoplasma* antigen**
- **Urine**
  - ***Blastomyces* and/or *Histoplasma* antigen**
  - ***Legionella* and pneumococcal antigen**
- **Respiratory/bronchoscopy specimens**
  - ***Aspergillus* galactomannan**
  - ***Pneumocystis jiroveci* PCR or DFA**
  - **Mycobacterial PCR**
  - ***Mycoplasma* and *Chlamydia* PCR**
  - **Nucleic acid amplification testing (NAAT) for viral respiratory pathogens via nasopharyngeal swab or BAL fluid**

→

- **Consider adding coverage for atypical bacteria (azithromycin, doxycycline,[s] or fluoroquinolone) (see General Recommendations for Vaccination in Patients with Cancer: Pneumococcal Vaccination [INF-7])**
- **Consider adding:**
  - **Mold-active antifungal agent [see Patients at Intermediate to High Risk on (INF-1)]**
  - **Antiviral therapy during influenza season in local area[t]**
  - **TMP/SMX if possible *Pneumocystis jirovecii* etiology**
  - **Vancomycin or linezolid if MRSA suspected**
- **Re-evaluate for ability to de-escalate**

[g] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.
[l] See Antifungal Agents (FEV-B) for dosing, spectrum, and specific comments/cautions.
[m] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions. The antivirals are not equal in terms of efficacy, side effects, and resistance.
[p] Other diagnoses to consider include pulmonary edema, hemorrhage, and drug toxicities.
[q] Assess for health care-acquired pneumonia and/or resistant pathogens.
[r] Rapid immunofluorescent viral antigen tests may be negative for H1N1.
[s] Doxycycline can be used for *Mycoplasma* or *Chlamydia*, but is not recommended for legionellosis as some non-pneumophila *Legionella* species can be resistant to tetracyclines.
[t] Antiviral susceptibility of influenza strains is variable and cannot be predicted based on prior influenza outbreaks. In cases of seasonal influenza and pandemic strains (eg, H1N1), it is necessary to be familiar with susceptibility patterns and guidelines on appropriate antiviral treatment.

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Follow-up (FEV-11)

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评价可能发生尿路感染的患者

初始临床呈现
(DAY 0 个)

评价

治疗修改

所有发热性中性粒细胞减少症患者应接受广谱抗生素（FEV–5）

Lung
浸润

潜在的诊断和有用的测试是广泛的。CT 扫描以确定渗透的性质和程度以及血和痰（如有）应进行。考虑支气管肺泡灌洗（BAL），特别是如果对初始治疗无反应。根据放射学表现的性质、流行病学、危险因素、病史和临床表现，考虑：诊断性检查 r:·血液 曲霉半乳甘露聚糖

隐球菌抗原 球孢子菌血清学 母细胞瘤和/或组织胞浆菌抗原·尿液 母细胞瘤和/或组织胞浆菌抗原

军团菌和肺炎球菌抗原·呼吸道/支气管镜检查标本 曲霉半乳甘露聚糖

Pneumocystis jiroveci PCR or DFA
重试 错误原因
Mycobacterial PCR
重试 错误原因
Mycoplasma and Chlamydia PCR
重试 错误原因
Nucleic acid amplification testing (NAAT) for viral respiratory pathogens via nasopharyngeal swab or BAL fluid
重试 错误原因
重试 错误原因

• Consider adding coverage for atypical bacteria (azithromycin, doxycycline,or fluoroquinolone) (see General Recommendations for Vaccination in Patients with Cancer: Pneumococcal Vaccination [INF–7]) • Consider adding: Mold–active antifungal agent [see Patients at Intermediate to High Risk on (INF–1)]
重试 错误原因

流感季节的抗病毒治疗
局域
TMP/SMX（如可能）
病因 如果怀疑 MRSA，则使用万古霉素或利奈唑胺·重新评估降级的能力

g 有关抗菌剂的剂量、谱和具体注释/注意事项，请参见抗菌剂（FEV–A）。l 参见抗真菌药（FEV–B）的剂量、谱和具体注释/注意事项。

m 有关剂量、谱和具体注释/注意事项，请参见抗病毒药物（FEV–C）。抗病毒药物在疗效、副作用和耐药性方面是不平等的。其他需要考虑的诊断包括肺水肿、出血和药物毒性。

q 评估卫生保健获得性肺炎和/或耐药病原体。

r 快速免疫荧光病毒抗原检测可能对 H1N1 呈阴性。强力霉素可用于支原体或衣原体，但不推荐用于军团菌病，因为一些非嗜肺军团菌物种可对四环素耐药。流感病毒株的抗病毒敏感性是可变的，不能根据以前的流感爆发来预测。在季节性流感和大流行性毒株（如 H1N1）的情况下，有必要熟悉敏感性模式和适当抗病毒治疗的指南。

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随访（FEV–11）

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TREATMENT OF *CLOSTRIDIOIDES DIFFICILE* INFECTIONS (CDI) IN PATIENTS WITH CANCER

TREATMENT

CDI confirmed →
- Reduce unnecessary antibiotic treatment
- Vancomycin (125 mg PO QID for 10 days) OR Fidaxomicin (200 mg BID for 10 days)

→ **Ongoing/worsening CDI[v]** →
- Reduce unnecessary antibiotic treatment
- Consider switching to fidaxomicin if initially treated with vancomycin
- Consider fecal transplant[x] (avoid in patients with neutropenia)

→ **Fulminant disease** →
- Oral vancomycin 500 mg QID (or via nasogastric [NG] tube) with IV metronidazole 500 mg TID
- Consider vancomycin via rectal instillation if ileus present

→ **CDI resolved[w]**

Relapse/recurrent CDI[u] →
- Reduce unnecessary antibiotic treatment
- Fidaxomicin if not previously received OR
- Vancomycin taper (125 mg 4 QID for 10–14 days, BID for 7 days, then QD for 7 days, and then every 2 or 3 days for 2–8 weeks)
- With appropriate consultation, consider fecal transplant (avoid in patients with neutropenia) or bezlotoxumab

[u] Recurrent CDI is defined as symptom onset and positive assay result following an episode with positive assay result in previous 2–8 weeks.
[v] For subsequent treatment options for ongoing CDI, also see the *Clostridium difficile* Practice Guidelines provided by the Infectious Diseases Society of America: https://www.idsociety.org/practice-guideline/clostridioides-difficile-2021-focused-update.
[w] If continuing antibiotics, may consider secondary prophylaxis.
[x] This treatment has not been proven to be effective in this patient population.

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Follow-up (FEV-11)

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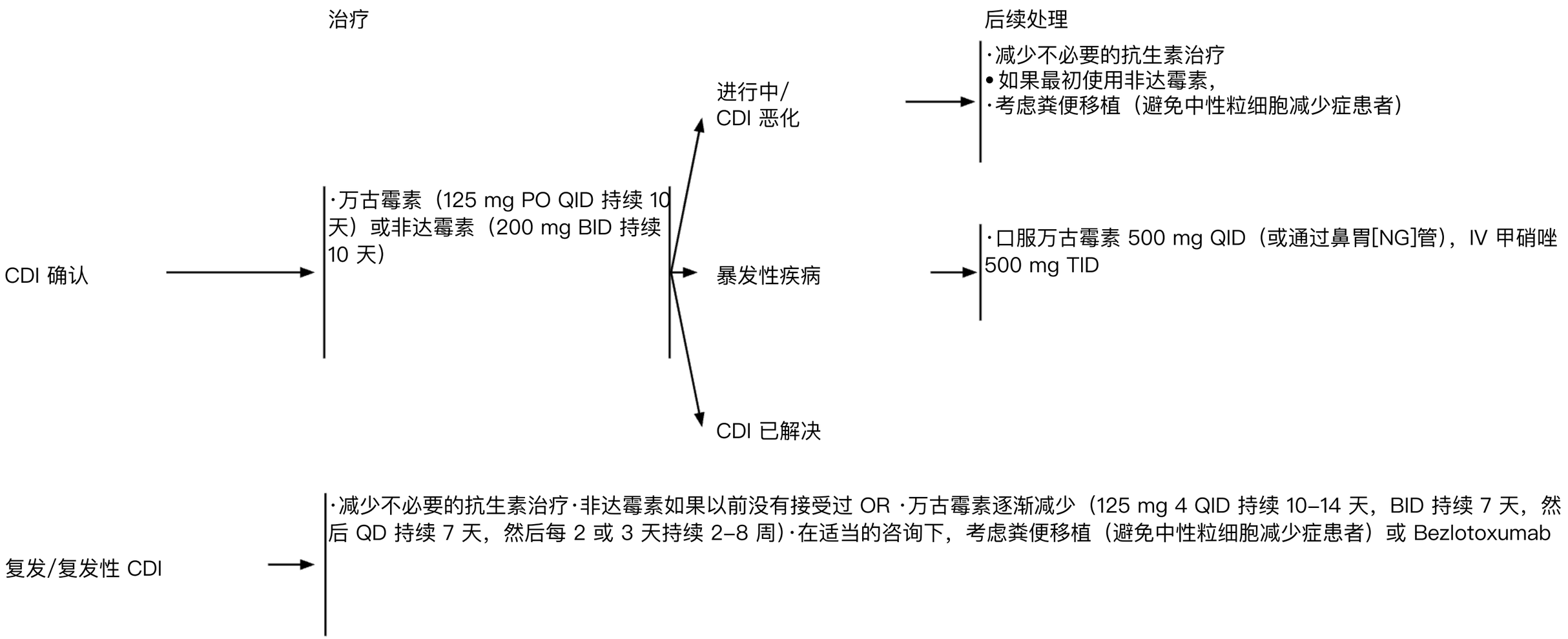
恶性肿瘤患者难治性类梭菌感染的治疗



[u] 复发性 CDI 定义为在前 2–8 周内检测结果为阳性的发作后出现症状和阳性检测结果。
[v] 有关持续 CDI 的后续治疗选择，请参阅美国传染病学会提供的艰难梭菌实践指南：https：www.idsociety.org/practice-guideline/clostridioides-difficile-2021-focused-update。

如果继续使用抗生素，可考虑二级预防。
[x] 该治疗尚未被证明在该患者人群中有效。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

随访（FEV–11）

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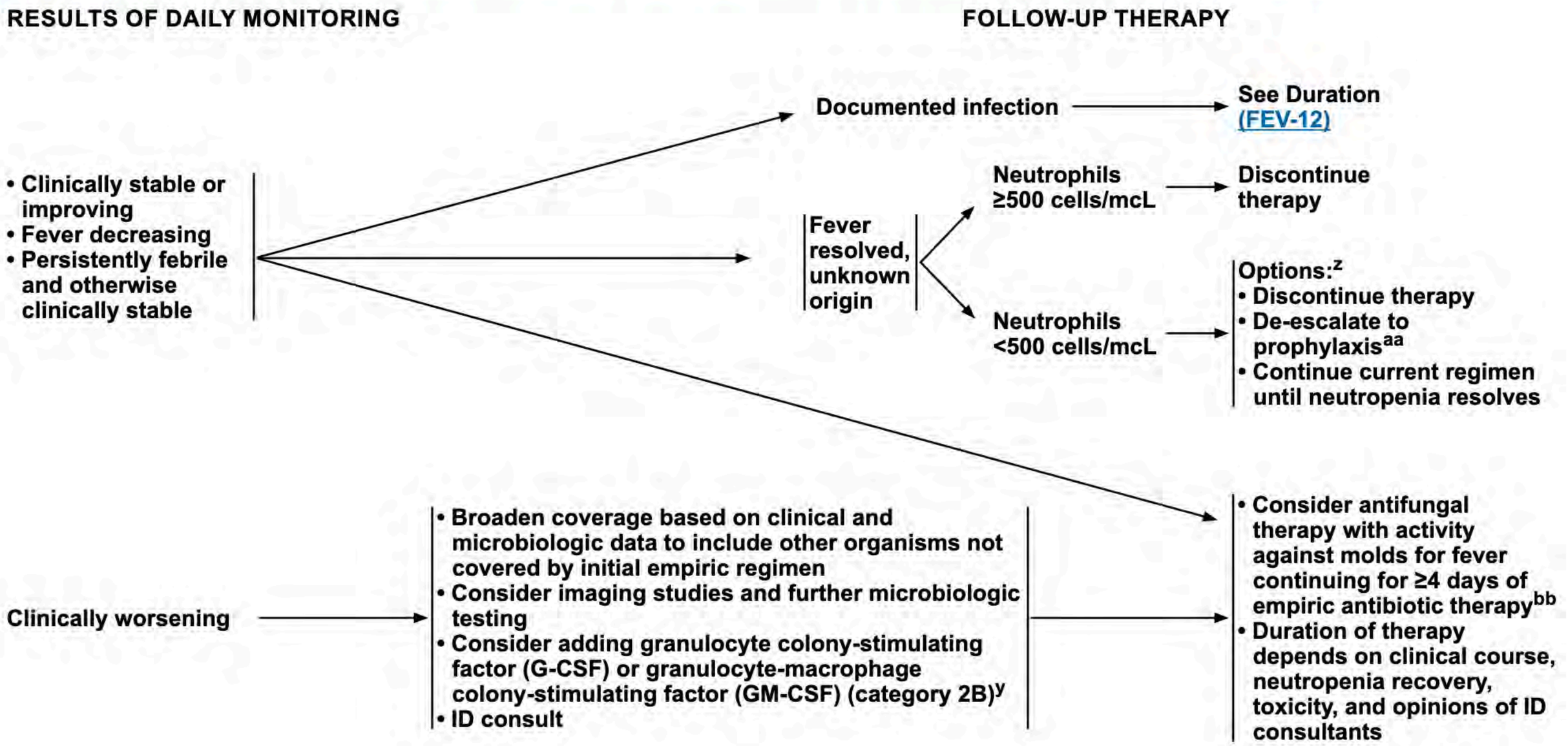

y NCCN Guidelines for Hematopoietic Growth Factors.

z The choice will depend on particular patient details; see Discussion for additional information.

aa In patients who defervesce for at least 48 hours, it may be appropriate in some cases to de-escalate to fluoroquinolone.

bb The timing to add empiric antifungal therapy varies with the risk of invasive mold infection but generally ranges between 4–7 days of neutropenic fever. In patients at high risk for mold infection (ie, neutropenia >10 days, allogeneic HCT recipients, high-dose corticosteroids), the panel recommends adding empiric antifungal therapy after the fourth day unless patient is receiving prophylaxis directed against molds.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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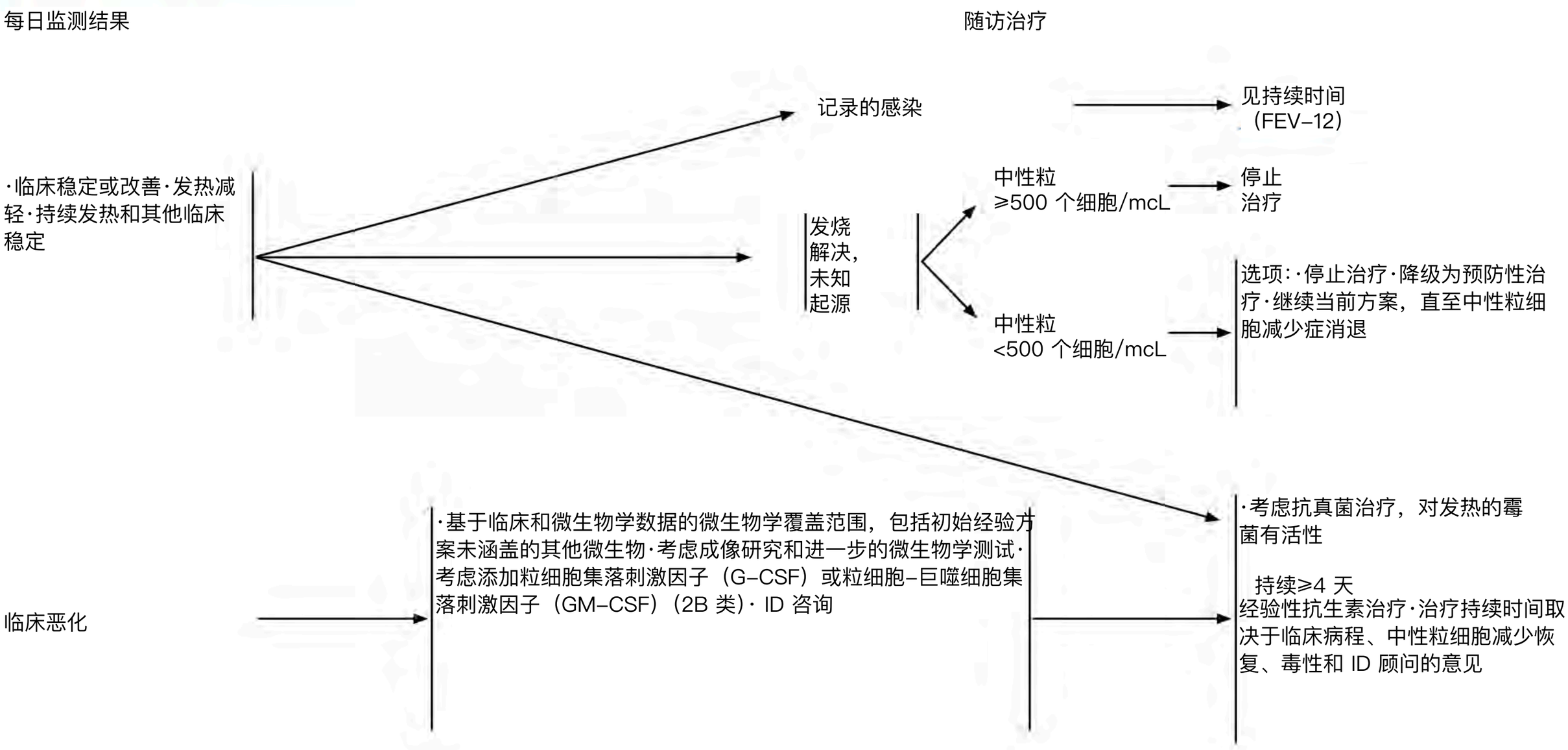

[y] NCCN 造血生长因子指南。
[z] 选择将取决于特定患者的详细信息;更多信息见讨论。

[aa] 在退热至少 48 小时的患者中，在某些情况下可能适合降级为氟喹诺酮。

[bb] 增加经验性抗真菌治疗的时机因侵袭性霉菌感染的风险而异，但通常在贫血性发热 4–7 天之间。在霉菌感染高危患者（即中性粒细胞减少>10 天，同种异体 HCT 接受者，高剂量皮质类固醇），专家组建议在第四天后增加经验性抗真菌治疗，除非患者正在接受针对霉菌的预防治疗。

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临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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FOLLOW-UP THERAPY FOR RESPONDING DISEASE	SUGGESTED MINIMUM DURATION OF THERAPY FOR DOCUMENTED INFECTION[g,l,m]

These are general guidelines for patients with uncomplicated disease and may need to be revised for individual patients. Treatment duration can be modified depending on infection severity and patient factors.

Documented infection →

- Targeted treatment of documented infections should be done
- Reassessment of empiric broad-spectrum therapy
- De-escalation and duration of antimicrobial therapy may be individualized based on:
  - Neutrophil recovery
  - Rapidity of defervescence
  - Specific site of infection
  - Infecting pathogen
  - Patient's underlying illness
- Catheter removal for septic phlebitis, tunnel infection, or port pocket infection
  - Catheter removal highly recommended if persistent positivity

→

- Skin/soft tissue: 5–14 days
- Bloodstream infection
  - Gram-negative: 7–14 days
  - Gram-positive: 7–14 days
  - *S. aureus*: typically requires 4 weeks (some institutions may use a shorter duration based on ID consultation) after first negative blood culture; ID consult strongly recommended (ID consult is associated with decreased mortality)
  - Yeast: ≥2 weeks after first negative blood culture
  - Catheter removal favored for bloodstream infections with *Candida* or other yeasts, *S. aureus*, *Pseudomonas aeruginosa*, *Corynebacterium jeikeium*, *Acinetobacter* spp., nontuberculous mycobacteria, molds, vancomycin-resistant enterococci (VRE), *Stenotrophomonas maltophilia*, and other MDROs
- Bacterial sinusitis: 7–14 days
- Bacterial pneumonia: 5–14 days
- Fungal (mold and yeast):
  - *Candida*: minimum of 2 weeks after first negative blood culture
  - Mold: minimum of 12 weeks
- Viral:
  - HSV/VZV: 7–10 days (category 1); acyclovir, valacyclovir, or famciclovir (uncomplicated, localized disease to the skin)
  - Influenza: a minimum 5-day course of oseltamivir[cc]

[g] See Antibacterial Agents (FEV-A) for dosing, spectrum, and specific comments/cautions.
[l] See Antifungal Agents (FEV-B) for dosing, spectrum, and specific comments/cautions.
[m] See Antiviral Agents (FEV-C) for dosing, spectrum, and specific comments/cautions. The antivirals are not equal in terms of efficacy, side effects, and resistance.
[cc] A minimum 5-day course is standard based on data from ambulatory and otherwise healthy individuals with intact immune systems; some centers consider longer courses or higher doses (eg, 150 mg) for the highly immunocompromised, but there is no proven benefit to prolonged therapy.

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FOLLOW-UP THERAPY FOR RESPONDING DISEASE

重试 错误原因

SUGGESTED MINIMUM DURATION OF THERAPY FOR DOCUMENTED INFECTION

重试 错误原因

These are general guidelines for patients with uncomplicated disease and may need to be revised for individual patients. Treatment duration can be modified depending on infection severity and patient factors.

重试 错误原因

重试 错误原因

记录感染

·应对有记录的感染进行靶向治疗·重新评估经验性广谱治疗·抗菌治疗的降级和持续时间可根据以下因素进行个体化： 中性粒细胞恢复 退热速度

特定感染部位
侵染病原 患者的基础疾病·因脓毒性静脉炎、隧道感染或输液港袋感染而移除导管
如果持续阳性，强烈建议移除导管

·皮肤/软组织：5–14 天 革兰氏阴性：7–14 天 革兰氏阳性：7–14 天

S.金黄色：首次血培养阴性后通常需要 4 周（某些机构可能根据 ID 咨询使用更短的持续时间）；ID 咨询
强烈推荐（ID 咨询与死亡率降低相关） 酵母菌：首次血培养阴性后≥2 周 导管移除有利于念珠菌或其他血流感染

酵母、S.金黄色葡萄球菌，铜绿假单胞菌，杰氏棒状杆菌，不动杆菌属，非结核分枝杆菌、霉菌、万古霉素耐药肠球菌（VRE）、嗜麦芽窄食单胞菌和其他MDRO·细菌性鼻窦炎：7–14 天·细菌性肺炎：5–14 天·真菌（霉菌和酵母菌）： 念珠菌：首次血培养阴性后至少 2 周 霉菌：至少 12 周·病毒： HSV/VZV：7–10 天（1 类）;阿昔洛韦、伐昔洛韦或泛昔洛韦（无并发症，皮肤局部疾病）

流感：至少 5 天的奥司他韦疗程

[g] 有关剂量、光谱和具体注释/注意事项，请参见抗菌剂（FEV–A）。
[l] 有关剂量、谱和具体注释/注意事项，请参见抗真菌药（FEV–B）。
[m] 有关剂量、谱和具体注释/注意事项，请参见抗病毒药物（FEV–C）。抗病毒药物在疗效、副作用和耐药性方面是不平等的。
[cc] 基于来自具有完整免疫系统的非卧床和其他健康个体的数据，最少 5 天的疗程是标准的;一些中心考虑为高度免疫功能低下的患者提供更长的疗程或更高的剂量（例如，150 mg），但没有证明延长治疗的益处。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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ANTIBACTERIAL AGENTS: GRAM-POSITIVE ACTIVITY ONLY[a]

Agents[b]	Dose[c]	Spectrum[e]	Comments/Precautions
Vancomycin	15 mg/kg IV every 12 hours, Loading dose[c] may be considered	Gram-positive organisms, with exception of VRE and a number of rare Gram-positive organisms	IV formulation • Should not be considered as routine therapy for neutropenia and fever unless certain risk factors are present • Dosing individualized with therapeutic drug monitoring (TDM)
Daptomycin	6–10 mg/kg/day IV[d] with higher doses indicated for specific infections	• Gram-positive organisms • Has in vitro activity against VRE	• Weekly creatine phosphokinase (CPK) to monitor for rhabdomyolysis • Not indicated for pneumonia due to inactivation by pulmonary surfactant
Linezolid	600 mg PO/IV every 12 hours	Gram-positive organisms, including VRE	• Hematologic toxicity (typically with prolonged cases, >2 weeks) may occur; thrombocytopenia most common (0.3%–10%) • Serotonin syndrome is rare; use cautiously with selective serotonin reuptake inhibitors (SSRIs)[1] • Treatment option for VRE and MRSA • Peripheral/optic neuropathy with long-term use
Tedizolid	200 mg QD	Gram-positive organisms, including VRE	Less hematologic toxicity compared to linezolid with prolonged use

Footnotes

[a] Drug resistance or clinical failure may dictate the use of newer restricted antibiotics, and an ID consult is recommended.

[b] These drugs are not recommended as monotherapy for fever in the setting of neutropenia and should only be added if there is high suspicion of infection with resistant Gram-positive organisms or if certain risk factors are present (FEV-D).

[c] These are standard dosing recommendations for adult patients; there are other situations where dose adjustments are required. Consult pediatric guidelines for recommended dosing in pediatric patients. Adjustments should also be made for patients with renal insufficiency and obesity according to institutional guidelines.

[d] Higher doses of daptomycin (8–12 mg/kg) are recommended for certain bloodstream infections (eg, enterococci). ID consult is strongly recommended.

[e] Once culture data are available, directed therapy may be initiated following an ID consult as appropriate for Gram-positive pathogens.

References

[1] Boyer EW, Shannon M. The serotonin syndrome. N Engl J Med 2005;352:1112-1120.

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抗生素：仅革兰氏阳性活性

剂	Dose	频谱	评论/注意事项
万古霉素	15 mg/kg IV，每 12 小时一次，可考虑负荷剂量	革兰氏阳性菌，VRE 和一些罕见革兰氏阳性菌除外	IV 制剂·不应被视为中性粒细胞减少症和发热的常规治疗，除非存在某些风险因素 • 通过治疗药物监测（TDM）进行个体化给药
达托霉素	6–10 mg/kg/天 IV 给药，适用于 特定感染	·对 VRE 具有体外活性	·每周肌酸磷酸激酶（CPK）监测横纹肌溶解
Linezolid	600 mg PO/IV，每 12 小时一次	革兰氏阳性菌，包括 VRE	• 可能发生血液学毒性（通常为长期病例，>2 周）;血小板减少症最常见（0.3%–10%） • 血清素综合征是罕见的;谨慎使用选择性 5–羟色胺再摄取抑制剂（SSRIs）·VRE 和 MRSA 的治疗选择·长期使用的周围/视神经病变
特地唑胺	200 mg QD	革兰氏阳性菌，包括 VRE	与利奈唑胺相比，长期使用血液学毒性更低

脚注
[a] 耐药性或临床失败可能要求使用新的限制性抗生素，建议进行 ID 咨询。
[b] 这些药物不推荐作为中性粒细胞减少症发热的单药治疗，只有在高度怀疑耐药革兰氏阳性菌感染或存在某些风险因素（FEV–D）时才应添加。

[c] 这些是成人患者的标准剂量建议;还有其他需要调整剂量的情况。咨询儿科指南，儿科患者的推荐剂量。对于肾功能不全和肥胖患者，也应根据机构指南进行调整。
d 对于某些血流感染（如肠球菌），建议使用较高剂量的达托霉素（8–12 mg/kg）。强烈建议咨询 ID。
e 一旦获得培养数据，可在 ID 咨询后开始针对革兰氏阳性病原体的定向治疗。引用

[1] 放大图片作者：Shannon M.血清素综合症。新英格兰医学杂志 2005;352：1112–1120。

注：除非另有说明，否则所有建议均为 2A 类。
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ANTIBACTERIAL AGENTS: ANTI-PSEUDOMONAL[f]

Agents	Dose[c]	Spectrum[g]	Comments/Precautions[h]
Cefepime	2 g IV every 8 hours	• Broad-spectrum activity against most Gram-positive and Gram-negative organisms • Not active against most anaerobes and *Enterococcus* spp.	• Use for suspected/proven CNS infection with susceptible organism • Empiric therapy for neutropenic fever (category 1) • Mental status changes may occur, especially in the setting of renal dysfunction
Ceftazidime	2 g IV every 8 hours	• Poor Gram-positive activity ▸ Breakthrough streptococcal infections reported; add Gram-positive agent to empiric neutropenic fever treatment • Not active against most anaerobes and *Enterococcus* spp.	• Use for suspected/proven CNS infection with susceptible organism • Empiric therapy for neutropenic fever (category 2B; due to resistance among certain Gram-negative rods)
Imipenem/ cilastatin sodium	500 mg IV every 6 hours	• Broad-spectrum activity against most Gram-positive, Gram-negative, and anaerobic organisms • Preferred against extended-spectrum beta-lactamase (ESBL)–producing organisms and serious *Enterobacter* infections • Carbapenem-resistant Gram-negative rod infections are an increasing problem at a number of centers	• Use for suspected intra-abdominal source • Meropenem is preferred over imipenem for suspected/proven CNS infection • Carbapenems may lower seizure threshold in patients with CNS malignancies or infection or with renal insufficiency • Ertapenem does not have anti-pseudomonal activity • Empiric therapy for neutropenic fever (category 1) • Data are limited, but it is expected that doripenem, like meropenem, would be efficacious
Meropenem	1–2 g IV every 8 hours or 500 mg IV every 6 hours		
Piperacillin/ tazobactam	3.375 g IV every 6 hours (mild-moderate infections) or 4.5 g IV every 6 hours (severe infections including fever and neutropenia) Administered over 30 min (Some institutions use extended infusion: 3.375 g or 4.5 g every 8 hours administered over 4 hours)	• Broad-spectrum activity against most Gram-positive, Gram-negative, and anaerobic organisms	• Use for suspected intra-abdominal source • Not recommended for meningitis • Empiric therapy for neutropenic fever (category 1)

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

Footnotes on FEV-A 3 OF 3

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抗生素：抗假单胞菌

剂	Dose	频谱	备注/注意事项
Cefepime	2 g IV every 8 hours 重试 错误原因	• Broad–spectrum activity against most Gram–positive and Gramnegative organisms • Not active against most anaerobes and Enterococcus spp. 重试 错误原因	• Use for suspected/proven CNS infection with susceptible organism • Empiric therapy for neutropenic fever (category 1) • Mental status changes may occur, especially in the setting of renal dysfunction 重试 错误原因
头孢他	每 8 小时 2 g IV	·革兰氏阳性活动不良 突破性链球菌 感染报告;添加革兰氏– ·对大多数厌氧菌和肠球菌属没有活性。	·用于疑似/证实的易感微生物的中枢神经系统感染·中性粒细胞减少性发热的经验性治疗（2B 类;由于某些革兰氏阴性杆菌的耐药性）
亚胺培南/ 西司他丁钠	500 mg IV，每 6 小时一次	·对大多数革兰氏阳性、革兰氏阴性和厌氧微生物具有广谱活性·优选对产超广谱β–内酰胺酶（ESBL）的微生物和严重	·对于疑似/确诊的 CNS 感染，美罗培南优于亚胺培南 • 碳青霉烯类可降低癫痫发作阈值在 CNS 恶性肿瘤患者中，或 ·厄他培南无抗假单胞菌活性·经验性治疗血小板减少性发热（1 类）·数据有限，但预计多利培南与美罗培南一样，
美罗培南	1–2 g IV 每 8 小时或 500 mg IV 每 6 小时	肠杆菌感染 ·耐碳青霉烯类革兰氏阴性杆菌感染是许多中心日益严重的问题	有效
氯唑西林/ 唑巴坦	3.375 g IV 每 6 小时一次（轻度–中度感染）或 4.5 g IV 每 6 小时一次（重度感染，包括发热和中性粒细胞减少症）给药时间超过 30 分钟（某些机构 使用延长输注：3.375 g 或 4.5 g 每 8 小时给药一次，持续 4 小时）	·对大多数革兰氏阳性、革兰氏阴性和厌氧微生物具有广谱活性	·用于可疑的腹腔内来源·不推荐用于脑膜炎·经验疗法用于血小板减少性发热（1 类）

注：除非另有说明，否则所有建议均为 2A 类。
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脚注见

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ANTIBACTERIAL AGENTS: OTHER

Agents[h]	Dose[c]	Spectrum	Comments/Cautions
Aminoglycosides • Amikacin • Gentamicin • Tobramycin	Consider extended interval dosing for patients with normal renal function (eg, 5–7 mg/kg every 24 hours)	Activity primarily against Gram-negative organisms	Sometimes used as part of combination therapy in patients who are seriously ill or hemodynamically unstable
Ciprofloxacin	500–750 mg PO every 12 hours or 400 mg IV every 8–12 hours	• Ciprofloxacin has good activity against Gram-negative and atypical organisms (eg, *Legionella* spp.) but less activity than levofloxacin or moxifloxacin against Gram-positive organisms • Ciprofloxacin alone has no activity against anaerobes	• Avoid for empiric therapy if patient recently treated with fluoroquinolone prophylaxis • Increasing Gram-negative resistance in many centers • Fluoroquinolone side effects should be taken into consideration (see the FDA warnings)
Levofloxacin	500–750 mg PO or IV daily	• Good activity against Gram-negative and atypical organisms (eg, *Legionella* spp.) • Improved Gram-positive activity compared to ciprofloxacin • Levofloxacin has no activity against anaerobes • Moxifloxacin is more active against anaerobes than other fluoroquinolones, but has insufficient activity against some gram-negative organisms, including *Pseudomonas*	• Prophylaxis may increase bacterial resistance and superinfection[2] • Limited studies as empiric therapy in patients with fever and neutropenia • Prophylaxis in patients with neutropenia[3,4] • Data support fluoroquinolones for prophylaxis; fluoroquinolone side effects should be taken into consideration (see the FDA warnings)
Moxifloxacin	400 mg PO or IV daily		
Metronidazole	500 mg PO (preferred) every 8–12 hours	Good activity against anaerobic organisms	• Associated with peripheral neuropathy with prolonged use (>4 weeks)
Trimethoprim/ sulfamethoxazole (TMP/SMX)	Prophylaxis: Single strength daily or double strength 3 times per week Therapy: 15 mg/kg/d in divided doses every 6–8 hours based on the trimethoprim component	Activity against *P. jirovecii* and other relevant pathogens, including *Toxoplasma gondii* and *Nocardia*	• Highly effective as prophylaxis against *P. jirovecii* in patients at high risk (INF-6) • Monitor for renal insufficiency, myelosuppression, hepatotoxicity, and hyperkalemia • Interactions with methotrexate

Footnotes

[c] These are standard dosing recommendations for adult patients; there are other situations where dose adjustments are required. Consult pediatric guidelines for recommended dosing in pediatric patients. Adjustments should also be made for renal insufficiency and patients with obesity according to institutional guidelines.

[f] Emerging data may support extended or continuous infusion of beta-lactam therapies. For highly resistant infections, see Discussion for recommendations regarding alternative antibiotics with restricted availability.

[g] No agents listed are active against MRSA or VRE.

[h] There are multiple new agents that may be useful against multiply drug-resistant bacteria including ceftolozane-tazobactam, ceftazidime/avibactam, meropenem-vaborbactam, imipenem/cilastatin/relebactam, and cefiderocol. These agents have variable spectrum and activity and should only be used with expert consultation.

References

[2] Baden LR. Prophylactic antimicrobial agents and the importance of fitness. N Engl J Med 2005;353:1052-1054.

[3] Bucaneve G, Micozzi A, Menichetti F, et al. Levofloxacin to prevent bacterial infection in patients with cancer and neutropenia. N Engl J Med 2005;353:977-987.

[4] Cullen M, Billingham SN, Gaunt C, et al. Antibacterial prophylaxis after chemotherapy for solid tumors and lymphomas. N Engl J Med 2005;353:988-998.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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抗生素：其他

药剂剂量		频谱	备注/注意事项
氨基糖苷 ·阿米卡星 ·庆大霉素 ·妥布霉素	对于肾功能正常的患者，考虑延长给药间隔（例如，每 24 小时 5–7 mg/kg）	主要针对革兰氏阴性菌的活性	有时作为联合治疗的一部分用于重病或血流动力学不稳定的患者
Ciprofloxacin	500–750 mg PO 每 12 小时一次或 400 mg IV 每 8–12 小时一次	• 环丙沙星对革兰阴性杆菌有较好的抗菌活性和非典型微生物（如军团菌属）但不太比左氧氟沙星或丙氧氟沙星对革兰氏阴性杆菌的活性更高，阳性生物体 • 单独使用环丙沙星对厌氧菌没有活性	·如果患者最近接受过氟喹诺酮预防治疗，应避免经验性治疗·许多中心的革兰氏阴性耐药性增加 • 氟喹诺酮类药物的副作用应考虑到注意事项（参见 FDA 警告）
左氧	每日 500–750 mg PO 或 IV	·对革兰氏阴性和非典型生物体（例如军团菌属）具有良好的活性·与环丙沙星相比，改善的革兰氏阳性活性	·预防可能增加细菌耐药性和双重感染·发热和中性粒细胞减少患者经验性治疗的研究有限·中性粒细胞减少患者的预防
莫西沙星	每日 400 mg PO 或 IV	• 左氧氟沙星对厌氧菌无活性 • 莫西沙星对厌氧菌的活性高于其他氟喹诺酮类药物，但活性不足针对一些革兰氏阴性菌，包括假单	• 数据支持氟喹诺酮类药物用于预防;应考虑氟喹诺酮类药物的副作用（见 FDA警告）
甲硝唑	500 mg PO（首选），每 8–12 小时一次	对厌氧微生物具有良好的活性	·与长期使用（>4 周）的周围神经病变相关
甲氧苄啶/磺胺甲(TMP/SMX)	预防：每日单剂量或双倍强度每周 3 次 治疗：15 mg/kg/d，分次给药 根据甲氧苄啶组分，每 6–8 小时给药一次	对耶氏疟原虫和其他相关病原体的活性，包括弓形虫和诺卡氏菌	• 在以下患者中预防吉罗维氏疟原虫非常有效：高风险（INF–6） • 监测肾功能不全，骨髓抑制，肝毒性和高钾血症·与甲氨蝶呤的相互作用

脚注

c 这些是成人患者的标准剂量建议;还有其他需要调整剂量的情况。关于儿科患者的推荐剂量，请参阅儿科指南。
还应根据机构指南对肾功能不全和肥胖患者进行调整。
f 新出现的数据可能支持β–内酰胺治疗的延长或连续输注。对于高度耐药的感染，请参阅讨论，了解关于可用性受限的替代抗生素的建议。

g 所列药剂均不对 MRSA 或 VRE 具有活性。
h 有多种新药剂可用于对抗多重耐药细菌，包括头孢洛扎–他唑巴坦、头孢他啶/阿维巴坦、美罗培南–戊硼巴坦、亚胺培南/西司他丁/曲马巴坦和头孢地罗可。这些药物具有不同的光谱和活性，只能在专家咨询后使用。

引用

2 Baden LR.预防性抗菌药物及健身的重要性。新英格兰医学杂志 2005;353：1052–1054。
3 Bucaneve G，Micozzi A，Menichetti F，et al.左氧氟沙星预防癌症和中性粒细胞减少患者细菌感染的研究。新英格兰医学杂志 2005;353：977–987。
4 Cullen M，比灵厄姆 SN，Gaunt C，et al.实体瘤和淋巴瘤化疗后的抗菌预防。新英格兰医学杂志 2005;353：988–998。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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ANTIFUNGAL AGENTS: AZOLES

Azoles as a class have important drug interactions, especially with the newer therapeutic agents. Please carefully review. Drug-drug interactions are common and need to be closely monitored (consult package inserts for details).

Azoles[a]	Dose	Spectrum	Comments/Cautions
Fluconazole	In adults with normal renal function: typical dosing is 400 mg IV/PO daily. May vary depending on indication and *Candida* susceptibility.	• Active against most *Candida* species • Active against coccidioidomycosis and *C. neoformans* • *Candida* susceptibility testing is recommended and should guide treatment decisions	• *C. glabrata* is associated with variable resistance in vitro, *C. krusei* is intrinsically resistant, and *C. auris* is typically resistant • Inactive against molds (eg, *Aspergillus* spp., *Mucorales*)
Isavuconazonium sulfate[b]	Loading dose 372 mg IV/PO every 8 hours x 6 doses then maintenance dose 372 mg IV/PO daily	• Active against invasive aspergillosis and mucormycosis in patients with cancer and in HCT recipients[1,2,3]	• Can be considered in patients intolerant or refractory to first-line anti-mold therapy • May shorten QTc interval • Moderate inhibitor of CYP3A4, may be less clinically significant than voriconazole, itraconazole, or posaconazole
Itraconazole[b]	Loading dose 200 mg PO TID x 3 days, then maintenance dose 200 mg PO BID	• Active against *Candida, Aspergillus* spp., and some of the rarer molds • Active against dimorphic fungi and *C. neoformans* • *Candida* susceptibility testing is recommended and should guide treatment decisions	• Itraconazole has negative inotropic properties and is contraindicated in patients with significant cardiac systolic dysfunction • H2 blockers and proton pump inhibitors (PPIs) may inhibit absorption of capsule formulation. Oral liquid formulation is preferred for improved absorption. • SUBA-itraconazole has improved absorption

[a] Azoles inhibit fungal cell membrane synthesis and inhibit cytochrome P450 isoenzymes that may lead to impaired clearance of other drugs metabolized by this pathway. Fluconazole is a less potent inhibitor of cytochrome P450 isoenzymes than the mold-active azoles. QTc prolongation has been reported with all azole antifungals except isavuconazole. With broad use, antimicrobial-resistant organisms may emerge and may have implications for future activity of these agents.
[b] TDM is routinely used in managing itraconazole, posaconazole, and voriconazole. TDM is not routinely used for isavuconazole.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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抗真菌剂：唑类

唑类化合物具有重要的药物相互作用，特别是与新的治疗药物。请仔细阅读。药物间相互作用很常见，需要密切监测（详情请参阅药品说明书）。唑类剂量谱备注/注意事项

氟康唑	在肾功能正常的成人中， 功能：典型剂量为每日 400 mg IV/PO 可能因适应症和念珠菌敏感性而异。	·对大多数念珠菌属有活性·对球孢子菌病和 C./新生 • 建议进行念珠菌敏感性试验，并应指导治疗决策	C. glabrata 在体外与可变抗性相关，C. krusei 具有内在抗性，C.·对霉菌（例如，曲霉属，毛霉目）
Isavuconazonium 的苏岩	负荷剂量 372 mg IV/ PO，每 8 小时一次，6 次给药，然后维持剂量 372 mg IV/PO，每日一次	·对癌症患者和 HCT 接受者的侵袭性曲霉病和毛霉菌病有效	·可能缩短 QTc 间期·CYP 3A 4 的中度抑制剂，临床意义可能低于伏立康唑、伊曲康唑或泊沙康唑
Itraconazole	负荷剂量 200 mg PO TID x 3 天，然后维持剂量 200 mg PO BID	• Active against Candida, Aspergillus spp., and some of the rarer molds • Active against dimorphic fungi and C. neoformans • Candida susceptibility testing is recommended and should guide treatment decisions	• Itraconazole has negative inotropic properties and is contraindicated in patients with significant cardiac systolic dysfunction • H2 blockers and proton pump inhibitors (PPIs) may inhibit absorption of capsule formulation. Oral liquid formulation is preferred for improved absorption. • SUBA–伊曲康唑改善了吸收

[a] 唑类抑制真菌细胞膜合成并抑制细胞色素 P450 同工酶，这可能导致通过该途径代谢的其他药物的清除受损。氟康唑对细胞色素 P450 同工酶的抑制作用弱于霉菌活性唑类药物。除艾沙康唑外，所有唑类抗真菌药均报告了 QTc 延长。随着广泛使用，可能会出现耐药性微生物，并可能对这些药物的未来活性产生影响。

[b] TDM 通常用于管理伊曲康唑、泊沙康唑和伏立康唑。艾沙康唑未常规使用 TDM。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

引用

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ANTIFUNGAL AGENTS: AZOLES

Azoles as a class have important drug interactions, especially with the newer therapeutic agents. Please carefully review. Drug-drug interactions are common and need to be closely monitored (consult package inserts for details).

Azoles[a]	Dose	Spectrum	Comments/Cautions
Posaconazole[b]	Prophylaxis or treatment: ▸ IV injection and delayed-release (DR) tablet: Loading dose 300 mg DR tablet PO BID OR 300 mg IV BID on Day 1 and then maintenance dose 300 mg PO daily	• Effective as prophylaxis in patients with neutropenia with myelodysplastic syndrome and AML,[6] and in HCT recipients with significant GVHD[7] • Active against *Candida*, *Aspergillus* spp., some *Mucorales* spp., and some of the rarer molds • Active against dimorphic fungi and *C. neoformans* • Limited data for histoplasmosis	• Used for treatment of refractory infection (but not FDA-approved) in several invasive fungal diseases • Tablets are better absorbed and preferred, except in circumstances where alternative dosing is needed • IV posaconazole or alternative antifungal therapy should be considered for patients who cannot eat a full meal or tolerate an oral nutritional supplement • PPIs decrease posaconazole plasma concentration with oral suspension. Liquid formulation should be administered with a full meal or liquid nutritional supplement or an acidic carbonated beverage. • IV formulation should be used with caution in patients with significant renal dysfunction • FDA-approved for invasive aspergillosis
Voriconazole[b]	• Treatment of invasive aspergillosis[4] ▸ Loading dose: 6 mg/kg IV or 400 mg PO BID x 2 doses on Day 1 ▸ Maintenance: 4 mg/kg IV BID OR the following oral maintenance dosing ◊ ≥40 kg: 200 mg PO BID ◊ <40 kg: 100 mg PO BID • Treatment of candidemia in patients without neutropenia[5] ▸ Loading dose: 6 mg/kg IV or 400 mg PO BID x 2 doses on Day 1 ▸ Maintenance: 3–4 mg/kg IV BID OR the following oral maintenance dosing ◊ ≥40 kg: 200 mg PO BID ◊ <40 kg: 100 mg PO BID	• Active against *Candida, Aspergillus* spp., and some of the rarer molds • Active against dimorphic fungi and *C. neoformans* • Standard of care as primary therapy for invasive aspergillosis (category 1)[4,8] • Effective in candidemia in patients without neutropenia[5]	• Poor activity against *Mucorales* • Long-term complications may include increased risk for squamous cell carcinoma and hyperphosphatemia • Fluorosis may occur with prolonged use and is associated with bone/muscle pain • Evidence for combination therapy with an echinocandin remains limited[9] • IV formulation should be used with caution in patients with significant renal dysfunction • Visual disturbances and hallucinations may occur on therapy

[a] Azoles inhibit fungal cell membrane synthesis and inhibit cytochrome P450 isoenzymes that may lead to impaired clearance of other drugs metabolized by this pathway. Fluconazole is a less potent inhibitor of cytochrome P450 isoenzymes than the mold-active azoles. QTc prolongation has been reported with all azole antifungals except isavuconazole. With broad use, antimicrobial-resistant organisms may emerge and may have implications for future activity of these agents.

[b] TDM is routinely used in managing itraconazole, posaconazole, and voriconazole. TDM is not routinely used for isavuconazole.

Note: All recommendations are category 2A unless otherwise indicated.
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抗真菌剂：唑类

唑类化合物具有重要的药物相互作用，特别是与新的治疗药物。请仔细阅读。药物间相互作用很常见，需要密切监测（详情请参阅药品说明书）。

唑类	Dose	频谱	备注/注意事项
泊沙康唑 [b]	预防或治疗： IV 注射剂和延迟释放（DR）片剂：负荷剂量 300 第 1 天口服 BID 或 300 mg IV BID，然后维持剂量 300 mg PO 每日一次	• 有效预防 中性粒细胞减少伴骨髓增生异常综合征和 AML 的患者，6 和 HCT 接受者中具有显著的 GVHD。一些毛霉菌属，·对二型真菌和 C.组织胞浆菌病的数据有限	·用于治疗几种侵袭性真菌疾病中的难治性感染（但未获得 FDA 批准）·片剂吸收更好，并且是首选，除非需要替代剂量的情况 • 静脉泊沙康唑或替代抗真菌治疗应 考虑用于不能进食全餐或耐受口服营养补充剂的患者 • PPI 可降低泊沙康唑口服混悬液的血药浓度。液体制剂应与全餐或液体营养补充剂或酸性碳酸饮料一起给药。 ·严重肾功能不全患者应谨慎使用 IV 制剂
Voriconazole [b]	·侵袭性曲霉病的治疗 负荷剂量：第 1 天 6 mg/kg IV 或 400 mg PO BID x 2 次给药 维持：4 mg/kg IV BID 或以下经口维持给药 Dad ≥40 kg：200 mg PO BID Dad <40 kg：100 mg PO BID ·治疗无中性粒细胞减少症的患者中的念珠菌血症 负荷剂量：第 1 天 6 mg/kg IV 或 400 mg PO BID x 2 次给药 维持：3–4 mg/kg IV BID 或以下经口 维持给药 Dad ≥40 kg：200 mg PO BID Dad <40 kg：100 mg PO BID	·对念珠菌、曲霉属有效，和一些罕见的霉菌·活性对二型真菌和 C.新生 ·标准治疗作为侵袭性曲霉病（1 类）的主要治疗方法 [4,8] • 对念珠菌血症有效 患者无 细胞减少	·对毛霉菌的活性差·长期并发症可能包括鳞状细胞癌和高磷血症的风险增加·长期使用可能发生氟中毒，并与骨/肌肉疼痛有关·与棘白菌素联合治疗的证据仍然有限 9 ·严重肾功能不全患者应谨慎使用 IV 制剂·治疗时可能出现视觉障碍和幻觉参考文献

a 唑类抑制真菌细胞膜合成，并抑制细胞色素 P450 同工酶，这可能导致通过该途径代谢的其他药物的清除受损。氟康唑对细胞色素 P450 同工酶的抑制作用弱于霉菌活性唑类药物。除艾沙康唑外，所有唑类抗真菌药均报告了 QTc 延长。随着广泛使用，可能会出现耐药性微生物，并可能对这些药物的未来活性产生影响。

B TDM 通常用于管理伊曲康唑、泊沙康唑和伏立康唑。艾沙康唑未常规使用 TDM。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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ANTIFUNGAL AGENTS: AMPHOTERICIN B FORMULATIONS[c]

Amphotericin B Formulations	Dose	Spectrum	Comments/Cautions
Amphotericin B deoxycholate (AmB-D)[d]	Varies by indication, generally 0.5–1.5 mg/kg IV daily	• Broad spectrum of antifungal activity including *Candida*, *Aspergillus* spp. (excluding *A. terreus*), *Mucorales*, rarer molds, *C. neoformans*, and dimorphic fungi • Several species of fungi may be intrinsically resistant to amphotericin (see Discussion) (eg, *Scedosporium, Lomentospora*)	• Substantial infusional and renal toxicity including electrolyte wasting • Saline loading may reduce nephrotoxicity • Infusional toxicity may be managed with antipyretics, an antihistamine, and meperidine (for rigors) • Slowing the rate of infusion is an additional way to manage amphotericin infusion reactions
Amphotericin B lipid complex (ABLC)	5 mg/kg IV daily		Reduced infusional and renal toxicity compared to AmB-D
Liposomal amphotericin B (L-AMB)	3–5 mg/kg IV daily[10,e]		

[c] Can be considered for prophylaxis with ID consult for appropriate dosing recommendations.
[d] AmB-D is not preferred whenever L-AMB or ABLC is available.
[e] In patients who are highly immunocompromised, 3 mg/kg liposomal amphotericin B was just as effective against aspergillosis compared to 10 mg/kg with significantly less toxicities. Optimal dosing for mucormycosis may require higher dosing based on other literature.

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抗真菌药：两性霉素 B 制剂

两性霉素 B 制剂	Dose	频谱	备注/注意事项
两性霉素 B 脱氧胆酸（AmB–D）	因适应症而异，通常为每日 0.5–1.5 mg/kg IV	·广谱抗真菌活性，包括念珠菌属、曲霉属。（不包括 A。terreus），毛霉目，rarer molds，C.新形菌和二型真菌 ·几种真菌可能固有地对阿氏菌素具有抗性（参见讨论）（例如，Scedosporium，长孢子虫）	·大量输注和肾毒性，包括电解质浪费·生理盐水负荷可减少肾毒性·输注毒性可使用退烧药、抗组胺药和哌替啶（用于寒战）进行管理·减慢输注速率是管理阿霉素输注反应的另一种方法
两性霉素 B 脂质复合物（ABLC）	5 mg/kg IV，每日一次		与 AmB–D 相比，输注和肾毒性降低
脂质体阿霉素 B（L–AMB）	3–5 mg/kg，每日 IV		

c 可以考虑进行预防，并咨询 ID 以获得适当的剂量建议。
d 当 L–AMB 或 ABLC 可用时，AmB–D 不是首选。
[e] 在高度免疫功能低下的患者中，与 10 mg/kg 相比，3 mg/kg 脂质体阿替霉素 B 对曲霉病同样有效，但毒性显著降低。根据其他文献，毛霉菌病的最佳剂量可能需要更高的剂量。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

引用

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ANTIFUNGAL AGENTS: ECHINOCANDINS

Echinocandins[9,f]	Dose	Spectrum	Comments/Cautions
Anidulafungin	200 mg IV x 1 dose, then 100 mg/IV daily	• Primary therapy for candidemia and invasive candidiasis (category 1)[11] • *C. auris* may be resistant to echinocandins • May be used as part of a second-line or subsequent regimen for invasive aspergillosis • Not reliable or effective against most other fungal pathogens (eg, *Trichophyton, Cryptococcus, Mucorales*)	• Echinocandins have poor CNS, urinary tract, and eye penetration • Excellent safety profile
Caspofungin	• 70 mg IV x 1 dose, then 50 mg IV daily (35 mg IV daily for patients with moderate liver disease) • Some investigators use 70 mg IV daily as therapy for aspergillosis in second-line therapy		
Micafungin	• 100 mg IV daily for candidemia and 50–100 mg/d IV as prophylaxis • 150 mg IV daily used at some centers for *Aspergillus* spp. infection as second-line therapy		

[f] A number of centers use combination voriconazole and an echinocandin for invasive aspergillosis based on clinical data. Evidence for combination therapy remains limited.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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抗真菌药：棘白菌素类

棘白菌素剂量		频谱	备注/注意事项
芬净 Caspofungin	200 mg IV x 1 剂，然后 100 mg/IV 每日一次 · 70 mg IV x 1 剂量，然后每天 50 mg IV（中度肝病患者每天 35 mg IV）	·念珠菌血症和侵袭性念珠菌病的主要治疗（1 类） • C.耳可能对棘白菌素有抗性 ·可作为侵袭性曲霉病二线或后续治疗方案的一部分	·棘白菌素具有较差的 CNS、泌尿道和眼睛穿透性 • 出色的安全性
米卡芬	·每天 100 mg IV 用于念珠菌血症，50–100 mg/d IV 用于预防感染作为二线治疗	• 不可靠的或对…无效的 大多数其他真菌病原体 （eg、发癣菌属、隐球菌属、毛霉目）	

[f] 许多中心根据临床数据联合使用伏立康唑和棘白菌素治疗侵袭性曲霉病。联合治疗的证据仍然存在有限公司

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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ANTIVIRAL AGENTS[a]

Agent	Typical Dosing Based on Indication[c]	Spectrum	Comments/Cautions
Acyclovir[b]	• Prophylaxis[d]: HSV (400–800 mg PO BID); VZV in allogeneic HCT recipients (400–800 mg PO BID)[1] • Post-VZV exposure prophylaxis: 800 mg PO 5 times daily • Treatment: Significant mucocutaneous HSV (5 mg/kg IV every 8 h for 7–10 d); single dermatomal VZV (800 mg PO 5 times daily or 10 mg/kg IV every 8 h for 7–10 d); disseminated HSV or VZV including viral encephalitis (10 mg/kg IV every 8 h)[2]	HSV VZV	• Hydration to avoid crystal nephropathy with high dose • Dosing based on ideal body weight
Famciclovir	• Prophylaxis: HSV or VZV (250 mg PO BID) • Treatment: HSV (250 mg PO TID) or VZV (500 mg PO TID)[3,4]	HSV VZV	No data for oncologic-related prophylaxis
Ganciclovir	• Preemptive therapy for CMV: 5 mg/kg every 12 h; if CMV remains detectable, further ID evaluation may be required • Treatment: CMV disease (5 mg/kg every 12 h for induction followed by 5 mg/kg daily for maintenance and resolution of all symptoms)	CMV HSV VZV	• May cause bone marrow suppression • Clinical data are limited for HHV-6 and HHV-8
Valacyclovir[b]	• Prophylaxis[d]: HSV or VZV (500 mg PO BID) preferred over oral acyclovir for VZV • Treatment: HSV or VZV (1 g PO TID)[2] preferred over oral acyclovir for HSV or VZV	HSV VZV	
Valganciclovir	• Preemptive therapy and treatment for CMV: Induction with 900 mg PO BID for induction and until negative test; consider additional 900 mg PO daily for maintenance after a negative test	CMV HSV VZV	• May cause bone marrow suppression • Clinical data are limited for HHV-6 and HHV-8

Note: All recommendations are category 2A unless otherwise indicated.
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Footnotes and References FEV-C (4 of 4)

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抗病毒剂

剂	基于适应症的典型给药	Spectrum 注释/注意事项	
阿昔洛韦	·预防：HSV（400–800 mg PO BID）;同种异体 HCT 接受者中的 VZV（400– 800 mg PO BID） • VZV 暴露后预防：800 mg PO，每日 5 次 • 治疗：显著的皮肤粘膜 HSV（5 mg/kg IV，每 8 h 一次，持续 7–10 d）；单次皮节 VZV（800 mg PO，每日 5 次或 10 mg/kg IV，每 8 h，持续 7–10 d）;播散性 HSV 或 VZV，包括病毒性脑炎（10 mg/kg IV，每 8 h 一次）	HSV VZV	·水化以避免高剂量晶体肾病·根据理想体重给药
泛昔	·预防：HSV 或 VZV（250 mg PO BID） • 治疗：HSV（250 mg PO TID）或 VZV（500 mg PO TID）	HSV VZV	无肿瘤相关预防的数据
更昔	• CMV 的抢先治疗：每 12 小时 5 mg/kg;如果 CMV 仍然可检测，可能需要进一步的 ID 评估 • 治疗：CMV 疾病（5 mg/kg，每 12 h 诱导一次，随后 5 mg/kg 每日维持和解决所有症状）	CMV HSV VZV	·HHV–6 和 HHV–8 的临床数据有限
伐昔	·预防：HSV 或 VZV（500 mg PO BID）优于口服阿昔洛韦用于 VZV • 治疗：HSV 或 VZV（1 g PO TID）优于口服阿昔洛韦治疗 HSV 或 VZV	HSV VZV	
缬更昔洛韦	• 预防性治疗和巨细胞病毒治疗：采用 900 mg PO BID 进行诱导，直至检测呈阴性;考虑每天额外 900 mg PO，阴性试验后的维护	CMV HSV VZV	·HHV–6 和 HHV–8 的临床数据有限

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ANTIVIRAL AGENTS[a]

Agent	Typical Dosing Based on Indication[c]	Spectrum	Comments/Cautions
Cidofovir	• Treatment: Cidofovir 5 mg/kg IV every wk for 2 wks, followed by cidofovir 5 mg/kg every 2 wks with probenecid 2 gm PO 3 h before the dose, followed by 1 gm PO 2 h after the dose and 1 gm PO 8 h after the dose and IV hydration. • Evidence is limited for treatment of adenovirus; when used, ID consult is strongly recommended.	CMV HSV VZV Adenovirus	• Hydration and probenecid required to reduce nephrotoxicity • Ocular toxicity, bone marrow toxicity
Foscarnet	• Prophylaxis for CMV: 60 mg/kg IV every 8–12 h for induction, followed by 90–120 mg/kg IV daily for maintenance after HCT.[5,6] • Preemptive therapy for CMV: Induction; either 60 mg/kg IV every 8 h or 90 mg/kg IV every 12 h. • Therapy: Acyclovir-resistant HSV (40 mg/kg every 8 h for 7–10 days); CMV disease (90 mg/kg every 12 h for induction followed by 90–120 mg/kg daily for maintenance and resolution of all symptoms).	HSV VZV CMV HHV-6	Drug of choice for acyclovir-resistant HSV and VZV and ganciclovir-resistant CMV • Nephrotoxic; monitor electrolytes Clinical data are limited for HHV-6 and HHV-8. Treatment should be reserved for clinically documented disease; ID consult is highly recommended.
Letermovir	• Primary prophylaxis for allogeneic HCT recipients who are CMV seropositive (R+): 480 mg PO daily or daily IV infusion over 1 h post-transplantation. Reduce dose to 240 mg PO/IV daily if co-administered with cyclosporine.	CMV	• Has not been studied as an agent for treatment • Has multiple drug interactions, including azoles, cyclosporine, and tacrolimus; see package insert (TDM is important) • Not active against other herpes group viruses. Acyclovir is also needed for HSV and VZV.
Maribavir	• Treatment: 400 mg PO BID	CMV	• Indicated for post-transplant CMV infection refractory to ganciclovir/valganciclovir, foscarnet, and cidofovir. ID consult is highly recommended. • No activities against HSV, VZV, or HHV-6 • Inhibitor of HCMV-encoded kinase UL97 • Virologic failure due to resistance can occur and cross-resistance between maribavir and ganciclovir/valganciclovir has been observed • Not recommended to be co-administered with ganciclovir/valganciclovir • Monitor for drug interactions (may increase level of immunosuppressants such as cyclosporine, tacrolimus, sirolimus, etc.) • May cause dysgeusia
Baloxavir[e]	• Treatment: 40 mg or 80 mg PO based on weight	Influenza A & B	• There are limited data for use in patients who are immunosuppressed. • Data show an emergence of resistance in people who are healthy.
Oseltamivir[f]	• Prophylaxis: 75 mg PO daily[g,7] • Treatment: 75 mg BID (typically for 5 days)	Influenza A & B	• May cause nausea (improved when taken with food)
Zanamivir[f]	• Prophylaxis: 2 oral inhalations (5 mg/inhalation) daily • Treatment: 2 oral inhalations (5 mg/inhalation) BID	Influenza A & B	• Duration influenced by nature of exposure (ongoing vs. time limited); may cause bronchospasm

Note: All recommendations are category 2A unless otherwise indicated.
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抗病毒剂

剂	基于适应症的典型给药	频谱	备注/注意事项
Cidofovir	• 治疗：西多福韦 5 mg/kg IV，每周一次，持续 2 周，随后西多福韦 5 mg/kg，每 2 周一次 在给药前 3 小时给予丙磺舒 2 gm PO，随后在给药后 2 小时给予 1 gm PO，在给药后 8 小时给予 1 gm PO，并 IV 水合。 • 治疗腺病毒的证据有限;使用时，强烈建议咨询 ID。	CMV HSV VZV 腺病毒	·需要水化和丙磺舒以减少肾毒性 ·眼部毒性、骨髓毒性
膦甲酸	• CMV 预防：60 mg/kg IV，每 8–12 h 一次，用于诱导，然后每天 IV 90–120 mg/kg，用于 HCT 后的维持。5,6 • CMV 的抢先治疗：诱导; 60 mg/kg IV 每 8 小时一次或 90 mg/kg IV 每 12 小时一次。 • 治疗：阿昔洛韦耐药 HSV（40 mg/kg，每 8 小时一次，持续 7–10 天）; CMV 疾病（90 mg/kg，持续 7–10 天）; kg 每 12 小时用于诱导，随后每天 90–120 mg/kg 用于维持和缓解所有症状）。	HSV VZV CMV HHV-6	抗阿昔洛韦 HSV 和 VZV 以及抗更昔洛韦 CMV 的首选药物 • 肾毒性;监测电解质 HHV–6 和 HHV–8 的临床数据有限。治疗应保留用于临床记录的疾病;强烈建议进行 ID 咨询。
Letermovir	• CMV 血清阳性（R+）的同种异体 HCT 接受者的一级预防：480 mg 移植后 1 小时内每日 PO 或每日 IV 输注。如果与环孢菌素联合给药，则将剂量降至 240 mg PO/IV 每日一次。	CMV	·尚未作为治疗药物进行研究 有多种药物相互作用，包括唑类，环孢素和他克莫司;见包装说明书（TDM 是重要的）·对其他疱疹病毒没有活性。阿昔洛韦也需要 HSV 和 VZV。
Maribavir	·治疗：400 mg PO BID	CMV	·适用于更昔洛韦/缬更昔洛韦、膦甲酸和西多福韦难治的移植后 CMV 感染。强烈推荐 ID 咨询。·对 HSV、VZV 或 HHV–6 无活性·HCMV 编码激酶 UL 97 的抑制剂·可能发生耐药导致的病毒学失败，并且已观察到 Maribavir 和更昔洛韦/缬更昔洛韦之间的交叉耐药·不建议与更昔洛韦/缬更昔洛韦联合给药·监测药物相互作用（可能增加免疫抑制剂如环孢素、他克莫司、西罗莫司等的水平）·可能导致味觉障碍
Baloxavir	·治疗：基于体重 40 mg 或 80 mg PO	流感 A & B	·用于免疫抑制患者的数据有限。·数据显示，健康人群中出现了耐药性。
Oseltamivir [f]	·预防：75 mg PO 每日 g，7 • 治疗：75 mg BID（通常持续 5 天）	流感 A & B	·可能引起恶心（与食物同服时改善）
Zanamivir	·预防：每日 2 次口服吸入（5 mg/吸入） • 治疗：2 次经口吸入（5 mg/吸入）BID	流感 A & B	• Duration influenced by nature of exposure (ongoing vs. time limited); may cause bronchospasm 重试 错误原因

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Footnotes and References
FEV–C (4 of 4)
重试 错误原因

FEV–C
2/4

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ANTIVIRAL AGENTS

Agent	Common Indication[c]	Spectrum	Comments/Cautions
Intravenous immuno-globulin (IVIG)	Doses of IVIG vary among different studies and different viral illnesses. Some data exist for use in the following: • Parvovirus B19,[8] 400–500 mg/kg IV daily for 5 days • Adjunctive therapy for CMV and RSV pneumonitis, 400 mg/kg IV every other day for 3–5 doses	RSV Parvovirus B19 CMV	• Pathogen-specific immunoglobulin or monoclonal antibodies may be considered. • CMV-specific IVIG is not more efficacious than standard IVIG. IVIG use as an antiviral is controversial.
Ribavirin (category 3)	Consider for treatment of lower respiratory tract RSV disease[h,9,10]: • 600–800 mg PO BID or TID • 6 gm administered by continuous inhalation via SPAG-2 nebulizer over 12–18 h daily or 2 g over 2 h TID	RSV	• Limit to patients undergoing HCT or with leukemia. • Experience in adults who are immunocompromised with RSV disease is limited. Ribavirin is teratogenic; precautions are required during administration (see package insert).
Entecavir	0.5 mg PO daily (nucleoside-treatment-naïve with compensated liver disease); or 1 mg PO daily (lamivudine-refractory or known lamivudine-resistant mutations or decompensated liver disease)	HBV	• Entecavir and tenofovir monotherapy are generally preferred. Choice of agent is heavily influenced by the overall condition of the patient, renal insufficiency, and the type of chemotherapy planned. Combination therapy is not generally recommended unless viral load is significantly elevated. • Potential for HBV resistance: ▸ Lamivudine: high (especially as monotherapy) ▸ Tenofovir: none reported to date ▸ Entecavir: low • Dose adjustment recommended for renal impairment • Lactic acidosis and severe hepatomegaly with steatosis reported with nucleoside analogues • Tenofovir (TDF more than TAF) has potential for nephrotoxicity; monitor for renal function
Lamivudine	100 mg PO daily		
Tenofovir	Tenofovir disoproxil fumarate (TDF) 300 mg PO daily Tenofovir alafenamide (TAF) 25 mg PO daily		

Footnotes and References
FEV-C (4 of 4)

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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抗病毒剂

剂	常见适应症	频谱	备注/注意事项
静脉 免疫的，免疫的 球蛋白（IVIG）	IVIG 的剂量因不同的研究和不同的病毒性疾病而异。 有些数据可用于以下方面： ·细小病毒 B19，400–500 mg/kg IV，每日一次，持续 5 天	RSV 细小病毒 B19 CMV	• 可考虑病原体特异性免疫球蛋白或单克隆抗体。 • CMV 特异性 IVIG 并非更有效 比标准的 IVIG IVIG 作为抗病毒药物的使用是有争议 的。
利巴韦林 （第 3 类）	考虑治疗下呼吸道 RSV 疾病：· 600–800 mg PO BID 或 TID · 6 gm 通过 SPAG–2 雾化器持续吸入给药，每日 12–18 小时或 2 g，每日 2 小时 TID	RSV	·仅限于接受 HCT 或白血病患者。·患有 RSV 疾病的免疫功能低下的成人的经验有限。利巴韦林有致畸作用； 给药期间需要注意事项（参见药品说明书）。
替卡韦	0.5 mg PO 每日一次（核苷类药物初治伴代偿治疗） 肝病）；或每日 1 mg PO（拉米夫定难治性或 已知拉米夫定耐药突变或失代偿性肝病）	HBV	·恩替卡韦和替诺福韦单药治疗通常是首选。药物的选择受患者的总体状况、肾功能和预后的影响。 以及计划的化疗类型。 除非病毒载量显著升高，否则一般不推荐联合治疗。 HBV 耐药的可能性： 拉米夫定：高（特别是作为单药治疗） 替诺福韦：迄今未报告 恩替卡韦：低剂量·建议对肾损害进行剂量调整·使用核苷类似物报告的乳酸酸中毒和重度肝肿大伴脂肪变性·替诺福韦（TDF 大于 TAF）有潜在肾毒性;监测肾功能
Lamivudine	每日 100 mg PO		
Tenofovir	富马酸替诺福韦酯（TDF）300 mg PO 每日一次 替诺福韦艾拉酚胺（TAF）25 mg PO 每日		

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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ANTIVIRAL AGENTS – FOOTNOTES

[a] Requires dose adjustment in patients with renal insufficiency.
[b] High-dose acyclovir and valacyclovir have been used as prophylaxis for CMV. Because these agents have weak activity against CMV, a strategy of CMV surveillance and preemptive therapy with ganciclovir, valganciclovir, or foscarnet is required among patients at high risk for CMV disease.
[c] Dosing is for adult patients. Consult pediatric guidelines for recommended dosing in these patients.
[d] Antiviral prophylaxis should be targeted to specific patients at high risk (INF-3). In patients who are at high risk and not receiving transplant, prophylaxis should be administered to patients seropositive for HSV or VZV (or with a history of chicken pox). In HCT recipients, prophylaxis is only indicated if either the donor or recipient is seropositive for the virus in question. The indicated doses for antiviral agents are for adults with normal renal function; consult package insert for dose modification in pediatric patients, in patients with renal impairment, and in patients with obesity. Prophylactic antiviral doses may be higher than those routinely used in persons who are immunocompetent (eg, for recurrent cold sores). There is substantial variability in the prophylactic doses of acyclovir used in different clinical trials in patients with hematologic malignancies and in HCT recipients.
[e] Not routinely recommended by the CDC
[f] Consider IV peramivir for patients who cannot have oral oseltamivir or inhaled zanamivir.
[g] During community and nosocomial outbreaks of influenza A, prophylaxis among persons who are highly immunocompromised should be considered.
[h] Inhaled ribavirin is only FDA approved for infants and young children who are hospitalized with severe lower respiratory tract RSV disease.

ANTIVIRAL AGENTS – REFERENCES

[1] Boeckh M, Kim HW, Flowers MED, et al. Long-term acyclovir for prevention of varicella zoster virus disease after allogeneic hematopoietic cell transplantation -- a randomized double-blind placebo-controlled study. Blood 2006;107:1800-1805.
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[6] Reusser P, Einsele H, Lee J, et al. Randomized multicenter trial of foscarnet versus ganciclovir for preemptive therapy of cytomegalovirus infection after allogeneic stem cell transplantation. Blood 2002;99:1159-1164.
[7] Vu D, Peck AJ, Nichols WG, et al. Safety and tolerability of oseltamivir prophylaxis in hematopoietic stem cell transplant recipients: a retrospective case-control study. Clin Infect Dis 2007;45:187-193.
[8] Heegaard Ed, Brown KE. Human parvovirus B19. Clin Microbial Rev 2002;15:485-505.
[9] Whimbey E, Champlin RE, Englund JA, et al. Combination therapy with aerosolized ribavirin and intravenous immunoglobulin for respiratory syncytial virus disease in adult bone marrow transplant recipients. Bone Marrow Transplant 1995;16:393-399.
[10] Marcelin JR, Wilson JW, Razonable RR.Oral ribavirin therapy for respiratory syncytial virus infections in moderately to severely immunocompromised patients. Transpl Infect Dis 2014;16:242-250.

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抗病毒剂–脚注

[a] Requires dose adjustment in patients with renal insufficiency.

[b] High–dose acyclovir and valacyclovir have been used as prophylaxis for CMV. Because these agents have weak activity against CMV, a strategy of CMV surveillance and preemptive therapy with ganciclovir, valganciclovir, or foscarnet is required among patients at high risk for CMV disease.

[c] Dosing is for adult patients. Consult pediatric guidelines for recommended dosing in these patients.

[d] Antiviral prophylaxis should be targeted to specific patients at high risk (INF–3). In patients who are at high risk and not receiving transplant, prophylaxis should be administered to patients seropositive for HSV or VZV (or with a history of chicken pox). In HCT recipients, prophylaxis is only indicated if either the donor or recipient is seropositive for the virus in question. The indicated doses for antiviral agents are for adults with normal renal function; consult package insert for dose modification 在儿科患者、肾损害患者和肥胖患者中。预防性抗病毒剂量可能高于免疫功能正常的人（如复发性唇疱疹）常规使用的剂量。在不同的临床试验中，在血液系统恶性肿瘤患者和 HCT 接受者中使用的阿昔洛韦预防剂量存在很大差异。

[e] 疾控中心不推荐

[f] 对于不能口服奥司他韦或吸入扎那米韦的患者，考虑静脉注射帕拉米韦。

[g] 在甲型流感社区和医院暴发期间，应考虑在免疫力高度低下的人群中进行预防。

[h] 吸入型利巴韦林仅被 FDA 批准用于因严重下呼吸道 RSV 疾病住院的婴幼儿。

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[8] Heegaard 艾德，Brown KE.人细小病毒 B19。临床微生物综述 2002;15：485–505。

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RISK ASSESSMENT RESOURCES

Using the Multinational Association for Supportive Care in Cancer (MASCC) Risk-Index Score

- Using the visual analogue score, estimate the patient's burden of illness at the time of initial clinical evaluation. No signs or symptoms or mild signs or symptoms are scored as 5 points; moderate signs or symptoms are scored as 3 points. These are mutually exclusive. No points are scored for severe signs or symptoms or moribund.
- Based on the patient's age, past medical history, present clinical features, and site of care (input/output when febrile episode occurred), score the other factors in the model and total the sum.

Burden of Illness

How sick is the patient at presentation?

No signs or symptoms	Mild signs or symptoms	Moderate signs or symptoms	Severe signs or symptoms	Moribund

Estimate the burden of illness considering all comorbid conditions

MASCC Risk-Index Score/Model[1,2]

Characteristic	Weight
• Burden of illness	
▸ No or mild symptoms	5
▸ Moderate symptoms	3
• No hypotension	5
• No COPD	4
• Solid tumor or hematologic malignancy with no previous fungal infection	4
• No dehydration	3
• Outpatient status	3
• Age <60 years	2

CISNE Score/Model[3]

Characteristic	Points
ECOG PS ≥2	2
Stress-induced hyperglycemia	2
COPD	1
Chronic cardiovascular disease	1
Mucositis NCI grade ≥2	1
Monocytes <200/µL	1

[1] The MASCC Risk-Index Score is for adults only. It does not apply to pediatric patients.

[2] Klastersky J, Paesmans M, Rubenstein EJ, et al. The Multinational Association for Supportive Care in Cancer risk index: a multinational scoring system for identifying low-risk febrile neutropenic cancer patients. J Clin Oncol 2000;18:3038-3051.

[3] Carmona-Bayonas A, Jimenez-Fonesca P, Virizuela Echaburu J, et al. Prediction of serious complications in patients with seemingly stable febrile neutropenia: validation of the Clinical Index of Stable Febrile Neutropenia in a prospective cohort of patients from the FINITE study. J Clin Oncol 2015;33:465-471.

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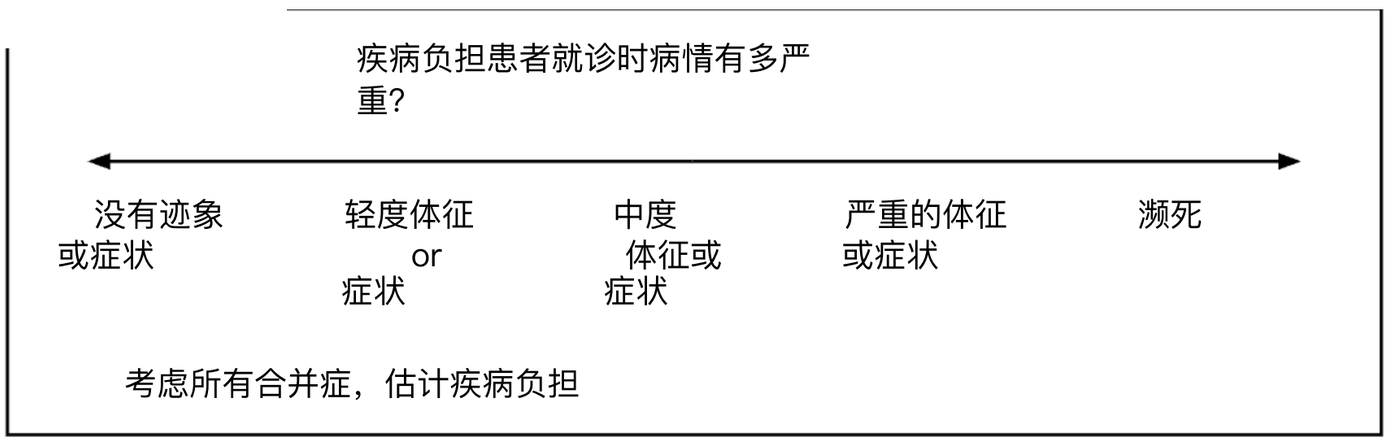
风险评估资源使用多国癌症支持治疗协会（MASCC）风险指数评分·使用视觉模拟评分，估计患者在初始临床评估时的疾病负担。无体征或症状或轻度体征或症状记为 5 分;中度体征或症状记为 3 分。这些是相互排斥的。严重体征、症状或濒死状态不计分。·基于患者的年龄、既往病史、目前的临床特征和护理部位（发热发作发生时的输入/输出），对模型中的其他因素进行评分并求和。



MASCC 风险指数评分/模型特征权重·疾病负担	
无或轻微症状	5
中度症状	3
·无低血压	5
·无 COPD	4
·实体瘤或恶性血液病，既往无真菌感染	4
·无脱水	3
·门诊状态	3
·年龄<60 岁	2

CISNE 评分/模型	
特性	点
ECOG PS ⩾2	2
应激性高血糖 2 慢性阻塞性肺疾病 1 慢性心血管疾病 1	
粘膜炎 NCI ⩾2 级	1
单核细胞<200/μL	1

[1] MASCC 风险指数评分仅适用于成年人。它不适用于儿科患者。
[2] Klastersky J，Paesmans M，Rubenstein EJ 等，癌症风险指数支持性护理多国协会：用于识别低风险发热性贫血癌症患者的多国评分系统。临床肿瘤学杂志 2000;18：3038–3051.

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TABLE 1: SARS-COV-2 TESTING INTERPRETATION AND INFECTIVITY IN PATIENTS WITH CANCER

The following table is recommended for interpreting SARS-CoV-2 PCR/antigen testing for patients considered to be moderately/severely immunocompromised. These patients may produce replication-competent virus beyond 20 days. Ending isolation in conjunction with consultation of an ID specialist is recommended.

Asymptomatic	Discontinue isolation at **Day 20** OR Two negative consecutive respiratory specimens collected ≥24 hours apart if within 20 days
Symptomatic at time of original COVID-19 diagnosis	Resolution of fever for at least 24 hours without fever-reducing medication + Improvement of symptoms + **20 days** since symptom onset OR two negative consecutive respiratory specimens collected ≥24 hours
Continued symptoms on or after day 20 OR symptoms worsening after ending isolation (reactivation of symptoms)	Recommend repeat SARS-CoV-2 testing and consider consultation with an ID specialist

- **Detection of sub-genomic SARS-CoV-2 RNA or recovery of replication-competent virus has been reported in patients who are moderately or severely immunocompromised beyond 20 days, and as long as >140 days after a positive SARS-CoV-2 test result. Patients who recover from COVID-19 can continue to have detectable SARS-CoV-2 RNA and upper respiratory symptoms for up to 3 months after illness onset. However, prolonged detection of viral RNA may not indicate higher infectious risk and risk of transmission. If a patient has persistently positive nucleic acid amplification tests beyond 30 days, additional testing could include molecular studies, determination of PCR cycle threshold (Ct), or attempt to identify replication of the competent virus in conjunction with ID consultation.**
- **Immunocompromising conditions that have been associated with shedding of replication-competent virus beyond 20 days include active treatment for solid tumor and hematologic malignancies, solid organ transplant and taking IST, receipt of CAR T-cell therapy or HCT (within 2 years of transplantation or taking immunosuppression therapy), moderate or severe primary immunodeficiency, and active treatment with high-dose corticosteroids (ie, ≥20 mg prednisone or equivalent per day when administered for ≥2 weeks), alkylating agents, antimetabolites, transplant-related immunosuppressive drugs, cancer chemotherapeutic agents classified as severely immunosuppressive, and other biologic agents that are immunosuppressive or immunomodulatory.**

Table 1 References (COV-A 1 of 3)

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表 1：SARS-COV-2 检测判读和癌症患者的感染性

建议使用下表解释被认为是中度/重度免疫功能低下患者的 SARS-CoV-2 PCR/抗原检测结果。这些患者可能在 20 天后产生可复制的病毒。建议在咨询 ID 专家的同时结束隔离。

无症状	在第 20 天停止隔离，或如果在 20 天内，则间隔≥24 小时采集两份阴性连续呼吸道标本
最初 COVID-19 诊断时的症状	Resolution of fever for at least 24 hours without fever-reducing medication + Improvement of symptoms + 20 days since symptom onset OR two negative consecutive respiratory specimens collected ≥24 hours 重试 错误原因
Continued symptoms on or after day 20 OR symptoms worsening after ending isolation (reactivation of symptoms) 重试 错误原因	Recommend repeat SARS-CoV-2 testing and consider consultation with an ID specialist 重试 错误原因

• Detection of sub-genomic SARS-CoV-2 RNA or recovery of replication-competent virus has been reported in patients who are moderately or severely immunocompromised beyond 20 days, and as long as >140 days after a positive SARS-CoV-2 test result. Patients who recover from COVID-19 can continue to have detectable SARS-CoV-2 RNA and upper respiratory symptoms for up to 3 months after illness onset. However, prolonged detection of viral RNA may not indicate higher infectious risk and risk of transmission. If a patient has persistently positive nucleic acid amplification tests beyond 30 天内，额外的检测可能包括分子研究、PCR 循环阈值（Ct）测定或尝试结合 ID 咨询确定主管病毒的复制。

·与有复制能力的病毒脱落超过 20 天相关的免疫损害状况包括实体瘤和血液恶性肿瘤的积极治疗、实体器官移植和服用 IST、接受 CAR T 细胞疗法或 HCT（移植后 2 年内或接受免疫抑制治疗）、中度或重度原发性免疫缺陷，以及高剂量皮质类固醇（即，≥20 mg 泼尼松或等效药物/天，给药≥2 周）、烷化剂、抗代谢药、移植相关免疫抑制药物、归类为重度免疫抑制的癌症化疗药物和其他免疫抑制或免疫调节生物制剂的积极治疗。

表 1 参考文献（COV-A 1/3）

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TABLE 2: CONSIDERATIONS FOR CANCER-DIRECTED THERAPY IN PATIENTS WITH POSITIVE SARS-COV-2

- **Duration of delaying the cancer-directed therapy depends on the severity of clinical SARS-CoV-2 infection (ie, mild, moderate, severe, asymptomatic), type and status of malignancy, risk of cancer relapse and progression as a result of delaying therapy, comorbidities, type and intensity of treatment, and adverse effects of treatment regimen.**
- **If cancer-directed therapy is urgently required due to uncontrolled cancer, it should be administered at the judgment of the oncologist.**

COVID-19 Severity[a]	High Risk for Progression of COVID-19	Cancer-Directed Therapy	General Recommendations for Timing of Initiation or Resumption of Cancer-Directed Therapy[c]
Patients who are hospitalized with severe to critical COVID-19	N/A	Any	• If feasible, hold therapy for at least 20 days and until improvement of symptoms and at least 24 hours have passed since resolution of fever without use of fever-reducing medications.
Mild to moderate COVID-19 or asymptomatic positive SARS-CoV-2[b]	• Prolonged neutropenia • T-cell deficiency (lymphopenia) or dysfunction • Hematologic malignancy • Tumor pulmonary involvement See complete listing of underlying medical conditions posing higher risk for severe COVID-19 at: Underlying Medical Conditions Associated with Higher Risk for Severe COVID-19.	Cytotoxic therapy directed at lymphocytes	• If feasible, hold therapy for at least 14 days and until improvement of symptoms and at least 24 hours have passed since resolution of fever without use of fever-reducing medications. • If the patient remained asymptomatic, hold therapy for at least 10 days after the date of the first positive test.
		• Prior to planned HCT • Prior to planned CAR T-cell therapy	• If feasible, hold therapy for at least 14 days and until improvement of symptoms and at least 24 hours have passed since resolution of fever without use of fever-reducing medications; or if the patient remained asymptomatic, hold for at least 14 days after the date of the first positive test.
		• Targeted therapy • Long-acting biologic therapy • Immune checkpoint inhibitors • Radiation therapy • Immune therapy • Hormonal therapy	• Consider holding therapy for at least 10 days and until improvement of symptoms and at least 24 hours have passed since resolution of fever without use of fever-reducing medications. • If the patient remained asymptomatic, hold therapy for 10 days after the date of the first positive test.

[a] Mild illness is defined as patients with signs and symptoms of COVID-19 (eg, fever, cough, sore throat, malaise, headache, muscle pain, nausea, vomiting, diarrhea, loss of taste and smell) but who do not have shortness of breath, dyspnea, or abnormal chest imaging. Moderate illness is defined as patients with lower respiratory disease during clinical assessment or imaging and an oxygen saturation (SpO_2) ≥94% on room air. Severe illness is defined as patients with SpO_2 <94% on room air, a ratio of arterial partial pressure of oxygen to fraction of inspired oxygen (PaO_2/FiO_2) <300 mm Hg, a respiratory rate >30 breaths/min, or lung infiltrates >50%, including patients on supplemental oxygen, oxygen through a high-flow device, or noninvasive ventilation. Critical illness is defined as patients with respiratory failure, septic shock, and/or multiple organ dysfunction, including patients on mechanical ventilation and extracorporeal mechanical oxygenation (ECMO) and end-organ dysfunction.

[b] Some providers delay cancer-directed therapy for shorter periods of time among patients who are asymptomatic who test positive for SARS-CoV-2.

[c] If feasible, consider doing test-based strategy with two negative tests separated by 24 hours.

Table 2 References (COV-A 1 of 3 and COV-A 2 of 3)

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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表 2：SARS–COV–2 阳性患者的癌症定向治疗考虑因素

·延迟癌症定向治疗的持续时间取决于临床 SARS–CoV–2 感染的严重程度（即轻度、中度、重度、无症状）、恶性肿瘤的类型和状态、延迟治疗导致的癌症复发和进展的风险、合并症、治疗的类型和强度以及治疗方案的不良反应。

• 如果由于不受控制的癌症而迫切需要癌症导向治疗，则应根据肿瘤学家的判断进行治疗。

COVID–19 严重程度	COVID–19 进展的高风险	癌症导向治疗	启动或恢复癌症导向治疗时间的一般建议
的患者 因严重 到关键的 COVID–19	N/A	Any	·如果可行，暂停治疗至少 20 天，直到症状改善，并且自发热消退以来至少 24 小时没有使用退烧药物。
轻度至中度 COVID–19 或 SARS–CoV–2 无症状阳性	·长期中性粒细胞减少症 • T 细胞缺乏（淋巴细胞减少） ·血液恶性肿瘤·肿瘤肺部受累参见导致严重肺功能障碍风险较高的基础医学状况的完整列表 COVID–19 at：基础 与严重 COVID–19 风险较高相关的医疗状况。	针对淋巴细胞的细胞毒性治疗	·如果可行，暂停治疗至少 14 天，直到症状改善，并且自发热消退以来至少 24 小时没有使用退烧药物。·如果患者仍无症状，则在首次阳性检测日期后至少 10 天内停止治疗。
		·在计划的 HCT 之前·在计划的 CAR T 细胞疗法之前	·如果可行，暂停治疗至少 14 天，直到症状改善，并且自解决以来至少经过 24 小时 发烧而不使用退烧药;或如果 患者保持无症状，在首次阳性检测日期后保持至少 14 天。
		·靶向治疗 ·长效生物治疗 ·免疫检查点抑制剂 ·放射治疗 ·免疫治疗 ·激素疗法	·考虑暂停治疗至少 10 天，直到症状改善，并且在不使用退烧药物的情况下，发热消退至少 24 小时。·如果患者仍无症状，则在首次阳性检测日期后暂停治疗 10 天。

[a] 轻度疾病定义为具有 COVID–19 体征和症状（如发热、咳嗽、咽喉痛、不适、头痛、肌肉疼痛、恶心、呕吐、腹泻、味觉和嗅觉丧失）但没有呼吸短促、呼吸困难或胸部影像学异常的患者。中度疾病定义为下呼吸道疾病患者
在临床评估或成像期间患有疾病，并且在室内空气中氧饱和度（SpO）≥94%。重度疾病定义为患者在室内空气中 SpO<94%，动脉氧分压与吸入氧分数之比（PaO /FiO）<300 mm Hg，呼吸频率>30 次/min，或肺浸润> 50%，包括接受补充供氧、通过高流量装置供氧或无创通气的患者。危重疾病定义为呼吸衰竭、感染性休克和/或多器官功能障碍患者，包括接受机械通气和体外机械氧合（ECMO）和终末器官功能障碍的患者。

[b] 对于 SARS–CoV–2 检测呈阳性的无症状患者，一些提供者将癌症导向治疗推迟较短的时间。
[c] 如果可行的话，考虑进行基于测试的策略，两次阴性测试间隔 24 小时。

表 2 参考文献（COV–A 1/3 和 COV–A 2/3）

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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TABLE 2: CONSIDERATIONS FOR CANCER-DIRECTED THERAPY IN PATIENTS WITH POSITIVE SARS-COV-2 (CONTINUED)

COVID-19 Severity[a]	High Risk for Progression of COVID-19	Cancer-Directed Therapy	General Recommendations for Timing of Initiation or Resumption of Cancer-Directed Therapy
Mild to moderate COVID-19 or asymptomatic positive SARS-CoV-2 (cont.)	No high-risk factors	• Targeted therapy • Long-acting biologic therapy • Immune checkpoint inhibitors • Radiation therapy • Immune therapy • Hormonal therapy	• Consider holding therapy for at least 10 days and until improvement of symptoms and at least 24 hours have passed since resolution of fever without use of fever-reducing medications. • If the patient remained asymptomatic, hold therapy for 10 days after the date of the first positive test.

[a] Mild illness is defined as patients with signs and symptoms of COVID-19 (eg, fever, cough, sore throat, malaise, headache, muscle pain, nausea, vomiting, diarrhea, loss of taste and smell) but who do not have shortness of breath, dyspnea, or abnormal chest imaging. Moderate illness is defined as patients with lower respiratory disease during clinical assessment or imaging and an SpO_2 ≥94% on room air. Severe illness is defined as patients with SpO_2 <94% on room air, a ratio of PaO_2/FiO_2 <300 mm Hg, a respiratory rate >30 breaths/min, or lung infiltrates >50%, including patients on supplemental oxygen, oxygen through a high-flow device, or noninvasive ventilation. Critical illness is defined as patients with respiratory failure, septic shock, and/or multiple organ dysfunction, including patients on mechanical ventilation and ECMO and end-organ dysfunction.

Table 2 References (COV-A 1 of 3 and COV-A 2 of 3)

Note: All recommendations are category 2A unless otherwise indicated.
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表 2：SARS–COV–2 阳性患者的癌症定向治疗考虑因素（续）

COVID–19 严重程度	COVID–19 进展的高风险	癌症导向治疗	启动或恢复癌症导向治疗时间的一般建议
轻度至中度 COVID–19 或无症状阳性 SARSCoV–2（SARSCoV–2）	无高危因素	·靶向治疗 ·长效生物治疗 ·免疫检查点抑制剂 ·放射治疗 ·免疫治疗 ·激素疗法	·考虑暂停治疗至少 10 天，直到症状改善，并且在不使用退烧药物的情况下，发热消退至少 24 小时。·如果患者仍无症状，则在首次阳性检测日期后暂停治疗 10 天。

[a] 轻度疾病定义为具有 COVID–19 体征和症状（如发热、咳嗽、咽喉痛、不适、头痛、肌肉疼痛、恶心、呕吐、腹泻、味觉和嗅觉丧失）但没有呼吸短促、呼吸困难或胸部影像学异常的患者。中度疾病定义为下呼吸道疾病患者
在临床评估或成像期间患有疾病，并且在室内空气中 SpO ≥94%。重度疾病定义为患者在室内空气中 SpO<94%，PaO/ FiO<300 mm Hg，呼吸频率>30 次/min，或肺浸润> 50%，包括患者接受辅助供氧，通过高流量装置供氧或无创通气。危重病定义为呼吸衰竭、感染性休克和/或多器官功能障碍患者，包括接受机械通气和 ECMO 治疗的患者和终末器官功能障碍患者。 表 2 参考文献（COV–A 1/3 和 COV–A 2/3）

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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TABLE 3: CONSIDERATIONS FOR CANCER-DIRECTED THERAPY IN PATIENTS WITH SIGNIFICANT EXPOSURE TO SARS-COV-2

- **The exact risk of viral transmission after significant exposure to SARS-CoV-2 is unknown and depends upon many variables (eg, symptoms of infected person, duration and proximity of contact, room ventilation, host susceptibility, viral variant). Household contacts pose the highest risk of SARS-CoV-2 transmission.**
- **If viral transmission occurs to the patient, the upper bound of COVID-19 incubation period is 14 days.**
- **The duration of cancer-directed therapy delay depends on the type and status of malignancy and risk of cancer relapse and progression as a result of delaying therapy. If cancer-directed therapy is urgently required due to uncontrolled cancer, it should be administered at the judgment of the oncologist.**

Significant Exposure to SARS-CoV-2[d] Recommendations
• Cancer-directed therapy of patients who are asymptomatic who have had a significant exposure to SARS-CoV-2 should be delayed for 14 days since exposure. • While the Centers for Disease Control and Prevention (CDC) does not recommend quarantine or routine empiric transmission-based isolation precautions of people who have been exposed, patients who are immunocompromised who are at high risk for severe COVID-19 (Table 2) and have had a significant exposure to a person with known SARS-CoV-2 infection should consider quarantining for 14 days after last exposure. • During the quarantine period, these patients should be masked and closely monitored for development of symptoms.

Table 2 References (COV-A 1 of 3 and COV-A 2 OF 3)

[d] Per CDC definition, significant SARS-CoV-2 exposure is defined as a patient who has had a close contact (within 6 feet for a total of 15 minutes or more in 24 hours) with a person known to be infected with SARS-CoV-2. More infectious agents are likely to require lesser exposure time for transmission.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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表 3：对严重暴露于 SARS–COV–2 的患者进行癌症导向治疗的考虑

- 严重暴露于 SARS–CoV–2 后病毒传播的确切风险尚不清楚，并取决于许多变量（例如，症状感染者的感染率、接触的持续时间和距离、房间通风、宿主易感性、病毒变异）。家庭接触是 SARS–CoV–2 传播的最高风险。

·如果患者发生病毒传播，COVID–19 潜伏期的上限为 14 天。
·癌症定向治疗延迟的持续时间取决于恶性肿瘤的类型和状态以及延迟治疗导致的癌症复发和进展的风险。如果由于不受控制的癌症而迫切需要癌症导向治疗，则应根据肿瘤学家的判断进行治疗。

严重暴露于 SARS–CoV–2 的建议
• 对于无症状且严重暴露于 SARS–CoV–2 的患者，癌症导向治疗应延迟至暴露后 14 天。 • 虽然疾病控制和预防中心（CDC）不建议隔离或常规经验性传播隔离已暴露人群的预防措施，免疫功能低下、严重 COVID–19 高风险患者（表 2），以及与已知 SARS–CoV–2 感染者有显著接触的患者，应考虑在最后一次暴露后 14 天内避免接触。 • 在隔离期间，这些患者应被掩盖并密切监测症状的发展。

表 2 参考文献（COV–A 1/3 和 COV–A 2/3）

[d] 根据 CDC 的定义，严重的 SARS–CoV–2 暴露被定义为与已知感染 SARS–CoV–2 的人有过密切接触（24 小时内 6 英尺内共 15 分钟或以上）的患者。更多的传染性病原体可能需要更短的接触时间进行传播。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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COVID-19 MANAGEMENT IN PATIENTS WITH CANCER

- **The heterogeneity of cancers, the complexity and number of different cancer treatment regimens, and variability in the COVID-19 clinical course among patients preclude a single management approach of COVID-19 in all patients with cancer.**
- **Treatment recommendations for COVID-19 among patients with cancer are largely similar to those without cancer (see Table 3); however, several new therapies have become available that demonstrate corresponding benefits for patients with cancer and/or other risk factors for more severe disease.**
- **Table 4 lists currently available COVID-19 treatment options, dosing, and clinical indications.**
- **Comprehensive information on COVID-19–based testing, infection control measures, and evidence-based data for current treatment recommendations can be accessed at: The Centers for Disease Control and Prevention and Infectious Disease Society of America, COVID-19 Guidelines.**
- **For unresolved COVID-19, ID consult is recommended.**

Note: All recommendations are category 2A unless otherwise indicated.
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癌症患者的 COVID–19 管理

- 癌症的异质性、不同癌症治疗方案的复杂性和数量，以及患者之间 COVID–19 临床病程的变异性，排除了所有癌症患者的 COVID–19 单一管理方法。

·癌症患者的 COVID–19 治疗建议与非癌症患者的治疗建议基本相似（见表 3）;然而，已有几种新的治疗方法可用于癌症患者和/或其他更严重疾病风险因素的患者。

·表 4 列出了目前可用的 COVID–19 治疗方案、剂量和临床适应症。
·有关基于 COVID–19 的检测、感染控制措施和当前治疗建议的循证数据的全面信息可访问：美国疾病控制与预防中心和传染病学会，COVID–19 指南。

·对于未解决的 COVID–19，建议咨询 ID。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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TABLE 4: COVID-19 TREATMENT IN PATIENTS WITH CANCER[e]

Clinical Scenario	Antiviral Options	Comments
• Outpatient with acute infection, recent symptom onset, and high risk of progression: ▸ Prolonged neutropenia ▸ T-cell deficiency (lymphopenia) or dysfunction ▸ Hematologic malignancy ▸ Tumor pulmonary involvement ▸ See complete list of underlying medical conditions posing higher risk for severe COVID-19 by the CDC: Underlying Medical Conditions Associated with Higher Risk for Severe COVID-19 • See Table 2: Considerations for Cancer-Directed Therapy in Patients with Positive SARS-CoV-2	Preferred • Nirmatrelvir/ritonavir[1] • Remdesivir[2] Other • Molnupiravir[3] • High-titer COVID-19 convalescent plasma[4,f,g,h]	• Nirmatrelvir/ritonavir is the favored oral treatment for outpatients with mild to moderate COVID-19 symptoms who are at highest risk of progressing to severe disease and who do not have adverse drug-drug interactions with ritonavir. ▸ Nirmatrelvir/ritonavir use has been shown to reduce risk of hospitalization and death by up to 88%.[5] ▸ Must review potential drug interactions with ritonavir before use. • IV remdesivir is favored when oral nirmatrelvir/ritonavir is not available or suitable because of adverse drug interactions. ▸ 3-day course IV remdesivir has been shown to reduce the risk of hospitalization and death by up to 87%.[2] ▸ Requires IV administration in an infusion center, emergency department, or outpatient clinic; may limit feasibility in outpatient setting. • Oral molnupiravir has decreased efficacy for reducing hospitalization and death (30%)[3] compared with other treatment options, and concerns exist for potential mutagenicity in animal studies. ▸ It is unclear if molnupiravir is a teratogen; however, it is not recommended for use in patients who are pregnant. ▸ Please see Table 5 for recommendations regarding the use of contraception. • High-titer COVID-19 convalescent plasma is under evaluation in the outpatient setting and has been shown to reduce outpatient hospitalizations by >50%.[4] ▸ High-titer COVID-19 convalescent plasma against prevalent circulating viral variants may not be uniformly available and requires transfusion capacity by a local center. • Post-treatment recurrence of symptoms should be treated as a new possible infection with an appropriate evaluation.

[e] This is a rapidly changing field. For links to routinely updated guidance, see the Centers for Disease Control and Prevention and Infectious Diseases Society of America guidelines for updated information: CDC and IDSA.

[f] Antibody therapy should not be used as an alternative to COVID-19 vaccination. COVID-19 vaccination can be given at any interval following receipt of passive antibody therapy. Note that COVID-19 vaccination status should not affect decisions regarding the use or timing of antibody therapy for treatment of breakthrough COVID-19 disease. All mAb FDA approvals have been suspended due to emergence of viral resistance as availability of mAb for treatment depends on susceptibility to circulating strains.[5,6]

[g] COVID-19 convalescent plasma obtained from those who have recovered from the recent circulating variants and have been previously vaccinated is preferred. COVID-19 convalescent plasma can be acquired via the Blood Centers of America.

[h] There is emerging evidence that high-titer COVID-19 convalescent plasma may be beneficial in patients who are immunocompromised (particularly with B-cell impairment) with persistent SARS-Cov-2 infection.

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

Table 4 References (COV-A 3 of 3)

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表 4：癌症患者的 COVID-19 治疗

临床场景	抗病毒选择	评论
·门诊急性感染，近期症状发作，高风险 进展： 长期中性粒细胞减少 T 细胞缺乏症（淋巴细胞减少症）或功能障碍 恶性血液病 肿瘤肺部受累 请参阅 CDC 发布的导致严重 COVID-19 高风险的基础医疗状况的完整列表：与严重 COVID-19 高风险相关的基础医疗状况 ·参见表 2：SARS-CoV-2 阳性患者的癌症导向治疗考虑因素	首选· Nirmatrelvir/ritonavir · Remdesivir 其他 ·莫努匹拉韦·高滴度 COVID-19 恢复期血浆	· Nirmatrelvir/ritonavir 是轻度至中度 COVID-19 症状的门诊患者的首选口服治疗，这些患者进展为严重疾病的风险最高，并且与 ritonavir 没有不良药物相互作用。 Nirmatrelvir/ritonavir 的使用已被证明可将住院和死亡风险降低高达 88%。 使用前必须审查与利托那韦的潜在药物相互作用。 ·当口服 Nirmatrelvir/利托那韦因不良药物相互作用而不可用或不适合时，IV Remdesivir 是有利的。 3-日疗程 IV Remdesivir 已被证明可将住院和死亡风险降低高达 87%。 需要在输液中心、急诊室或门诊进行 IV 给药;可能限制门诊环境的可行性。 • 与其他治疗方案相比，口服莫努匹韦减少住院和死亡的疗效降低（30%），并且在动物研究中存在潜在致突变性的担忧。 目前尚不清楚莫努匹韦是否是一种致畸剂;但是，不建议怀孕患者使用。 关于使用避孕措施的建议，请参见表 5。·高滴度 COVID-19 恢复期血浆正在门诊环境中进行评估，并已被证明可将门诊住院人数减少> 50%。 针对流行循环病毒变种的高滴度 COVID-19 恢复期血浆可能无法统一提供，并且需要当地中心的输血能力。·治疗后症状复发应视为新的可能感染，并进行适当的评估。

f This is a rapidly changing field. For links to routinely updated guidance, see the Centers for Disease Control and Prevention and Infectious Diseases Society of America guidelines for updated information: CDC and IDSA. f Antibody therapy should not be used as an alternative to COVID-19 vaccination. COVID-19 vaccination can be given at any interval following receipt of passive antibody therapy. Note that COVID-19 vaccination status should not affect decisions regarding the use or timing of antibody therapy for treatment of breakthrough COVID-19 disease. All mAb FDA approvals have been suspended due to emergence of viral resistance as availability of mAb for treatment depends on susceptibility to circulating strains.

重试 错误原因

g COVID-19 convalescent plasma obtained from those who have recovered from the recent circulating variants and have been previously vaccinated is preferred. COVID-19 convalescent plasma can be acquired via the Blood Centers of America.

重试 错误原因

h 有新的证据表明，高滴度 COVID-19 恢复期血浆可能对持续 SARS-Cov-2 感染的免疫功能低下（特别是 B 细胞受损）患者有益。

表 4 参考文献
（COV-A 3/3）

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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TABLE 4: COVID-19 TREATMENT IN PATIENTS WITH CANCER (CONTINUED)[e]

Clinical Scenario	Antiviral Options	Comments
Patient hospitalized for non–COVID-19 indication but with acute symptomatic COVID-19	Preferred • IV remdesivir[2] Other • COVID-19 convalescent plasma[4,f,g,h]	• IV remdesivir x 3 days (extending to 5 days can be considered for patients with more significant disease or concurrent immunosuppression). • Consider high-titer COVID-19 convalescent plasma. Pre-BA[4,5] plasma may not be as effective.[f] ▸ High-titer COVID-19 convalescent plasma active against circulating viral variants may not be uniformly available and requires transfusion capacity by a local center. • Although a mechanistic rationale exists for use of oral antivirals, they are currently not authorized by the FDA for use in patients who are hospitalized.
Patient hospitalized for acute symptomatic COVID-19	Preferred • IV remdesivir x 5 days[6,7] Other • Consider COVID-19 convalescent plasma if meets criteria per treatment benefit index (TBI)[8] or hematologic malignancy.[f] There is emerging evidence that high-titer COVID-19 convalescent plasma may be beneficial in patients who are immunocompromised (particularly with B-cell impairment) with persistent SARS-Cov-2 infection.[f]	• Mild to moderate COVID-19 disease (Table 2) ▸ IV remdesivir x 5 days • Severe COVID-19 disease (Table 2) ▸ IV remdesivir x 5 days with dexamethasone ◊ Although investigational (not standard of care), consider extending remdesivir duration to 10 days for patients hospitalized for COVID-19 if PCR Ct is still low after 5 days and the patient remains symptomatic or is not improving. ▸ The benefit of adding IV remdesivir to dexamethasone in patients who require mechanical ventilation or ECMO is unclear, although completion of 5 days of remdesivir is favored if already started before admission to intensive care unit (ICU). • A second immunomodulatory agent (eg, IL-6 inhibitor, JAK inhibitor) is often added for patients with rapidly or progressively declining oxygen saturation (SpO_2) (Table 2). ▸ Use of IL-6 inhibitor and JAK inhibitor is generally avoided in combination, in patients with uncontrolled active infection (bacterial, fungal, mycobacterial, or non-SARS-CoV-2 viral), or in patients with significant concurrent immunosuppression (eg, neutropenia, antineoplastic chemotherapy). ▸ Further details of immunomodulatory therapeutic options and indications for patients with moderate to severe COVID-19 are available on the NIH and IDSA COVID-19 websites.[5,9] • Additional information for COVID-19 convalescent plasma use is available at: https://covid-convalescentplasma-tbi-calc.org.

Table 4 References (COV-A 3 of 3)
Table 4 Footnotes (COV-6)

Note: All recommendations are category 2A unless otherwise indicated.
Clinical Trials: NCCN believes that the best management of any patient with cancer is in a clinical trial. Participation in clinical trials is especially encouraged.

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表 4：癌症患者的 COVID–19 治疗（续）

临床场景	抗病毒选择	评论
患者因非 COVID–19 适应症住院，但患有急性症状性 COVID–19	优选 · IV 瑞德西韦 其他· COVID–19 恢复期血浆	· IV remdesivir x 3 天（对于患有更严重疾病或并发免疫抑制的患者，可以考虑延长至 5 天）。·考虑高滴度 COVID–19 恢复期血浆。BA 前血浆可能不那么有效。 高滴度 COVID–19 恢复期血浆对循环病毒变体具有活性 可能不是统一可用的，并且需要当地中心的输血能力。 ·虽然存在使用口服抗病毒药物的机械原理，但 FDA 目前尚未批准将其用于住院患者。
患者因急性症状性 COVID–19 住院	首选· IV remdesivir x 5 天 其他·如果符合每次治疗的标准，则考虑 COVID–19 恢复期血浆 获益指数（TBI）或血液学 有新的证据表明，高滴度 COVID–19 恢复期 血浆可能对患者有益 免疫功能低下（特别是 B 细胞受损）并持续感染 SARS–Cov–2 的患者。	·轻度至中度 COVID–19 疾病（表 2） IV Remdesivir x 5 天·严重 COVID–19 疾病（表 2） IV remdesivir x 5 days with dexamethasone dad 虽然是研究性的（非标准治疗），但考虑延长 remdesivir 对于因 COVID–19 住院的患者，如果 PCR Ct 在 5 天后仍然较低，并且患者仍有症状或没有改善，则持续时间为 10 天。 在需要地塞米松治疗的患者中， 机械通气或 ECMO 尚不清楚，但如果在进入重症监护室（ICU）之前已经开始使用瑞德西韦，则最好完成 5 天的瑞德西韦治疗。·对于血氧饱和度（SpO）快速或进行性下降的患者，通常添加第二种免疫调节剂（例如，IL–6 抑制剂，JAK 抑制剂）（表 2）。 在患有不受控制的活动性感染（细菌、真菌、分枝杆菌或非 SARS–病毒）的患者中，通常避免联合使用 IL–6 抑制剂和 JAK 抑制剂。 CoV–2 病毒），或伴有显著并发免疫抑制的患者（例如，中性粒细胞减少症、化疗）。 有关中重度 COVID–19 患者的免疫调节治疗选择和适应症的更多详情，请访问 NIH 和 IDSA COVID–19 网站。 • 有关 COVID–19 恢复期血浆使用的更多信息，请访问：https：//covid–convalescentplasma–tbi–calc.org。

表 4 参考文献（COV–A 3/3）

表 4 脚注（COV–6）

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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TABLE 4: COVID-19 TREATMENT IN PATIENTS WITH CANCER (CONTINUED)[e]

Clinical Scenario	Antiviral Options	Comments
Persistent symptomatic COVID-19 infection; particularly B-cell impairment	Remdesivir or nirmatrelvir/ritonavir. Immunotherapy (COVID-19 convalescent plasma) may be added in combination.	• There are no uniform treatment recommendations, but clinical investigational approaches include use of: ▸ High-titer COVID-19 convalescent plasma[10,11] • To determine potential benefit of COVID-19 convalescent plasma, some providers (via clinical investigational approach) will first check: ▸ SARS-CoV-2 antibodies to nucleocapsid antigens to confirm lack of adequate humoral response post-infection ▸ Viral load burden by PCR after ID consult • Consider avoiding molnupiravir due to concerns for risk of producing escape viral mutants.
Persistent asymptomatic SARS-CoV-2–positive testing	Unclear if therapy indicated	• Clinical significance and role for supplemental therapy remain unclear. • SARS-CoV-2 RT-PCR testing does not distinguish replication-competent and infectious virus (eg, growth in cell-line culture) from inactive virus. ▸ In a prior review of 28 studies, the pooled median duration of RNA shedding from respiratory sources was 18.4 days with wide heterogeneity (range, 1–63 days) and relatively little difference based on disease severity.[12,13] ▸ Prolonged SARS-CoV-2 detection of replication of the competent virus (>100 days) has been reported in patients who are immunocompromised and has often been associated with a weak or absent antibody response to the virus.[14,15] • Clinical decisions can be influenced by viral load interpretation (PCR Ct), patient immunologic status, cancer response/lack of response to current therapeutics, etc.
Pre-exposure prophylaxis	None	• Please see the CDC for Use of COVID-19 Vaccines in the United States for more information.

Table 4 References (COV-A 3 of 3)

Note: All recommendations are category 2A unless otherwise indicated.
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表 4：癌症患者的 COVID-19 治疗（续）

临床场景	抗病毒选择	评论
持续症状性 COVID-19 感染;特别是 B 细胞受损	Remdesivir 或 Nirmatrelvir/ritonavir。免疫治疗（COVID-19 恢复期血浆）可联合使用。	没有统一的治疗建议，但临床研究方法包括使用： 高滴度 COVID-19 恢复期血浆 •为了确定 COVID-19 恢复期血浆的潜在益处，一些提供者（通过临床研究方法）将首先检查：SARS-CoV-2 抗体的核衣壳抗原，以确认缺乏足够的体液 感染后反应 ID 咨询后通过 PCR 测定的病毒载量负荷·考虑避免莫努匹拉韦，因为担心产生逃逸病毒突变体的风险。
持续无症状 SARS-CoV-2 阳性检测	Unclear if therapy indicated	• Clinical significance and role for supplemental therapy remain unclear. • SARS-CoV-2 RT-PCR testing does not distinguish replication-competent and infectious virus (eg, growth in cell-line culture) from inactive virus. In a prior review of 28 studies, the pooled median duration of RNA shedding from respiratory sources was 18.4 days with wide heterogeneity (range, 1—63 days) and relatively little difference based on disease severity. Prolonged SARS-CoV-2 detection of replication of the competent virus (>100 days) has been reported in patients who are immunocompromised and has often been associated with a weak or absent antibody response to the virus. • Clinical decisions can be influenced by viral load interpretation (PCR Ct), patient 免疫状态、癌症反应/对当前治疗缺乏反应等。
暴露前预防	None	·请参阅美国疾病控制和预防中心关于 COVID-19 疫苗使用的更多信息。

表 4 参考文献（COV-A 3/3）

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临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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TABLE 5: COVID-19 TREATMENT OPTIONS[i]

Treatment	Dosing/Duration	Comments
Antiviral Agents		
Remdesivir	• 200 mg IV on day 1; followed by 100 mg IV daily on days 2–5	• A nucleoside analogue that inhibits SARS-CoV-2 replication by interfering with the viral RNA-dependent RNA polymerase • 5-day treatment for patients who are hospitalized with symptomatic COVID-19 • 3-day treatment for patients who are not hospitalized with mild to moderate disease and high risk of disease progression within 7 days of symptom onset • If the patient is hospitalized for reasons other than severe COVID-19, give a 3-day course of remdesivir to inpatients incidentally diagnosed with COVID-19 who are at high risk for disease progression. • Avoid if ALT >10 x ULN or ALT elevated with signs of active hepatitis. • Can be extended up to 10 days
Nirmatrelvir/ ritonavir	• 300 mg nirmatrelvir / 100 mg ritonavir orally twice daily for 5 days • Renal impairment (estimated glomerular filtration rate [eGFR] eGFR <60 to ≥30 mL/min): 150 mg nirmatrelvir / 100 mg ritonavir twice daily for 5 days • Avoid with severe renal impairment (eGFR <30 mL/min) • Avoid with severe hepatic (Child-Pugh Class C) impairment	• A SARS-CoV-2 protease inhibitor; ritonavir boosts plasma nirmatrelvir concentrations through hepatic cytochrome 3A inhibition • Review potential for drug-drug interactions. Co-administration of ritonavir is contraindicated for many drugs. • For outpatient use only • Start within 5 days of symptom onset. • Ritonavir is a weak HIV protease inhibitor and may lead to HIV protease inhibitor resistance in uncontrolled HIV infection.
Molnupiravir	800 mg orally twice daily for 5 days	• A nucleoside analogue that causes replication failure of SARS CoV-2 replication by lethal mutagenesis • For outpatient use only • Start within 5 days of symptom onset. • Use only when nirmatrelvir and remdesivir are not available. • May cause fetal harm. Patients who may become pregnant should use reliable contraception during therapy and for 4 days after the last dose of molnupiravir. Sexually active males with partners who may become pregnant should also use effective contraception during therapy and for at least 3 months after the last molnupiravir dose.

[i] Dosing is for adults only. For pediatric dosing, consult with pharmacist. See CDC and American Academy of Pediatrics (AAP).

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表 5：COVID-19 治疗方案

治疗	给药/持续时间	评论
抗病毒剂		
Remdesivir	·第 1 天 200 mg IV;随后第 2-5 天每天 100 mg IV	·一种核苷类似物，通过干扰病毒 RNA 依赖性 RNA 聚合酶来抑制 SARS-CoV-2 复制 • 5-为有症状的 COVID-19 住院患者提供日间治疗 • 3-针对未因轻度至中度疾病住院的患者的日间治疗，症状发作后 7 天内疾病进展风险较高 • 如果患者因严重 COVID-19 以外的原因住院，则给予 3 天对偶然诊断为 COVID-19 的住院患者进行 Remdesivir 疗程，这些患者具有疾病进展的高风险。 ·如果 ALT >10 x ULN 或 ALT 升高伴活动性肝炎体征，则应避免。 ·可延长至 10 天
Nirmatrelvir/ ritonavir	· 300 mg 尼尔马瑞韦/ 100 mg 利托那韦，口服，每日两次，持续 5 天 min)：150 mg 尼马曲韦/ 100 mg 利托那韦 ·避免重度肾损害（eGFR <30 mL/min)·避免重度肝损害（Child-Pugh C 级）	• 一种 SARS-CoV-2 蛋白酶抑制剂;利托那韦提高血浆 Nirmatrelvir 浓度 ·审查药物相互作用的可能性。许多药物禁忌与利托那韦联合给药。·仅限门诊患者使用·症状发作后 5 天内开始使用。利托那韦是一种弱的 HIV 蛋白酶抑制剂，在不受控制的 HIV 感染中可能导致 HIV 蛋白酶抑制剂耐药性。
Molnupiravir	800 mg 口服，每日两次，持续 5 天	·一种核苷类似物，通过致死诱变导致 SARS CoV-2 复制失败 ·仅限门诊使用 ·在症状发作后 5 天内开始。 ·仅在 nirmatrelvir 和 remdesivir 不可用时使用。 ·可能对胎儿造成伤害。可能怀孕的患者应在治疗期间和末次莫努匹韦给药后 4 天内采取可靠的避孕措施。性 有伴侣的活跃男性也应该使用有效的 在治疗期间和末次莫努匹韦给药后至少 3 个月内避孕。

[i] 剂量仅适用于成人。关于儿科用药，请咨询药剂师。参见 CDC 和美国儿科学会（AAP）。

注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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注：除非另有说明，否则所有建议均为 2A 类。
临床试验：NCCN 认为，任何癌症患者的最佳管理是在临床试验中。特别鼓励参加临床试验。

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ABBREVIATIONS

ACIP	Advisory Committee on Immunization Practices
ADV	adenovirus
ALL	acute lymphoblastic leukemia
ALT	alanine aminotransferase
AML	acute myeloid leukemia
BAL	bronchoalveolar lavage
BCMA	B-cell maturation antigen
CAR	chimeric antigen receptor
CBC	complete blood count
CDI	*Clostridioides difficile* infection
CISNE	clinical index of stable febrile neutropenia
CLL	chronic lymphocytic leukemia
CMV	cytomegalovirus
CNS	central nervous system
COPD	chronic obstructive pulmonary disease
CPK	creatine phosphokinase
CrCl	creatinine clearance
CRS	cytokine release syndrome
Ct	cycle threshold
CTLA-4	cytotoxic T-lymphocyte–associated antigen 4
DFA	direct fluorescence antibody
DR	delayed release
DTaP	diphtheria, tetanus, and acellular pertussis
ECMO	extracorporeal mechanical oxygenation
eGFR	estimated glomerular filtration rate
ENT	ear, nose, and throat
ESBL	extended-spectrum beta-lactamase
FiO_2	fraction of inspired oxygen
FRα	folate receptor alpha
G-CSF	granulocyte colony-stimulating factor
GM-CSF	granulocyte-macrophage colony-stimulating factor
GI	gastrointestinal
GVHD	graft-versus-host disease
HBcAb	hepatitis B core antibody
HBsAg	hepatitis B surface antigen
HBV	hepatitis B virus
HCT	hematopoietic cell transplant
HCV	hepatitis C virus
HHV	human herpes virus
Hib	haemophilus influenzae type b
HIV	human immunodeficiency virus
HPV	human papillomavirus
HSV	herpes simplex virus
HZ	herpes zoster
H&P	history and physical
ICU	intensive care unit
ID	infectious disease
IL-6	interleukin-6
irAE	immune-related adverse event
IST	immunosuppressive therapy
IVIG	intravenous immunoglobulin
mAb	monoclonal antibody
MDRO	multidrug-resistant organism
MDS	myelodysplastic syndromes
MenACWY	meningococcal conjugate vaccine, quadrivalent
MMR	measles, mumps, rubella
MRSA	methicillin-resistant *Staphylococcus aureus*
NAAT	nucleic acid amplification testing
NG	nasogastric
PaO2	partial pressure of oxygen
PCR	polymerase chain reaction
PCV	pneumococcal conjugate vaccine
PD-1	programmed cell death protein 1
PD-L1	programmed death ligand 1
PJP	*pneumocystis jirovecii* pneumonia
PML	progressive multifocal leukoencephalopathy
PPI	proton pump inhibitor
PPSV23	pneumococcal polysaccharide vaccine
RCC	renal cell carcinoma
RSV	respiratory syncytial virus
RT-PCR	reverse transcriptase polymerase chain reaction
RZV	recombinant zoster vaccine
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SSRI	selective serotonin reuptake inhibitor
STI	sexually transmitted infection
SpO_2	oxygen saturation
TB	tuberculosis
TBI	treatment benefit index

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缩写

ACIP	免疫实践咨询委员会
ADV	腺病毒
ALL	急性淋巴细胞白血病
ALT	丙氨酸氨基转移酶
AML	急性髓性白血病
BAL	支气管肺泡灌
BCMA	b 细胞成熟抗原
CAR	嵌合抗原受体
CBC	全血细胞计数
CDI	艰难梭菌感染
CISNE	稳定性发热性中性粒细胞减少症的临床指标
CLL	慢性淋巴细胞白血病
CMV	细胞病毒
CNS	中枢神经系统
COPD	慢性阻塞性肺疾病
CPK	肌酸磷酸激酶

CrCl 肌酐清除率 CRS 细胞因子释放综合征 Ct 循环阈值 CTLA–4 细胞毒性 T 淋巴细胞–相关抗原 4

DFA	直接荧光抗体
DR	延迟释放
DTaP	白喉、破伤风和无细胞百日咳
ECMO	体外机械氧合
eGFR	估计的肾小球滤过 rate
ENT	耳鼻喉
ESBL	超广谱β–内酰胺酶

吸入氧的 FiO 2 分数
FRα叶酸受体α G–CSF 粒细胞集落刺激因子

GM–CSF	粒细胞–巨噬细胞集落刺激因子
GI	胃肠
GVHD	graft–versus–host disease

HBcAb 肝炎 B 核心抗体 HBsAg 肝炎 B 表面抗原 HBV 肝炎 B 病毒 HCT 造血细胞移植 HCV 丙型肝炎病毒 HHV 人类疱疹病毒 HiB 嗜血杆菌流感 B 型 HIV 人类免疫缺陷病毒 HPV 人类乳头状瘤病毒 HSV 单纯疱疹病毒 HZ 带状疱疹 H&P 病史及体检 ICU 重症监护室 ID 传染病 IL–6 白细胞介素–6 irAE 免疫–相关不良事件 IST 免疫抑制治疗 IVIG 静脉注射免疫球蛋白 mAb 单克隆抗体 MDRO 多药耐药微生物 MDS 骨髓增生异常综合征

MenACWY	脑膜炎球菌结合四价疫苗
MMR	麻疹、腮腺炎、风疹
MRSA	耐甲金黄色葡萄球菌
NAAT	核酸扩增检测
NG	鼻胃
PaO2	氧分压
PCR	聚合酶链反应
PCV	肺炎球菌结合疫苗
PD–1	程序性细胞死亡蛋白 1
PD–L1	程序性死亡配体 1
PJP	耶氏肺孢子虫肺炎
PML	进行性多灶性白质脑病
PPI	质子泵抑制剂
PPSV23	肺炎球菌多糖疫苗
RCC	肾细胞癌
RSV	呼吸道合胞病毒
RT–PCR	逆转录聚合酶链反应
RZV	重组带状疱疹疫苗
SARS–CoV–2	严重急性呼吸综合征冠状病毒 2
SSRI	羟色胺再摄取抑制剂
STI	性传播感染

SpO 氧饱和度 TB 结核病 TBI 处理效益指数

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ABBREVIATIONS

Td	**tetanus-diphtheria**
Tdap	**tetanus, diphtheria, pertussis**
TDM	**therapeutic drug monitoring**
ULN	**upper limit of normal**
VADs	**vascular access devices**
VEGF	**vascular endothelial growth factor**
VOD	**veno-occlusive disease**
VRE	**vancomycin-resistant enterococci**
VZV	**varicella zoster virus**
ZVL	**herpes zoster vaccine**

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缩写

Td 破伤风–白喉 Tdap 破伤风、白喉、百日咳 TDM
治疗药物监测 ULN 正常上限 VAD 血管通路装置
VEGF 血管内皮生长

	因子
VOD	静脉闭塞病
VRE	耐万古霉素肠球菌
VZV	水痘带状疱疹病毒
ZVL	带状疱疹疫苗

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence, there is uniform NCCN consensus that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

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NCCN 证据和共识类别
第 1 类基于高级别证据，NCCN 一致认为干预是适当的。
2A 类基于较低级别的证据，NCCN 一致认为干预是适当的。
2B 类基于较低级别的证据，NCCN 一致认为干预是适当的。
第 3 类基于任何级别的证据，NCCN 对干预是否适当存在重大分歧。

除非另有说明，所有建议均为 2A 类。

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National Comprehensive Cancer Network®

NCCN Guidelines Version 3.2024
Prevention and Treatment of Cancer-Related Infections

Discussion

This discussion corresponds to the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections.
Last updated September 23, 2024.

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讨论

该讨论对应于 NCCN 癌症相关感染预防和治疗指南。
最后更新于 2024 年 9 月 23 日。

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NCCN Guidelines Version 3.2024
Prevention and Treatment of Cancer-Related Infections

Overview

There is an increased risk of infection in patients with cancer that results in higher morbidity and mortality. In certain instances, the malignancy itself can predispose patients to severe or recurrent infections. Neutropenia has been recognized as a major risk factor for the development of infections in patients with cancer undergoing chemotherapy. Effective strategies to anticipate, prevent, and manage these infectious complications have led to improved outcomes.[1-4] Due to advances in antimicrobial therapy, it is less common for patients with acute leukemia or patients undergoing hematopoietic cell transplantation (HCT) to die from infections during the neutropenic period.

Although neutropenia remains a key risk factor for infections, other immunocompromised states pose at least equal risk. Allogeneic HCT recipients with neutrophil recovery who require intensive immunosuppressive therapy (IST) for graft-versus-host disease (GVHD) are an example of patients who do not have neutropenia, but are at great risk for common bacterial, viral, and opportunistic infections.[5-8] The spectrum of infectious diseases in allogeneic HCT recipients with GVHD is distinct from neutropenia. The NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Prevention and Treatment of Cancer-Related Infections discuss infections in patients who are neutropenic and immunocompromised non-neutropenic with cancer. In addition to corticosteroids and purine analogs, the increased use of monoclonal antibodies, proteasome inhibitors, and other emerging cancer therapeutics like CAR-T cell therapy has generated an ever more complex assessment of those who are immunocompromised. The scope of these guidelines is to address infections that may be seen in all of these immunocompromised populations.

The NCCN Guidelines® for the Prevention and Treatment of Cancer-Related Infections characterize the major pathogens to which patients with cancer are susceptible, with a focus on the prevention, diagnosis, and treatment of major common and opportunistic infections. The guidelines are largely divided into 5 sections comprising discussions on the following: 1) risk factors for infection (major host factors that predispose patients to infectious diseases); 2) prevention of infectious complications (including the use of antimicrobial prophylaxis and preemptive therapy); 3) infection concerns and recommendations regarding immune and targeted treatments; 4) management of neutropenic fever; and 5) management of site-specific infections (eg, pneumonia, abdominal infections, catheter-associated infections). These guidelines provide a framework for prevention and treatment of infections that should be applied in conjunction with careful, individual patient evaluation and with an understanding of both the host factors that predispose patients to specific infectious diseases and antimicrobial susceptibility patterns. Additionally, the guidelines are based primarily on studies with adult patients and application of these recommendations to pediatric patients may differ. Consultation with an infectious disease expert is highly recommended.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of this version of the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections, an electronic search of the PubMed database was performed to obtain key literature in Prevention and Treatment of Cancer-Related Infections published since the previous Guidelines updates, using the following search terms: cancer related infections; cancer infections; cancer induced infections; prevention of cancer related infections; cancer and virus; cancer and bacterial; cancer and fungal; cancer and microbial; cancer and hepatitis; cancer and influenza; cancer and candida; cancer and aspergillus; cancer and

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NCCN 指南第 3.2024 版癌症相关感染的预防和治疗

概述

癌症患者感染的风险增加，导致发病率和死亡率升高。在某些情况下，恶性肿瘤本身可使患者易患严重或复发性感染。中性粒细胞增多症已被认为是接受化疗的癌症患者发生感染的主要风险因素。有效的策略来预测，预防和管理这些感染并发症，导致改善的 outcome.Due 的抗菌治疗的进步，它是罕见的急性白血病患者或接受造血细胞移植（HCT）的患者死于感染期间的贫血。

Although neutropenia remains a key risk factor for infections, other immunocompromised states pose at least equal risk. Allogeneic HCT recipients with neutrophil recovery who require intensive immunosuppressive therapy (IST) for graft–versus–host disease (GVHD) are an example of patients who do not have neutropenia, but are at great risk for common bacterial, viral, and opportunistic infections.The spectrum of infectious diseases in allogeneic HCT recipients with GVHD is distinct from neutropenia. The NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines) for Prevention and Treatment of Cancer–Related Infections discuss infections in patients who are neutropenic and immunocompromised non–neutropenic with cancer. In addition to corticosteroids and purine analogs, the increased use of monoclonal antibodies, proteasome inhibitors, and other emerging cancer therapeutics like CAR–T cell therapy has generated an ever more complex assessment of those who are immunocompromised. The scope of these guidelines is to address infections that may be seen in all of these immunocompromised populations.

重试 错误原因

NCCN 癌症相关感染的预防和治疗指南描述了癌症患者感染的主要病原体。癌症是易感的，重点是预防，诊断和治疗主要的常见和机会性感染。该指南主要分为 5 个部分，包括对以下内容的讨论：1）感染的风险因素（使病人易患传染病的主要宿主因素）；2）预防传染性并发症（包括使用抗菌预防和先发制人的治疗）；3）感染问题和有关免疫和靶向治疗的建议; 4）贫血热的管理;和 5）部位特异性感染（例如肺炎、腹部感染、导管相关感染）的管理。这些指南为预防和治疗感染提供了一个框架，应与仔细的个体患者评估结合使用，并了解使患者易患特定传染病的宿主因素和抗菌药物敏感性模式。 此外，该指南主要基于成人患者的研究，这些建议对儿科患者的应用可能有所不同。

强烈建议咨询传染病专家。

准则更新方法

NCCN 指南的制定和更新的完整细节可在 www.NCCN.org 上查阅。

文献检索标准

在更新此版本的 NCCN 癌症相关感染预防和治疗指南之前，使用以下检索词对 PubMed 数据库进行了电子检索，以获得自上次指南更新以来发表的癌症相关感染预防和治疗方面的关键文献：癌症相关感染;癌症感染;癌症诱导的感染;癌症相关感染的预防;癌症和病毒;癌症和细菌;癌症和真菌;癌症和微生物;癌症和肝炎;癌症和流感;癌症和念珠菌;癌症和曲霉菌;癌症和

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clostridium; cancer and staphylococcus; cancer and pseudomonas; cancer and pneumocystis; cancer and herpes; cancer and varicella zoster; or cancer and HIV. The PubMed database was chosen as it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[9]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Meta-Analysis; Randomized Controlled Trial; Systematic Reviews; Validation Studies; and Practice Guidelines.

The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines and discussed by the Panel have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[10] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Host Factors That Predispose Patients to Infectious Complications

Immunodeficiencies Associated with Primary Malignancy

Certain malignancies are inherently associated with immune deficits. Patients with hematologic malignancies (eg, chronic and acute leukemias, non-Hodgkin lymphomas [NHL], myelodysplastic syndromes [MDS]) may be leukopenic due to infiltration of the marrow with malignant cells or due to a dysfunctional marrow. Patients with chronic lymphocytic leukemia (CLL) frequently have hypogammaglobulinemia leading to increased susceptibility to encapsulated bacteria, principally *Streptococcus pneumoniae*.[11] Such patients may have recurrent sinopulmonary infections and septicemia. Patients with multiple myeloma are often functionally hypogammaglobulinemic; the total level of immunoglobulin production may be elevated, but the repertoire of antibody production is restricted. Savage et al[12] noted a biphasic pattern of infection among patients with multiple myeloma. Infections by *Streptococcus pneumoniae* and *Haemophilus influenzae* occurred early in the disease and in patients with disease that responds to chemotherapy, whereas infections by *Staphylococcus aureus* and gram-negative pathogens occurred more commonly in advanced disease and during neutropenia.

Patients with advanced or refractory malignancy have a greater risk of infectious complications than those on earlier lines of therapy. Refractory hematologic malignancies can be associated with marrow failure caused by the underlying disease or from the multiple lines of prior cytotoxic therapy or IST. Patients with CLL who receive multiple chemotherapeutic

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梭菌属;癌症和乳腺癌;癌和假性囊肿;癌症和肺囊虫病;癌症和疱疹;癌症和水痘带状疱疹;或者癌症和艾滋病选择 PubMed 数据库是因为它仍然是最广泛使用的医学文献和同行评议生物医学文献索引资源。

通过选择以英文发表的人类研究缩小了搜索结果。结果仅限于以下文章类型：临床试验，III 期;临床试验，IV 期;指南;荟萃分析;随机对照试验;系统综述;验证研究;和实践指南。

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敏感/包容性语言使用

NCCN 指南努力使用语言，推进公平，包容和代表性的目标。NCCN 指南奋进使用语言，以人为本;不污名化;反种族主义，反阶级，反厌女症，反年龄歧视，反能力歧视和反体重偏见;并包括所有性取向和性别认同的个人。NCCN 指南包含非性别语言，而不是专注于器官特定的建议。这种语言更准确，更具包容性，可以帮助充分满足所有性取向和性别认同的个人的需求。NCCN 指南在引用统计数据、建议或来自不使用包容性术语的组织或来源的数据时，将继续使用男性、女性、女性和男性。大多数研究没有报告性和性别数据是如何收集的，并且互换使用这些术语或不一致。

如果来源不区分性别与出生时指定的性别或存在的器官，则假定信息主要代表顺性个体。NCCN 鼓励研究人员在未来的研究中收集更具体的数据，并鼓励组织在未来的分析中使用更具包容性和准确性的语言。

使患者易患感染性并发症的宿主因素

与原发性胰腺炎相关的免疫缺陷

某些恶性肿瘤与免疫缺陷固有相关。恶性血液病患者（如慢性和急性白血病、非霍奇金淋巴瘤[NHL]、骨髓增生异常综合征[MDS]）可能因恶性细胞浸润骨髓或骨髓功能障碍而出现白细胞减少。慢性淋巴细胞性白血病（CLL）患者经常患有低丙种球蛋白血症，

对包裹的细菌，主要是肺炎链球菌敏感。这类患者可能有反复的鼻窦炎感染和败血症。多发性骨髓瘤患者通常是功能性低丙种球蛋白血症;免疫球蛋白产生的总水平可能升高，但抗体产生的库受到限制。Savage 等人注意到多发性骨髓瘤患者感染的双相模式。肺炎链球菌和流感嗜血杆菌的感染发生在疾病早期和对化疗有反应的患者中，而金黄色葡萄球菌和革兰氏阴性病原体的感染更常见于晚期疾病和中性粒细胞减少症期间。

晚期或难治性恶性肿瘤患者发生感染并发症的风险高于早期治疗的患者。难治性恶性血液病可能与基础疾病或既往多线细胞毒性治疗或 IST 引起的骨髓衰竭相关。接受多种化疗的 CLL 患者

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regimens are at a significantly increased risk of developing severe infections.[13] A retrospective study showed that nearly 90% of patients who are heavily pretreated (median number of prior regimens, 3; range, 1–8) with fludarabine-refractory CLL experienced serious infectious complications requiring hospitalization.[14] These infections resulted from bacterial, viral, fungal, and opportunistic pathogens, including *Pneumocystis jirovecii* (formerly called *Pneumocystis carinii)*.[14]

Solid tumors may predispose patients to infection because of anatomic factors. Tumors that overgrow their blood supply become necrotic, thus forming a nidus for infection. Endobronchial tumors may cause recurrent postobstructive pneumonias. Abdominal tumors may obstruct the genitourinary or hepatobiliary tracts, predisposing patients to pyelonephritis and cholangitis, respectively. Direct invasion through the colonic mucosa is associated with local abscess formation and sepsis by enteric flora. Patients undergoing surgery for malignancies may be at high risk for infectious complications as a result of the type of surgery (eg, esophagectomy, hepatobiliary reconstruction), the extent of tumor burden, their preoperative performance status, and any previous surgery, chemotherapy, or radiation therapy. Patients with advanced malignancy are also commonly malnourished, which further increases the risk of infection.

Neutropenia

Factors that predispose the patients who are neutropenic to infection include the absence of granulocytes; the disruption of the integumentary, mucosal, and mucociliary barriers; and the inherent microbial flora shifts that accompany severe illness and antimicrobial usage. The signs and symptoms of infection are often absent or muted in the absence of neutrophils, but fever remains an early, although nonspecific, sign.[2] Approximately 50% to 60% of patients who become febrile have an established or occult infection.[15] Roughly 10% to 20% of patients with neutrophil counts <100 cells/mcL will develop a bloodstream infection.[16] Primary sites of infection are the alimentary tract (ie, mouth, pharynx, esophagus, large and small bowel, rectum), sinuses, lungs, and skin.

Initial infections early in the course of fever and neutropenia are primarily bacterial, whereas antibiotic-resistant bacteria, yeast, other fungi, and viruses are common causes of subsequent infections.[17,18] Coagulase-negative staphylococci, *S aureus,* viridans group streptococci, and enterococci are the major gram-positive pathogens. Coliforms (eg, *Escherichia coli*, *Klebsiella* species*, Enterobacter* species) and *Pseudomonas aeruginosa* (*P aeruginosa*) are the most common gram-negative infections complicating neutropenia.[17] Herpes simplex virus (HSV), respiratory syncytial virus (RSV), parainfluenza, and influenza A and B are occasionally initial pathogens.[18] Infections due to *Candida* species may occur later in the course of neutropenia, particularly as a consequence of gastrointestinal (GI) mucositis. *Aspergillus* species and other filamentous fungi are important causes of morbidity and mortality in patients with severe and prolonged neutropenia.[17,19] Deaths resulting from infections identified at the onset of fever during neutropenia remain uncommon, and most infection-associated deaths result from subsequent infections during the course of neutropenia.

A seminal study demonstrated that as the neutrophil count decreases to <500 cells/mcL (defined as *neutropenia*), the susceptibility to infection increases.[20] The frequency and severity of infection are inversely proportional to the neutrophil count. The risks of severe infection and bloodstream infection are greatest when the neutrophil count is <100 cells/mcL. The rate of decline of the neutrophil count and the duration of neutropenia are also critical factors that measure bone marrow reserve and are highly correlated with the severity of infection and clinical outcome.

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一项回顾性研究显示，近 90%的氟达拉滨难治性 CLL 患者接受了大量预治疗（既往治疗方案的中位数为 3;范围为 1–8），发生了需要住院治疗的严重感染并发症。

卡氏肺孢子虫（Pneumocystis carinii）

由于解剖因素，实体瘤可能使患者易于感染。肿瘤生长超过其血液供应变得坏死，从而形成感染病灶。支气管内肿瘤可引起复发性阻塞后肺炎。腹部肿瘤可阻塞泌尿生殖道或肝胆道，使患者分别易患肾盂肾炎和胆管炎。通过结肠粘膜的直接侵袭与局部脓肿形成和肠道植物群引起的脓毒症有关。由于手术类型（例如，食管切除术、肝胆重建术）、肿瘤负荷程度、术前体能状态以及既往任何手术、化疗或放疗，接受恶性肿瘤手术的患者感染并发症的风险可能较高。患有晚期恶性肿瘤的患者通常也营养不良，这进一步增加了感染的风险。

细胞减少

导致血小板减少患者易感染的因素包括粒细胞缺乏;表皮、粘膜和粘膜纤毛屏障破坏;以及伴随严重疾病和抗菌药物使用的固有微生物植物群变化。在中性粒细胞缺乏的情况下，感染的体征和症状通常不存在或减弱，但发热仍然是一种早期的非特异性体征。大约 50%至 60%的发热患者有一个

确诊或隐匿性感染。大约 10%至 20%的

中性粒细胞计数<100 个细胞/mcL 将发展为血流感染。感染的主要部位是消化道（即口、咽、食道、大肠和小肠、直肠）、鼻窦、肺和皮肤。

在发热和中性粒细胞减少症过程的早期初始感染主要是细菌性的，而耐药性细菌、酵母菌、其他真菌和病毒是随后感染的常见原因。革兰阳性菌主要为凝固酶阴性葡萄球菌、金黄色葡萄球菌、草绿色链球菌和肠球菌。大肠菌（例如，大肠埃希菌、克雷伯菌属、肠杆菌属）和铜绿假单胞菌（铜绿假单胞菌）是最常见的革兰阴性菌感染并发中性粒细胞减少症。单纯疱疹病毒（HSV）、呼吸道合胞病毒（RSV）、副流感病毒以及甲型和 B 型流感病毒偶尔是初始病原体。念珠菌属的感染可能在中性粒细胞减少症的后期发生，特别是胃肠（GI）粘膜炎的结果。 曲霉菌属和其他丝状真菌是严重和长期中性粒细胞减少症患者发病和死亡的重要原因。中性粒细胞减少症期间发热时发现的感染导致的死亡仍然不常见，大多数感染相关死亡是中性粒细胞减少症过程中的后续感染所致。

一项开创性研究表明，当中性粒细胞计数降至<500 个细胞/mcL（定义为中性粒细胞减少症）时，感染的易感性增加。感染的频率和严重程度与中性粒细胞计数成反比。当中性粒细胞计数<100 个细胞/mcL 时，严重感染和血流感染的风险最大。中性粒细胞计数的下降速率和中性粒细胞减少的持续时间也是测量骨髓储备的关键因素，并且与感染的严重程度和临床结果高度相关。

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Disruption of Mucosal Barriers

The mucosal linings of the GI, sinopulmonary, and genitourinary tracts constitute the first line of host defense against a variety of pathogens. Mucosal immunity is impaired by chemotherapy and radiation therapy. When the physical protective barrier conferred by the epithelial lining is compromised, local flora may invade. Furthermore, neutropenia and loss of the epithelial cell anatomic barrier may predispose patients to typhlitis (neutropenic enterocolitis). Chemotherapy-related GI mucositis predisposes patients to blood stream infections by viridans group streptococci,[21-24] gram-negative rods, and *Candida* species.[25,26]

Splenectomy and Functional Asplenia

In the spleen, rapid antigen presentation occurs, which leads to the production of opsonizing antibodies by B cells. The removal of non-opsonized bacteria protects against encapsulated bacteria to which the patient is not yet immune. Splenic irradiation results in functional asplenia, which predisposes patients to pneumococcal sepsis. Functional asplenia is also a late complication of severe GVHD.[27] Thus, in allogeneic HCT recipients, fever in the late transplant period must be evaluated promptly (similar to patients with asplenia) because of the risk of overwhelming infection by encapsulated pathogens.

Overwhelming sepsis by encapsulated bacteria is also the principal risk factor for infection in patients who are asplenic. The most common pathogen is *Streptococcus pneumoniae*, but other pathogens include *H influenzae* and *Neisseria meningitidis.* The NCCN Guidelines provide recommendations for immunization with the pneumococcal polysaccharide and meningococcal vaccines (see *Vaccination*).

Corticosteroids and Other Immunosuppressive Agents

While many agents administered to patients with cancer can cause some degree of immunosuppression, certain agents (detailed in this section) are more likely to put patients at a risk for serious infection. When assessing a possible infection, it is important to note that many of the newer immunotherapies (eg, nivolumab, ipilimumab, pembrolizumab) can cause inflammation-related side effects that may be mistaken for infection.

Corticosteroids

High-dose corticosteroids (>20 mg prednisone daily) have profound effects on the distribution and function of neutrophils, monocytes, and lymphocytes. In patients with cancer, corticosteroids are seldom the only immunosuppressive agents administered, and it is therefore difficult to delineate the degree of impairment in host defense elicited by the corticosteroid regimen alone. The risk of infections is a function of the dose and duration of corticosteroids, coexisting immunodeficiencies (such as neutropenia and use of other immunosuppressive agents), and the status of the malignancy. Corticosteroids blunt fever and local signs of infection, such as peritonitis.

Purine Analogue Therapies

Purine analogues (including fludarabine, clofarabine, nelarabine, and cladribine [2-CdA]) are used to treat a variety of hematologic malignancies. These therapies are lymphocytotoxic, primarily affecting CD4+ lymphocytes. In previously treated patients with CLL, fludarabine treatment (especially in combination with other IST) was associated with infections such as listeriosis, mycobacterial infections, and opportunistic fungal and viral infections.[28] Additionally, fludarabine was associated with infections caused by *Pneumocystis jirovecii,* which is the causative agent of pneumocystis pneumonia (PJP), also known as pneumocystosis. When used alone, purine analogs are associated with an increased risk for infection; risk of infection is further escalated when purine analogs are combined with other immunosuppressive or cytotoxic agents.[29] The combination of fludarabine and corticosteroids is more immunosuppressive than either agent alone.[30] Fludarabine plus

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粘液屏障破坏

胃肠道、鼻窦和泌尿生殖道的粘膜衬里构成了宿主抵抗各种病原体的第一道防线。

化疗和放疗会损害粘膜免疫力。当由上皮衬里赋予的物理保护屏障受损时，局部植物群可能侵入。此外，中性粒细胞减少症和上皮细胞解剖屏障的丧失可能使患者易患盲肠炎（贫血性小肠结肠炎）。化疗相关的胃肠道粘膜炎使患者易于受到草绿色链球菌、革兰氏阴性杆菌和念珠菌属的血流感染。

脾切除术与功能性无脾

在脾脏中，发生快速抗原呈递，这导致 B 细胞产生调理抗体。非调理素化细菌的去除可保护患者免受尚未免疫的包囊细菌的侵害。脾照射导致功能性无脾，使患者易患肺炎球菌败血症。功能性无脾也是严重 GVHD 的晚期并发症。因此，在同种异体 HCT 受者中，由于包裹病原体的压倒性感染风险，必须及时评估移植后期的发热（类似于无脾患者）。

被包裹的细菌引起的压倒性脓毒症也是无脾患者感染的主要危险因素。最常见的病原体是肺炎链球菌，但其他病原体包括流感嗜血杆菌和脑膜炎奈瑟菌。NCCN 指南提供了肺炎球菌多糖疫苗和脑膜炎球菌疫苗的免疫接种建议（见疫苗接种）。

皮质类固醇和其他免疫抑制剂

虽然给予癌症患者的许多药剂可引起一定程度的免疫抑制，但某些药剂（在本节中详述）是可接受的。

更有可能使患者面临严重感染的风险。在评估可能的感染时，重要的是要注意许多较新的免疫疗法（例如，nivolumab，ipilimumab，pembrolizumab）可能会导致炎症相关的副作用，这些副作用可能被误认为是感染。

皮质类固醇

大剂量皮质类固醇（>20 mg 泼尼松每日）对中性粒细胞、单核细胞和淋巴细胞的分布和功能有深远的影响。在癌症患者中，皮质类固醇很少是唯一的免疫抑制剂，因此很难描述单独使用皮质类固醇方案引起的宿主防御受损程度。感染的风险是皮质类固醇的剂量和持续时间，共存的免疫缺陷（如中性粒细胞减少症和使用其他免疫抑制剂）和恶性肿瘤的状态的函数。皮质类固醇钝热和局部感染体征，如腹膜炎。

嘌呤抗肿瘤治疗

嘌呤类似物（包括氟达拉滨、氯法拉滨、奈拉滨和克拉屈滨[2-CdA]）用于治疗各种血液恶性肿瘤。这些疗法具有淋巴细胞毒性，主要影响 CD 4+淋巴细胞。在既往接受过治疗的 CLL 患者中，氟达拉滨治疗（特别是与其他 IST 联合治疗）与感染相关，如肺孢子虫病、分枝杆菌感染、机会性真菌和病毒感染。此外，氟达拉滨与耶氏肺孢子虫引起的感染相关，耶氏肺孢子虫是肺孢子虫肺炎（PJP）的病原体，也称为肺孢子虫病。当单独使用时，嘌呤类似物与感染的风险增加相关;当嘌呤类似物与其他免疫抑制剂或细胞毒性剂联合使用时，感染的风险进一步增加。氟达拉滨和皮质类固醇的联合使用比单独使用任何一种药物更具有免疫抑制作用。

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prednisone results in a uniform depression of CD4+ cells that may persist for several months after completion of therapy.[31] In one series, 14 of 264 patients (5%) with CLL developed either PJP or listeriosis, and 3 cases occurred >1 year after therapy in patients who were in remission.[31]

Alemtuzumab

An increasing number of allogeneic HCT recipients and patients with hematologic malignancies are being treated with novel monoclonal antibodies that cause a depletion of lymphocyte subsets. Alemtuzumab is a humanized monoclonal antibody that targets CD52, which is abundantly expressed on most normal and malignant B and T lymphocytes. This agent has been used most extensively in patients with CLL who have disease that has progressed on fludarabine therapy. Alemtuzumab has been associated with grade 3 or 4 neutropenia in about 40% of patients with previously untreated CLL and in 56% to 78% of patients with fludarabine-refractory disease.[32-34] Alemtuzumab is associated with prolonged and severe lymphopenia in most patients. Prescribing information indicates that 4 weeks after initiation of alemtuzumab, the median CD4+ count was 0 cells/mcL and 6 months after discontinuation, the count was 238 cells/mcL in previously untreated patients. The CD8+ cell counts changed in a similar manner. In patients who are previously treated and are receiving alemtuzumab, CD4+ and CD8+ counts may not recover to baseline levels until >1 year after completion of therapy. Infections pose a concern for morbidity and/or mortality in alemtuzumab recipients, particularly for patients with heavily pretreated, fludarabine-refractory disease.[14,33,35] Bacterial, viral, fungal, mycobacterial, and *Pneumocystis jirovecii* infections have been reported with alemtuzumab.[33,35,36]

Anti-infective prophylaxis against herpes viruses and PJP is recommended in patients receiving alemtuzumab treatment (see *Antiviral Prophylaxis and Preemptive Antiviral Therapy* and *Prophylaxis for Pneumocystis jirovecii*). Several studies have shown that patients treated with alemtuzumab have increased susceptibility to cytomegalovirus (CMV) reactivation and disease.[32,33,37-39] In the absence of a large randomized controlled trial, the Infectious Diseases Working Party of the German Society of Hematology and Medical Oncology does not currently recommend CMV surveillance in alemtuzumab recipients.[40] Conversely, both the Working Group of the UK CLL Forum on behalf of the British Committee for Standards in Haematology and the International Workshop on CLL on behalf of the National Cancer Institute (NCI) recommend routine monitoring for CMV in patients with CLL who have therapies associated with the potential for CMV reactivation (eg, alemtuzumab or HCT).[41,42] The NCCN Panel recommends that surveillance for CMV reactivation is conducted at least weekly using polymerase chain reaction (PCR) in alemtuzumab recipients (see *Antiviral Prophylaxis and Preemptive Antiviral Therapy: Cytomegalovirus)*. Other compounds known to cause lymphopenia (eg, proteasome inhibitors) are associated with an increased risk of herpes zoster reactivation; therefore, prophylaxis with acyclovir, famciclovir, or valacyclovir is recommended.

Anti-CD20 Monoclonal Antibodies

Anti-CD20 monoclonal antibodies (eg, rituximab, ofatumumab, obinutuzumab) are widely used in the treatment of patients with B-cell lymphoid malignancies. The use of these monoclonal antibodies has been associated with increased risks for hepatitis B virus (HBV) reactivation, which can lead to fulminant hepatitis, liver failure, and/or death.[43-48] Antiviral prophylaxis is generally recommended for patients who test positive for HBV surface antigen (see *Antiviral Prophylaxis and Preemptive Antiviral Therapy: Hepatitis B virus*).

The use of anti-CD20 monoclonal antibodies in patients with B-cell malignancies has been associated with rare instances of progressive multifocal leukoencephalopathy (PML). PML is a demyelinating disease of

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在一个系列中，264 名 CLL 患者中有 14 名（5%）发展为 PJP 或骨髓增生，3 例发生在治疗后>1 年的缓解患者中。

Alemtuzumab

越来越多的同种异体 HCT 接受者和恶性血液病患者正在接受新型单克隆抗体治疗，这些抗体导致淋巴细胞亚群耗竭。Alemtuzumab 是一种靶向 CD 52 的人源化单克隆抗体，CD 52 在大多数正常和恶性 B 和 T 淋巴细胞上大量表达。该药物已广泛用于在氟达拉滨治疗期间疾病进展的 CLL 患者。阿仑单抗与约 40%未经治疗的 CLL 患者和 56%至 78%的氟达拉滨难治性疾病患者的 3 级或 4 级中性粒细胞减少症有关。阿仑单抗与大多数患者的长期和严重淋巴细胞减少症有关。处方信息表明，在既往未接受过治疗的患者中，开始 Alemtuzumab 治疗后 4 周，中位 CD 4+计数为 0 个细胞/mcL，停药后 6 个月，计数为 238 个细胞/mcL。CD 8+细胞计数以类似方式变化。 在既往接受过治疗并正在接受 Alemtuzumab 治疗的患者中，CD 4+和 CD 8+计数可能在完成治疗后>1 年才恢复至基线水平。感染是 Alemtuzumab 接受者的发病率和/或死亡率问题，特别是对于接受过大量预治疗、氟达拉滨难治性疾病的患者。Alemtuzumab 治疗期间曾报告细菌、病毒、真菌、分枝杆菌和耶氏肺孢子虫感染。

在接受 Alemtuzumab 治疗的患者中，建议对疱疹病毒和 PJP 进行抗感染预防（参见抗病毒预防和预防性抗病毒治疗和预防）。

肺孢子虫）。几项研究表明，接受 Alemtuzumab 治疗的患者对巨细胞病毒（CMV）再激活和疾病的易感性增加。在缺乏大型随机对照试验的情况下，德国血液学和医学肿瘤学会传染病工作组目前不建议在 Alemtuzumab 接受者中进行 CMV 监测。相反，代表英国血液学标准委员会的英国 CLL 论坛工作组和代表国家癌症研究所（NCI）的 CLL 国际研讨会建议在接受与 CMV 再激活可能性相关治疗的 CLL 患者中进行 CMV 常规监测（例如，NCCN 专家组建议至少每周使用聚合酶链反应（PCR）进行 CMV 再活化监测在 Alemtuzumab 接受者中（参见抗病毒预防和预防性抗病毒治疗：巨细胞病毒）。 其他已知引起淋巴细胞减少症的化合物（如蛋白酶体抑制剂）与带状疱疹再激活风险增加有关;因此，建议使用阿昔洛韦、泛昔洛韦或伐昔洛韦进行预防。

抗 CD20 单克隆抗体

抗 CD20 单克隆抗体（如利妥昔单抗、奥法木单抗、奥滨尤妥珠单抗）广泛用于治疗 B 细胞淋巴恶性肿瘤患者。这些单克隆抗体的使用与 B 型肝炎病毒（HBV）再激活的风险增加相关，这可能导致暴发性肝炎、肝衰竭和/或死亡。抗病毒预防通常推荐用于 HBV 表面抗原检测阳性的患者（参见抗病毒预防和预防性抗病毒治疗：B 型肝炎病毒）。

在 B 细胞恶性肿瘤患者中使用抗 CD20 单克隆抗体与罕见的进行性多灶性白质脑病（PML）相关。PML 是一种脱髓鞘疾病，

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the central nervous system (CNS) resulting from reactivation of the John Cunningham (JC) virus, and occurs in severely immunocompromised individuals. Though rare, PML is most often fatal. In reports of PML potentially associated with rituximab treatment in patients with B-cell malignancies, rituximab was typically given in combination with chemotherapy regimens or in patients who had received prior immunosuppressive regimens.[49-56] Moreover, patients who developed PML often presented with low CD4+ counts or abnormal (low) CD4+/CD8+ ratio,[49,51,54,56] which points to a critical role of T-cell immunity in suppressing reactivation of the JC virus.

Other Immunosuppressive Therapies

In addition to the agents mentioned above, there are other immunosuppressive therapies associated with a greater risk of infection in patients with cancer. For example, temozolomide (often administered in conjunction with radiation therapy) is associated with an increased risk of infection, particularly with *Pneumocystis jirovecii*, the causative agent for PJP.[57] Likewise, idelalisib with or without rituximab is associated with an increased risk of infections including *Pneumocystis jirovecii.* Treatment with other therapies, including ibrutinib and bendamustine, have also been reported to increase susceptibility to infection, including *Pneumocystis jirovecii*.[58,59]

Hematopoietic Cell Transplantation

Autologous HCT

Autologous HCT recipients generally have fewer infectious complications than allogeneic transplant recipients. Most infections in autologous HCT recipients occur during neutropenia or within the first few months after transplantation before reconstitution of cellular immunity. However, compared to unmanipulated autologous HCTs, CD34+ cell enrichment leads to a substantial reduction in T cells, natural killer cells, and monocytes, which delays immune reconstitution.[60] Recipients of CD34+ cell-enriched autologous HCT appear to have a similar level of risk as allogeneic HCT recipients for contracting CMV and other opportunistic infections.[60] Severe or ulcerative mucositis, which develops as a result of myeloablative high-dose therapy administered prior to HCT, is associated with the occurrence of bacteremia in autologous HCT recipients.[61-63]

A multicenter prospective study evaluated the potential role of granulocyte-colony stimulating factor (G-CSF) responsiveness in predicting the occurrence of infections in patients with hematologic malignancies undergoing high-dose therapy and autologous HCT.[64] Responsiveness to G-CSF was determined by the administration of a single dose of G-CSF after completion of high-dose therapy (but prior to HCT), and measuring the induced leukocyte peak occurring 12 to 14 hours after the G-CSF dose. G-CSF responsiveness showed a significant inverse correlation with incidences of febrile neutropenia and infections (ie, higher responsiveness associated with lower infection rates), and was shown to be the only independent predictor of infections based on multivariate analysis.[64]

Allogeneic HCT

The spectrum of pathogens to which allogeneic HCT recipients are most susceptible follows a timeline corresponding to the predominant immune defects. In the first month after allogeneic HCT (pre-engraftment period), neutropenia and breakdown of the mucocutaneous barrier comprise the principal host defense defect, which predisposes patients to bacterial and fungal infections.[65,66] In addition, reactivation of HSV can often occur during this period. After myeloid engraftment, qualitative dysfunction of phagocytes persists due to corticosteroid and other immunosuppressive agents. The risk of infection by opportunistic viruses and filamentous fungi (molds) during this period is strongly associated with the severity of GVHD and with the requirement for potent immunosuppressive regimens.

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中枢神经系统（CNS）由约翰坎宁安（JC）病毒的再激活引起，并发生在严重免疫功能低下的个体中。虽然罕见，但 PML 通常是致命的。在与利妥昔单抗治疗 B 细胞恶性肿瘤患者可能相关的 PML 报告中，利妥昔单抗通常与化疗方案联合使用或用于既往接受过免疫抑制方案的患者。此外，发生 PML 的患者通常表现为低 CD 4+计数或异常（低）CD 4 +/CD 8+比值，这表明 T 细胞免疫在抑制 JC 病毒再活化中起关键作用。

其他免疫抑制治疗

除了上述药物外，还有其他免疫抑制疗法与癌症患者感染风险增加相关。例如，替莫唑胺（通常与放射治疗联合施用）与感染风险增加相关，特别是与 PJP 的病原体耶氏肺孢子虫（Pneumocystis jirovecii）相关。同样，艾代拉里斯与或不与利妥昔单抗一起与包括耶氏肺孢子虫在内的感染风险增加相关。据报道，使用其他疗法（包括伊曲替尼和苯达莫司汀）治疗也会增加对感染（包括肺孢子虫）的易感性。

造血细胞移植

自动 HCT

自体 HCT 受者通常比同种异体移植受者具有更少的感染并发症。自体 HCT 受者中的大多数感染发生在中性粒细胞减少期间或移植后细胞免疫重建前的最初几个月内。然而，与未经操作的自体 HCT 相比，CD34+细胞富集导致 T 细胞、自然杀伤细胞和单核细胞的大量减少，从而延迟免疫重建。细胞富集的自体 HCT 似乎具有与同种异体 HCT 接受者相似的感染 CMV 和其他机会性感染的风险水平。严重或溃疡性粘膜炎是由于在 HCT 之前给予清髓性高剂量治疗而发生的，与自体 HCT 接受者中菌血症的发生相关。

一项多中心前瞻性研究评价了粒细胞集落刺激因子（G-CSF）反应性在预测接受高剂量治疗和自体 HCT 的恶性血液病患者发生感染方面的潜在作用。通过在完成高剂量治疗后（但在 HCT 之前）给予单次剂量的 G-CSF，并测量 G-CSF 给药后 12 至 14 小时出现的诱导白细胞峰值，确定对 G-CSF 的反应性。G-CSF 反应性与发热性中性粒细胞减少症和感染的发生率呈显著负相关（即，反应性越高，感染率越低），基于多变量分析，G-CSF 反应性被证明是感染的唯一独立预测因素。

同种异体 HCT

同种异体 HCT 接受者最易感的病原体谱遵循对应于主要免疫缺陷的时间轴。在同种异体 HCT 后的第一个月（植入前期），中性粒细胞减少和皮肤粘膜屏障的破坏是主要的宿主防御缺陷，使患者易于发生细菌和真菌感染。此外，HSV 的再激活也经常发生

在此期间骨髓移植后，由于皮质类固醇和其他免疫抑制剂，吞噬细胞的定性功能障碍持续存在。在此期间机会性病毒和丝状真菌（霉菌）感染的风险与 GVHD 的严重程度以及对强效免疫抑制方案的需求密切相关。

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Susceptibility to infections during the early post-engraftment period is primarily due to defects in cell-mediated immunity that can persist for several months even in uncomplicated allogeneic HCT recipients, predisposing them to common bacterial and viral infections and to multiple opportunistic infections (eg, molds, viruses, atypical bacteria). In particular, the dominant pathogens during this early post-engraftment period can include herpes viruses (especially CMV), *Pneumocystis jirovecii*, and invasive molds such as *Aspergillus*.[65,66] Prophylaxis against pneumococcal infection is advised in allogeneic HCT recipients (see *Prophylaxis for Pneumococcal Infection*).

Allografts from human leukocyte antigen (HLA)–matched unrelated donors, partially mismatched related donors, and cord blood are associated with a higher risk of GVHD. T-cell depletion delays immune reconstitution and, consequently, carries a greater risk of infectious complications, most notably by opportunistic viral[67] and fungal[68-70] pathogens. Cord blood transplant recipients may have a higher risk of infections than other allograft recipients during the early transplant period because of slower myeloid engraftment.

Guidelines from the Centers for Disease Control and Prevention (CDC) recommend that allogeneic HCT recipients with severe hypogammaglobulinemia (IgG <400 mg/dL) and with recurrent infections receive intravenous immunoglobulin (IVIG) prophylaxis; IVIG is not routinely recommended in other patient groups or in autologous HCT recipients.[8] The 2009 guidelines on the prevention of infections in HCT recipients (jointly sponsored by the CDC, Infectious Diseases Society of America [IDSA], American Society for Blood and Marrow Transplantation, and European Society for Blood and Marrow Transplantation, among other organizations) reported similar recommendations on the use of IVIG.[66]

Chronic GVHD

Whereas mature and cooperative T- and B-cell functions are usually reconstituted by 1 to 2 years following engraftment, chronic GVHD is associated with persistently depressed cell-mediated and humoral immunity. Defective reconstitution of humoral immunity is a major factor contributing to increased infection susceptibility in the late post-engraftment transplant period. Winston et al[71] noted a high frequency of pneumococcal infections between 7 and 36 months after transplantation, associated with serum opsonic deficiency for *Streptococcus pneumoniae*. Kulkarni et al[72] reported that pneumococcal sepsis occurred a median of 10 months after transplant (range, 3–187 months) and was significantly more frequent in patients with chronic GVHD.

CAR T-Cell Therapy

Several chimeric antigen receptor (CAR) T-cell products targeting CD19 (e.g., axicabtagene ciloleucel, tisagenlecleucel, lisocabtagene maraleucel) or B-cell maturation antigen (BCMA) (e.g., idecabtagene vicleucel, ciltacabtagene autoleucel) have been approved for hematologic malignancies. However, CAR T-cell products confer a risk of potentially life-threatening immunological toxicities, like neutropenia, lymphopenia, and hypogammaglobulinemia, and thus may increase the risk of infection in patients.[73] The highest risk for infection occurs within the first 30 days of administration of CAR T-cell therapy, and bacterial infections predominate during this period. Opportunistic infections can be expected, and prophylaxis, particularly for the prevention of *Pneumocystis jirovecii*, herpes simplex virus (HSV), and varicella zoster virus (VZV), is warranted until immune reconstitution. Pre-infusion risk factors for infection include certain underlying malignancies, prior lines of therapy with or without autologous/allogeneic HCT, antecedent infection, and neutropenia. Risk factors post-infusion may include prolonged neutropenia, cytokine release syndrome/immune effector cell-associated

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在移植后早期对感染的易感性主要是由于细胞介导的免疫缺陷，即使在无并发症的同种异体 HCT 接受者中也可持续数月，使其易受常见细菌和病毒感染以及多种机会性感染（例如，霉菌、病毒、非典型细菌）。特别是，在移植后早期，主要的病原体包括疱疹病毒（特别是 CMV）、耶氏肺孢子虫和侵袭性霉菌（如曲霉菌）。建议异基因 HCT 接受者预防肺炎球菌感染（见预防肺炎球菌感染）。

来自人类白细胞抗原（HLA）匹配的非亲缘供者、部分不匹配的亲缘供者和脐带血的同种异体移植物与更高的 GVHD 风险相关。T 细胞耗竭延迟免疫重建，因此，感染并发症的风险更大，最明显的是机会性病毒和真菌病原体。由于骨髓移植较慢，在移植早期，脐带血移植受者可能比其他同种异体移植受者有更高的感染风险。

Guidelines from the Centers for Disease Control and Prevention (CDC) recommend that allogeneic HCT recipients with severe hypogammaglobulinemia (IgG <400 mg/dL) and with recurrent infections receive intravenous immunoglobulin (IVIG) prophylaxis; IVIG is not routinely recommended in other patient groups or in autologous HCT recipients.The 2009 guidelines on the prevention of infections in HCT recipients (jointly sponsored by the CDC, Infectious Diseases Society of America [IDSA], American Society for Blood and Marrow Transplantation, and European Society for Blood and Marrow Transplantation, among other organizations) reported similar recommendations on the use of IVIG.

重试 错误原因

慢性 GVHD

而成熟和合作的 T–和 B–细胞功能通常在移植后 1 至 2 年重建，慢性 GVHD 与持续抑制的细胞介导和体液免疫有关。体液免疫重建缺陷是导致植活后移植后期感染易感性增加的主要因素。Winston 等注意到移植后 7 – 36 个月肺炎球菌感染的高频率，与肺炎链球菌的血清调理素缺乏有关。Kulkarni 等报道，肺炎球菌败血症发生在移植后 10 个月（中位数，范围，3–187 个月），并且在慢性 GVHD 患者中更常见。

CAR t 细胞疗法

几种靶向 CD19 的嵌合抗原受体（CAR）T 细胞产物（例如，axicabtagene ciloleucel、tisagenlecleucel、lisocabtagene maraleucel）或 B 细胞成熟抗原（BCMA）（例如，idecabtagene vicleucel、ciltacabtagene autoleucel）已被批准用于血液恶性肿瘤。然而，CAR T 细胞产品具有潜在危及生命的免疫毒性的风险，如中性粒细胞减少症、淋巴细胞减少症和低丙种球蛋白血症，因此可能增加患者感染的风险。感染的最高风险发生在 CAR T 细胞治疗给药的前 30 天内，在此期间细菌感染占主导地位。预期会发生细菌性感染，需要进行预防，特别是预防耶氏肺孢子虫、单纯疱疹病毒（HSV）和水痘带状疱疹病毒（VZV），直至免疫重建。 输注前感染风险因素包括某些基础恶性肿瘤、既往接受过有或无自体/同种异体 HCT 的治疗、既往感染和中性粒细胞减少症。输注后的风险因素可能包括长期中性粒细胞减少症、细胞因子释放综合征/免疫效应细胞相关性

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neurotoxicity syndrome (CRS/ICANS) and associated treatment (eg, high-dose steroids, IL-6 inhibitors), lymphopenia, and hypogammaglobulinemia. Beyond day +30, respiratory viral infections predominate. Fungal infection risk remains low but may vary based on prior therapies, degree of immunosuppression, and other relevant risk factors.

Relevant serologic screening for human immunodeficiency virus (HIV), HBV, and hepatitis C virus (HCV) is recommended. Moreover, CMV and additional screening may be considered based on epidemiologic risks. Antibacterial and antifungal prophylaxis can be considered for patients who are neutropenic. PJP and HSV/VZV prophylaxis are recommended. The specific risk profile of the patient (duration of neutropenia, prior allogeneic HCT, previous infections, and local antibiotic resistance profiles) should guide diagnostic workup and selection of antimicrobial agents. In patients with additional risk factors, such as prolonged neutropenia, prior allogeneic HCT, or augmented immunosuppressive therapy (IST) for CRS/ICANS, mold-active antifungal prophylaxis can be considered (see *Immune And Targeted Treatments: Chimeric Antigen Receptor-Engineered T-Cell Therapy* in the algorithm).[73-76] Patients should also be monitored for CRS, which may mimic sepsis. The incidence of CRS depends on the CAR T-cell product, disease characteristics, and CRS grading system used.[73-76]

NCCN Recommendations for Categories of Infection Risk

The Panel acknowledges that there are multiple definitions of risk related to infection in patients with cancer.[36,77,78] This section is specific to the overall risk of developing infection and recommendations for prophylaxis are based on this risk characterization. The NCCN Guidelines provide a summary of infection risk categories (low, intermediate, and high risk) in patients with cancer, which are based on factors such as the underlying malignancy, disease status (eg, active disease or disease in remission), duration of neutropenia, prior exposure to chemotherapy, and intensity of IST. Development of the categories of risk is further based on the expert opinion of the Panel. An overview of the antimicrobial recommendations based on risk for infection is presented below. For more details, refer to the *Prevention of Infectious Diseases* section in the discussion and *Antimicrobial Prophylaxis Based on Overall Infection Risk in Patients with Cancer* in the algorithm.

Briefly, patients with solid tumors receiving standard chemotherapy regimens and who have an anticipated duration of neutropenia <7 days are generally considered at low risk for infectious complications; thus, antimicrobial prophylaxis is not routinely recommended in these patients.[17] For patients with HSV-positive serology who are otherwise at low risk for infections, prophylaxis with antivirals can be considered.

Patients with an anticipated duration of neutropenia of 7 to 10 days are considered to be at intermediate risk for infections. In addition, patients with lymphoma, multiple myeloma, CLL, autologous HCT recipients, or those receiving CAR T-cell therapy or treatment with purine analog-containing regimens (most often for hematologic malignancies such as NHL or CLL) are also considered to be at intermediate risk. For those who are at intermediate risk, prophylaxis with antibacterials (eg, fluoroquinolones) should be considered during neutropenia. Antivirals should be given during periods of neutropenia, and for autologous HCT recipients, until at least 30 days following transplant (however, consider antiviral prophylaxis for VZV for at least 6–12 months after HCT). Antifungals should be considered during periods of neutropenia and for anticipated mucositis (with the latter pertaining to autologous HCT) for intermediate risk. PJP prophylaxis should be considered in patients with intermediate risk.

Patients with anticipated duration of neutropenia >10 days, those undergoing intensive induction/consolidation therapy for acute leukemias

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神经毒性综合征（CRS/ICANS）和相关治疗（例如，高剂量类固醇、IL-6 抑制剂）、淋巴细胞减少症和低丙种球蛋白血症。超过 30 天，呼吸道病毒感染占主导地位。真菌感染的风险仍然很低，但可能会因既往治疗、免疫抑制程度和其他相关风险因素而异。

建议进行人类免疫缺陷病毒（HIV）、HBV 和丙型肝炎病毒（HCV）的相关血清学筛查。此外，根据流行病学风险，可考虑 CMV 和额外筛查。对于血小板减少的患者，可以考虑进行抗菌和抗真菌预防。推荐 PJP 和 HSV/VZV 预防。患者的特定风险特征（中性粒细胞减少持续时间、既往同种异体 HCT、既往感染和局部抗生素耐药特征）应指导诊断检查和抗菌药物的选择。在有其他风险因素的患者中，如长期中性粒细胞减少症、既往同种异体 HCT 或 CRS/ICANS 的强化免疫抑制治疗（IST），可以进行霉菌活性抗真菌预防。

（参见免疫和靶向治疗：算法中的嵌合抗原受体工程化 T 细胞疗法）。还应监测患者的 CRS，这可能与脓毒症相似。CRS 的发生率取决于 CAR T 细胞产品，疾病特征和所使用的 CRS 分级系统。

NCCN 对感染风险类别的建议

专家组承认，癌症患者感染相关风险有多种定义，本节专门针对发生感染的总体风险，预防建议基于这一风险特征。NCCN 指南提供了癌症患者感染风险类别（低、中和高风险）的总结，这些风险类别基于以下因素：基础恶性肿瘤、疾病状态（例如，活动性疾病或缓解中的疾病），

中性粒细胞减少持续时间、既往化疗暴露和 IST 强度。风险类别的确定还以小组的专家意见为依据。基于感染风险的抗菌药物建议概述如下。更多详情请参见

讨论中的传染病预防部分和算法中基于癌症患者总体感染风险的抗菌剂预防。

简单地说，接受标准化疗方案的实体瘤患者和预期中性粒细胞减少持续时间<7 天的患者通常被认为感染并发症的风险较低;因此，对这些患者不推荐常规预防性抗生素治疗。对于 HSV 血清学阳性的感染风险较低的患者，可以考虑使用抗病毒药物进行预防。

中性粒细胞减少症预期持续时间为 7 – 10 天的患者被认为具有中度感染风险。此外，淋巴瘤、多发性骨髓瘤、CLL、自体 HCT 接受者或接受 CAR T 细胞治疗或嘌呤治疗的患者

含有类似物的方案（最常用于血液恶性肿瘤如 NHL 或 CLL）也被认为处于中等风险。对于中度风险的患者，在中性粒细胞减少期间应考虑使用抗菌药物（如氟喹诺酮类）预防。抗病毒药物应在中性粒细胞减少期间给予，对于自体 HCT 接受者，应至少在移植后 30 天内给予（但是，考虑在 HCT 后至少 6–12 个月内对 VZV 进行抗病毒预防）。中性粒细胞减少期间和预期粘膜炎（后者与自体 HCT 有关）的中等风险期间应考虑使用抗真菌药。中度风险患者应考虑 PJP 预防。

预期中性粒细胞减少持续时间>10 天的患者，正在接受急性白血病强化诱导/巩固治疗的患者

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(ie, acute lymphoblastic leukemia [ALL] or acute myeloid leukemia [AML]), patients undergoing treatment with alemtuzumab-containing regimens, allogeneic HCT recipients, and those with GVHD following allogeneic HCT are considered at high risk for infectious complications. Patients with NHL (in particular, for T-cell malignancy subtypes) or CLL treated with alemtuzumab-containing regimens are considered at high risk for infections. For these patients who are at high risk for infections, prophylaxis with antibacterials (eg, fluoroquinolones) should be considered during neutropenia. These patients should receive antiviral prophylaxis during periods of neutropenia, and antiviral prophylaxis for VZV for at least 1 year after HCT. In addition, prophylaxis with antifungals can be considered for patients with ALL and for patients with neutropenia with AML/MDS.[17] For allogeneic HCT recipients or those with chronic GVHD receiving IST, antifungal prophylaxis can also be considered during periods of neutropenia and until resolution of GVHD. PJP prophylaxis should be considered in those who are at high risk for infections.

Prevention of Infectious Diseases

Preventive measures against infections in patients with cancer include upfront prophylaxis or preemptive therapy using broad-spectrum antimicrobial agents directed against the most common infecting pathogens (including bacterial, viral, and fungal) in patients at high risk for infections. Vaccination and minimization of potential exposures to opportunistic pathogens that may be harmful to patients who are immunocompromised due to cancer are additional components of infectious disease prevention.

Antibacterial Prophylaxis

During Neutropenia

Patients with cancer and chemotherapy-induced neutropenia are at risk for severe bacterial infections. Fluoroquinolones are the most commonly used prophylactic antibacterial agents in adults with chemotherapy-induced neutropenia.

Two large, randomized, placebo-controlled studies showed the benefit of levofloxacin prophylaxis in neutropenia at different levels of risk for infectious complications.[79,80] The main advantage of levofloxacin prophylaxis (in patients with intermediate and higher risk of infections) with chemotherapy-induced neutropenia was a reduction in clinically significant bacterial infections, including gram-negative rod bacteremia.[79] In contrast, the main advantage of prophylaxis in lower risk neutropenia was a small, but statistically significant, reduction in fever and hospitalization for neutropenic fever.[80] Neither study conducted a systematic long-term evaluation of antimicrobial resistance. The NCCN Guidelines Panel considers that reduction in the incidence of significant infections is a more clinically meaningful endpoint than reduction in the incidence of neutropenic fever. Using prevention of neutropenic fever as the primary endpoint in this study by Cullen et al,[80] 1000 hypothetical patients with low risk of infections would have to receive prophylaxis during each cycle of chemotherapy-induced neutropenia to benefit only 44 patients.

In a single-center randomized study in patients undergoing high-dose therapy followed by autologous HCT (N = 157), patients were randomized to receive prophylaxis (with oral ciprofloxacin 500mg twice daily and intravenous [IV] vancomycin 1000mg once daily) or no prophylaxis; all patients received antifungal prophylaxis with fluconazole.[81] Empiric therapy (comprising amikacin, ceftazidime, and full-dose vancomycin) was initiated when neutropenic fever developed. The use of antibacterial prophylaxis significantly reduced the incidences of neutropenic fever (56% vs. 91%; $P < .001$) and bacteremia (6% vs. 35%; $P = .005$) compared with no prophylaxis, but at the expense of decreased response to first-line empiric therapy (66% vs. 84%; $P = .025$). Among the patients who received prophylaxis and developed neutropenic fever, 34% required

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(ie 急性淋巴细胞白血病[ALL]或急性髓性白血病[AML])、接受含 Alemtuzumab 方案治疗的患者、异基因 HCT 接受者和异基因 HCT 后发生 GVHD 的患者被认为具有感染并发症的高风险。认为接受含 Alemtuzumab 方案治疗的 NHL（特别是 T 细胞恶性肿瘤亚型）或 CLL 患者感染风险较高。对于这些感染风险高的患者，应使用抗菌药物（如氟喹诺酮类）进行预防。

在中性粒细胞减少症期间考虑。这些患者应在中性粒细胞减少期间接受抗病毒预防，并在 HCT 后至少 1 年内接受 VZV 抗病毒预防。此外，预防与

对于 ALL 患者和伴有 AML/MDS 的中性粒细胞减少症患者，可以考虑使用抗真菌药物，对于异基因 HCT 接受者或接受 IST 的慢性 GVHD 患者，在中性粒细胞减少症期间和 GVHD 消退之前，也可以考虑使用抗真菌药物预防。PJP 预防应考虑在那些谁是感染的高风险。

预防传染病

预防癌症患者感染的措施包括预先预防或使用广谱抗菌药物进行先发制人的治疗，这些抗菌药物直接针对最常见的感染。

病原体（包括细菌、病毒和真菌）感染高危患者。接种疫苗和尽量减少可能对因癌症而免疫功能低下的患者有害的机会性病原体的潜在暴露是传染病预防的额外组成部分。

抗菌预防

中性化期间

患有癌症和化疗诱导的中性粒细胞减少症的患者存在严重细菌感染的风险。氟喹诺酮类药物是最常用的

预防性抗菌药物在成人化疗诱导的中性粒细胞减少症。

两项大型、随机、安慰剂对照研究显示，左氧氟沙星预防治疗在不同感染并发症风险水平的中性粒细胞减少症中获益。（在具有中等和较高感染风险的患者中）化疗诱导的中性粒细胞减少症是临床上显著的细菌感染（包括革兰氏阴性杆菌血症）的减少。在低风险中性粒细胞减少症中预防的主要优势是发热和因中性粒细胞减少症而住院的减少，但幅度较小，但具有统计学意义。NCCN 指南专家组认为，显著感染发生率的降低是比血小板减少性发热发生率的降低更有临床意义的终点。 在 Cullen et al 的这项研究中，使用预防血小板减少性发热作为主要终点，假设 1000 例感染风险低的患者必须在化疗诱导的中性粒细胞减少症的每个周期接受预防治疗，只有 44 例患者受益。

在接受高剂量治疗后接受自体 HCT 的患者中进行的一项单中心随机研究（N = 157）中，患者随机接受预防性治疗（口服环丙沙星 500 mg 每日两次和静脉注射[IV]万古霉素 1000 mg 每日一次）或无预防;所有患者均接受氟康唑预防性抗真菌治疗。当出现血小板减少性发热时，开始使用阿米卡星、头孢他啶和全剂量万古霉素。与未预防相比，预防性使用抗菌药物显著降低了血小板减少性发热（56% vs. 91%; P <0.001）和菌血症（6% vs. 35%; P = 0.005）的发生率，但其代价是一线经验性治疗的反应降低（66% vs. 84%; P = 0.025）。在接受预防性治疗并出现贫血性发热的患者中，34%需要

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second-line therapy that included a carbapenem, suggesting that these patients developed infections resistant to the prophylactic regimen. Duration of hospitalization and overall survival rates were similar between study arms. These results led the study investigators to conclude that routine antibacterial prophylaxis was not recommended in patients undergoing high-dose therapy and autologous HCT.[81] It should be noted, however, that the prophylactic regimen in this study included vancomycin (albeit at a lower dose), is not supported by the NCCN or IDSA Panels for use as either antimicrobial prophylaxis or standard initial empiric therapy for fever and neutropenia. [17]

In a systematic review and meta-analysis by Gafter-Gvili et al (based on 109 trials, N = 13,579 patients) comparing antibacterial prophylaxis with placebo, no intervention, or prevention with another agent in patients with afebrile neutropenia, the use of antibacterial prophylaxis was found to significantly reduce the risk of all-cause mortality (risk ratio, 0.66; 95% CI, 0.55–0.79) as well as infection-related deaths (risk ratio, 0.61; 95% CI, 0.48–0.77) compared with placebo or no intervention.[82] The use of prophylaxis also significantly reduced the incidence of fever and clinically or microbiologically documented infections. Although no significant differences in all-cause or infection-related mortality were seen between prophylactic quinolones or trimethoprim/sulfamethoxazole (TMP/SMX), the use of quinolones was associated with decreased drug resistance and fewer adverse events that subsequently reduced the incidence of drug discontinuation.[82]

A systematic review and meta-analysis evaluated the risks associated with colonization and infections by fluoroquinolone-resistant bacteria.[83] Most of the studies (48 of 56 trials) included patients with hematologic malignancies or HCT recipients. Results of the analysis (based on 56 trials, N = 7878 patients; data on colonization by resistant bacteria based on 27 trials) showed that quinolone prophylaxis was associated with an increase in colonization with quinolone-resistant organisms compared with placebo or no intervention, although the increase was not statistically significant (relative risk [RR], 1.68; 95% CI, 0.71–4.00). However, no difference was observed in the incidence of infections caused by quinolone-resistant organisms (RR, 1.04; 95% CI, 0.73–1.50), regardless of whether these were resistant gram-negative or gram-positive bacteria.[83] Moreover, in an analysis of trials comparing quinolones with TMP/SMX (11 trials), prophylaxis with quinolones was associated with fewer incidents of colonization and infections by resistant bacteria (those resistant to the prophylactic agents) compared with the use of TMP/SMX.[83] This analysis suggests that prophylaxis with quinolones does not appear to increase the rate of infections by resistant organisms.

A 2017 prospective intercontinental study assessed gram-negative rod (GNR) resistance to fluoroquinolones, non-carbapenem β-lactams (including ceftazidime, cefepime or β-lactam/β-lactamase inhibitors), and carbapenems (meropenem/imipenem/doripenem) in HCT recipients. Of 655 GNR bacteremia episodes, half were found to be fluoroquinolone and non-carbapenem resistant. The total resistance rates were higher in patients with allogeneic HCT than in those with autologous HCT. Risk factors for fluoroquinolone resistance in allogeneic HCT included prolonged neutropenia and breakthrough on fluoroquinolones. Mortality was found to be significantly more common in infections caused by resistant bacteria.[84]

A 2018 meta-analysis of studies published in the years 2006–2014 assessed the role of fluoroquinolone prophylaxis during neutropenia in consideration of increasing antibiotic resistance globally.[85] The meta-analysis found no effect of the background rate of fluoroquinolone resistance on the efficacy of fluoroquinolone prophylaxis. However, in few studies, fluoroquinolone prophylaxis resulted in an increased colonization or infection with fluoroquinolone- or multidrug-resistant strains.

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second–line therapy that included a carbapenem, suggesting that these patients developed infections resistant to the prophylactic regimen. Duration of hospitalization and overall survival rates were similar between study arms. These results led the study investigators to conclude that routine antibacterial prophylaxis was not recommended in patients undergoing high–dose therapy and autologous HCT.It should be noted, however, that the prophylactic regimen in this study included vancomycin (albeit at a lower dose), is not supported by the NCCN or IDSA Panels for use as either antimicrobial prophylaxis or standard initial empiric therapy for fever and neutropenia.

重试 错误原因

在 Gafter–Gvili 等人的系统综述和荟萃分析中，（基于 109 项试验，N = 13，579 例患者）在无发热性中性粒细胞减少症患者中比较抗菌预防与安慰剂、无干预或用另一种药物预防，发现使用抗菌预防可显著降低全因死亡风险与安慰剂组或无干预组相比，预防性治疗显著降低了发热和临床或微生物学记录的感染的发生率（风险比，0.66; 95%CI，0.55–0.79）以及感染相关死亡（风险比，0.61; 95%CI，0.48–0.77）。虽然预防性喹诺酮类或甲氧苄啶/磺胺甲恶唑（TMP/SMX）之间的全因或感染相关死亡率无显著差异，但喹诺酮类的使用与耐药性降低和不良事件减少相关，随后降低了停药的发生率。

A systematic review and meta–analysis evaluated the risks associated with colonization and infections by fluoroquinolone–resistant bacteria.Most of the studies (48 of 56 trials) included patients with hematologic malignancies or HCT recipients. Results of the analysis (based on 56 trials, N = 7878 patients; data on colonization by resistant bacteria based on 27 trials) showed that quinolone prophylaxis was associated with an

重试 错误原因

与安慰剂或无干预相比，喹诺酮类耐药微生物定植增加，尽管增加无统计学显著性（相对风险[RR]，1.68; 95% CI，0.71–4.00）。然而，无论是耐药革兰氏阴性菌还是革兰氏阳性菌，喹诺酮类耐药微生物引起的感染发生率均无差异（RR，1.04; 95% CI，0.73–1.50）。此外，在一项比较喹诺酮类药物与 TMP/SMX 的试验分析（11 项试验）中，与使用 TMP/SMX 相比，使用喹诺酮类药物预防与耐药细菌（对预防药物耐药的细菌）定植和感染事件较少相关。该分析表明，使用喹诺酮类药物预防似乎不会增加耐药微生物的感染率。

2017 年的一项前瞻性洲际研究评估了 HCT 接受者中革兰氏阴性杆菌（GNR）对氟喹诺酮类、非碳青霉烯类β–内酰胺类（包括头孢他啶、头孢吡肟或β–内酰胺/β–内酰胺酶抑制剂）和碳青霉烯类（美罗培南/亚胺培南/多利培南）的耐药性。在 655 例 GNR 菌血症发作中，发现一半是氟喹诺酮和非碳青霉烯耐药。异基因 HCT 组总耐药率高于自体 HCT 组。异基因 HCT 中氟喹诺酮类耐药的危险因素包括长期中性粒细胞减少和氟喹诺酮类药物的突破。死亡率在耐药菌引起的感染中明显更常见。

2018 年一项对 2006–2014 年发表的研究进行的荟萃分析评估了氟喹诺酮类药物预防在中性粒细胞减少症中的作用，考虑到全球抗生素耐药性的增加。荟萃分析发现氟喹诺酮类药物耐药性的背景发生率对氟喹诺酮类药物预防的疗效没有影响。然而，在少数研究中，氟喹诺酮类药物预防导致氟喹诺酮类或多药耐药菌株的定植或感染增加。

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Fluoroquinolone prophylaxis had no effect on mortality, but it reduced the rate of bloodstream infections and episodes of fever.[85]

An important consideration for those with low risk with short durations of neutropenia is whether fluoroquinolone prophylaxis is of greater benefit than outpatient fluoroquinolone treatment for fever and neutropenia, should it occur. Both the NCCN Guidelines and IDSA[17] Panels recommend oral fluoroquinolone-based regimens as outpatient empiric therapy for neutropenic fever in adults who meet criteria for a low risk of complications. Fluoroquinolone prophylaxis may preclude its subsequent use as empiric therapy for neutropenic fever in the same patient. The modest difference in rates of hospitalization for suspected infection in levofloxacin compared to placebo recipients (15.7% vs. 21.6%, respectively) may be offset by exclusion of outpatient oral empiric therapy in patients receiving fluoroquinolone prophylaxis. To limit antibacterial use, Cullen et al[86] have suggested prophylaxis with levofloxacin on cycle 1 of myelosuppressive cancer chemotherapy and only in subsequent cycles if a febrile episode occurs.

The decision whether to use antibacterial prophylaxis and the selection of the specific agent requires a balance between expected benefit and risk (explained in greater detail in a review by Hoffman et al[87]). The concept of risk applies to immediate adverse effects of the drug (eg, rash, GI intolerance), the potential for selection for resistant pathogens that can harm the individual receiving prophylaxis, and the risk of resistant organisms to a specific population of patients (eg, those being treated at a cancer center). Thus, the emergence of multidrug-resistant organisms (MDROs), disruption of the microbiome, and antibiotic toxicities must be taken into consideration when choosing an antimicrobial prophylactic agent. The link between fluoroquinolone use and severe *Clostridium difficile* as well as methicillin-resistant *S aureus* (MRSA) infections provides an additional cautionary note regarding excess use of fluoroquinolones.[88-91]

NCCN Recommendations for Antibacterial Prophylaxis

Antibacterial prophylaxis is not recommended for patients with a low risk of overall infection (see *Antimicrobial Prophylaxis Based On Overall Infection Risk In Patients With Cancer* in the algorithm). In patients with neutropenia who are at lower risk of infectious complications (a category that includes most patients with solid tumor malignancies), the main benefit of antibacterial prophylaxis is a reduction in fever rather than in documented infections. In patients with neutropenia expected to last <7 days who are not receiving immunosuppressive regimens (eg, systemic corticosteroids), the Panel suggests no antibiotic prophylaxis.[17] In patients deemed at intermediate or high risk, the NCCN Guidelines Panel advises that fluoroquinolone prophylaxis (levofloxacin is preferred) be considered in patients with an expected duration of neutropenia (absolute neutrophil count [ANC] ≤500 neutrophils/mcL or ≤1000 neutrophils/mcL and a predicted decline to ≤500/ mcL over the next 48 hours) for >7 days. This is in agreement with the recommendations of the IDSA guidelines for the use of antimicrobial agents in patients with neutropenia with cancer.[17] For patients who are intolerant to fluoroquinolone, TMP/SMX or an oral third-generation cephalosporin may be considered.

Clinical practice guidelines have also been developed for systemic antibacterial prophylaxis in pediatric patients with cancer and those undergoing HCT.[92] The Panel for these guidelines recommends consideration of antibacterial prophylaxis in children with acute myeloid leukemia (AML) and relapsed acute lymphoblastic leukemia (ALL) receiving intensive chemotherapy expected to result in severe neutropenia (ANC <500/mcL) for at least 7 days. The Panel recommends against the routine use of antibacterial prophylaxis for children undergoing induction chemotherapy for ALL, autologous HCT and allogeneic HCT. A strong

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Fluoroquinolone prophylaxis had no effect on mortality, but it reduced the rate of bloodstream infections and episodes of fever.

重试 错误原因

An important consideration for those with low risk with short durations of neutropenia is whether fluoroquinolone prophylaxis is of greater benefit than outpatient fluoroquinolone treatment for fever and neutropenia, should it occur. Both the NCCN Guidelines and IDSAPanels recommend oral fluoroquinolone–based regimens as outpatient empiric therapy for neutropenic fever in adults who meet criteria for a low risk of complications. Fluoroquinolone prophylaxis may preclude its subsequent use as empiric therapy for neutropenic fever in the same patient. The modest difference in rates of hospitalization for suspected infection in levofloxacin compared to placebo recipients (15.7% vs. 21.6%, respectively) may be offset by exclusion of outpatient oral empiric therapy in patients receiving fluoroquinolone prophylaxis. To limit antibacterial use, Cullen et alhave suggested prophylaxis with levofloxacin on cycle 1 of myelosuppressive cancer chemotherapy and only in subsequent cycles if a febrile episode occurs.

重试 错误原因

决定是否使用抗菌预防和选择特定药物需要在预期获益和风险之间取得平衡（在霍夫曼等人的综述中进行了更详细的解释）。风险的概念适用于药物的即时不良反应（例如皮疹、GI 不耐受）、选择可能损害接受预防治疗的个体的耐药病原体的可能性以及耐药微生物对特定患者人群（例如在癌症中心接受治疗的患者）的风险。因此，在选择抗微生物预防剂时，必须考虑多重耐药微生物（MDRO）的出现，微生物组的破坏和抗生素毒性。氟喹诺酮类药物使用与严重艰难梭菌和耐甲氧西林金黄色葡萄球菌（MRSA）感染之间的联系

提供了关于过量使用氟喹诺酮类药物的额外警示说明。

NCCN 抗菌预防建议

不建议对低风险的患者进行抗菌预防

总体感染（见基于总体感染的抗菌预防

算法中的癌症患者风险）。在感染并发症风险较低的中性粒细胞减少症患者（包括大多数恶性实体瘤患者）中，抗菌预防的主要获益是减少发热而不是记录的感染。在未接受免疫抑制方案的中性粒细胞减少症预期持续<7 天的患者中（例如，全身性皮质类固醇），专家组建议不使用抗生素预防。在被认为有中度或高度风险的患者中，NCCN 指南专家组建议氟喹诺酮类药物预防（首选左氧氟沙星）在中性粒细胞减少预期持续时间的患者中考虑（中性粒细胞绝对计数[ANC] ⩽500 个中性粒细胞/mcL 或⩽1000 个中性粒细胞/mcL，预计在接下来的 48 小时内下降至⩽500 个/ mcL）>7 天。 这与 IDSA 关于中性粒细胞减少伴癌症患者抗菌药物使用指南的建议一致。对于氟喹诺酮类药物不耐受的患者，可考虑使用 TMP/SMX 或口服第三代头孢菌素。

对于患有癌症和接受 HCT 的儿童患者，也已经制定了全身性抗菌预防的临床实践指南。这些指南的专家组建议，对于患有急性髓性白血病（AML）和复发性急性淋巴细胞白血病（ALL）的儿童，在接受预计会导致严重中性粒细胞减少（ANC <500/mcL）的强化化疗至少 7 天后，应考虑进行抗菌预防。专家组建议，对于接受 ALL 诱导化疗、自体 HCT 和同种异体 HCT 的儿童，不应常规使用抗菌预防。一个强大

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recommendation against its routine use was made for children whose therapy is not expected to result in prolonged severe neutropenia. If used, prophylaxis with levofloxacin is recommended as a preferred option during severe neutropenia.[92]

Prophylaxis for Pneumococcal Infection

Prophylaxis for pneumococcal infection is advised in allogeneic HCT recipients. Patients undergoing allogeneic HCT are at an increased risk for pneumococcal sepsis due to functional asplenia and impaired B-cell immunity. Pneumococcal sepsis is most common in the late transplant period, between 3 months to several years after HCT.[72,93] IST for GVHD delays reconstitution of B-cell immunity and significantly increases the risk of post-transplant pneumococcal sepsis.[72,94]

The NCCN Guidelines Panel advises that penicillin prophylaxis be initiated at 3 months after HCT and be continued until at least 1 year following transplant in patients with GVHD. Patients should receive prophylaxis regardless of prior administration of pneumococcal vaccines.[95] Prophylaxis should be continued in patients with chronic GVHD until IST has been discontinued. Post-transplant pneumococcal infection is generally community-acquired, and the frequency of resistance to antibiotics reflects regional susceptibility patterns. In some regions as many as 35% of pneumococcal isolates have intermediate- or high-level resistance to penicillin,[96] and cross-resistance to other classes of antibiotics is common. Breakthrough pneumococcal sepsis in HCT recipients receiving penicillin prophylaxis is well described.[97] Thus, in areas with a significantly higher frequency of penicillin-resistant pneumococcal isolates, alternative agents should be considered based on local susceptibility patterns. Daily TMP/SMX used as prophylaxis for PJP is likely to be protective against pneumococcal disease, but depends on local susceptibility patterns. In high-risk populations (e.g., allogeneic HCT recipients with GVHD), prophylaxis with penicillin and TMP/SMX should be considered.

Vaccination with pneumococcal vaccine is also strongly recommended (see *Vaccination*) 3 to 6 months post-HCT.

Antifungal Prophylaxis

Antifungal prophylaxis should not be used routinely in all patients with neutropenia. The rationale for antifungal prophylaxis is to prevent fungal infections in a targeted group of patients with high risk, especially those with longer durations of neutropenia or with GVHD after allogeneic HCT.[17] Selection of an antifungal agent is determined by the disease or therapy and includes azoles, amphotericin B products, and echinocandins.

Azoles

Azoles are among the most commonly used medications for the prevention and treatment of fungal infections. Early-generation azoles such as ketoconazole are used less commonly now because of toxicity, drug interactions, and limited spectrum of activity. Some “first-generation” triazoles (ie, fluconazole and itraconazole) are used widely due to their low cost and minimal toxicity but are limited by increasing resistance among *Candida* species and lack of activity against most molds. Several “second-generation” triazoles (eg, voriconazole and posaconazole) have been subsequently developed. These drugs extend the spectrum of activity of triazoles to include potent activity against many molds (importantly, activity differs within the class) but can also have complicated drug interactions and distinct toxicities and remain extremely costly with extended use.

Prophylaxis with voriconazole was compared with fluconazole in a large, randomized, double-blind study that included serum galactomannan surveillance in allogeneic HCT recipients (N = 600).[98] Patients were randomized to receive study drugs for 100 days or for 180 days in the higher risk cohort of patients. No difference was noted in the primary endpoint (invasive fungal infection-free survival rate at 180 days) between the fluconazole and voriconazole prophylaxis arms (75% vs. 78%,

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建议不将其常规用于治疗预期不会导致长期严重中性粒细胞减少的儿童。如果使用左氧氟沙星预防性治疗，建议作为重度中性粒细胞减少症的首选。

预防肺炎球菌感染

对于同种异体 HCT 接受者，建议预防肺炎球菌感染。由于功能性无脾和 B 细胞免疫受损，接受同种异体 HCT 的患者发生肺炎球菌败血症的风险增加。肺炎球菌脓毒症在移植后期最常见，在 HCT 后 3 个月至数年之间。用于 GVHD 的 IST 延迟 B 细胞免疫的重建，并显著增加移植后肺炎球菌脓毒症的风险。

NCCN 指南专家组建议，对于 GVHD 患者，应在 HCT 后 3 个月开始青霉素预防治疗，并持续至移植后至少 1 年。患者应接受预防性治疗

无论既往是否接种过肺炎球菌疫苗，慢性 GVHD 患者应继续预防直至 IST 停止。移植后肺炎球菌感染通常是社区获得性的，对抗生素的耐药性频率反映了区域敏感性模式。在某些地区，多达 35%的肺炎球菌分离株对青霉素具有中度或高度耐药性，对其他类别抗生素的交叉耐药性很常见。在接受青霉素预防的 HCT 受者中发生突破性肺炎球菌脓毒症已有很好的描述，因此，在青霉素耐药肺炎球菌分离株频率明显较高的地区，应根据当地的敏感性模式考虑替代药物。每日使用 TMP/SMX 预防 PJP 可能对肺炎球菌疾病有保护作用，但取决于当地的易感性模式。 在高危人群中（例如，异基因 HCT 受者的 GVHD），应考虑使用青霉素和 TMP/SMX 预防。

也强烈建议在 HCT 后 3 至 6 个月接种肺炎球菌疫苗（见疫苗接种）。

预防性抗真菌治疗

抗真菌预防不应常规用于所有中性粒细胞减少症患者。抗真菌预防的基本原理是预防高风险目标患者组的真菌感染，特别是那些中性粒细胞减少持续时间较长或同种异体 HCT 后发生 GVHD 的患者。抗真菌剂的选择取决于疾病或治疗，包括唑类、阿替霉素 B 产品和棘白菌素。

唑类

唑类是预防和治疗真菌感染的最常用药物之一。早期的唑类药物，如酮康唑，由于毒性、药物相互作用和有限的活性谱，现在不太常用。一些“第一代”三唑类药物（即氟康唑和伊曲康唑）因其成本低、毒性小而被广泛使用，但受到念珠菌耐药性增加和对大多数霉菌缺乏活性的限制。随后开发了几种“第二代”三唑（如伏立康唑和泊沙康唑）。这些药物扩展了三唑类药物的活性范围，包括对许多霉菌的有效活性（重要的是，活性在类别内不同），但也可能具有复杂的药物相互作用和明显的毒性，并且在长期使用中仍然非常昂贵。

在一项大型、随机、双盲研究中，将伏立康唑预防与氟康唑预防进行了比较，该研究包括对异基因 HCT 受者（N = 600）的血清半乳甘露聚糖监测。患者随机接受研究药物 100 天或接受高风险队列患者 180 天。氟康唑和伏立康唑预防组之间的主要终点（180 天无侵袭性真菌感染生存率）无差异（75% vs. 78%，

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respectively), but a trend for reduced incidence of *Aspergillus* infections (17% vs. 9%), reduced incidence of invasive fungal infections (11% vs. 7%), and less frequent use of empiric antifungal treatment (30% vs. 24%) was noted in the voriconazole arm, although the differences were not statistically significant. No differences in relapse-free and overall survival rates, nor incidence of severe adverse events were seen between treatment arms.[98]

In a multicenter randomized trial, prophylaxis with posaconazole in patients with neutropenia with AML or myelodysplastic syndromes (MDS) receiving induction or re-induction chemotherapy significantly reduced the rate of invasive fungal infections during the treatment period (2% vs. 8%; $P < .001$) and during the 100 days following randomization (5% vs. 11%; $P = .003$). Posaconazole prophylaxis also reduced the incidence of invasive aspergillosis (1% vs. 7%; $P < .001$) and was associated with a significant survival benefit ($P = .04$) compared with the fluconazole/itraconazole arm.[99] Data from a prospective, randomized study showed that posaconazole was as effective as fluconazole in allogeneic HCT recipients with severe GVHD and reported reduced incidence of invasive aspergillosis and overall invasive fungal infections in patients receiving posaconazole compared to those receiving fluconazole.[100] Posaconazole is equally effective compared to fluconazole as primary therapy for oropharyngeal candidiasis.[101] Moreover, in a 2021 phase 3 randomized controlled trial, posaconazole was found to be non-inferior to voriconazole as primary therapy for invasive aspergillosis, and participants had fewer treatment-related adverse events in the posaconazole group than in the voriconazole group.[102]

A 2020 systematic review and meta-analysis of 69 randomized clinical trials compared prophylaxis with various antifungal agents in patients with hematologic disease or who received HCT. Posaconazole prophylaxis significantly reduced invasive fungal infections (RR, 0.57; 95% CI, 0.42–0.79) and invasive aspergillosis (RR, 0.36; 95% CI, 0.15–0.85) compared with placebo. Additionally, voriconazole was associated with a significant reduction in invasive candidiasis (RR, 0.15; 95% CI, 0.09–0.26) compared with placebo. However, patients taking posaconazole had a higher incidence of withdrawal due to adverse effects.[103]

Isavuconazonium sulfate is a second-generation azole that was approved in March 2015 for the treatment of invasive aspergillosis and invasive mucormycosis. Isavuconazonium sulfate may be considered for antifungal prophylaxis when standard therapy is contraindicated, due to drug interactions or the risk of QTc prolongation.[104]

Toxicities and Drug-Drug Interactions of Azoles

Experience to date suggests that fluconazole and posaconazole are generally well tolerated and serious adverse events, primarily liver toxicity, are rare. Toxicities for voriconazole include neurologic and ophthalmic adverse events. In the context of renal dysfunction, accumulation of cyclodextrin vehicle may occur when the IV formulation is used. . Data suggest that long-term use of voriconazole may be associated with severe photosensitivity and other adverse events including cutaneous malignancies, elevated serum fluoride levels, and periosteitis.[105-109] Itraconazole may be associated with hepatic toxicity and GI intolerance[110] and is contraindicated in patients with a decreased cardiac ejection fraction or a history of congestive heart failure based on its negative inotropic properties. It can also increase cyclophosphamide metabolites, which in turn are associated with hyperbilirubinemia and nephrotoxicity during the early transplant period.[111] Fluconazole, itraconazole, posaconazole, and voriconazole may cause QTc prolongation. Conversely, isavuconazonium sulfate has been associated with dose-dependent QTc shortening in healthy individuals.

Azole-associated drug-drug interactions are common clinical occurrences. Both the addition and withdrawal of azoles can result in either increased

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伏立康唑组中曲霉菌感染发生率降低（17% vs. 9%），侵袭性真菌感染发生率降低（11% vs. 7%），经验性抗真菌治疗使用频率降低（30% vs. 24%），但差异无统计学意义。治疗组间无复发生存率和总生存率以及严重不良事件的发生率均无差异。

在一项多中心随机试验中，在接受诱导或再诱导化疗的 AML 或骨髓增生异常综合征（MDS）中性粒细胞减少患者中，泊沙康唑预防治疗显著降低了治疗期间（2% vs. 8%; P <0.001）和随机分组后 100 天内（5% vs. 11%; P = 0.003）侵袭性真菌感染的发生率。泊沙康唑预防也降低了侵袭性曲霉病的发生率（1% vs. 7%; P < .001），并与显著的生存获益相关（P = 0.04）与氟康唑/伊曲康唑组相比。一项随机研究显示，在患有严重 GVHD 的同种异体 HCT 受者中，泊沙康唑与氟康唑一样有效，并且报告了与接受氟康唑的患者相比，接受泊沙康唑的患者中侵袭性曲霉病和总体侵袭性真菌感染的发生率降低。与氟康唑作为口咽念珠菌病的主要治疗方法同样有效。此外，在 2021 年的一项 3 期随机对照试验中，泊沙康唑作为侵袭性曲霉病的主要治疗方法不劣于伏立康唑，

泊沙康唑组较伏立康唑组有显著性差异。

2020 年，一项对 69 项随机临床试验进行的系统性综述和荟萃分析比较了血液病患者或接受 HCT 的患者使用各种抗真菌药物进行预防。泊沙康唑预防治疗显著降低了侵袭性真菌感染（RR，0.57; 95%CI，0.42– 0.55）。

0.79）和侵袭性曲霉病（RR，0.36; 95% CI，0.15–0.85）与安慰剂相比。此外，与安慰剂相比，伏立康唑与侵袭性念珠菌病的显著减少相关（RR，0.15; 95%CI，0.09–0.26）。然而，服用泊沙康唑的患者因不良反应而停药的发生率较高。

硫酸异沙唑氮铵是第二代唑类药物，于 2015 年 3 月获批用于治疗侵袭性曲霉病和侵袭性毛霉菌病。当由于药物相互作用或 QTc 间期延长的风险而禁忌标准治疗时，可考虑硫酸异沙康唑铵用于抗真菌预防。

唑类化合物的毒性和药物相互作用

迄今为止的经验表明，氟康唑和泊沙康唑通常耐受性良好，严重不良事件（主要是肝毒性）罕见。伏立康唑的毒性包括神经系统和眼科不良事件。在肾功能不全的情况下，使用 IV 制剂时可能发生环糊精溶媒蓄积。.数据表明，长期使用伏立康唑可能与严重的光敏性和其他不良事件，包括皮肤恶性肿瘤，血清氟水平升高，骨膜炎。伊曲康唑可能与肝毒性和 GI 不耐受相关，基于其负性肌力特性，心脏射血分数降低或有充血性心力衰竭病史的患者禁用。 它也可以增加环磷酰胺的代谢产物，这反过来又与高胆红素血症和肾毒性在早期移植期间。氟康唑，伊曲康唑，泊沙康唑，伏立康唑可能会导致 QTc 延长。相反，硫酸艾沙康氮铵与健康个体的剂量依赖性 QTc 缩短相关。

唑类药物相关的药物相互作用是常见的临床事件。唑类的添加和撤销都可以导致增加

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uptake of these other drugs or sub-therapeutic exposure and potential transplant rejection or GVHD. Several studies demonstrate the interaction of azoles with hepatic enzymatic pathways. Administration of itraconazole with medications that are metabolized by the 3A4 isoenzyme can increase plasma concentrations causing QTc prolongation and ventricular tachyarrhythmias.[112] These findings reinforce a note of caution about itraconazole (and by extension, fluconazole, voriconazole, isavuconazonium sulfate, and posaconazole), with regard to potential serious drug-drug interactions through inhibition of the cytochrome P450 3A4 isoenzyme. Additionally, fluconazole and voriconazole have demonstrated inhibition of CYP2C9 and CYP2C19 enzymes and high interpatient variability of genetic CYP2C19 polymorphisms that may also affect dosing.

The potential for QTc prolongation is a concern exacerbated by the combination of azoles and other drugs (eg, fluoroquinolones, macrolides, ondansetron) and with some chemotherapies (eg, nilotinib for chronic myeloid leukemia [CML], panobinostat for myeloma). Itraconazole and posaconazole are also known inhibitors of gastric P-glycoprotein, which can increase systemic levels of drugs that are affected by this transport system. The list of drug-drug interactions is expansive and continues to grow. While azoles may be necessary for antifungal therapy, they should only be incorporated into treatment following consultation with an infectious disease expert.

Therapeutic Drug Monitoring of Azoles

Therapeutic drug monitoring (TDM) for the pharmacokinetic evaluation of antifungal agents provides guidance for achieving adequate plasma drug concentration while reducing toxicity.

TDM should be considered for patients receiving triazoles; there is no current evidence to support the use of TDM for the evaluation of polyenes or echinocandins. Fluconazole and isavuconazonium sulfate are the two triazoles that do not require TDM. Fluconazole has linear pharmacokinetics that eliminate the need for TDM,[113-117] though patients in renal failure should receive a modified dose. Studies intended to define a therapeutic range for isavuconazonium sulfate have not been performed; thus, TDM is not routinely recommended for isavuconazonium sulfate. TDM should be considered for posaconazole, voriconazole, and itraconazole. Variability of therapeutic drug levels may be affected by the route of drug administration, drug formulation, timing of monitoring, location of the infection, and intrinsic patient factors (ie, age, weight).

There are three formulations of posaconazole: oral suspension, delayed-release tablet, and IV solution. Pharmacokinetic studies with the oral suspension of posaconazole in healthy individuals showed that administration with or after a high-fat meal, or with any meal or nutritional supplement, greatly enhanced its absorption up to 400%.[118] The plasma concentration of posaconazole can be reduced by proton pump inhibitors (PPIs) due to the increase in gastric pH when given orally.[118] Subtherapeutic concentrations and breakthrough fungal infections have been reported.[119,120] As reviewed by Brüggemann et al,[121] a substantial list of drug interactions with azole antifungal drugs can result in subtherapeutic effects or toxicity. The delayed-release (DR) tablet formulation of posaconazole was approved in 2013 and has improved absorption and more predictable bioavailability. Gastric pH does not affect plasma concentration of DR posaconazole,[122] nor does it have the same interaction with PPIs or metoclopramide.[123] The IV formulation has also demonstrated similar pharmacokinetics and safety compared with the DR tablet.[124] A target concentration of posaconazole for prophylactic TDM of >0.7 mcg/mL is supported by individual studies[120,125,126] as well as two phase III studies;[99,100] however, doses as low as 0.5 mcg/mL have also been reported as effective.[120,126-130] TDM may not be necessary when using either the DR tablet or IV formulation in the prophylactic setting as data indicate that a dose of 300 mg/day results in at least 0.5 ug/mL in

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这些其他药物的摄取或亚治疗暴露和潜在的移植排斥或 GVHD。几项研究证明了唑类与肝酶途径的相互作用。伊曲康唑与经 3A 4 同工酶代谢的药物联合给药可增加血浆浓度，导致 QTc 间期延长和心室率降低。

这些发现加强了对伊曲康唑（以及扩展的氟康唑、伏立康唑、硫酸艾沙康唑和泊沙康唑）的注意，关于通过抑制细胞色素 P450 3A 4 同工酶的潜在严重药物–药物相互作用。此外，已证明氟康唑和伏立康唑可抑制 CYP 2C 9 和 CYP 2C 19 酶，且 CYP 2C 19 遗传多态性的患者间变异性较高，这也可能影响给药。

QTc 间期延长的可能性是一个问题，唑类药物与其他药物（例如，氟喹诺酮类、大环内酯类、昂丹司琼）联合使用以及与某些化疗（例如，尼洛替尼治疗慢性髓性白血病[CML]，帕比司他治疗骨髓瘤）联合使用会加剧这一问题。伊曲康唑和泊沙康唑也是已知的胃 P–糖蛋白抑制剂，其可以增加受该转运系统影响的药物的全身水平。药物相互作用的清单是广泛的，并继续增长。虽然唑类药物可能是抗真菌治疗所必需的，但只有在咨询了

传染病专家

唑类药物的治疗药物监测

治疗药物监测（TDM）的抗真菌药物的药代动力学评价提供了指导，以达到足够的血浆药物浓度，同时降低毒性。

接受三唑类药物治疗的患者应考虑 TDM;目前没有证据支持使用 TDM 评价多烯或棘白菌素。氟康唑和硫酸沙武康唑不需要 TDM 的三唑。氟康唑具有线性药代动力学，无需 TDM，但肾衰竭患者应接受修改剂量。尚未进行旨在确定硫酸艾沙康氮铵治疗范围的研究;因此，不建议硫酸艾沙康氮铵常规进行 TDM。泊沙康唑、伏立康唑和伊曲康唑应考虑 TDM。治疗药物水平的变异性可能受到给药途径、药物制剂、监测时间、感染部位和患者内在因素（即年龄、体重）的影响。

泊沙康唑有三种剂型：口服混悬剂、缓释片和静脉注射液。泊沙康唑口服混悬液在健康人体内的药代动力学研究表明，与高脂餐同服或同餐后服用，或与任何膳食或营养补充剂同时服用，泊沙康唑的吸收可达 400%;口服时，质子泵抑制剂（PPI）可使胃内 pH 升高，从而降低泊沙康唑的血药浓度。亚治疗浓度和突破性真菌感染已被报道，如 Br Boggemann 等人所述，大量药物与唑类抗真菌药物的相互作用可导致亚治疗效应或毒性。泊沙康唑的延迟释放（DR）片剂于 2013 年获得批准，具有更好的吸收和更可预测的生物利用度。 胃内 pH 值不影响泊沙康唑的血药浓度，也不与 PPI 或甲氧氯普胺发生相同的相互作用。与 DR 片剂相比，IV 制剂也显示出相似的药代动力学和安全性。泊沙康唑用于预防性 TDM 的目标浓度>0.7 mcg/mL 得到了个体研究和两项 III 期研究的支持;当在预防性环境中使用 DR 片剂或 IV 制剂时，可能不需要 TDM，因为数据表明 300 mg/天的剂量导致至少 0.5 ug/mL 的

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>95% of patients. Treatment of an established infection is recommended to have a trough concentration >1 mcg/mL with potentially higher doses based on the pathogen minimal inhibitor concentrations (MICs) and concerns for resistance.[131,132]

Studies of itraconazole demonstrate a significant rate of breakthrough infections when plasma drug concentrations are <0.5 mcg/mL;[133,134] however, increased mortality was observed at plasma drug concentrations >5 mcg/mL.[135,136] Targeting a lower itraconazole plasma concentration for prophylaxis and a higher dose if an active infection is being treated may be beneficial. Studies suggest that trough concentrations of itraconazole between 1 and 2 mcg/mL have shown the best therapeutic responses for invasive infections,[137-140] while a trough concentration of >0.5 mcg/mL may be sufficient for prophylaxis. Currently, an upper limit of 17 mcg/mL measured by bioassay has been suggested,[141] but studies for the upper limit have not been extensive. Itraconazole solution should be given either 1 hour before or 1 hour after meals based on the 43% increase in bioavailability in patients who fasted[142], while the capsule formulation should be given with a full meal. An alternative formulation, SUBA-itraconazole (for "super bioavailability"), addresses the absorption concerns of conventional itraconazole formulations with enhanced bioavailability that allows for lower dosing, while achieving similar serum concentrations. SUBA-itraconazole is FDA approved for the treatment of blastomycosis, histoplasmosis and aspergillosis.[143,144]

Target voriconazole trough values between 0.5 and 4 mcg/mL have been proposed in clinical studies.[145-152] While 0.5 mcg/mL is a suggested target for prophylaxis, a higher range of 1 to 4 mcg/mL may be necessary for active disease and for patients with disease that has a poor prognosis. Higher concentrations may also benefit the patients who are immunocompromised by reducing breakthrough infection.[153,154] Trough concentrations ≥4 mcg/mL have correlated with toxicity in various studies.[145,149,152,155-159] Voriconazole bioavailability was lowered by about 22% when taken with food and by 34% when given with a high-fat meal.[160] Therefore, voriconazole should be given either 1 hour before or 1 hour after meals.

Studies have shown a general consensus regarding a minimal level of plasma concentration necessary for the triazoles, though the lack of prospective studies has limited the adoption of formal monitoring standards. The Society of Infectious Diseases Pharmacists (SIDP) has published its guidelines for the use of TDM of antifungal agents based on available literature.[161] Consideration of TDM is recommended by the NCCN Panel in conjunction with involvement of an infectious disease expert.

Amphotericin B Formulations

Amphotericin B formulations are broad-spectrum antifungal agents that binds to ergosterol in the fungal cell membrane, leading to the formation of pores in the membrane and subsequent cell death. The original formulation, amphotericin B deoxycholate (AmB-D), was associated with dose-limiting toxicities including infusion-related reactions and nephrotoxicity. Three lipid-associated formulations, amphotericin B lipid complex (ABLC), liposomal amphotericin B (L-AmB), and amphotericin B colloidal dispersion (ABCD), have since been developed to have reduced toxicity. When available, ABLC and L-AmB are preferred over AmB-D.

Low-dose amphotericin B formulations have been studied in patients at high risk and have been shown to provide protection against invasive molds, although no survival benefit in randomized studies was seen when compared with fluconazole.[110,162,163] Based on the toxicity of amphotericin B products and the availability of safer and equally effective alternative agents, amphotericin B products are considered a category 2B recommendation for prophylaxis. If an amphotericin B product is used, a lipid formulation is generally preferred because of less infusional and renal

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>95%的患者。根据病原体最小抑制剂浓度（MIC）和耐药性问题，建议对已确定感染的治疗应使谷浓度>1 mcg/mL，并可能使用更高的剂量。

Studies of itraconazole demonstrate a significant rate of breakthrough infections when plasma drug concentrations are <0.5 mcg/mL; however, increased mortality was observed at plasma drug concentrations >5 mcg/mL.Targeting a lower itraconazole plasma concentration for prophylaxis and a higher dose if an active infection is being treated may be beneficial. Studies suggest that trough concentrations of itraconazole between 1 and 2 mcg/mL have shown the best therapeutic responses for invasive infections,while a trough concentration of >0.5 mcg/mL may be sufficient for prophylaxis. Currently, an upper limit of 17 mcg/mL measured by bioassay has been suggested,but studies for the upper limit have not been extensive. Itraconazole solution should be given either 1 hour before or 1 hour after meals based on the 43% increase in bioavailability in patients who fasted, while the capsule formulation should be given with a full meal. An alternative formulation, SUBAitraconazole (for “super bioavailability”), addresses the absorption concerns of conventional itraconazole formulations with enhanced bioavailability that allows for lower dosing, while achieving similar serum concentrations. SUBA–itraconazole is FDA approved for the treatment of blastomycosis, histoplasmosis and aspergillosis.

重试 错误原因

在临床研究中，伏立康唑谷值的目标值在 0.5 – 4 mcg/mL 之间。虽然 0.5 mcg/mL 是预防的建议目标值，但对于活动性疾病和预后不良的疾病患者，可能需要 1 – 4 mcg/mL 的更高范围。高浓度也可能通过减少突破性感染而使免疫功能低下的患者受益。

伏立康唑的生物利用度在与食物同服时降低约 22%，在与高脂肪食物同服时降低 34%。

因此，伏立康唑应在饭前 1 小时或饭后 1 小时给药。

研究表明，关于三唑类药物所需的最低血药浓度水平已达成普遍共识，但缺乏前瞻性研究限制了正式监测标准的采用。传染病学会药学委员会（SIDP）已根据现有文献发表了抗真菌药物 TDM 使用指南，NCCN 专家组建议在传染病专家参与下考虑 TDM。

两性霉素 B 制剂

两性霉素 B 制剂是广谱抗真菌剂，其与真菌细胞膜中的麦角固醇结合，导致膜中孔的形成和随后的细胞死亡。最初的制剂，阿替西霉素 B 脱氧胆酸盐（AmB–D），与剂量限制性毒性相关，包括输注相关反应和肾毒性。已经开发了三种脂质相关制剂，即阿替霉素 B 脂质复合物（ABLC）、脂质体阿替霉素 B（L–AmB）和阿替霉素 B 胶体分散体（ABCD），以降低毒性。当可用时，ABLC 和 L–AmB 优于 AmB–D。

低剂量阿替西霉素 B 制剂已在高危患者中进行了研究，并已被证明可提供对侵袭性霉菌的保护，尽管与氟康唑相比，在随机研究中没有观察到生存益处。基于阿替西霉素 B 产品的毒性以及更安全和同样有效的替代药物的可用性，阿替西霉素 B 产品被认为是预防性的第 2 类 B 推荐。如果使用阿替霉素 B 产品，则脂质制剂通常是优选的，因为较少的输注和肾毒性。

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toxicity compared to conventional amphotericin B. Use of the lipid formulation is particularly important for patients at high risk for renal failure, such as patients with pre-existing renal disease, HCT recipients, and patients who are concurrently receiving other nephrotoxic agents.[164,165]

Aerosolized delivery of amphotericin B products has been considered for several years with the advantage of local delivery to the lungs while simultaneously avoiding systemic toxicity. A randomized, placebo-controlled trial found that aerosolized L-AmB was useful for preventing invasive pulmonary aspergillosis in patients with prolonged neutropenia.[166] Limitations to the use of aerosolized amphotericin B for prophylaxis relate to the variability of this treatment due to different nebulizers and amphotericin B formulations, the lack of dosing optimization, and a dearth of direct comparative data with systemically administered mold-active azoles or echinocandins.[167]

Echinocandins

Echinocandins are a class of antifungal agents that disrupt the integrity of the fungal cell wall through noncompetitive inhibition of β-(1,3)-D-glucan synthase, an enzyme involved in the biosynthesis of β-(1,3)-D-glucan, which is a component specific to the cell wall of many fungi. Echinocandins have fungicidal activity against *Candida* species and are fungistatic towards *Aspergillus* species. Combination therapy with amphotericin B or triazoles has been proposed to improve activity against molds; however, clinical evidence for this remains quite limited. Advantages of this family of antifungals are the relatively low toxicity profiles and limited drug-drug interactions. Though echinocandins demonstrate activity against *Candida* species that are resistant to other antifungal agents,[168] there is limited or no activity against dimorphic fungi. Four echinocandins are approved for use: caspofungin, micafungin, anidulafungin, and rezafungin. Caspofungin, micafungin, and anidulafungin are all approved for the treatment of esophageal candidiasis, while rezafungin is approved for the treatment of candidemia and invasive candidiasis in adults with limited or no alternative treatment options. Caspofungin and anidulafungin have additional indications for the treatment of candidemia and other infections caused by *Candida* species. Caspofungin is indicated for treatment of candidal pleural space infections, empiric treatment of fungal infections in neutropenia, and treatment of invasive aspergillosis in patients who are refractory to or intolerant of other antifungal agents. Micafungin has the additional indication for prophylaxis of candidal infections in patients receiving HCT.

Caspofungin was evaluated in a double-blind study including 128 patients with esophageal candidiasis.[169] Patients received either caspofungin or AmB-D. Two doses of caspofungin were evaluated (50 mg or 70 mg IV once daily) with a greater response in the patients given the higher dose (96% vs. 85%). Both groups treated with caspofungin had a better response than patients receiving amphotericin B (78%). At the 2-week follow-up, a greater percentage of patients remained negative for candidiasis with the caspofungin treatment (89% in the 70 mg group, 74% in the 50 mg group, and 63% in the amphotericin B group). Furthermore, drug-related adverse events were lower with caspofungin (7%, 4%, and 24%, respectively). Several studies have evaluated the role of caspofungin in the treatment of invasive aspergillosis in patients refractory to or intolerant of other antifungals, supporting its recommendation in this capacity.[170,171]

Micafungin is an echinocandin approved for prophylaxis against *Candida* infections in patients undergoing HCT. In a randomized, double-blind trial of autologous and allogeneic HCT recipients, the success rate with micafungin was superior to fluconazole (80% vs. 73.5%; absolute difference +6.5%; 95% CI, 0.9–12%; $P = .03$) based on prespecified criteria for treatment success (absence of suspected, proven, or probable

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与常规的阿替霉素 B 相比，脂质制剂的使用对于肾衰竭高风险患者尤其重要，例如既存肾病患者、HCT 接受者和同时接受其他肾毒性药物的患者。

几年来一直在考虑雾化递送阿替西霉素 B 产品，其优点是局部递送至肺，同时避免全身毒性。一项随机、安慰剂对照试验发现，雾化 L–AmB 可有效预防长期中性粒细胞减少患者的侵袭性肺曲霉病。使用雾化阿替霉素 B 进行预防的局限性与这种治疗的可变性有关，这是由于不同的雾化器和阿替霉素 B 制剂、缺乏剂量优化以及缺乏与全身给药的霉菌活性唑类或棘白菌素的直接比较数据。

棘白

棘白菌素是一类通过非竞争性抑制β–（1，3）–β–葡聚糖合酶破坏真菌细胞壁完整性的抗真菌剂，β–（1，3）–β–葡聚糖合酶是参与β–（1，3）–β–葡聚糖生物合成的酶，β–（1，3）–β–葡聚糖是许多真菌细胞壁的特异性组分。棘白菌素对念珠菌属具有杀真菌活性，对曲霉属具有抑真菌活性。已经提出了与阿替霉素 B 或三唑类的联合治疗来提高抗霉菌的活性;然而，这方面的临床证据仍然非常有限。该抗真菌药家族的优点是毒性相对较低，药物相互作用有限。尽管棘白菌素对耐其他抗真菌剂的念珠菌属物种表现出活性，但对二型真菌的活性有限或没有活性。四种棘白菌素类药物被批准用途：卡泊芬净、米卡芬净、阿尼芬净和利扎芬净。卡泊芬净、米卡芬净和阿尼芬净都被批准用于治疗食管念珠菌病，而利扎芬净被批准用于治疗念珠菌血症和侵袭性念珠菌病，成人的替代治疗选择有限或没有替代治疗选择。卡泊芬净和阿尼芬净还有治疗念珠菌血症和念珠菌属引起的其他感染的其他适应症。卡泊芬净适用于治疗念珠菌性胸膜间隙感染、经验性治疗中性粒细胞减少症患者的真菌感染以及治疗其他抗真菌药物难治或不耐受患者的侵袭性曲霉病。米卡芬净还有预防接受 HCT 患者念珠菌感染的其他适应症。

在一项双盲研究中对卡泊芬净进行了评估，该研究包括 128 例食管念珠菌病患者。评价了两种剂量的卡泊芬净（50 mg 或 70 mg IV，每日一次），给予较高剂量的患者反应更大（96% vs. 85%）。卡泊芬净治疗的两组患者的反应均优于阿替霉素 B 治疗的患者（78%）。在 2 周随访时，卡泊芬净治疗的念珠菌病阴性患者比例更高（70 mg 组为 89%，50 mg 组为 74%，阿替霉素 B 组为 63%）。此外，卡泊芬净的药物相关不良事件较低（分别为 7%、4%和 24%）。几项研究评价了卡泊芬净在治疗其他抗真菌药难治或不耐受的侵袭性曲霉病患者中的作用，支持其在这方面的建议。

米卡芬净是一种棘白菌素，获批用于预防接受 HCT 的患者的念珠菌感染。在一项自体和同种异体 HCT 受者的随机、双盲试验中，基于预先设定的治疗成功标准（无可疑、证实或很可能的异常），米卡芬净的成功率上级优于氟康唑（80% vs. 73.5%;绝对差异+6.5%; 95%CI，0.9–12%; P = 0.03

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invasive fungal infections during treatment period and absence of proven or probable infection during the 4-week period after treatment).[172] The duration of this study encompassed the neutropenic period, but not the period after neutrophil recovery when GVHD would be expected to occur. The frequency of breakthrough candidemia was similar in both arms, but there was a trend to fewer episodes of invasive aspergillosis in allogeneic HCT recipients receiving micafungin. Survival and drug-related toxicity were similar between treatment arms.[172] Micafungin has shown activity in the treatment of aspergillosis in patients refractory to or intolerant of other antifungal agents.[173-175]

Anidulafungin has been shown to be an effective antifungal agent against *Candida* infection in several studies. A randomized double-blind study in 601 patients with esophageal candidiasis demonstrated noninferiority of IV anidulafungin to oral fluconazole (97.2% vs. 98.8%, respectively) and lower adverse effects (9.3% vs. 12.0%) and recurring infections at the 2-week follow-up (64.4% vs. 89.5%).[176] In a smaller study of 19 patients with triazole-refractory mucosal candidiasis, anidulafungin treatment resolved infection in 18 of the patients.[177] A larger phase III trial similarly showed superiority of anidulafungin compared to fluconazole in the treatment of candidemia and invasive candidiasis (75.6% vs. 60.2%).[178] The response at 2-week follow-up was 64.6% in the anidulafungin group versus 49.2% in the fluconazole group.

Rezafungin was evaluated in a phase 3 multicentre, double-blind randomized trial.[179] 199 patients with candidemia or invasive candidiasis were randomly assigned to either the rezafungin group (50%) or to the caspofungin group (50%). Rezafungin was found to be non-inferior to caspofungin in the treatment of candidemia and invasive candidiasis (59% vs. 61%, respectively) as well as for 30-day all-cause mortality (24% vs. 21%, respectively).

NCCN Recommendations for Antifungal Prophylaxis

CYP3A4 inhibition by azoles can lead to toxicity when administered with several classes of drugs used in cancer therapy, including proteasome inhibitors, tyrosine kinase inhibitors, and vinca alkaloids.[180] Thus, mold-active azoles should be stopped several days before the potential interacting drug is given. These azoles should also not be started until the other agent has been discontinued and sufficient time has elapsed for the drug to be eliminated. Due to variations in drug pharmacokinetics, firm recommendations regarding a minimum time from drug discontinuation to azole administration cannot be made, though some institutions consider waiting at least 10 days following administration of these classes of drugs. Use of echinocandin prophylaxis may be considered in the place of azoles. Consultation with pharmacology and infectious disease experts is recommended.

The NCCN Guidelines Panel recommends posaconazole (category 1) for antifungal prophylaxis in neutropenia with AML and MDS receiving induction or re-induction chemotherapy (see *Prevention Of Fungal Infections* in the algorithm).[17] The role of antifungal prophylaxis in patients with acute leukemia receiving consolidation chemotherapy has not been adequately evaluated. Voriconazole, isavuconazole, fluconazole, echinocandin, or amphotericin B products are all category 2B recommendations in this disease setting. Antifungal prophylaxis should be continued until resolution of neutropenia.

In patients with mucositis receiving autologous HCT, antifungal prophylaxis with fluconazole or echinocandin (both category 1) is recommended until resolution of neutropenia. No prophylaxis is recommended in autologous HCT recipients without mucositis.

The NCCN Guidelines Panel recognizes that strong evidence exists for the use of fluconazole or echinocandin as prophylaxis in neutropenic allogeneic HCT recipients (category 1).[17] However, it should be noted that

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在治疗期间没有侵袭性真菌感染，并且在治疗后 4 周期间没有证实或可能的感染）。该研究的持续时间包括中性粒细胞减少期，但不包括中性粒细胞恢复后预期发生 GVHD 的时期。两组中突破性念珠菌血症的发生率相似，但接受米卡芬净的同种异体 HCT 受者中侵袭性曲霉菌病的发生率有减少的趋势。两个治疗组的生存率和药物相关毒性相似。米卡芬净在治疗其他抗真菌药物难治或不耐受的曲霉病患者中显示出活性。

阿尼芬净已被证明是一种有效的抗真菌剂对念珠菌感染的几项研究。一项在 601 例食管念珠菌病患者中进行的随机双盲研究证明，静脉注射阿尼芬净非劣效于口服氟康唑（分别为 97.2%和 98.8%）和较低的不良反应（9.3% vs. 12.0%）和 2 周随访时复发感染（64.4%对 89.5%）。在一项对 19 例三唑难治性粘膜念珠菌病患者的小型研究中，阿尼芬净治疗缓解了 18 例患者的感染。念珠菌血症和侵袭性念珠菌病的治疗（75.6% vs. 60.2%）。2 周随访时，阿尼芬净组的有效率为 64.6%，氟康唑组为 49.2%。

在一项 3 期多中心、双盲随机试验中对瑞扎芬净进行了评估。199 例念珠菌血症或侵袭性念珠菌病患者被随机分配至瑞扎芬净组（50%）或卡泊芬净组（50%）。研究发现，在治疗念珠菌血症和侵袭性念珠菌病（分别为 59%和 61%）以及 30 天全因死亡率（分别为 24%和 61%）方面，瑞扎芬净不劣于卡泊芬净。

21%）。

NCCN 抗真菌预防建议

当与几类用于癌症治疗的药物（包括蛋白酶体抑制剂、酪氨酸激酶抑制剂和长春花生物碱）同时给药时，唑类药物对 CYP3A4 的抑制作用可导致毒性。因此，应在给予潜在相互作用药物前几天停用具有模塑活性的唑类药物。这些唑类药物也不应该开始，直到其他药物已经停止，并有足够的时间来消除药物。由于药物药代动力学的变化，无法就从停药到唑类药物给药的最短时间提出明确建议，尽管一些机构考虑在这些药物给药后至少等待 10 天。可以考虑使用棘白菌素预防性治疗来代替唑类药物。建议咨询药理学和传染病专家。

The NCCN Guidelines Panel recommends posaconazole (category 1) for antifungal prophylaxis in neutropenia with AML and MDS receiving induction or re–induction chemotherapy (see Prevention Of Fungal Infections in the algorithm).The role of antifungal prophylaxis in patients with acute leukemia receiving consolidation chemotherapy has not been adequately evaluated. Voriconazole, isavuconazole, fluconazole, echinocandin, or amphotericin B products are all category 2B recommendations in this disease setting. Antifungal prophylaxis should be continued until resolution of neutropenia.

重试 错误原因

In patients with mucositis receiving autologous HCT, antifungal prophylaxis with fluconazole or echinocandin (both category 1) is recommended until resolution of neutropenia. No prophylaxis is recommended in autologous HCT recipients without mucositis.

重试 错误原因

NCCN 指南小组认识到，有强有力的证据表明，氟康唑或棘白菌素可用于血小板减少的同种异体 HCT 接受者（1 类）的预防。

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fluconazole use can predispose patients to colonization and bloodstream infection by fluconazole-resistant *Candida* strains.[69,181] Posaconazole as prophylaxis has not been evaluated during the neutropenic period following conditioning in allogeneic HCT recipients; thus, the safety of this approach is unknown. Drug-drug interactions during conditioning for HCT, specifically with posaconazole, itraconazole, or voriconazole, complicate treatment of fungal infections in these patients. Prophylaxis may need to be tailored following consultation with an infectious disease expert. Posaconazole, isavuconazole, voriconazole, echinocandins, and amphotericin B products are all considered category 2B recommendations. Antifungal prophylaxis should be considered until at least day 75 or count recovery (ANC >500) after allogeneic HCT.[17,182-184]

Patients with hematologic malignancies are at high risk of developing invasive fungal disease. The German Society of Haematology and Medical Oncology (DGHO) strongly recommends administration of antifungal prophylaxis in patients with hematologic malignancies with prolonged neutropenia (ie, <500 cells/mcL for >7 days) with posaconazole as the drug of choice for mold-active prophylaxis.[184]

Although many centers reasonably use antifungal prophylaxis in non-neutropenic allogeneic HCT recipients with GVHD, this practice was only evaluated in a single, properly designed study. In the prospective, randomized, double-blind study, posaconazole was compared with fluconazole as prophylaxis in allogeneic HCT recipients with severe GVHD requiring intensive IST.[100] Inclusion criteria included grade II to IV GVHD, chronic extensive GVHD, or intensive IST consisting of either high-dose corticosteroids, antithymocyte globulin, or a combination of two or more immunosuppressive agents or types of treatment. Prophylaxis with posaconazole resulted in reduced incidences of invasive aspergillosis, total invasive fungal infections while on treatment, and deaths attributed to fungal infection.[100] Posaconazole is recommended (category 1) as prophylaxis in patients with GVHD receiving intensive IST, as defined by the inclusion criteria in this trial, although the benefit/risk ratio of mold-active prophylaxis in patients receiving less intensive IST has not been established.

Patients with chronic severe neutropenia (ANC <500 neutrophils/mcL) due to an underlying disease (such as aplastic anemia) are at substantial risk for invasive aspergillosis.[185] Although this population has not been evaluated in clinical trials of antifungal prophylaxis, some Panel members advise the use of a prophylactic mold-active agent (eg, posaconazole or voriconazole).

Secondary antifungal prophylaxis is defined as administration of antifungal therapy in a patient with a prior fungal infection to prevent recrudescence. The Panel recommends secondary prophylaxis with an appropriate antifungal agent in patients with prior chronic disseminated candidiasis[186] or with invasive filamentous fungal infection[187] during subsequent cycles of chemotherapy or HCT. In patients with invasive aspergillosis before HCT, antifungal therapy for >1 month and resolution of radiologic abnormalities correlate with a lower likelihood of post-transplant recurrence of infection.[188] Secondary prophylaxis with a mold-active agent is advised for the entire period of immunosuppression.

Antiviral Prophylaxis and Preemptive Antiviral Therapy

Herpes Simplex Virus

HSV is an important pathogen in patients who develop neutropenia and mucositis. HSV infections primarily result from reactivation of latent virus. The presence of latent HSV can be determined by pretreatment HSV serology. Reactivation and infection with HSV occur in 60% to 80% of HCT recipients and patients (without prophylaxis) with acute leukemia undergoing induction or re-induction therapy who are seropositive for HSV.[189-191] Among allogeneic HCT recipients, HSV disease is most likely

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氟康唑的使用可使患者易受氟康唑耐药念珠菌的定植和血流感染。泊沙康唑在异基因 HCT 受者预处理后的血小板减少期作为预防性治疗尚未得到评价;因此，该方法的安全性尚不清楚。HCT 预处理期间的药物相互作用，特别是泊沙康唑、伊曲康唑或伏立康唑，使这些患者的真菌感染治疗复杂化。预防措施可能需要在咨询传染病专家后进行调整。

泊沙康唑、艾沙康唑、伏立康唑、棘白菌素和阿替霉素 B 产品均被视为 2B 类推荐产品。应考虑抗真菌预防，直至同种异体 HCT 后至少第 75 天或计数恢复（ANC >500）。

恶性血液病患者发生侵袭性真菌病的风险很高。德国血液学和肿瘤内科学会（DGHO）强烈建议对长期中性粒细胞减少（即<500 个细胞/mcL>7 天）的恶性血液病患者进行抗真菌预防，并将泊沙康唑作为霉菌活性预防的首选药物。

虽然许多中心合理地使用抗真菌药物预防非血小板减少的异基因造血干细胞移植物抗宿主病受者，这种做法只在一个单一的，适当设计的研究进行了评估。在前瞻性、随机、双盲研究中，泊沙康唑与氟康唑作为预防性治疗需要强化 IST 的严重 GVHD 的异基因 HCT 受者，纳入标准包括 II 至 IV 级 GVHD、慢性广泛 GVHD 或强化 IST，包括高剂量皮质类固醇、抗胸腺细胞球蛋白或两种或多种免疫抑制剂或治疗类型的组合。泊沙康唑预防治疗可降低侵袭性曲霉病、治疗期间侵袭性真菌感染的发生率以及真菌感染导致的死亡。

在接受强化 IST 的 GVHD 患者中，根据本试验的入选标准，Moldactive 预防的获益/风险比尚未确定，但在接受较低强化 IST 的患者中，Moldactive 预防的获益/风险比尚未确定。

由于基础疾病（如再生障碍性贫血）导致的慢性重度中性粒细胞减少症（ANC <500 中性粒细胞/mcL）患者有发生侵袭性曲霉病的显著风险。尽管这一人群尚未在抗真菌预防的临床试验中进行评估，但一些专家组成员建议使用预防性霉菌活性药物（如泊沙康唑或伏立康唑）。

二级抗真菌预防定义为对既往有真菌感染的患者进行抗真菌治疗以防止复发。专家组建议，在随后的化疗或 HCT 周期中，既往患有慢性播散性念珠菌病或侵袭性丝状真菌感染的患者应使用适当的抗真菌药物进行二级预防。对于 HCT 前有侵袭性曲霉病的患者，抗真菌治疗>1 个月和放射学异常消退与移植后感染复发的可能性较低相关。建议在整个免疫抑制期间使用霉菌活性剂进行二级预防。

抗病毒预防和预防性抗病毒治疗

单纯疱疹病毒

HSV 是发生中性粒细胞减少症和粘膜炎的患者的重要病原体。HSV 感染主要由潜伏病毒的再激活引起。潜伏 HSV 的存在可以通过治疗前 HSV 血清学来确定。在 60%–80%的 HCT 接受者和接受诱导或再诱导治疗的 HSV 血清阳性的急性白血病患者（未预防）中，HSV 再活化和感染发生。

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to occur within the first month post-transplant, but may occur in later stages during intense immunosuppression.[65,66] Although disseminated HSV infection is uncommon, infection from viral reactivation is frequently associated with increased mucosal damage, resulting in increased pain, limited ability to maintain oral hydration and nutrition, and an increased risk of bacterial and fungal superinfections.

NCCN Recommendations for HSV Prophylaxis

Antiviral prophylaxis against HSV is advised during the period of neutropenia in patients who are HSV-seropositive who are receiving chemotherapy (induction or consolidation) for acute leukemia, and during neutropenia and possibly longer in allogeneic and autologous HCT recipients depending on the degree of immunosuppression (see *Prevention Of Herpes Simplex Virus And Varicella Zoster Virus Reactivation Or Disease* in the algorithm). A longer period of prophylaxis should be considered in allogeneic HCT recipients with GVHD or with frequent HSV reactivations before transplantation.[8] Acyclovir, valacyclovir or famciclovir are the initial agents of choice for HSV prophylaxis.[17,192] Foscarnet is typically reserved for patients with acyclovir-resistant HSV infection.[17,192] In patients receiving antiviral prophylaxis with ganciclovir or foscarnet for prevention of CMV reactivation, additional prophylaxis with acyclovir is not necessary given that these agents are active against HSV.[192] However, it is required in patient receiving CMV prophylaxis with letermovir as this agent lacks activity against HSV.

HSV and herpes zoster infections are common in patients with CLL treated with the CD52 monoclonal antibody alemtuzumab. For these patients, antiviral prophylaxis is advised until at least 2 months after completion of alemtuzumab therapy and until CD4+ cell counts are ≥200 cells/mcL.[193]

Prophylaxis against HSV should be considered in other patients at intermediate risk for HSV reactivation, including those with hematologic malignancies with prolonged neutropenia or those receiving high-dose corticosteroids or T-cell–depleting agents (eg, fludarabine). Once a patient has had HSV reactivation requiring treatment, the Panel recommends HSV prophylaxis for that patient during all future episodes of neutropenia induced by cytotoxic therapy. HSV prophylaxis is also indicated in children who are seropositive for the virus.

Varicella Zoster Virus

Impaired cellular immunity is the principal risk factor for VZV disease. In allogeneic HCT recipients with a history of VZV infection, about 30% have reactivation of VZV disease without antiviral prophylaxis.[194] In patients with a history of chicken pox, oral acyclovir administered from 1 to 2 months until 1 year after allogeneic HCT significantly decreased the incidence of VZV disease compared to placebo (5% vs. 26%, respectively).[194] The frequency of VZV disease in the post-prophylactic period was similar between the groups and predominantly occurred in patients who required systemic immunosuppression. This prolonged course of acyclovir prophylaxis is likely to also prevent HSV reactivations. Subsequent studies have consistently demonstrated the benefit of long-term antiviral prophylaxis against VZV disease in recipients of allogeneic HCT. Patients who received anti-VZV prophylaxis with acyclovir or valacyclovir for 1 year post-HCT had significantly reduced VZV disease compared with those who did not receive long-term prophylaxis (9% vs. 25%; $P < .001$); no evidence of rebound VZV disease was observed.[195] Long-term (1 year post-allogeneic HCT) prophylaxis with lower doses of acyclovir or valacyclovir was associated with a higher cumulative incidence of VZV reactivation.[196,197]

NCCN Recommendations for VZV Prophylaxis

The NCCN Guidelines Panel recommends prophylaxis against VZV for at least 1 year after allogeneic HCT in patients seropositive for VZV pretransplant (see *Prevention Of Herpes Simplex Virus And Varicella*

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虽然播散性 HSV 感染并不常见，但病毒再激活引起的感染通常与粘膜损伤增加有关，导致疼痛增加，维持口服水化和营养的能力有限，以及细菌和真菌双重感染的风险增加。

NCCN 对 HSV 预防的建议

在接受急性白血病化疗（诱导或巩固）的 HSV 血清阳性患者的中性粒细胞减少期间，以及在异基因和自体 HCT 接受者的中性粒细胞减少期间和可能更长的时间（取决于免疫抑制的程度），建议针对 HSV 的抗病毒预防（参见

单纯疱疹病毒和水痘带状疱疹病毒

算法中的再激活或疾病）。对于移植前有 GVHD 或 HSV 频繁再激活的异基因 HCT 受者，应考虑更长时间的预防，阿昔洛韦、伐昔洛韦或泛昔洛韦是预防 HSV 的首选药物。膦甲酸钠通常用于对阿昔洛韦耐药的 HSV 感染患者。在接受更昔洛韦或膦甲酸钠预防 CMV 再激活的抗病毒预防治疗的患者中，由于这些药物对 HSV 有活性，因此不需要使用阿昔洛韦进行额外的预防治疗。然而，在接受莱特莫韦预防 CMV 的患者中，由于这种药物缺乏对 HSV 的活性，因此需要使用阿昔洛韦进行预防治疗。

HSV 和带状疱疹感染在接受 CD 52 单克隆抗体 Alemtuzumab 治疗的 CLL 患者中很常见。对于这些患者，建议进行抗病毒预防治疗，直至完成 Alemtuzumab 治疗后至少 2 个月，直至 CD 4+细胞计数≥200 个细胞/mcL。

对于其他 HSV 再激活中度风险的患者，应考虑预防 HSV，包括血液学异常的患者。

恶性肿瘤伴长期中性粒细胞减少或接受高剂量皮质类固醇或 T 细胞耗竭剂（如氟达拉滨）的患者。一旦患者发生 HSV 再激活需要治疗，专家组建议在细胞毒性治疗诱导的中性粒细胞减少症的所有未来发作期间对该患者进行 HSV 预防。HSV 预防也适用于病毒血清反应阳性的儿童。

水痘带状疱疹病毒

细胞免疫功能受损是 VZV 病的主要危险因素。在有 VZV 感染史的异基因 HCT 受者中，约 30%的人在没有抗病毒预防的情况下 VZV 疾病复发。在有水痘史的患者中，与安慰剂相比，异基因 HCT 后 1 至 2 个月至 1 年口服阿昔洛韦可显著降低 VZV 疾病的发生率预防后阶段 VZV 疾病的频率在两组之间相似，并且主要发生在需要全身免疫抑制的患者中。这种延长的阿昔洛韦预防疗程也可能预防 HSV 再激活。随后的研究一致表明，长期抗病毒预防对 VZV 疾病的异基因 HCT 受体的好处。 与未接受长期预防治疗的患者相比，HCT 后 1 年接受阿昔洛韦或伐昔洛韦抗 VZV 预防治疗的患者 VZV 疾病显著减少（9%对 25%; P < .001）;没有观察到 VZV 病反弹的证据。（同种异体 HCT 后 1 年）预防低剂量阿昔洛韦或伐昔洛韦与 VZV 再激活的累积发生率较高。

NCCN 对 VZV 预防的建议

NCCN 指南小组建议 VZV 血清阳性患者在同种异体 HCT 后至少预防 VZV 1 年

移植前（见单纯疱疹病毒和水痘的预防

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Zoster Virus Reactivation Or Disease in the algorithm), and recommends considering the extension of prophylaxis in patients who continue to receive systemic IST. Although higher doses are necessary, the same agents used as HSV prophylaxis are also active against VZV. For pediatric patients, prophylaxis for VZV should not be routinely given unless there is a history of recurrent zoster infections or incidence of first zoster infection while on myelosuppressive therapy, even if they are seropositive or vaccinated.

Among autologous HCT recipients, HSV reactivation is more likely to occur in the early neutropenic phase, whereas the risk of VZV reactivation extends through the first year.[198] Thus, VZV prophylaxis for at least 6 to 12 months post-transplant should be considered in autologous HCT recipients. Prophylaxis against VZV should be considered in other patients at intermediate risk for viral reactivation, including patients with hematologic malignancies with prolonged neutropenia or those receiving T-cell–depleting agents (eg, fludarabine, alemtuzumab). Bortezomib is associated with an increased risk of VZV reactivation during active therapy[199-202]; carfilzomib may also be associated with VZV reactivation. Prophylaxis with acyclovir, valacyclovir or famciclovir should be protective and can be considered in these settings.[203,204] As previously discussed, among patients with CLL receiving alemtuzumab treatment, antiviral prophylaxis is recommended until 2 months after completion of treatment and until the CD4+ cell counts reach 200 cells/mcL or more.[193]

Cytomegalovirus

CMV infections most frequently occur in patients with cancer who undergo allogeneic HCT or who receive alemtuzumab therapy or other T-cell depleting therapies. CMV is a common cause of opportunistic infections during the early post-engraftment phase following allogeneic HCT, but can also occur in the late post-engraftment phase (particularly for patients with GVHD during the latter phase).[65,66] Infection can result from viral reactivation (in patients who are immunocompromised CMV-seropositive) or primary infection (in CMV-seronegative). The risk for CMV reactivation and disease is highest among HCT recipients with CMV-seropositive status prior to transplant.[205] Among patients who are CMV-seropositive undergoing allogeneic HCT (with graft sources from peripheral blood, bone marrow, or umbilical cord blood), the incidence of CMV reactivation ranged from 50% to 60% (with CMV disease in about 10%–30% of seropositive recipients) even with routine surveillance and antiviral prophylaxis or preemptive therapy.[205-208]

In a randomized phase II prophylaxis trial, oral letermovir was administered to allogeneic HCT recipients with CMV-seropositive status daily at 60 mg, 120 mg, or 240 mg for 12 weeks post-transplant. The incidence of prophylaxis failure (defined as discontinuation of the study drug because of CMV detection, end-organ disease, or any other cause) was significantly lower in the letermovir groups than in the placebo group (32% for the 120 mg group, 29% for the 240 mg group vs. 64%). Moreover, letermovir was well tolerated, with no indication of hematologic toxicity or nephrotoxicity.[209] In a randomized, double-blind, phase III trial, prophylaxis with daily oral or IV dose of letermovir, 480 mg/day (or 240 mg/day in patients taking ciclosporin), was administered to allogeneic HCT recipients for 14 weeks after transplantation. Fewer patients developed clinically significant CMV infection in the letermovir group than in the placebo group (38% vs. 61% in placebo; $P < .001$). The frequency of adverse events, like vomiting and edema, were mildly higher in the letermovir group.[210] Despite the lower toxicity profile of letermovir compared to other CMV-targeted drugs, rapid emergence of resistant mutants under letermovir has been described in cases where treatment is interrupted or underdosed,[211] or in patients with other risk factors.[212,213]

Valganciclovir and ganciclovir are the agents of choice for first-line preemptive therapy; foscarnet is more commonly used for patients who

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带状疱疹病毒再激活或疾病的算法），并建议

考虑在继续接受系统性 IST 的患者中延长预防。虽然需要更高的剂量，但用作 HSV 预防的相同药物也对 VZV 有效。对于儿科患者，不应常规给予 VZV 预防，除非有复发性带状疱疹感染史或在骨髓抑制治疗期间发生首次带状疱疹感染，即使他们是血清阳性或接种了疫苗。

在自体 HCT 受者中，HSV 再激活更可能发生在早期血小板减少期，而 VZV 再激活的风险持续到第一年，因此，自体 HCT 受者应考虑移植后至少 6 至 12 个月的 VZV 预防。在其他具有病毒再活化中等风险的患者中，应考虑预防 VZV，包括长期中性粒细胞减少的恶性血液病患者或接受 T 细胞耗竭剂（例如，氟达拉滨、阿仑单抗）的患者。硼替佐米与积极治疗期间 VZV 再激活风险增加相关;卡非佐米也可能与 VZV 再激活相关。 如前所述，在接受 Alemtuzumab 治疗的 CLL 患者中，建议抗病毒预防直至治疗完成后 2 个月，直至 CD 4+细胞计数达到 200 个细胞/mcL 或以上。

细胞病毒

CMV 感染最常发生在接受同种异体 HCT 或接受 Alemtuzumab 治疗或其他 T 细胞耗竭治疗的癌症患者中。CMV 是同种异体 HCT 后早期移植后阶段期间机会性感染的常见原因，但也可发生在晚期移植后阶段（特别是对于在后一阶段期间患有 GVHD 的患者）。

再激活（在免疫功能低下的 CMV 血清阳性患者中）或原发性感染（在 CMV 血清阴性患者中）。在移植前 CMV 血清阳性的 HCT 受者中，CMV 再激活和疾病的风险最高。（移植物来源于外周血、骨髓或脐带血），CMV 再激活的发生率为 50%~60（在约 10%–30%的血清阳性受者中患有 CMV 疾病），即使常规监测和抗病毒预防或抢先治疗也是如此。

在一项随机 II 期预防性试验中，将口服莱特莫韦给予 CMV 血清阳性状态的同种异体 HCT 接受者

移植后 12 周，每天 60 mg、120 mg 或 240 mg。的

莱特莫韦组预防失败（定义为由于 CMV 检测、终末器官疾病或任何其他原因而停用研究药物）的发生率显著低于安慰剂组

(32 120 mg 组为 29%，240 mg 组为 64%)。

在一项随机、双盲、III 期临床试验中，在移植后 14 周内，对异基因 HCT 受者给予每日口服或 IV 剂量的莱特莫韦 480 mg/天（或服用环孢素的患者 240 mg/天）进行预防。莱特莫韦组发生临床显著 CMV 感染的患者少于安慰剂组（38% vs.安慰剂组 61%; P < .001）。不良事件的频率，如呕吐和水肿，在莱特莫韦组中略高。尽管莱特莫韦的毒性较低，与其他 CMV 靶向药物相比，莱特莫韦的耐药突变体迅速出现在治疗中断或剂量不足的情况下，或在有其他风险因素的患者中。

缬更昔洛韦和更昔洛韦是一线预防治疗的首选药物;膦甲酸更常用于以下患者：

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cannot tolerate ganciclovir or for second-line preemptive therapy.[192] Foscarnet and ganciclovir had similar efficacy as preemptive CMV therapies in allogeneic HCT recipients, but ganciclovir was associated with a higher rate of early discontinuation because of neutropenia or thrombocytopenia.[214] Although ganciclovir had a higher rate of early discontinuation, there remains a paucity of data to recommend foscarnet as first-line treatment for CMV..[215-217]

Pharmacokinetic studies have demonstrated the feasibility and safety of using oral valganciclovir, a pro-drug of ganciclovir, in place of ganciclovir in patients who underwent allogeneic HCT.[218,219] Oral valganciclovir used as preemptive anti-CMV therapy was shown to have acceptable oral bioavailability and was safe and effective in controlling CMV infection in allogeneic HCT recipients, including patients with grades I and II GI GVHD.[218,220-222] Thus, valganciclovir is a highly acceptable oral option for preemptive therapy for CMV in the absence of substantial GI GVHD. Reports of higher rates of CMV disease with oral valganciclovir compared to IV ganciclovir in patients with hepatic dysfunction restricted approval for solid tumor transplant patients by specifically excluding liver transplant patients.[223,224] It is postulated that hepatic dysfunction allows bioabsorption of valganciclovir but decreases cleavage of the valine ester, thereby limiting conversion to the active form.[224]

Cidofovir has been evaluated as both primary and secondary preemptive therapy in allogeneic HCT recipients.[225-228] In a retrospective study of allogeneic HCT recipients (N = 82) treated for CMV disease (n = 20), primary preemptive therapy (n = 24) or secondary preemptive therapy (n = 38) with cidofovir demonstrated an observed response in 50% of patients treated for CMV disease (mainly CMV pneumonia) and in 62% of patients treated with primary preemptive therapy.[227] Moreover, secondary preemptive therapy with cidofovir resulted in a response rate of 66% in patients where disease progressed or relapse occurred (defined as continued presence or recurrence of pp65 antigenemia or viral DNA after at least 1 week of antivirals) following initial preemptive therapy with ganciclovir, foscarnet, or a combination of these agents.[227]

For CMV infections that are refractory or resistant to available antivirals (valganciclovir, ganciclovir, foscarnet, cidofovir), maribavir has shown promise in some studies. In a phase II, randomized, double-blind trial, ≥400 mg of maribavir twice daily was found to be active against refractory or resistant CMV infections in transplant recipients, with a higher proportion of patients achieving undetectable CMV DNA within 6 weeks of treatment compared to placebo.[229] Dysgeusia was the most common treatment-related adverse event. In another phase III randomized trial, maribavir was superior to other anti-CMV agents in HCT and solid organ transplant recipients, with 56% of patients achieving CMV viremia clearance under maribavir compared to 24% under other agents (P < .001). Maribavir was also associated with less acute kidney injury versus foscarnet (8.5% vs. 21.3%) and with less neutropenia versus valganciclovir/ganciclovir (9.4% vs. 33.9%).[230] On the other hand, in another phase III randomized double-blind trial, noninferiority of maribavir to valganciclovir for the primary endpoint of confirmed CMV viremia clearance was not achieved based on the prespecified noninferiority margin (69.6% vs 77.4%).[231] However, maribavir demonstrated comparable CMV viremia clearance to valganciclovir during post-treatment follow-up (52.7% vs 48.5%), with fewer discontinuations due to neutropenia.

Late CMV disease, defined as occurring after day 100 of HCT, remains a persistent problem in the era of CMV prophylaxis and preemptive therapy. CMV infections are also common among patients with lymphoproliferative malignancies, patients receiving T-cell suppressive therapy with purine analogues, and patients receiving alemtuzumab therapy, and occurs most frequently between 3 to 6 weeks after initiation of therapy when T-cell

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不能耐受更昔洛韦或二线预防治疗。膦甲酸钠和更昔洛韦作为 CMV 的预防性治疗在异基因 HCT 受者中具有相似的疗效，但更昔洛韦与更高的早期停药率相关，因为中性粒细胞减少或血小板减少。尽管更昔洛韦有更高的早期停药率，但仍缺乏数据推荐膦甲酸钠作为 CMV 的一线治疗。

药代动力学研究已经证明了在接受同种异体 HCT 的患者中使用更昔洛韦的前药口服缬更昔洛韦代替更昔洛韦的可行性和安全性。口服缬更昔洛韦作为抢先抗 CMV 治疗被证明具有可接受的口服生物利用度，并且在控制同种异体 HCT 接受者（包括 I 级和 II 级 GI GVHD 患者）的 CMV 感染方面是安全有效的。因此，缬更昔洛韦是一种高度可接受的口服选择，用于在不存在实质性 GI GVHD 的情况下对 CMV 进行预先治疗。在肝功能障碍患者中，口服缬更昔洛韦的 CMV 疾病发生率高于静脉注射更昔洛韦，这一报道限制了实体瘤移植患者的批准，特别是排除了肝移植患者。

西多福韦已被评估为同种异基因 HCT 接受者的主要和次要抢先治疗。在一项针对治疗 CMV 疾病（n = 20）的同种异基因 HCT 接受者（N = 82）的回顾性研究中，用西多福韦进行一次抢先治疗（n = 24）或二次抢先治疗（n = 38），在 50%的 CMV 疾病治疗患者中观察到应答（主要是 CMV 肺炎）和 62%的患者接受了主要的抢先治疗。

在用更昔洛韦、膦甲酸或这些药物的组合进行初始抢先治疗后，用西多福韦进行第二抢先治疗导致疾病进展或复发（定义为在至少 1 周的抗病毒药物治疗后继续存在或复发 pp65 抗原血症或病毒 DNA）的患者的应答率为 66%。

对于现有抗病毒药（缬更昔洛韦、更昔洛韦、膦甲酸、西多福韦）难治或耐药的 CMV 感染，Maribavir 在一些研究中显示出前景。在一项 II 期、随机化、双盲试验中，发现 Maribavir ⩾400 mg 每日两次对移植受者的难治性或耐药性 CMV 感染具有活性，与安慰剂相比，治疗 6 周内达到 CMV DNA 不可检测的患者比例更高。味觉障碍是最常见的治疗相关不良事件。在另一项 III 期随机试验中，Maribavir 在 HCT 和实体器官移植受者中上级其他抗 CMV 药物，Maribavir 组 56%的患者实现 CMV 病毒血症清除，而其他药物组为 24%（P < .001）。与膦甲酸相比，Maribavir 的急性肾损伤更少（8.5% vs. 21.3%），与缬更昔洛韦/更昔洛韦相比，中性粒细胞减少更少（9.4% vs. 另一方面，在另一项 III 期随机双盲试验中，基于预先规定的非劣效性界值，Maribavir 在确认的 CMV 病毒血症清除率主要终点方面未达到非劣效于缬更昔洛韦（69.6% vs 77.4%）。然而，在治疗后随访期间，Maribavir 的 CMV 病毒血症清除率与缬更昔洛韦相当（52.7% vs 48.5%），因中性粒细胞减少而停药的患者较少。

晚期 CMV 疾病，定义为 HCT 100 天后发生，在 CMV 预防和抢先治疗时代仍然是一个持续存在的问题。CMV 感染在淋巴增生性恶性肿瘤患者、接受嘌呤类似物 T 细胞抑制治疗的患者和接受阿仑单抗治疗的患者中也很常见，并且当 T 细胞抑制治疗开始后 3 – 6 周最常见。

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counts reach a nadir.[33,37-39,232,233] Several studies of alemtuzumab in patients with CLL have demonstrated the effectiveness of using routine CMV monitoring coupled with preemptive anti-CMV therapy with ganciclovir in preventing overt CMV disease.[33,37,38,234] A small randomized study in patients with lymphoproliferative disease treated with alemtuzumab-containing regimens (N = 40) showed that upfront CMV prophylaxis with oral valganciclovir significantly reduced the incidence of CMV reactivation compared with oral valacyclovir (0% vs. 35%; $P = .004$).[39]

NCCN Recommendations for CMV Prophylaxis

Based on the available data that predict the risk of CMV disease, the NCCN Guidelines Panel recommends routine surveillance for CMV reactivation after allogeneic HCT, consisting of weekly monitoring by PCR, especially during alemtuzumab therapy and at least 2 months after completion of treatment.[235] Upon confirmation of CMV viremia (defined as PCR positivity for CMV in ≥2 consecutive samples obtained 1 week apart), the Panel recommends preemptive therapy with oral valganciclovir, IV ganciclovir or IV foscarnet for at least 2 weeks and until CMV is no longer detectable (see *Prevention of Cytomegalovirus Reactivation or Disease* in the algorithm).

In cases of ganciclovir-resistant CMV or when ganciclovir is not tolerated (eg, ganciclovir-induced myelosuppression), IV foscarnet or IV cidofovir may be used. Additionally, in cases of post-transplant CMV infection that is refractory to ganciclovir/valganciclovir, foscarnet, or cidofovir, oral maribavir may be used. However, infectious disease consult is highly recommended due to the possibility of drug resistance or interaction. Primary prophylaxis with oral or IV letermovir may be considered for CMV-seropositive recipients who undergo allogeneic HCT, with a dose reduction recommended if co-administered with cyclosporine due to drug interactions. In patients receiving allogeneic HCT, the strategy of CMV surveillance testing by PCR followed by preemptive anti-CMV therapy for a positive result is favored over universal long-term prophylaxis. Some centers consider the use of letermovir through day 100 post-HCT and continue CMV surveillance for patients at high risk for CMV reactivation. In certain circumstances, up to day 200 can be considered.[236] Letermovir lacks HSV and VZV coverage and HSV/VZV prophylaxis should be continued.

Surveillance should typically occur for at least 3 to 6 months post-transplant and during chronic GVHD requiring IST. Higher risk transplant subgroups may exist and require different management strategies. Note that the CD4+ count will be reduced by systemic corticosteroids and by other lymphocyte-depleting agents. The majority of cases of late CMV disease occur within the first year of transplant and <5% occur after the second year.[237] Therefore, the value of CMV surveillance beyond 2 years after HCT is unknown but can be considered in patients with significant chronic GVHD. There is debate about how to treat patients after a negative test for CMV. There are not enough data to determine whether patients should be transitioned to surveillance or continue with chronic maintenance therapy, and if so, for how long. The benefits must be weighed against the potential toxicity associated with long-term antiviral use. Ganciclovir and valganciclovir are associated with bone marrow suppression that may increase the risk of common opportunistic infections. Foscarnet can cause nephrotoxicity and electrolyte abnormalities but is tolerated.[214,238] Cidofovir can be associated with substantial nephrotoxicity; although less frequent, ocular toxicity has been reported.[227,228] Acyclovir and valacyclovir have excellent safety profiles but are only weakly active against CMV and are not recommended as prophylaxis or treatment of CMV infection.

For the prevention and treatment of CMV pneumonitis, adjunctive IVIG can be administered; however, IVIG is generally not recommended for

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在数达到缓解点进行的几项 Alemtuzumab 研究已经证明了常规 CMV 监测联合更昔洛韦抢先抗 CMV 治疗在预防明显 CMV 疾病方面的有效性。一项在淋巴增生性疾病患者中进行的含 Alemtuzumab 方案治疗的小型随机研究（N = 40）结果表明，与口服伐昔洛韦相比，口服伐更昔洛韦预先预防 CMV 可显著降低 CMV 再激活的发生率（0%对 35%; P = .004）。

NCCN 对 CMV 预防的建议

基于预测 CMV 疾病风险的可用数据，NCCN 指南小组建议对同种异体 HCT 后的 CMV 再活化进行常规监测，包括每周通过 PCR 进行监测，特别是在阿仑单抗治疗期间和治疗完成后至少 2 个月。

在间隔 1 周获得的≥2 份连续样本中 CMV PCR 阳性），

专家组建议预先口服缬更昔洛韦、静脉注射更昔洛韦或静脉注射膦甲酸，至少治疗 2 周，直到 CMV 不再

可检测（参见算法中的预防巨细胞病毒再激活或疾病）。

在更昔洛韦耐药 CMV 或更昔洛韦不耐受（例如更昔洛韦诱导的骨髓抑制）的情况下，可使用 IV 膦甲酸或 IV 西多福韦。此外，在更昔洛韦/缬更昔洛韦、膦甲酸或西多福韦难治的移植后 CMV 感染的情况下，可以使用口服 Maribavir。然而，由于耐药性或相互作用的可能性，强烈建议进行传染病咨询。对于接受同种异体 HCT 的 CMV 血清阳性受试者，可以考虑口服或 IV 莱特莫韦进行一级预防，如果由于药物相互作用而与环孢素联合给药，则建议降低剂量。在接受同种异体 HCT 的患者中，CMV 策略

通过 PCR 监测检测，然后对阳性结果进行先发制人的抗 CMV 治疗，优于普遍的长期预防。一些中心考虑使用莱特莫韦直至 HCT 后 100 天，并继续对 CMV 再激活高风险患者进行 CMV 监测。在某些情况下，可以考虑长达 200 天。莱特莫韦缺乏 HSV 和 VZV 覆盖率，应继续 HSV/VZV 预防。

监测通常应在移植后至少 3 至 6 个月和需要 IST 的慢性 GVHD 期间进行。可能存在更高风险的移植亚组，需要不同的管理策略。请注意，全身性皮质类固醇和其他淋巴细胞清除剂会降低 CD4+计数。大多数晚期 CMV 病病例发生在移植后第一年，<5%发生在第二年后，因此，HCT 后 2 年后 CMV 监测的价值尚不清楚，但可考虑用于严重慢性 GVHD 患者。关于如何治疗 CMV 检测阴性的患者存在争议。没有足够的数据来确定患者是否应该过渡到监测或继续慢性维持治疗，如果是，持续多久。这些益处必须与长期使用抗病毒药物的潜在毒性相权衡。 更昔洛韦和缬更昔洛韦与骨髓抑制相关，可能增加常见机会性感染的风险。膦甲酸可引起肾毒性和电解质异常，但可耐受。西多福韦可引起大量肾毒性;虽然不太常见，但已报道了眼毒性。阿昔洛韦和伐昔洛韦具有良好的安全性，但对 CMV 只有微弱的活性，不推荐作为 CMV 感染的预防或治疗。

对于 CMV 肺炎的预防和治疗，可以给予连续性 IVIG;然而，IVIG 通常不推荐用于

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prophylactic use except in limited situations due to cost and the limited evidence of activity of this treatment. Although no optimal dosing regimen has been determined, IVIG is commonly administered every other day for 3 to 5 doses. CMV-specific IVIG has not been shown to be any more efficacious than standard IVIG.

Hepatitis B Virus

The risk factors for HBV infection include personal or parental history of an intermediate to high prevalence of HBV infection in one's birthplace (defined as a prevalence of hepatitis B surface antigen [HBsAg] positivity in >2% of the population); household and sexual contact with HBsAg+ persons; individuals with multiple sexual partners or history of sexually transmitted diseases; individuals who have been inmates of correctional facilities; patients with chronically elevated AST or ALT levels; patients with a history of injection drug use; males who have sex with other males (MSM); and patients positive for HCV or HIV.

A positive HBsAg is associated with active infection or a window period before the development of protective immunity in patients exposed to HBV. An individual who has been vaccinated for HBV typically has the following serology: negative HBsAg, positive hepatitis B surface antibody (HBsAb), and negative hepatitis B core antibody (HBcAb).[239] False-negative HBsAg results may occur in patients with chronic liver disease.[240] HBsAb positivity is generally equated with protective immunity, although reactivated HBV disease may occur in the setting of significant immunosuppression in individuals who are HBcAb-positive.[241] A patient with resolved hepatitis B infection will be HBcAb positive but HBsAg negative. As mentioned above, some patients with cancer are at increased risk for HBV reactivation due to profound immunosuppression stemming from cytotoxic regimens, high-dose corticosteroids, tyrosine kinase inhibitors, anti-CD20/CD52 monoclonal antibodies, and/or the underlying malignancy (eg, leukemia, lymphoma).

Patients with malignancies who are HBsAg positive and/or HBcAb positive are at risk for HBV reactivation with cytotoxic chemotherapy. Approximately 20% to 50% of patients with HBsAg positivity and 3% to 45% with HBcAb positivity develop HBV reactivation.[46,239,242-250] The risk of HBV reactivation for patients who are HBsAg negative, HBcAb positive varies widely based on the virological profile, disease, and immunosuppressive regimen. Serum HBV DNA testing prior to the initiation of therapy may help define their risk of reactivation. If viremic, they may receive similar prophylaxis as patients who are HBsAg positive.[251] Complications of HBV reactivation can range from self-limited hepatitis to fulminant hepatic failure and death.[250,252-256] HBV reactivation can lead to early discontinuation or delayed initiation of treatment.[257,258] A systematic review and meta-analysis by Zhang et al[259] revealed that patients with B-cell non-Hodgkin lymphoma (NHL) who are HBsAg+ had worse prognosis and higher incidence of hepatic dysfunction during chemotherapy. In a meta-analysis and evaluation of the U.S. Food and Drug Administration (FDA) safety reports, it was reported that HBcAb positivity correlated with increased incidence of rituximab-associated HBV reactivation.[243] After allogeneic HCT, loss of HBV-specific immunity may occur (ie, loss of HBsAb and development of HBsAg and HBV PCR positivity). This has been observed in up to 40% of susceptible individuals in one report[260] and may be confused with hepatic GVHD. A retrospective study showed that allogeneic HCT recipients who were HBsAg negative but HBcAb positive had a high risk of seroconversion to HBsAg positivity and HBV reactivation (subsequently leading to hepatitis) following allogeneic HCT.[261]

There are several nucleos(t)ide analogs approved by the FDA for the prevention and treatment of HBV. Historically, data supporting the use of these analogues have been based on lamivudine, a reverse transcriptase inhibitor. In a meta-analysis of clinical trials evaluating lamivudine prophylaxis in patients with HBsAg-positive lymphoma treated with IST,

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预防性使用，除非在有限的情况下，由于成本和有限的证据，这种治疗的活性。虽然没有确定最佳的给药方案，但 IVIG 通常每隔一天给药 3 至 5 次。CMV 特异性 IVIG 尚未显示比标准 IVIG 更有效。

B 型肝炎病毒

HBV 感染的危险因素包括个人或父母在出生地有中度至高度 HBV 感染的病史（定义为 B 型肝炎表面抗原[HBsAg]阳性的流行率> 2%的人口）;与 HBsAg+的人有家庭和性接触;有多个性伴侣或有性传播疾病史的人;曾在教养机构服刑的人; AST 或 ALT 水平长期升高的患者;有注射药物使用史的患者;与其他男性发生性关系的男性（MSM）;以及 HCV 或 HIV 阳性的患者。

HBsAg 阳性与活动性感染或暴露于 HBV 的患者保护性免疫形成前的窗口期有关。已接种 HBV 疫苗的个体通常具有以下血清学：HBsAg 阴性、B 肝炎表面抗体（HBcAb）阳性和 B 肝炎核心抗体（HBcAb）阴性。慢性肝病患者可能出现 HBsAg 假阴性结果。HBcAb 阳性通常等同于保护性免疫，尽管 HBcAb 阳性个体在显著免疫抑制的情况下可能发生再活化的 HBV 疾病。 如上所述，由于细胞毒性方案、高剂量皮质类固醇、酪氨酸激酶抑制剂、抗 CD 20/CD 52 单克隆抗体和/或基础恶性肿瘤（例如白血病、淋巴瘤）引起的严重免疫抑制，一些癌症患者的 HBV 再激活风险增加。

Patients with malignancies who are HBsAg positive and/or HBcAb positive are at risk for HBV reactivation with cytotoxic chemotherapy. Approximately 20% to 50% of patients with HBsAg positivity and 3% to 45% with HBcAb positivity develop HBV reactivation.The risk of HBV reactivation for patients who are HBsAg negative, HBcAb positive varies widely based on the virological profile, disease, and immunosuppressive regimen. Serum HBV DNA testing prior to the initiation of therapy may help define their risk of reactivation. If viremic, they may receive similar prophylaxis as patients who are HBsAg positive.Complications of HBV reactivation can range from self–limited hepatitis to fulminant hepatic failure and death.HBV reactivation can lead to early discontinuation or delayed initiation of treatment.A systematic review and meta–analysis by Zhang et alrevealed that patients with B–cell non–Hodgkin lymphoma (NHL) who are HBsAg+ had worse prognosis and higher incidence of hepatic dysfunction during chemotherapy. In a meta–analysis and evaluation of the U.S. Food and Drug Administration (FDA) safety reports, it was reported that HBcAb positivity correlated with increased incidence of rituximab–associated HBV reactivation.After allogeneic HCT, loss of HBV–specific immunity may occur (ie, loss of HBsAb and development of HBsAg and HBV PCR positivity). This has been observed in up to 40% of susceptible individuals in one reportand may be confused with hepatic GVHD. A retrospective study showed that allogeneic HCT recipients who were HBsAg negative but HBcAb positive had a high risk of seroconversion to HBsAg positivity and HBV reactivation (subsequently leading to hepatitis) following allogeneic HCT.

重试 错误原因

There are several nucleos(t)ide analogs approved by the FDA for the prevention and treatment of HBV. Historically, data supporting the use of these analogues have been based on lamivudine, a reverse transcriptase inhibitor. In a meta–analysis of clinical trials evaluating lamivudine prophylaxis in patients with HBsAg–positive lymphoma treated with IST,

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prophylaxis resulted in a significant reduction in HBV reactivation (risk ratio, 0.21; 95% CI, 0.13–0.35) and a trend for reduced HBV-related deaths (risk ratio, 0.68; 95% CI, 0.19–2.49) compared with no prophylaxis.[262] However, despite its initial effectiveness, virologic breakthrough was high, with reports of resistance in 80% of patients after 5 years of therapy.[263] Thus, lamivudine monotherapy has fallen out of favor. Studies suggest one of the newer agents (such as entecavir or tenofovir) may be preferable or combination therapy may have a possible role for patients with lamivudine-resistant HBV infections.[264-266]

As of 2015, tenofovir is available in two different pro-drug forms, tenofovir disoproxil fumarate (DF) and tenofovir alafenamide (AF). Tenofovir AF has greater plasma stability than tenofovir DF, allowing use of a lower dose and lesser systemic exposure to the drug.[267] Tenofovir DF has demonstrated superior antiviral efficacy compared with adefovir in a phase III randomized double-blind study in patients with chronic HBV infection, making tenofovir preferred over adefovir in this setting.[268] Two randomized, phase III, double-blind studies comparing tenofovir AF to tenofovir DF in patients with HBeAg-negative[269] or HBeAg-positive[270] chronic HBV infection showed that the efficacy of tenofovir AF was non-inferior to tenofovir DF, with better bone and renal safety for tenofovir AF. While these data support the use of tenofovir for HBV infection, limited data are available regarding its use in patient populations with cancer. A systematic review and meta-analysis showed that in patients with HBV-associated hepatocellular carcinoma (HCC), tenofovir was associated with better overall survival and reduced late recurrence compared to entecavir.[271] No detectable resistance to tenofovir DF was reported in patients with chronic hepatitis B after 6 years of treatment.[272] In another study, sequencing of the HBV polymerase/reverse transcriptase indicated sequence changes at polymorphic sites, though none resulted in drug resistance.[273] In total, there were only 16 cases of virologic breakthrough, 12 of which were associated with nonadherence to study medication. Resistance for tenofovir DF remained undetectable throughout a 5-year span. By comparison, lamivudine resistance was calculated to be 24% in the first year, and this number steeply climbed to 70% by year 5.[273]

Entecavir has shown improved antiviral activity compared to adefovir in randomized open-label studies in patients with chronic hepatitis B.[274] A few small case studies have evaluated entecavir in the prevention[275] or treatment of HBV in patients with cancer (reviewed by Liu et al[276]). Entecavir had a low drug resistance of 1.2% at 5 years[277] compared to adefovir, which had an intermediate resistance that increased from 0% in the first year to 29% by year 5.[268,278,279]

In addition to drug resistance, the safety profile of the nucleos(t)ide analogues should affect drug selection. Nephrotoxicity has been seen with adefovir[280,281] and tenofovir (specifically tenofovir disoproxil fumarate [TDF]).[282] No significant side effects have been reported with lamivudine or entecavir; however, it is recommended that all patients be monitored for lactic acidosis and severe hepatomegaly with steatosis.

NCCN Recommendations for HBV Prophylaxis

Risk-based screening is recommended by the American Society of Clinical Oncology (ASCO)[283] and the American Association for the Study of Liver Disease (AASLD). [284] Although it is possible that risk-based screening may be more cost-effective than universal screening, there are currently no validated risk tools that could easily be implemented into clinical practice. Furthermore, <60% of patients with HBV infection may have obvious risk factors,[285] and only 10% to 35% of these patients may be aware of their own HBV infection.[286] Therefore, any patient expected to receive IST or chemotherapy should be screened. Implementation of universal screening, as recommended by the CDC, should be considered. [287]

In patients undergoing intensive IST, including HCT, both patient and donor should be screened for HBV, HCV, and HIV prior to treatment.

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预防性治疗显著减少了 HBV 再激活（风险比，0.21; 95%CI，0.13–0.35）和 HBV 相关死亡减少的趋势（风险比，0.68; 95%CI，0.19–2.49）。然而，尽管其初始有效性，病毒学突破是高的，据报道，拉米夫定单药治疗 5 年后，80%的患者出现耐药。因此，拉米夫定单药治疗已经失宠。研究表明，一种较新的药物（如恩替卡韦或替诺福韦）可能是首选或联合治疗可能有拉米夫定耐药的 HBV 感染患者的作用。

截至 2015 年，替诺福韦有两种不同的前药形式，富马酸替诺福韦二异丙酯（DF）和替诺福韦艾拉酚胺（AF）。替诺福韦 AF 比替诺福韦 DF 具有更高的血浆稳定性，允许使用较低的剂量和较少的全身暴露于药物。在慢性 HBV 感染患者的 III 期随机双盲研究中，与阿德福韦相比，替诺福韦 DF 显示出上级抗病毒疗效，使得替诺福韦在这种情况下优于阿德福韦。两个随机，III 期，在 HBeAg 阴性或 HBeAg 阳性慢性 HBV 感染患者中比较替诺福韦 AF 与替诺福韦 DF 的双盲研究显示，替诺福韦 AF 的疗效不劣于替诺福韦 DF，虽然这些数据支持使用替诺福韦治疗 HBV 感染，但关于其在癌症患者人群中的使用的数据有限。 一项系统回顾和荟萃分析显示，在 HBV 相关的肝细胞癌（HCC）患者中，与恩替卡韦相比，替诺福韦与更好的总生存率和更低的晚期复发率相关。在治疗 6 年后，慢性 B 肝炎患者对替诺福韦 DF 没有检测到耐药性。在另一项研究中，HBV 聚合酶/逆转录酶的测序表明多态性位点的序列变化，尽管没有一个导致耐药性。总共只有 16 例病毒学突破，其中 12 例与不依从研究药物有关。

替诺福韦酯 DF 的耐药性在整个 5 年的跨度内仍然检测不到。相比之下，拉米夫定耐药率在第一年计算为 24%，到第 5 年这个数字急剧上升到 70%。

在慢性乙型肝炎患者的随机开放标签研究中，与阿德福韦相比，恩替卡韦显示出更好的抗病毒活性。一些小型病例研究评估了恩替卡韦在预防或治疗癌症患者 HBV 中的作用（由 Liu et al 综述）。

恩替卡韦在 5 年时的耐药性较低，为 1.2%，而阿德福韦的耐药性中等，从第一年的 0%增加到第 5 年的 29%。

除了耐药性，核苷（酸）类似物的安全性也会影响药物的选择。阿德福韦和替诺福韦（特别是富马酸替诺福韦酯[TDF]）可观察到肾毒性。拉米夫定或恩替卡韦未报告显著副作用;但是，建议监测所有患者的乳酸酸中毒和严重肝肿大伴脂肪变性。

NCCN 对 HBV 预防的建议

基于风险的筛查由美国临床肿瘤学会（ASCO）和美国肝病研究协会（AASLD）推荐。虽然基于风险的筛查可能比普遍筛查更具成本效益，但目前还没有经过验证的风险工具可以轻松地应用于临床实践。此外，<60%的 HBV 感染患者可能存在明显的危险因素，而这些患者中只有 10%~ 35%可能意识到自己的 HBV 感染，因此，任何预期接受 IST 或化疗的患者都应进行筛查。应考虑实施疾病预防控制中心建议的普遍筛查。

在接受强化 IST（包括 HCT）的患者中，患者和供体均应在治疗前进行 HBV、HCV 和 HIV 筛查。

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Evaluation of HBsAg, HBcAb, and HBsAb should be considered at baseline. [192,239] Vaccination against HBV should be strongly considered in patients who are HBV-naïve (ie, negative for HBsAg, HBsAb, and HBcAb) (see *Vaccination*).[192,239] In patients who are HBV-naïve undergoing allogeneic HCT, grafts from donors who are HBsAg-positive or HBV DNA-positive should be avoided wherever possible. Donors who have not been exposed to HBV should be considered for HBV vaccination before hematopoietic cell collection.

In individuals who are HBsAg-positive or HBcAb-positive, baseline quantitative PCR for HBV DNA should be obtained. In allogeneic HCT candidates with evidence of active HBV infection (chronic hepatitis based on biopsy or positive HBsAg or high levels of HBV DNA), transplant procedure should be delayed when possible, and antiviral therapy should be given for 3 to 6 months prior to conditioning.[192] In HCT candidates who are HBsAg-positive or HBcAb-positive but without evidence of active HBV replication, antiviral prophylaxis should be considered (starting shortly before the transplant procedure). All allogeneic HCT recipients should continue surveillance for at least 12 months after transplant or during GVHD (see *Management Of Hepatitis B Virus, Hepatitis C Virus, And Human Immunodeficiency Virus Reactivation Or Disease* in the algorithm). For details on the management of HBV infection in patients with B-cell NHL, see the NCCN Guidelines for B-Cell Lymphomas.

The optimal choice of antiviral agents for prophylaxis (or preemptive approaches) will primarily be driven by institutional standards. The NCCN Panel recommends consultation with an expert in hepatitis treatment to determine appropriate antiviral prophylaxis for patients who test positive for HBV. Preferred agents for HBV prophylaxis are entecavir and tenofovir. Although data were originally obtained with lamivudine, entecavir and tenofovir are preferred, especially when treating patients with active HBV infections due to the low threshold of resistance with lamivudine.

Monitoring of HBsAg and HBV DNA and transaminases should be considered for patients without active HBV infection who are not receiving prophylaxis.

Hepatitis C Virus

Studies for HCV reactivation in patients with cancer are not as expansive as studies for hepatitis B; however, an increase in mortality was reported in patients with cancer who had HCV infection compared to patients with cancer who were HCV negative.[288] A review by Yazici et al[289] summarized studies of HCV reactivation in patients receiving targeted therapies and the data correlated an increase in HCV reactivation with these therapies.[289] Differences in outcomes between patients who are HCV positive with cancer versus HCV positive without cancer were reported to include higher occurrence of occult infection, higher risk of developing early cirrhosis, higher rate of fibrosis progression, development of viral reactivation, and poorer virologic outcomes (reviewed by Borchardt et al).[290] The guidelines from the joint IDSA and AASLD Panels for the testing, management, and treatment of hepatitis C recommend that treatment for HCV be considered for patients with chronic HCV with a life expectancy of >12 months.[291]

NCCN Recommendations for HCV Screening and Management

All patients who are expected to receive chemotherapy or IST should be screened for HCV. The data are limited regarding the treatment of HCV in patients with cancer, but it is generally not recommended that HCV treatment and cancer therapy be given concurrently.[290] The IDSA/AASLD guidelines can provide additional guidance for antiviral therapy, but an infectious diseases consult is necessary to evaluate the use of concomitant or sequential anti-HCV and cancer therapy.[291] Monitoring of ALT levels and HCV viral load monthly, or as clinically indicated, should be initiated as part of surveillance (see *Management Of Hepatitis B Virus, Hepatitis C Virus, And Human Immunodeficiency Virus Reactivation Or*

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应在基线时考虑对 HBsAg、HBcAb 和 HBcAb 进行评价。对于 HBV 初治患者（即 HBsAg、HBcAb 和 HBcAb 阴性），应强烈考虑接种抗 HBV 疫苗（见疫苗接种）。对于接受同种异体 HCT 的 HBV 初治患者，应尽可能避免来自 HBsAg 阳性或 HBV DNA 阳性供体的移植物。未接触过 HBV 的献血者应考虑在采集造血细胞前接种 HBV 疫苗。

对于 HBsAg 阳性或 HBcAb 阳性的个体，应进行 HBV DNA 的基线定量 PCR。在有活动性 HBV 感染证据的同种异体 HCT 候选者中（根据活检或 HBsAg 阳性或 HBV DNA 高水平的慢性肝炎），移植程序应尽可能延迟，并应在预处理前给予抗病毒治疗 3 至 6 个月。在 HCT 候选人中，HBsAg 阳性或 HBcAb 阳性，但没有活动性 HBV 复制的证据，应考虑抗病毒预防（在移植手术前不久开始）。所有同种异体 HCT 受者应在移植后或移植期间继续监测至少 12 个月。

对于没有活动性 HBV 感染且未接受预防治疗的患者，应考虑监测 HBsAg 和 HBV DNA 以及转氨酶。

丙型肝炎病毒

癌症患者中 HCV 再激活的研究不像 B 型肝炎的研究那样广泛;但是，在此情况下，据报道，与 HCV 阴性的癌症患者相比，HCV 感染的癌症患者的死亡率增加。Yazici 等人的综述总结了接受靶向治疗的患者中 HCV 再激活的研究，并将这些治疗与 HCV 再激活增加的数据相关联。据报道，HCV 阳性伴癌症患者与 HCV 阳性不伴癌症患者之间的差异包括隐匿性感染的发生率较高，发生早期肝硬化的风险较高，纤维化进展的发生率较高，病毒再活化的发生率较高，病毒学结局较差（由 Borchardt 等人审查）。IDSA 和 AASLD 联合专家组关于检测、管理和丙型肝炎治疗建议，对于预期寿命>12 岁的慢性 HCV 患者，应考虑治疗 HCV 个月

GVHD（参见算法中的 B 型肝炎病毒、丙型肝炎病毒和人类免疫缺陷病毒再激活或疾病的管理）。

有关 B 细胞 NHL 患者 HBV 感染管理的详细信息，请参阅 NCCN B 细胞淋巴瘤指南。

用于预防（或先发制人的方法）的抗病毒药的最佳选择将主要由机构标准驱动。NCCN 专家组建议咨询肝炎治疗专家，以确定 HBV 检测阳性患者的适当抗病毒预防措施。用于 HBV 预防的优选药剂是恩替卡韦和替诺福韦。虽然最初获得的数据与拉米夫定，恩替卡韦和替诺福韦是首选，特别是当治疗活动性 HBV 感染的患者，由于拉米夫定的耐药阈值低。

NCCN 对 HCV 筛查和管理的建议

所有预期接受化疗或 IST 的患者都应进行 HCV 筛查。关于癌症患者 HCV 治疗的数据有限，但通常不建议同时给予 HCV 治疗和癌症治疗。IDSA/AASLD 指南可以为抗病毒治疗提供额外的指导，但必须进行传染病咨询，以评估同时或顺序使用抗 HCV 和癌症治疗。每月监测 ALT 水平和 HCV 病毒载量，或有临床指征时，应作为监测的一部分启动（见 B 型肝炎病毒、丙型肝炎病毒和人类免疫缺陷病毒再激活或

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Disease in the algorithm). For details on the management of HCV infection in patients with HCV-associated lymphomas, see the NCCN Guidelines for B-Cell Lymphomas.

Human Immunodeficiency Virus

The CDC surveillance report estimates that 1.1 million persons are living with HIV in the United States. This includes the estimated 166,000 persons whose infection has not yet been diagnosed.[292] There is support for HIV testing in all patients treated for cancer.[293] Patients who are HIV-positive and have cancer are classified as having either AIDS-defining cancer (ADC) or non–AIDS-defining cancer (NADC). ADC includes Kaposi sarcoma, NHL, and cervical cancer. There is a higher incidence of these cancers in those who are HIV-positive than in those who are HIV-negative.[294]

The incidence of NADC is increasing, likely due to the longer life expectancy of patients with HIV resulting from the advancement of treatment options.[295] Patients with HIV and NADC were shown to have an overall worse cancer outcome when compared to patients who are HIV-negative with the same cancer.[296] However, improvement in outcome was seen when patients with HIV received highly active antiretroviral therapy (HAART).[297] There should be caution regarding the concomitant administration of select antiretroviral therapies (including the protease inhibitors and non-nucleoside reverse transcriptase inhibitors) with cancer therapy as adverse events through cytochrome P450 3A4 have been documented.[298] A publication from MD Anderson Cancer Center retrospectively evaluated the use of HIV screening in patients prior to systemic cancer therapy.[299] Out of the 18,874 patients in this study, there were 3514 patients who tested positive for HIV at the initiation of systemic cancer therapy. Patient histories indicated a higher incidence in patients with sexually transmitted disease (37.7% vs. 18.5%; $P < .001$) or a history of illegal drug use (46.2% vs. 18.6%; $P < .001$). Patients screened for HIV included 12.1% of patients with NADC and 9.4% of patients with cervical cancer. Interestingly, a significantly higher percentage (88.4%) of patients with NHL were screened for HIV, which may be partially attributed to clinician education of the role of HIV in these patients.[299]

NCCN Recommendations for HIV Screening

In 2006, the CDC published recommendations for routine HIV testing in all patients (13–64 years of age) in the health care setting.[300] The testing is intended to be voluntary and conducted only with consent from patients. Under these guidelines, patients are informed either verbally or in written format that HIV testing will be conducted unless the patient declines testing (opt-out screening). The CDC recommends that patients at high risk for HIV infection be screened at least annually.[300] The implementation of these guidelines is largely dependent upon institutional practices and the prevalence of undiagnosed HIV infections in specific institutions. However, the NCCN Panel strongly encourages concordance with the CDC recommendations.

In addition to the CDC recommendations, the NCCN Panel emphasizes that all patients receiving chemotherapy or IST be screened for HIV.[293] Patients co-infected with hepatitis pose an additional complication. Select antiretroviral therapies including the integrase strand inhibitors and nucleoside/nucleotide reverse transcriptase inhibitors have demonstrated fewer drug-drug interactions compared with the protease inhibitors and non-nucleoside reverse transcriptase inhibitors. However, consultation with an infectious disease expert is necessary for treatment of HIV in patients with cancer as therapies continuously evolve. HIV viral load should be monitored monthly during therapy and then as clinically indicated (see *Management Of Hepatitis B Virus, Hepatitis C Virus, And Human Immunodeficiency Virus Reactivation Or Disease* in the algorithm and NCCN Guidelines for Cancer in People with HIV).

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算法中的疾病）。有关 HCV 相关淋巴瘤患者 HCV 感染管理的详细信息，请参见 NCCN B 细胞淋巴瘤指南。

人类免疫缺陷病毒

CDC 的监测报告估计，美国有 110 万人感染艾滋病毒。这包括尚未确诊感染的估计 166 000 人，支持对所有接受治疗的癌症患者进行艾滋病毒检测，艾滋病毒阳性的癌症患者被分为艾滋病界定癌症（ADC）或非艾滋病界定癌症（NADC）。ADC 包括卡波西肉瘤、NHL 和宫颈癌。这些癌症在艾滋病毒阳性者中的发病率高于艾滋病毒阴性者。

NADC 的发病率正在增加，这可能是由于治疗方案的进步导致 HIV 患者的预期寿命延长。与 HIV 阴性的相同癌症患者相比，HIV 和 NADC 患者的癌症结局总体上更差。然而，当 HIV 患者接受高效抗逆转录病毒治疗（HAART）时，。同时使用某些抗逆转录病毒疗法时应谨慎。（包括蛋白酶抑制剂和非核苷类逆转录酶抑制剂）MD 安德森癌症中心的一份出版物回顾性地评价了细胞色素 P450 3A4 在癌症治疗中的应用，在系统性癌症治疗之前对患者进行 HIV 筛查在本研究的 18，874 名患者中，有 3514 名患者在系统性癌症治疗开始时检测出 HIV 阳性。 患者病史表明，性传播疾病患者（37.7%对 18.5%; P <0.001）或非法药物使用史患者（46.2%对 18.6%; P <0.001）的发病率较高。接受艾滋病毒筛查的患者

包括 12.1%的 NADC 患者和 9.4%的宫颈癌患者。有趣的是，一个显着更高的百分比（88.4%）的 NHL 患者进行了艾滋病毒筛查，这可能部分归因于临床医生教育的作用，艾滋病毒在这些患者。

NCCN 对 HIV 筛查的建议

2006 年，疾病预防控制中心公布了对所有患者（13–64 岁）进行常规艾滋病毒检测的建议，该检测是自愿的，只有在患者同意的情况下才能进行。根据这些准则，以口头或书面形式告知患者，除非患者拒绝检测（选择退出筛查），否则将进行艾滋病毒检测。疾病预防控制中心建议，艾滋病毒感染的高风险患者至少每年进行一次筛查。这些指导方针的实施在很大程度上取决于机构的做法和特定机构中未确诊的艾滋病毒感染的流行情况。然而，NCCN 小组强烈鼓励与 CDC 的建议保持一致。

除了 CDC 的建议，NCCN 小组强调，所有接受化疗或 IST 的患者都要进行 HIV 筛查。合并感染肝炎的患者会造成额外的并发症。与蛋白酶抑制剂和非核苷逆转录酶抑制剂相比，包括整合酶链抑制剂和核苷/核苷酸逆转录酶抑制剂在内的精选抗逆转录病毒疗法已证明药物间相互作用较少。然而，随着治疗方法的不断发展，与传染病专家的协商对于癌症患者的艾滋病毒治疗是必要的。HIV 病毒载量应在治疗期间每月监测一次，

（参见算法中的“B 型肝炎病毒、丙型肝炎病毒和人类免疫缺陷病毒再激活或疾病的管理”和“NCCN HIV 感染者癌症指南”）。

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Screening for Other Viruses

Rapid PCR panels should be considered for detection of respiratory viruses including RSV, influenza, parainfluenza virus, adenovirus, rhinovirus, and metapneumovirus in patients with cough and/or shortness of breath that might indicate a viral infection. Ribavirin and IVIG have been proposed as antiviral therapies[301-305]; however, data are not sufficient to provide recommendations.

RSV is a major cause of severe infection in patients who are immunocompromised, with mortality rates ≤80% in HCT recipients.[306,307] Progression of RSV to the lower respiratory tract occurs in ≤50% of patients receiving HCT or chemotherapy.[308-310] The virulent nature of RSV requires hospitalization for treatment. Treatment options are limited to ribavirin and adjunctive IVIG. There is a diversity of practice among the institutions for the treatment of RSV disease. Based on limited data[311-313] and strong Panel disagreement regarding the use of ribavirin and the best method of delivery, ribavirin has been designated a category 3 recommendation. Recommendations for inhaled versus oral ribavirin should be based on the individual institution.

Rapid screening tests are available for detection of influenza. Clinical benefit is highest when treatment is initiated within the first 48 hours of influenza symptoms, although benefits can still be seen when initiated after the 48-hour window.[314] During the influenza season, consider empiric antiviral therapy for patients within 48 hours after symptoms develop that are suggestive of influenza (eg, high fever, coryza, myalgia, dry cough), especially during community outbreaks. Both the IDSA (2007) and CDC guidelines (2011) recommend antiviral treatment with the neuraminidase inhibitors oseltamivir or zanamivir, which are active against both influenza A and B viruses.[315,316] Both agents are approved by the FDA for the treatment of influenza within 48 hours of symptomatic onset; the indicated duration of treatment is 5 days. However, longer courses of treatment (eg, 10 days) and treatment until resolution of symptoms can be considered in those who are immunocompromised, though this is controversial. Some centers have used higher doses (eg, 150 mg BID) of oseltamivir in these patients with mixed results. Pandemic influenza does not have a predictable seasonal pattern and may spread in the community concurrently with a seasonal influenza strain. Antiviral susceptibility of influenza strains is variable and cannot be predicted based on previous influenza outbreaks. In cases of seasonal influenza and pandemic strains, it is necessary to be familiar with susceptibility patterns and guidelines on appropriate antiviral treatment.[317] There are limited data on the activity of peramivir; with comparable clinical outcomes observed in comparison with oseltamivir.[318,319] Peramivir, available as an IV injection, can be considered for patients who cannot absorb oral oseltamivir or tolerate oseltamivir or inhaled zanamivir.

A few small case studies showed limited efficacy of baloxavir, a polymerase inhibitor active against both influenza A and B viruses, in immunocompromised patients with oseltamivir- or peramivir-resistant influenza.[320-322] However, due to limited data and emergence of resistant strains,[322,323] this option is not routinely recommended by the CDC.

BK virus is a common polyomavirus that remains dormant in the kidney and urinary tract. In immunosuppressed individuals, BK virus can reactivate. Patients undergoing allogeneic HCT are particularly vulnerable to BK virus, and the development of hemorrhagic cystitis and additional complications such as ureteral stenosis.[324,325] Supportive care remains the mainstay of management. While cidofovir demonstrates effectiveness as a treatment option for BK virus, renal toxicity is a significant complication.[326] There is currently a lack of data to support recommendations on the treatment of BK virus. Expert consensus on management has been published for pediatric HCT recipients.[327]

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其他病毒的筛查

对于可能提示病毒感染的咳嗽和/或呼吸急促患者，应考虑使用快速 PCR 检测呼吸道病毒，包括 RSV、流感病毒、副流感病毒、腺病毒、鼻病毒和偏肺病毒。利巴韦林和 IVIG 已被提议作为抗病毒治疗;然而，数据不足以提供建议。

RSV 是免疫功能低下患者严重感染的主要原因，HCT 接受者的死亡率≤80%。≤50%接受 HCT 或化疗的患者发生 RSV 进展至下呼吸道。RSV 的毒性需要住院治疗。治疗选择仅限于利巴韦林和免疫球蛋白。在 RSV 疾病治疗机构中，实践存在多样性。根据有限的数据和专家小组对利巴韦林的使用和最佳给药方法的强烈分歧，利巴韦林被指定为第 3 类建议。吸入与口服利巴韦林的推荐应基于个体机构。

快速筛查测试可用于检测流感。在出现流感症状的前 48 小时内开始治疗的临床获益最高，但在 48 小时窗口期后开始治疗仍能看到获益。在流感季节，对于出现提示流感症状（如高热、鼻炎、肌痛、干咳）的患者，尤其是在社区暴发期间，应考虑在 48 小时内进行经验性抗病毒治疗。IDSA（2007）和 CDC 指南（2011）均推荐使用神经氨酸酶抑制剂奥司他韦或扎那米韦进行抗病毒治疗，这两种药物对甲型和 B 型流感病毒都有活性。FDA 批准这两种药物用于在症状发作后 48 小时内治疗流感;指示的治疗持续时间为 5 天。然而，较长的疗程（例如，10 天）和治疗，直到解决症状可以考虑在那些谁是免疫功能低下，虽然这是有争议的。一些中心在这些患者中使用更高剂量（例如，150 mg BID）的奥司他韦，结果好坏参半。大流行性流感没有可预测的季节性模式，可能与季节性流感毒株同时在社区传播。流感病毒株的抗病毒敏感性是可变的，不能根据以往的流感疫情进行预测。在季节性流感和大流行性流感病毒株的情况下，有必要熟悉敏感性模式和适当的抗病毒治疗指南。关于帕拉米韦活性的数据有限;与奥司他韦相比，观察到的临床结果相当。帕拉米韦可作为 IV 注射剂，可用于不能吸收口服奥司他韦或耐受奥司他韦或吸入扎那米韦的患者。

一些小型病例研究显示，baloxavir（一种对甲型和 B 型流感病毒都有活性的聚合酶抑制剂）对奥司他韦或帕拉米韦耐药流感的免疫功能低下患者的疗效有限。

BK 病毒是一种常见的多瘤病毒，在肾脏和泌尿道中保持休眠状态。在免疫抑制的个体中，BK 病毒可以重新激活。接受同种异体 HCT 的患者特别容易感染 BK 病毒，并发生出血性膀胱炎和其他并发症，如输尿管狭窄。支持性护理仍然是管理的主要支柱。虽然西多福韦作为 BK 病毒的治疗选择证明是有效的，但肾毒性是一种重要的并发症。目前缺乏数据来支持治疗 BK 病毒的建议。已发表了儿科 HCT 接受者的管理专家共识。

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COVID-19 screening, prophylaxis and treatment is a complex and rapidly evolving area. Please see the CDC guidelines for latest recommendations.

Prophylaxis for *Pneumocystis jirovecii*

TMP/SMX prophylaxis for *Pneumocystis jirovecii* is highly effective in preventing *Pneumocystis jirovecii* pneumonia (PJP).[328-331] In a systematic review and meta-analysis of 12 randomized studies (N = 1245; primarily in patients with acute leukemias or in HCT recipients), prophylaxis with TMP/SMX resulted in a significant reduction in PJP occurrence by 91% compared with placebo, no treatment, or treatment with non-PJP antibiotics (RR, 0.09; 95% CI, 0.02–0.32). In addition, TMP/SMX prophylaxis significantly reduced PJP-related mortality (RR, 0.17; 95% CI, 0.03–0.94).[328] TMP/SMX has the potential advantage of activity against other infectious complications (such as common bacterial infections, listeriosis, nocardiosis, and toxoplasmosis) that may afflict patients with severe T-cell depletion or impairment.[332] TMP/SMX is considered the treatment of choice for PJP prophylaxis (preferred, category 1; see *Prevention Of Pneumocystis Jirovecii Infection* in the algorithm). In cases of intolerance, TMP/SMX desensitization should be considered. Atovaquone, dapsone and aerosolized or IV pentamidine are alternatives to TMP/SMX in cases of intolerance. [333-335] For patients receiving dapsone, measurement of G6PD levels is recommended prior to the initiation of therapy. Patients who are G6PD deficient may have an increased risk for hemolytic adverse reactions. Methemoglobinemia can also occur with dapsone therapy.[336] Atovaquone appears to be equivalent to dapsone in HIV patients who cannot tolerate TMP/SMX.[337] In pediatric patients with acute leukemias who were intolerant of TMP/SMX, atovaquone was reported to be an effective strategy for PJPprophylaxis.[338]

Prophylaxis against PJP should be used in allogeneic HCT recipients (category 1) and patients receiving CAR T-cell therapy for at least 6 months and while receiving IST, as well as in patients with ALL (category 1) throughout anti-leukemic therapy.[339,340] Patients should receive prophylaxis against PJP for a minimum of 2 months after alemtuzumab and until the CD4 count is >200 cells/mcL. Other patients who should receive PJP prophylaxis at least through active treatment include: 1) those receiving treatment with select phosphatidylinositol-3-kinase (PI3K) inhibitors (copanlisib, idelalisib, or duvelisib) +/- rituximab; 2) patients with neoplastic diseases receiving intensive corticosteroid treatment (eg, the equivalent of ≥20 mg of prednisone daily for ≥4 weeks, also depending on the patient's overall immunologic status); and 3) patients receiving temozolomide (see *Prevention Of Pneumocystis Jirovecii Infection* in the algorithm).[340-343] Some panel members advise prophylaxis against PJP (category 2B) for patients receiving purine analog therapy (eg, fludarabine, cladribine [2-CdA]) and other T-cell–depleting agents until CD4 count is >200 cells/mcL and for autologous HCT recipients until 3 to 6 months post-transplant. Prophylaxis against PJP may also be considered in patients receiving certain bispecific antibodies. For more information on infection risks associated with bispecific antibodies, see the full NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections.

Vaccination

Vaccination in patients with cancer can reduce the morbidity and mortality associated with infection. In general, patients with hematologic malignancies have a greater risk for infection than patients with solid tumors. Patients receiving HCT may lose immunity to pathogens post-transplant. Therefore, vaccination recommendations for these patients are more expansive than the recommendations for the general population of patients with cancer. In any patient who is immunocompromised, live vaccines, including the live attenuated influenza vaccine (LAIV), have the potential to cause disease and should not be administered during chemotherapy or periods of significant immunosuppression such as treatment for GVHD. The safety of vaccines for patients receiving immunostimulatory drugs has not been established. Inactivated vaccines

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COVID–19 筛查、预防和治疗是一个复杂且快速发展的领域。请参阅 CDC 指南了解最新建议。

预防耶氏肺孢子虫

TMP/SMX 预防卡氏肺孢子虫肺炎（PJP）非常有效。在对 12 项随机研究的系统回顾和荟萃分析中，（N = 1245;主要在急性白血病患者或 HCT 接受者中），与安慰剂、未治疗相比，TMP/SMX 预防导致 PJP 发生率显著降低 91%，或使用非 PJP 抗生素治疗（RR，0.09; 95% CI，0.02–0.32）。此外，TMP/SMX 预防治疗显著降低了 PJP 相关死亡率（RR，0.17; TMP/SMX 具有对抗其他感染并发症的活性的潜在优势（如常见的细菌感染、弧菌病、诺卡氏菌病，TMP/SMX 被认为是预防 PJP 的首选治疗方法（首选，类别 1;参见算法中的肺孢子虫感染预防）。 在不耐受的情况下，应考虑 TMP/SMX 脱敏。在不耐受的情况下，阿托伐醌、氨苯砜和雾化或 IV 喷他脒是 TMP/SMX 的替代品。对于接受氨苯砜治疗的患者，建议在开始治疗前测量 G6 PD 水平。G6 PD 缺乏的患者可能会增加溶血性不良反应的风险。氨苯砜治疗也可引起高铁血红蛋白血症，阿托伐醌在不能耐受 TMP/SMX 的 HIV 患者中似乎与氨苯砜相当，在不能耐受 TMP/SMX 的急性白血病患儿中，阿托伐醌被报道是预防 PJP 的有效策略。

应在同种异体 HCT 接受者（1 类）和接受 CAR T 细胞治疗至少 6 个月且同时接受 IST 的患者以及 ALL 患者（1 类）中预防 PJP。

1)患者应在阿仑单抗后接受针对 PJP 的预防至少 2 个月，直至 CD 4 计数>200 个细胞/mcL。至少应通过积极治疗接受 PJP 预防的其他患者包括：1）接受选定磷脂酰肌醇–3–激酶（PI 3 K）抑制剂治疗的患者（copanlisib、idelalisib 或 duvelisib）+/–利妥昔单抗; 2）接受强化皮质类固醇治疗的肿瘤性疾病患者（例如，相当于每天≥20 mg 泼尼松，持续≥4 周，也取决于患者的总体免疫状态）;和 3）接受

替莫唑胺（见预防肺孢子虫感染，

一些专家组成员建议对于接受嘌呤类似物疗法（例如，氟达拉滨、克拉屈滨[2–CdA]）和其他 T 细胞耗竭剂的患者预防 PJP（2B 类），直到 CD 4 计数>200 个细胞/mcL，并且对于自体 HCT 接受者预防 PJP（2B 类），直到移植后 3 至 6 个月。在接受某些双特异性抗体的患者中也可以考虑预防 PJP。有关与双特异性抗体相关的感染风险的更多信息，请参阅完整的 NCCN 癌症相关感染预防和治疗指南。

疫苗接种

癌症患者接种疫苗可以降低与感染相关的发病率和死亡率。一般而言，血液恶性肿瘤患者的感染风险高于实体瘤患者。接受 HCT 的患者可能会在移植后失去对病原体的免疫力。因此，对这些患者的疫苗接种建议比对癌症患者的一般人群的建议更广泛。在任何免疫功能低下的患者中，活疫苗（包括减毒活流感疫苗（LAIV））有可能导致疾病，不应在化疗或显著免疫抑制期间（如 GVHD 治疗）接种。疫苗对接受免疫刺激药物的患者的安全性尚未确定。灭活疫苗

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can often be safely administered to patients with cancer. Although the immunogenicity of the vaccines may be reduced in patients who are immunocompromised, the potential for protection conferred by antigen-derived vaccines, even if incomplete, is better than no protection if the vaccine is withheld. While guidelines may provide general recommendations for vaccination schedules, the efficacy and safety of each vaccine should be evaluated to optimize the schedule on a case-by-case basis. For more information on vaccination in cancer survivors, see the NCCN Guidelines for Survivorship.

Influenza Vaccine

Influenza infections cause significant morbidity and mortality in patients with cancer. Among bone marrow transplant recipients, influenza accounts for about 10% to 40% of all community-acquired viral respiratory infections.[344-346] An increase in both the incidence and duration of influenza infections has been observed in patients with cancer who are immunosuppressed compared to healthy controls.[347,348] During community outbreaks, influenza infections may represent a significant proportion of fever and neutropenia episodes.[349] Influenza infections in patients with cancer who are severely immunocompromised are often associated with hospitalizations, delays in potentially life-saving chemotherapy, and occasionally death.[347-349] As a result, annual vaccination against influenza with the inactivated influenza virus is recommended for all individuals at increased risk due to immunosuppression.[350] A randomized study of 97 patients receiving cytotoxic chemotherapy (3-week cycles) for solid tumors found that the immunogenicity of the influenza vaccine was similar when administered at the time of chemotherapy administration (day 1) or within the cytopenic period (day 11).[351] The Advisory Committee on Immunization Practices (ACIP) for the CDC guidelines includes health care professionals and household members or caregivers in their target group for annual immunization to prevent transmission of influenza to patients at high risk.[350]

The intranasal vaccine should be avoided in patients with immunosuppression, because a LAIV is still capable of replication, which could theoretically lead to infection in immunocompromised individuals.[350,352] Because no data are available assessing the risk for person-to-person transmission of the LAIV from vaccine recipients to immunosuppressed contacts, the CDC recommends that inactivated influenza vaccine should be used in household contacts, health care workers, and others who have close contact with patients who are severely immunocompromised (ie, persons requiring a protected environment). Persons with close contact to patients with a lesser degree of immunosuppression (eg, patients receiving chemotherapy or corticosteroids, patients with HIV) may receive the LAIV.[350,352]

There are not yet sufficient data for the Panel to recommend the high-dose influenza vaccine over the standard-dose influenza vaccine. Preliminary data have shown that the high-dose influenza vaccine is safe for patients with cancer and may show more immunogenicity compared to the standard-dose influenza vaccine for this patient population.[353,354] A randomized, single-blind, controlled trial of influenza vaccine in autologous HCT recipients found comparable seroprotection and seroconversion rates against all influenza strains with both the high-dose and the standard-dose vaccines when used as part of a two-dose regimen.[355] Further data are needed to assess whether the high-dose influenza vaccine confers a clinical benefit compared to the standard-dose vaccine for patients with cancer.

Pneumococcal Vaccine

The pneumococcal conjugate vaccine can be given in newly diagnosed adults with hematologic or solid tumor malignancies following assessment of their immune status. The conjugate pneumococcal vaccine (PCV20) should be administered to newly diagnosed adults with cancer who are pneumococcal vaccine-naïve. Alternatively, PCV15 can be given, followed

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通常可以安全地用于癌症患者。尽管疫苗的免疫原性在免疫功能低下的患者中可能会降低，但抗原衍生疫苗所赋予的保护潜力，即使不完全，也比不接种疫苗时没有保护要好。虽然指南可能会对疫苗接种时间表提供一般性建议，但应评估每种疫苗的有效性和安全性，以根据具体情况优化时间表。有关癌症幸存者疫苗接种的更多信息，请参阅 NCCN 生存指南。

流感疫苗

流感感染在癌症患者中引起显著的发病率和死亡率。 在骨髓移植受者中，流感约占所有社区获得性病毒性呼吸道感染的 10%至 40%。与健康对照组相比，在免疫抑制的癌症患者中观察到流感感染的发病率和持续时间增加。在社区爆发期间，流感感染在发热和中性粒细胞减少症中占很大比例。严重免疫功能低下的癌症患者感染流感，与住院治疗，延迟可能挽救生命的化疗，偶尔死亡有关。因此，对于所有因免疫抑制而风险增加的个体，建议每年接种灭活流感病毒疫苗。(3 周周期）的研究发现，在化疗给药时（第 1 天）或在血细胞减少期（疾病预防控制中心的免疫实践咨询委员会（ACIP）将医疗保健专业人员和家庭成员或护理人员纳入其目标群体，每年进行免疫接种，以防止流感传播给高危患者。

免疫抑制的患者应避免使用鼻内疫苗，因为 LAIV 仍然能够复制，理论上可能导致免疫功能低下的个体感染。由于没有数据评估 LAIV 从疫苗接种者到免疫抑制接触者的人际传播风险，CDC 建议应将灭活流感疫苗用于家庭接触者，卫生保健工作者，和其他与病人有密切接触的人，

严重免疫功能低下（即需要保护环境的人）。与免疫抑制程度较低的患者（例如，接受化疗或皮质类固醇的患者、HIV 患者）密切接触的人员可接受 LAIV。

目前还没有足够的数据让专家小组建议使用高剂量流感疫苗而不是标准剂量流感疫苗。初步数据显示，高剂量流感疫苗对癌症患者是安全的，并且与标准剂量流感疫苗相比，对该患者群体可能显示出更高的免疫原性。在自体 HCT 接受者中进行的流感疫苗对照试验发现，高剂量和标准剂量的流感疫苗对所有流感毒株的血清保护和血清转换率相当，需要进一步的数据来评估高剂量流感疫苗与标准剂量疫苗相比是否能为癌症患者带来临床益处。

肺炎球菌疫苗

肺炎球菌结合疫苗可在评估免疫状态后用于新诊断的血液系统或实体瘤恶性肿瘤成人。肺炎球菌结合疫苗（PCV 20）应用于新诊断的未接种过肺炎球菌疫苗的癌症成人。或者，可以给予 PCV 15，然后

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by the polysaccharide pneumococcal vaccine (PPSV23) at least 8 weeks later. Additional PPSV23 is not needed for those receiving PCV20. For patients who have previously received PPSV23, PCV20 (preferred) or PCV15 can be given. Patients who have previously received PCV13 only can receive PCV20 (rather than PPSV23) at least 1 year later. Patients who have previously received PCV13 and 1 to 2 doses of PPSV23 can receive PCV20 at least 5 years later.

A phase 3 randomized controlled clinical trial found that PCV15 had a comparable safety profile to PCV13 in allo-HCT recipients, and induced comparable immune responses to PCV13.[356] Based on CDC Recommendations, vaccination with the conjugated 20-valent or 15-valent vaccine 3 to 6 months after HCT is recommended. If PCV20 is used, 4 doses should be administered. The first 3 doses are generally 1 to 2 months apart, with the fourth dose 6 months after the third dose. There is no need to give PPSV23. If PCV15 is used, 3 doses should be administered, followed by PPSV23 6 to 12 months post primary series. Following the primary series of 3 PCV doses, a dose of the PPSV23 to broaden the immune response might be given. For patients with chronic GVHD in whom PPSV23 might not be as effective, a fourth dose of PCV20 or PCV15 should be considered instead of PPSV23. Patients with asplenia should receive the pneumococcal vaccine. According to the ACIP Vaccine Recommendations and Guidelines for Altered Immunocompetence, the pneumococcal vaccine should be administered at least 2 weeks before elective splenectomy. Penicillin prophylaxis is advised in patients who are asplenic to prevent pneumococcal disease.[357,358]

Meningococcal Conjugate Vaccine

The meningococcal vaccine is recommended for patients with increased risk for meningococcal disease including patients with persistent complement component deficiency, patients taking eculizumab, and patients with anatomic or functional asplenia. The ACIP recommends that asplenic persons be immunized with the meningococcal vaccine. The meningococcal vaccine should be administered at least 2 weeks before elective splenectomy. Patients at increased risk for meningococcal disease should receive quadrivalent conjugate vaccine series (MenACWY) and monovalent meningococcal serogroup B vaccine series. Patients at risk include those with persistent complement component deficiencies, those taking a complement C5 inhibitor (eg, eculizumab, ravulizumab), or those with anatomic or functional asplenia. MenACWY vaccine is given in 2 doses ≥8 weeks apart; serogroup B vaccine is available in a 2- or 3-dose series, depending on the vaccine formulation used. The meningococcal vaccine is recommended 6 to 12 months after HCT. Booster doses can be considered in those that remain at risk for meningococcal disease. This includes revaccination every 5 years with MenACWY vaccine, while for MenB vaccine, 1 booster dose is recommended 1 year after primary series with revaccination every 2–3 years if risk remains.

Human Papillomavirus Vaccine

The human papillomavirus (HPV) vaccine is a recombinant 3-dose vaccine that is recommended for patients ≤26 years of age and can be considered for those up to age 45. The lower age limit for this vaccine is 9 years of age. While it doesn't treat existing HPV infection or disease, HPV vaccination is still recommended in individuals with prior HPV since it can still provide protection against HPV serotypes not already acquired.

Haemophilus Influenzae Type b Vaccine

Immunization of adults with the pediatric *H influenzae* type b (Hib) vaccine is considered optional because of limited data on efficacy in older children and adults, although studies suggest good immunogenicity in patients who are immunocompromised. The Hib vaccine is recommended 6 to 12 months post-HCT. For patients with planned splenectomy, immunization is ideally performed at least 2 weeks in advance. If this is not feasible,

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多糖肺炎球菌疫苗（PPSV 23）至少 8 周后。接受 PCV 20 的患者不需要额外的 PPSV 23。对于既往接受过 PPSV 23、PCV 20（首选）或 PCV 15 的患者。既往仅接受 PCV 13 的患者至少 1 年后可接受 PCV 20（而非 PSV 23）。既往接受过 PCV 13 和 1 – 2 剂 PPSV 23 的患者可在至少 5 年后接受 PCV 20。

一项 3 期随机对照临床试验发现，PCV 15 在 allo-HCT 受者中具有与 PCV 13 相当的安全性，并且诱导与 PCV 13 相当的免疫应答。根据 CDC 建议，建议在 HCT 后 3 至 6 个月接种 20 价或 15 价结合疫苗。如果使用 PCV 20，应给予 4 剂。前 3 剂通常间隔 1 至 2 个月，第 4 剂在第 3 剂后 6 个月。没有必要给出 PSV 23。如果使用 PCV 15，则应给予 3 剂，然后在主要系列后 6 至 12 个月给予 PSV 23。在 3 次 PCV 给药的主要系列后，可给予 PPSV 23 剂量以扩大免疫应答。对于 PPSV 23 可能不那么有效的慢性 GVHD 患者，应考虑使用第四剂 PCV 20 或 PCV 15 代替 PPSV 23。无脾患者应接种肺炎球菌疫苗。 根据 ACIP 疫苗建议和免疫能力改变指南，肺炎球菌疫苗应在选择性脾切除术前至少 2 周接种。无脾的患者建议预防青霉素以预防肺炎球菌疾病。

脑膜炎球菌结合疫苗

The meningococcal vaccine is recommended for patients with increased risk for meningococcal disease including patients with persistent complement component deficiency, patients taking eculizumab, and patients with anatomic or functional asplenia. The ACIP recommends that

重试 错误原因

asplenic persons be immunized with the meningococcal vaccine. The meningococcal vaccine should be administered at least 2 weeks before elective splenectomy. Patients at increased risk for meningococcal disease should receive quadrivalent conjugate vaccine series (MenACWY) and monovalent meningococcal serogroup B vaccine series. Patients at risk include those with persistent complement component deficiencies, those taking a complement C5 inhibitor (eg, eculizumab, ravulizumab), or those with anatomic or functional asplenia. MenACWY vaccine is given in

重试 错误原因

2 doses ≥8 weeks apart; serogroup B vaccine is available in a 2- or 3-dose

重试 错误原因

根据所使用的疫苗配方。建议在 HCT 后 6 至 12 个月接种脑膜炎球菌疫苗。加强剂量可以考虑在那些仍然有脑膜炎球菌病的风险。这包括 MenACWY 疫苗每 5 年复种一次，而对于 MenB 疫苗，如果风险仍然存在，建议在初级系列后 1 年进行 1 次加强接种，每 2–3 年复种一次。

人乳头瘤病毒疫苗

人乳头瘤病毒（HPV）疫苗是一种重组 3 剂疫苗，推荐用于≤26 岁的患者，可考虑用于 45 岁以下的患者。该疫苗的年龄下限为 9 岁。虽然它不治疗现有的 HPV 感染或疾病，但 HPV 疫苗接种仍然推荐用于先前患有 HPV 的个体，因为它仍然可以提供对尚未获得的 HPV 血清型的保护。

流感嗜血杆菌 B 型疫苗

尽管研究表明免疫功能低下患者的免疫原性良好，但由于年龄较大儿童和成人的有效性数据有限，因此认为儿童 B 型流感嗜血杆菌（HiB）疫苗对成人的免疫接种是可选的。建议在 HCT 后 6 至 12 个月接种 Hib 疫苗。对于计划性脾切除术的患者，最好至少提前 2 周进行免疫接种。如果这不可行，

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immunization is advisable after splenectomy, because such patients are still capable of mounting a protective antibody response.

Varicella/Zoster Vaccines

The recombinant zoster vaccine (RZV) should be given to patients 50 to 70 days after autologous HCT. RZV may be considered after allogeneic HCT; however, its efficacy in allogeneic HCT, in the presence of GVHD, or during ongoing immunosuppression has not been established. The CDC recommends administration of RZV for patients aged ≥50 years, and those ≥18 years who are at increased risk for herpes zoster. The RZV vaccine is given in 2 doses 2 to 6 months apart. For adults who are immunocompromised and aged ≥18 years, a second dose can be given 1 to 2 months after the first dose if they will benefit from a shorter vaccination schedule. For patients who have previously received the live-attenuated herpes zoster vaccine (ZVL), RZV should be given at least 2 months after the last ZVL dose.

COVID-19 Vaccine

The NCCN Guidelines recommend COVID-19 vaccination for all persons with cancer, or who have been previously treated for cancer. According to the CDC, individuals aged ≥6 months who are moderately or severely immunocompromised (including recipients of HCT) and not previously vaccinated against COVID-19 should get 2 or 3 doses of the same brand of updated COVID-19 vaccine. Additionally, updated vaccines should be administered at least 2 weeks before initiation or resumption of immunosuppressive therapies. The NCCN Guidelines recommend vaccination with COVID-19 vaccine 6 months post-HCT, with a consideration for early vaccination at 3 months during community outbreaks and high disease activity. For further details of the most up-to-date vaccination recommendations, refer to CDC guidance.

Tetanus/Diphtheria/Pertussis (Tdap) Vaccine

As per the CDC, individuals ≥7 years should receive one dose of Tdap every 10 years. The FDA/ACIP approved a 3-dose series of Tdap/Tdap/Tdap or Tdap/Td/Td (tetanus/diphtheria) for individuals ≥7 years. For recipients of HCT, NCCN recommends administration of the 3-dose vaccine series, with the first dose 6 to 12 months post-HCT.

Respiratory Syncytial Virus (RSV) Vaccine

The RSV vaccine is available for adults aged ≥60 years. However, its effectiveness in patients with cancer is unknown. The long-acting RSV monoclonal antibody (nirsevimab) is approved for infants <24 months of age to prevent RSV infection.

Travel Vaccines

Vaccines have variable risk and efficacy in patients receiving cancer care; therefore, the Panel recommends consultation with an infectious disease expert prior to the administration of travel vaccines (eg, typhoid, yellow fever). Additional information on travel vaccines may be found in the CDC Yellow Book.

Vaccine Summary

Although efficacy data are lacking for the use of vaccines in patients with cancer, recommendations for their use are based on the principles of immunization and safety data. Persons receiving chemotherapy or radiation therapy for malignancies should not receive live vaccines for at least 3 months after cessation of therapy and until they are presumed to be immunocompetent.[359] Live vaccines are contraindicated during treatment and for a period of at least 6 to 12 months in patients receiving IST (eg, blinatumomab, CAR T-cell therapy, monoclonal antibodies). These patients may also have a blunted response to inactivated vaccines. Certain live vaccines can be safely administered to household members of patients who are severely immunocompromised (eg, measles, mumps, rubella [MMR]), whereas others cannot (eg, smallpox vaccine) because of

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immunization is advisable after splenectomy, because such patients are still capable of mounting a protective antibody response.

重试 错误原因

Varicella/Zoster Vaccines

重试 错误原因

The recombinant zoster vaccine (RZV) should be given to patients 50 to 70 days after autologous HCT. RZV may be considered after allogeneic HCT; however, its efficacy in allogeneic HCT, in the presence of GVHD, or during ongoing immunosuppression has not been established. The CDC recommends administration of RZV for patients aged ≥50 years, and those ≥18 years who are at increased risk for herpes zoster. The RZV vaccine is given in 2 doses 2 to 6 months apart. For adults who are immunocompromised and aged ≥18 years, a second dose can be given 1 to 2 months after the first dose if they will benefit from a shorter vaccination schedule. For patients who have previously received the liveattenuated herpes zoster vaccine (ZVL), RZV should be given at least 2 months after the last ZVL dose.

重试 错误原因

COVID–19 疫苗

NCCN 指南建议所有癌症患者或既往接受过癌症治疗的人接种 COVID–19 疫苗。据 CDC 称，年龄≥6 个月、中度或严重免疫功能低下（包括 HCT 接受者）且之前未接种过 COVID–19 疫苗的个人应接种 2 或 3 剂相同品牌的更新 COVID–19 疫苗。此外，更新的疫苗应在开始或恢复免疫抑制治疗前至少 2 周接种。NCCN 指南建议在 HCT 后 6 个月接种 COVID–19 疫苗，并考虑在社区爆发和高疾病活动期间提前 3 个月接种疫苗。有关最新疫苗接种建议的更多详细信息，请参阅 CDC 指南。

破伤风/白喉/百日咳疫苗

根据 CDC，≥7 岁的个体应每 10 年接受一剂 Tdap。FDA/ACIP 批准了 Tdap/Tdap/Tdap 或 Tdap/Td/Td（破伤风/白喉）3 剂系列用于≥7 岁的个体。对于 HCT 接受者，NCCN 建议接种 3 剂疫苗系列，第一剂在 HCT 后 6 至 12 个月接种。

呼吸道合胞病毒（RSV）疫苗

RSV 疫苗适用于≥60 岁的成人。然而，它对癌症患者的有效性尚不清楚。长效 RSV 单克隆抗体（nirsevimab）被批准用于<24 个月的婴儿，以预防 RSV 感染。

旅行疫苗

疫苗在接受癌症治疗的患者中具有可变的风险和疗效;因此，专家组建议在接种旅行疫苗（例如伤寒、黄热病）之前咨询传染病专家。关于旅行疫苗的更多信息可以在 CDC 黄皮书中找到。

疫苗总结

虽然缺乏癌症患者使用疫苗的有效性数据，但其使用建议是基于免疫和安全性数据的原则。接受恶性肿瘤化疗或放疗的患者在停止治疗后至少 3 个月内，直至假定其具有免疫活性，不得接种活疫苗。IST 患者在治疗期间和至少 6 – 12 个月内禁用活疫苗（例如，Blinatumomab、CAR T 细胞治疗、单克隆抗体）。

这些患者也可能对灭活疫苗有迟钝的反应。某些活疫苗可以安全地接种给严重免疫功能低下患者的家庭成员（例如麻疹、腮腺炎、风疹[MMR]），而其他活疫苗则不能接种（例如天花疫苗），因为

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the potential risk of transmission. The package insert for the vaccine should be reviewed prior to administration. The NCCN Panel recommends that all household members be up to date on vaccinations.

Ideally, patients should be vaccinated at least 2 weeks before receiving cytotoxic therapy or IST; however, this timing is often not feasible in patients with cancer. In general, vaccination should not be given on the same day as cytotoxic therapy as cytotoxic therapy may reduce the proliferative lymphocytic responses required for protective immunity. In patients receiving chemotherapy, immunization between cytotoxic chemotherapy courses is likely to be associated with higher response rates than during chemotherapy administration.[360,361] Patients vaccinated <2 weeks before starting cytotoxic therapy or IST or while receiving these agents may have a limited response to vaccination. These patients should be revaccinated at least 3 months after therapy is discontinued and once immune competence has been restored.[359]

In summary, the NCCN Panel recommends that patients with cancer receive the influenza, COVID-19, pneumococcal, meningococcal, and HPV vaccines (see *General Recommendations For Vaccination In Patients With Cancer* in the algorithm). HCT recipients should also receive the inactivated vaccines for Tdap, Hib, hepatitis A and B, RZV, and polio (see *Recommended Vaccination Schedule After Autologous Or Allogeneic HCT* in the algorithm). The live vaccines for MMR and varicella may be given if no GVHD or ongoing immunosuppression is seen 2 years post-transplant in patients who are seronegative. Consultation with an infectious disease expert is recommended prior to administration of travel vaccines.

Protected Environments

Although well-designed clinical trials have not validated the use of high-efficiency particulate air (HEPA) filtration, the CDC recommends that allogeneic HCT recipients be placed in rooms with HEPA filters.[8,362] It is also reasonable to use HEPA filtration in patients who are nontransplant recipients with prolonged neutropenia. The principal benefit of HEPA filtration is likely to be related to the prevention of mold infections. In a retrospective analysis, HEPA filters were protective in those who are highly immunocompromised with hematologic malignancies in the setting of an outbreak of aspergillosis.[363] The value of laminar airflow in preventing infections is unclear and generally is not recommended.[364]

Management of Neutropenia and Fever

The definitions of fever and neutropenia in the NCCN Guidelines are consistent with those developed by the IDSA and FDA for evaluating antimicrobial therapy for fever and neutropenia.[1] *Fever* is defined as a single oral temperature of ≥38.3°C (or equivalent) or ≥38.0°C over 1 hour in the absence of an obvious cause. Axillary or rectal temperature measurements should be avoided.[17] Although uncommon, a patient with neutropenia and signs or symptoms of infection (eg, abdominal pain, severe mucositis, perirectal pain) without fever should be considered to have an active infection. The concomitant administration of corticosteroids may blunt fever response and any localized signs of infection. The NCCN Guidelines define *neutropenia* as either: 1) an ANC ≤500 neutrophils/mcL; or 2) an ANC ≤1000 neutrophils/mcL and a predicted decline to ≤500 neutrophils/mcL over the next 48 hours.

Initial Evaluation

The initial evaluation should focus on determining the potential sites and causative organisms of infection and on assessing the patient's risk of developing an infection-related complication. A site-specific history and physical examination should be performed promptly, cultures should be obtained, and empiric antibiotics should be started soon after the time of presentation (see *Initial Evaluation of Fever and Neutropenia* in the

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传播的潜在风险。接种前应审查疫苗说明书。NCCN 小组建议所有家庭成员都要及时接种疫苗。

理想情况下，患者应在接受细胞毒性治疗或 IST 前至少 2 周接种疫苗;然而，这种时机在癌症患者中通常不可行。一般而言，疫苗接种不应与细胞毒性治疗在同一天进行，因为细胞毒性治疗可能会降低保护性免疫所需的增殖性淋巴细胞应答。在接受化疗的患者中，细胞毒性化疗疗程之间的免疫接种可能比化疗期间的应答率更高。在开始细胞毒性治疗或 IST 前<2 周或接受这些药物时接种疫苗的患者可能对疫苗接种的应答有限。这些患者应在治疗停止后至少 3 个月重新接种疫苗，一旦免疫能力恢复。

总之，NCCN 专家组建议癌症患者接受流感，COVID-19，肺炎球菌，脑膜炎球菌，

HPV 疫苗（见疫苗接种的一般建议）

算法中的癌症患者）。HCT 受者还应接种 Tdap、HiB、甲型和 B 型肝炎、RZV 和脊髓灰质炎灭活疫苗（参见算法中的自体或同种异体 HCT 后推荐的疫苗接种时间表）。如果血清反应阴性的患者在移植后 2 年内未出现 GVHD 或持续的免疫抑制，则可以接种 MMR 和水痘活疫苗。建议在接种旅行疫苗之前咨询传染病专家。

受保护环境

虽然精心设计的临床试验尚未验证高效微粒空气（HEPA）过滤器的使用，但 CDC 建议，

异基因 HCT 受者应被安置在有 HEPA 过滤器的房间里。对于长期中性粒细胞减少的非移植受者，使用 HEPA 过滤也是合理的。HEPA 过滤的主要好处可能与预防霉菌感染有关。在一项回顾性分析中，HEPA 过滤器在曲霉菌病爆发的背景下对那些高度免疫功能低下的恶性血液病患者具有保护作用。层流气流在预防感染方面的价值尚不清楚，通常不建议使用。

中性粒细胞增多症和发热的管理

NCCN 指南中对发热和中性粒细胞减少的定义与 IDSA 和 FDA 为评价发热和中性粒细胞减少的抗菌治疗而制定的定义一致。发热定义为在无明显原因的情况下，单次口腔温度≥38.3°C（或同等温度）或≥38.0°C 超过 1 小时。腋窝或直肠温度

虽然不常见，但如果患者出现中性粒细胞减少症和感染的体征或症状（例如腹痛、严重粘膜炎、直肠周围疼痛），但无发热，则应视为活动性感染。同时使用皮质类固醇可能会减弱发热反应和任何局部感染体征。NCCN 指南将中性粒细胞减少症定义为：1）ANC ≤500 个中性粒细胞/mcL;或 2）ANC ≤1000 个中性粒细胞/mcL，并且预计在未来 48 小时内下降至≤500 个中性粒细胞/mcL。

初评

最初的评估应侧重于确定潜在的感染部位和致病微生物，并评估患者发生感染相关并发症的风险。应立即进行特定部位的病史和体格检查，应获得培养物，并应在感染后立即开始使用经验性抗生素。

演示（参见《发热和中性粒细胞减少症的初步评估》

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algorithm). The common sites of infection for patients with fever and neutropenia (such as the alimentary tract, skin, lungs, sinus, ears, perivaginal/perirectal, urologic, neurologic, and intravascular access device sites) should be thoroughly assessed with special attention to any devices. Other important factors in patient history to consider include major comorbid illness, medications, type and time since last chemotherapy administration, recent antibiotic therapy/prophylaxis, use of invasive devices, and previously documented infections. Other epidemiologically relevant exposures that should be considered include marijuana use, cigarette smoking, infections from household members, pets, travel, and recent blood product administration (see *Initial Evaluation of Fever and Neutropenia* in the algorithm).

Initial laboratory/radiology evaluation should include a complete blood count with differential analysis and a comprehensive metabolic panel, including blood chemistry tests to assess liver function (eg, total bilirubin, albumin, alanine aminotransferase [ALT], aspartate aminotransferase [AST]) and renal function (eg, blood urea nitrogen [BUN], creatinine, electrolytes). Oxygen saturation and urinalysis should be considered, depending on symptoms. Chest radiographs should be done for all patients with respiratory signs or symptoms; however, radiographic findings may be absent in neutropenia with pulmonary infection.[365]

Cultures

Culture specimens should be collected during or immediately after completing the examination. Two blood samples should be cultured. When obtaining blood cultures, one set can be obtained peripherally and one can be obtained from a central venous catheter (preferred but not required) (see *Initial Evaluation of Fever and Neutropenia* in the algorithm). The positive predictive value (PPV) of a catheter culture is less than of a peripheral culture. Obtaining blood for culture from both the central venous catheter and peripherally may help determine whether the venous access device (VAD) is the source of a bloodstream infection based on the differential time to positivity (DTP).[366] However, some experts recommend that only blood from the VAD needs to be obtained for culture, without the requirement for a peripheral vein blood culture.[366] A meta-analysis has shown little clinical use for two-site culturing in patients with cancer who have a VAD, and poor patient acceptance of peripheral venipunctures when a VAD is in place.[367] The Panel consensus is that the volume of blood for culture is the most important aspect of blood culturing; however, the Panel recommends obtaining one peripheral and one catheter culture for distinguishing between catheter-related infections and those from secondary sources.

In the absence of lesions or clinical signs and symptoms, routine cultures of the anterior nares, oropharynx, urine, stool, and rectum are rarely helpful. Diarrheal stools suggestive of infection should be tested for the presence of *C difficile*.[368] In patients with diarrhea, consider screening for enteric pathogens including rotavirus and norovirus in winter months and during outbreaks. Symptoms of urinary tract infection should be evaluated with a urinalysis and culture. Vascular access site inflammation or drainage should be cultured. Biopsy with microbiologic and pathologic evaluation should be considered for new or undiagnosed skin lesions (see *Initial Evaluation of Fever and Neutropenia* in the algorithm). Viral cultures of vesicular or ulcerated mucosal or cutaneous lesions may identify HSV infections. In patients with symptoms of respiratory viral infection, viral cultures and rapid viral antigen testing of the nasopharyngeal secretions can be useful during local outbreaks of such infections.[369,370] However, note that rapid immunofluorescent viral antigen tests may still result in a false negative for H1N1 (swine flu).

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算法）。应彻底评估发热和中性粒细胞减少患者的常见感染部位（如消化道、皮肤、肺、鼻窦、耳朵、阴道周/直肠周、泌尿系统、神经系统和血管内通路器械部位），特别注意任何器械。患者病史中需要考虑的其他重要因素包括主要共病、药物、自上次化疗给药以来的类型和时间、近期抗生素治疗/预防、侵入性器械的使用和既往记录的感染。应考虑的其他流行病学相关暴露包括大麻使用、吸烟、家庭成员感染、宠物、旅行和近期血液制品给药（参见算法中的发热和中性粒细胞减少的初步评估）。

初始实验室/放射学评价应包括全血细胞计数和分类分析以及综合代谢检查，包括评估肝功能（例如，总胆红素、白蛋白、丙氨酸氨基转移酶[ALT]、天冬氨酸氨基转移酶[AST]）和肾功能（例如，血尿素氮[BUN]、肌酐、电解质）的血生化检查。根据症状，应考虑血氧饱和度和尿液分析。所有有呼吸系统体征或症状的患者均应进行胸片检查;然而，中性粒细胞减少症伴肺部感染患者可能无胸片检查结果。

文化

应在检查期间或检查完成后立即采集培养标本。两份血样应进行培养。当获得血培养时，一组可以从外周获得，一组可以从中心静脉导管获得（首选但不是必需的）

(see 算法中发热和中性粒细胞减少的初始评价）。的
导管培养的阳性预测值（PPV）小于外周培养。从中心静脉导管和外周静脉采集血液进行培养可能有助于确定静脉通路是否

根据不同的阳性时间（DTP），VAD 是血流感染的来源。然而，一些专家建议只需要从 VAD 获得血液进行培养，而不需要外周静脉血培养。荟萃分析显示，在患有 VAD 的癌症患者中，以及当配备 VAD 时患者对外周静脉穿刺的接受度较差。专家组的共识是，用于培养的血液体积是血液培养最重要的方面;然而，专家小组建议进行一次外周和一次导管培养，以区分导管相关感染和继发感染。

在没有病变或临床体征和症状的情况下，前鼻孔、口咽、尿液、粪便和直肠的常规培养很少有帮助。提示感染的腹泻粪便应检测艰难梭菌的存在。在腹泻患者中，考虑在冬季和爆发期间筛查肠道病原体，包括轮状病毒和诺如病毒。尿路感染的症状应该通过尿液分析和培养来评估。血管穿刺部位炎症或

排水系统应该培养。对于新的或未确诊的皮肤病变，应考虑进行活检，并进行微生物学和病理学评价（见

算法中发热和中性粒细胞减少的初始评价）。病毒培养物

水疱或溃疡性粘膜或皮肤病变可鉴别 HSV 感染。对于有呼吸道病毒感染症状的患者，在局部爆发此类感染时，鼻咽分泌物的病毒培养和快速病毒抗原检测可能有用。但请注意，快速免疫荧光病毒抗原检测仍可能导致 H1N1（猪流感）假阴性。

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Outpatient Therapy for Patients with Neutropenic Fever

Initial Evaluation of Risk

Patients with neutropenia may be categorized into either a high- or low-risk group using criteria derived either from validated clinical prediction rules based on risk models or from clinical trial eligibility criteria. [4,78,371-376] Risk assessment attempts to predict the probability that a patient with neutropenia will experience serious complications during a febrile episode. This assessment helps to determine whether a patient at low risk for serious complications could safely receive treatment outside of the hospital and which initial empiric therapy with oral antibiotics is appropriate.

Prospective trials have indicated that febrile neutropenia can be initially evaluated in the hospital, ambulatory clinic, or home and then treated effectively with broad-spectrum IV therapy, sequential IV then oral therapy, or oral therapy.[377-379] Only centers with the necessary infrastructure should treat patients at low risk in an outpatient setting, preferably in an investigational context.

Risk assessment should be performed as part of the initial evaluation (see *Initial Risk Assessment for Patients with Febrile Neutropenia* in the algorithm). A widely used and validated prediction rule to assess risk was developed by the Multinational Association of Supportive Care in Cancer (MASCC). The MASCC risk index is derived from a model that includes weighted scores based on burden of illness (eg, extent of febrile neutropenia), evidence of clinical instability or comorbid conditions (eg, hypotension, chronic obstructive pulmonary disease [COPD], dehydration), history of prior fungal infections, site of medical care (eg, inpatient, outpatient), and age (cut off of 60 years); patients with MASCC risk index scores <21 are considered at high risk for developing infectious complications (see *Risk Assessment Resources* in the algorithm).[380-383]

It is also acceptable to employ risk assessment criteria that have been identified in large clinical trials to distinguish between patients at low and high risk for complications during the course of neutropenia. For example, the Clinical Index of Stable Febrile Neutropenia (CISNE) is a prognostic model developed in the FINITE study from a retrospective cohort of patients with solid tumors and seemingly clinically stable febrile neutropenia episodes.[384] The CISNE model comprises six explanatory variables associated with serious complications: Eastern Cooperative Oncology Group (ECOG) performance status ≥2 (2 points); COPD (1 point); chronic cardiovascular disease (1 point); mucositis of grade ≥2 (NCI Common Toxicity Criteria; 1 point); monocytes <200 per µL (1 point); and stress-induced hyperglycemia (2 points). These factors were integrated into a score ranging from 0 to 8, which classifies patients into three prognostic classes: low (0 points), intermediate (1–2 points), and high risk (≥3 points).[384] There are risks of applying models to populations that differ from those in which they were originally validated. Thus, a two-step combination of a set of exclusion criteria (MASCC risk index) plus a validated triage-friendly scale (CISNE) might render better prognostic stratification than those of single-model approaches.

It is important to note that risk stratification generally, as well as the MASCC risk index and the CISNE index specifically, were validated in adults. No generalizable, cross-validated, risk-stratified management exists for pediatric patients with febrile neutropenia.

The MASCC prediction rule does not consider the duration of neutropenia to be a deciding factor that influences the clinical course of treatment;[382] however, the Panel acknowledges that the duration of anticipated neutropenia may be helpful in risk assessment. A patient with severe neutropenia (ANC ≤100 neutrophils/mcL) anticipated to last ≥7 days may be considered at high risk, regardless of the MASCC risk index score or other risk factors listed in the guidelines. This recommendation is also in

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Outpatient Therapy for Patients with Neutropenic Fever

重试 错误原因

Initial Evaluation of Risk

重试 错误原因

Patients with neutropenia may be categorized into either a high– or low–risk group using criteria derived either from validated clinical prediction rules based on risk models or from clinical trial eligibility criteria. Risk assessment attempts to predict the probability that a patient with neutropenia will experience serious complications during a febrile episode. This assessment helps to determine whether a patient at low risk for serious complications could safely receive treatment outside of the hospital and which initial empiric therapy with oral antibiotics is appropriate.

重试 错误原因

前瞻性试验表明，发热性中性粒细胞减少症最初可以在医院、门诊或家中进行评估，然后采用广谱静脉治疗、序贯静脉治疗然后口服治疗或口服治疗进行有效治疗。只有具备必要基础设施的中心才应在门诊环境中治疗低风险患者，最好是在研究背景下。

风险评估应作为初始评估的一部分进行（见

发热性中性粒细胞增多症患者的初步风险评估

算法）。一个广泛使用和验证的预测规则来评估风险是由多国癌症支持性护理协会（MASCC）开发的。MASCC 风险指数来自一个模型，该模型包括基于疾病负担的加权评分（例如，发热性中性粒细胞减少的程度）、临床不稳定或共病的证据（例如，低血压、慢性阻塞性肺疾病[COPD]、脱水）、既往真菌感染史、医疗护理地点（例如，住院患者，门诊患者）和年龄（截止年龄 60 岁）；MASCC 风险指数评分<21 的患者被认为是发生感染性并发症的高风险患者（参见算法中的风险评估资源）。

采用大型临床试验中确定的风险评估标准来区分中性粒细胞减少症过程中并发症低风险和高风险患者也是可以接受的。例如，稳定性发热性中性粒细胞减少症的临床指数（CISNE）是在 FINITE 研究中从具有实体瘤和看似临床稳定的发热性中性粒细胞减少症发作的患者的回顾性队列开发的预后模型 CISNE 模型包括与严重并发症相关的六个解释变量：（2 分）；COPD（1 分）;慢性心血管疾病（1 分）；≥2 级粘膜炎（NCI 常见毒性标准; 1 分）;单核细胞<200/μL（1 分）;和应激诱导的高血糖症（2 分）。 这些因素被整合到一个 0 – 8 分的评分中，将患者分为三个预后类别：低风险（0 分）、中等风险（1–2 分）和高风险（≥3 分）。因此，一组排除标准（MASCC 风险指数）加上有效的分类友好量表（CISNE）的两步组合可能比单一模型方法提供更好的预后分层。

重要的是要注意，一般的风险分层，以及 MASCC 风险指数和 CISNE 指数，特别是在成人中得到验证。对于发热性中性粒细胞减少症的儿科患者，尚无可推广的、交叉验证的、风险分层的管理方法。

MASCC 预测规则不认为中性粒细胞减少症的持续时间是影响临床治疗过程的决定性因素;但是，专家小组承认预期中性粒细胞减少症的持续时间可能有助于风险评估。预期持续≥7 天的重度中性粒细胞减少症（ANC ≤100 中性粒细胞/mcL）患者可能被视为高风险，无论 MASCC 风险指数评分或指南中列出的其他风险因素如何。这一建议也在

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agreement with those of the current IDSA guidelines on the management of neutropenia in patients with cancer.[17]

Duration of Neutropenia and Risk

For decades, clinicians have regarded depth and duration of neutropenia as critical determinants of a patient's risk for infection. Once the relationship between the ANC and incidence of infections was demonstrated, the importance of increased neutrophil counts for improved outcomes was evident. In the original study by Bodey et al,[20] the fatality rate was highest (80%) among patients with initial neutrophil counts <100 cells/mcL that did not change during the first week of infection compared to the lower rate (27%) seen in patients with initial neutrophil counts <1000 cells/mcL that rose to >1000 cells/mcL with treatment.[20] Subsequently, clinical trials have reported that response rates to antibiotic regimens are highly influenced by trends in the neutrophil count during febrile episodes. In one study, the overall response rate was 73% when the initial neutrophil count increased compared to 43% when it decreased or remained unchanged ($P < .0001$). The response rate in patients who recovered from neutropenia was 67%, compared to only 32% in patients who remained severely neutropenic ($P < .0001$).

In 1988, Rubin et al examined the influence of the duration of neutropenia on the response to empiric antimicrobial therapy in patients with fever of undetermined origin.[385] Patients with <7 days of neutropenia had a 95% response rate to initial antimicrobial therapy, compared to a 32% response rate in patients with >14 days of neutropenia ($P < .001$); however, intermediate durations between 7 and 14 days had response rates of 79%.[385]

Bone marrow recovery is an important factor that influences outcome during the febrile neutropenic episode. Delayed bone marrow recovery might be anticipated in certain patient subsets (eg, patients who have received multiple cycles of myelosuppressive chemotherapy, HCT recipients, patients with known bone marrow metastases, patients who have received radiation therapy to the pelvis, spine, or long bones). Most patients with solid tumors have neutropenia lasting <7 days and are generally lower risk. Several studies have demonstrated the ability of clinicians to predict a patient's anticipated duration of neutropenia. In prospective studies of patients identified as low risk for morbidity and mortality from febrile neutropenia, the expected duration of neutropenia was used as an eligibility criterion. Clinicians were correctly able to identify patients with an expected short duration of neutropenia (ie, <7–10 days) in >80% of the cases,[377,378,386] indicating that the duration of neutropenia can be one of several factors in selecting patients for outpatient management of neutropenic fever.

Evaluation of Patients for Outpatient Therapy for Neutropenic Fever

Outpatient therapy has become a common practice in those at low risk with neutropenic fever. Several single-center clinical trials generally support the shift in care for patients at low risk to the outpatient setting; the hospital is not necessarily a safer place for such patients, given the documented hazards of hospitalization.[387,388] However, not all centers are equipped to manage outpatient treatment, and some patients with fever are not appropriate candidates. Early success with this type of therapy has been predicated on the ability to accurately determine an individual patient's risk of developing complications associated with infection and on the presence of an adequate infrastructure for treatment and monitoring.

Once a level of risk has been identified, it can then be used to determine the appropriate site of care and route of broad-spectrum antibiotics administration. The Panel recommends that all patients at high risk for infections receive hospital care with broad-spectrum IV therapy (see *Initial Risk Assessment for Patients with Febrile Neutropenia* in the algorithm). Patients at low risk may be treated in the hospital with oral or IV antibiotics, in an ambulatory clinic, or at home if adequate follow-up care

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与当前 IDSA 指南中关于癌症患者中性粒细胞减少症的管理一致。

中性粒细胞持续时间和风险

几十年来，临床医生一直认为中性粒细胞减少症的深度和持续时间是患者感染风险的关键决定因素。一旦证实了 ANC 和感染发生率之间的关系，中性粒细胞计数增加对改善结局的重要性就显而易见了。在 Bodey 等人的原始研究中，在感染的第一周期间，初始中性粒细胞计数<100 个细胞/mcL 的患者的死亡率最高（80%），而在初始中性粒细胞计数<1000 个细胞/mcL 的患者中观察到的死亡率较低（27%），其在治疗后上升至>1000 个细胞/mcL。临床试验已经报道，对抗生素治疗方案的应答率受到发热发作期间中性粒细胞计数趋势的高度影响。 在一项研究中，当初始中性粒细胞计数增加时，总有效率为 73%，而当其减少或保持不变时，总有效率为 43%（P < .0001）。从中性粒细胞减少症中恢复的患者的反应率为 67%，而严重中性粒细胞减少症患者的反应率仅为 32%（P <0.0001）。

1988 年，Rubin 等研究了中性粒细胞减少持续时间对不明原因发热患者经验性抗菌治疗反应的影响，中性粒细胞减少<7 天的患者对初始抗菌治疗的反应率为 95%，而中性粒细胞减少>14 天的患者的反应率为 32（P <0.001）;然而，7 至 14 天的中间持续时间的反应率为 79%。

骨髓恢复是影响发热性贫血发作预后的重要因素。在某些患者亚组中（例如，接受多个周期骨髓抑制化疗（HCT）的患者），可能预期骨髓恢复延迟接受者、已知骨髓转移的患者、接受过骨盆、脊柱或长骨放射治疗的患者）。大多数实体瘤患者的中性粒细胞减少持续<7 天，通常风险较低。一些研究已经证明了临床医生预测患者中性粒细胞减少症的预期持续时间的能力。在对发热性中性粒细胞减少症的发病率和死亡率风险较低的患者进行的前瞻性研究中，中性粒细胞减少症的预期持续时间被用作合格性标准。临床医生能够在>80%的病例中正确识别预期中性粒细胞减少持续时间较短（即<7–10 天）的患者，表明中性粒细胞减少持续时间可能是选择患者进行贫血性发热门诊治疗的几个因素之一。

中性粒细胞减少性发热门诊治疗患者评价

Outpatient therapy has become a common practice in those at low risk with neutropenic fever. Several single–center clinical trials generally support the shift in care for patients at low risk to the outpatient setting; the hospital is not necessarily a safer place for such patients, given the documented hazards of hospitalization.However, not all centers are equipped to manage outpatient treatment, and some patients with fever are not appropriate candidates. Early success with this type of therapy has been predicated on the ability to accurately determine an individual patient’s risk of developing complications associated with infection and on the presence of an adequate infrastructure for treatment and monitoring.

重试 错误原因

一旦确定了风险水平，就可以用来确定适当的护理部位和广谱抗生素给药途径。专家小组建议所有感染高风险患者接受广谱 IV 治疗的医院护理（见算法中发热性中性粒细胞增多症患者的初始风险评估）。低风险的患者可以在医院接受口服或静脉注射抗生素治疗，在门诊或在家中接受适当的随访治疗

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can be provided (ie, 24 hours per day, 7 days per week). Outpatient therapy should be considered only for patients at low risk who consent to home care, have a telephone, have access to emergency facilities, have an adequate and supportive home environment, and are within 1-hour travel time of a medical center or physician's office. Outpatient therapy requires a period of early assessment and an observation period of 2 to 12 hours can be considered (category 2B) (see *Outpatient Therapy for Patients at Low Risk* in the algorithm). The assessment requires a careful examination, review of laboratory results, review of social criteria for home therapy (as described above), and assessment of whether oral antibiotics are feasible. The observation period is used to confirm that the patient is at low risk and to ensure the clinical stability of the patient; to administer the first dose of antibiotics and monitor for any reactions; to organize discharge plans for home and follow-up care; and to provide patient education. A telephone follow-up should be performed within 12 to 24 hours. This assessment and observation can be performed during a short hospital stay or in an ambulatory facility or office staffed with qualified health care professionals. Providers who perform the early assessment and follow-up should be well trained (eg, a physician, nurse, physician assistant, and/or nurse practitioner) and should have experience and expertise in treating patients with fever and neutropenia.

Outpatient Regimens

Outpatient antimicrobial treatment may consist of broad-spectrum IV antibiotics given at home or in the clinic, or an oral regimen for carefully selected patients.[389] For selected patients at low risk, the combination of ciprofloxacin with amoxicillin/clavulanate is considered the oral regimen of choice based on well-designed randomized trials (category 1) (see *Outpatient Therapy for Patients at Low Risk* in the algorithm). Although some of these trials were performed in an inpatient setting, they demonstrate the efficacy of the oral combination compared with standard IV therapy in the low-risk population.[371,390,391] Ciprofloxacin plus clindamycin is an acceptable alternative for those with penicillin allergies.[3,17] However, ciprofloxacin monotherapy is not considered by the Panel to be an adequate broad-spectrum agent because of the suboptimal coverage for gram-positive organisms and potential for serious breakthrough infections caused by viridans group streptococci.[392] Nonetheless, several small studies have used high-dose oral ciprofloxacin alone in patients at low risk of infection with fever and neutropenia.[393-395]

Moxifloxacin (category 1) is a newer-generation fluoroquinolone that was shown to be safe in patients at low risk with neutropenic fever.[396] In a double-blind, randomized trial, single-daily moxifloxacin was compared with twice-daily ciprofloxacin plus amoxicillin/clavulanic acid in the treatment of low-risk febrile neutropenia in patients with cancer.[397] Low risk was defined as an MASCC score >20 that is equivalent to a <10% complication rate. Of the 333 patients treated on this trial, 169 were given moxifloxacin, and 169 patients were treated with the ciprofloxacin combination. Therapy success was observed in 80% of patients treated with moxifloxacin compared with 82% of patients given ciprofloxacin combination therapy (95% CI, -10%–8%; *P* = NS). Despite similar therapy success rates, the reasons for disease progression differed between the two groups. Moxifloxacin-treated patients had greater microbial complications including persistent or breakthrough resistance, while patients given the ciprofloxacin combination had mostly drug intolerance or adverse events that resulted in disease progression. Rates of patients treated with moxifloxacin compared to ciprofloxacin combination with serious adverse events (6% vs. 8%; *P* = .23) or any adverse event (44% vs. 52%; *P* = .13) were similar. Moxifloxacin has a longer half-life, which allows for once-daily dosing. It is more active against gram-negative bacteria but has limited activity against *P aeruginosa* compared to ciprofloxacin. Therefore, both of these treatments are recommended for those with low risk for infections with febrile neutropenia, but the choice of regimen may be influenced by local resistance and infection patterns.

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可以提供（即，每天 24 小时，每周 7 天）。门诊治疗应仅考虑低风险患者，他们同意家庭护理，有电话，可以使用急救设施，有足够的支持性家庭环境，并且距离医疗中心或医生办公室不到 1 小时。门诊治疗需要一段时间的早期评估，可以考虑 2 – 12 小时的观察期（2B 类）（见算法中低风险患者的门诊治疗）。评估需要仔细检查，审查实验室结果，审查家庭治疗的社会标准（如上所述），并评估口服抗生素是否可行。 观察期用于确认患者处于低风险状态，并确保患者的临床稳定性;给予第一剂抗生素并监测任何反应;组织出院计划，以便进行家庭和后续护理;并提供患者教育。应在 12 至 24 小时内进行电话随访。这种评估和观察可以在短期住院期间或在配备有合格医疗保健专业人员的门诊设施或办公室进行。进行早期评估和随访的提供者应经过良好培训（例如，医生、护士、医生助理和/或执业护士），并应具有治疗发热和中性粒细胞减少患者的经验和专业知识。

门诊方案

门诊抗菌治疗可包括在家中或诊所给予的广谱 IV 抗生素，或精心选择的患者的口服方案。对于低风险的选定患者，基于精心设计的随机试验（第 1 类），环丙沙星与阿莫西林/克拉维酸的组合被认为是选择的口服方案（见

算法中低风险患者的门诊治疗）。虽然

其中一些试验是在住院患者中进行的，这些试验证明了口服复方制剂与标准 IV 治疗相比在低风险人群中的疗效。

克林霉素是青霉素过敏者可接受的替代药物，但是，专家组认为环丙沙星单一疗法不是一种足够的广谱药物，因为它对革兰氏阳性菌的覆盖率不够理想，而且有可能引起草绿色链球菌引起的严重突破性感染。尽管如此，一些小型研究已经在低感染风险伴发热和中性粒细胞减少的患者中单独使用高剂量口服环丙沙星。

莫西沙星（1 类）是新一代氟喹诺酮类药物，已被证明对低风险的甲流患者安全。在一项双盲随机试验中，在癌症患者中，比较了每日一次的诺氟沙星与每日两次的环丙沙星加阿莫西林/克拉维酸治疗低风险发热性中性粒细胞减少症的疗效。低风险定义为 MASCC 评分>20，相当于<10%。并发症发生率在该试验治疗的 333 名患者中，169 名患者接受了阿托沙星治疗，169 名患者接受了环丙沙星联合治疗。与环丙沙星联合治疗的 82%患者相比，在 80%的患者中观察到治疗成功（95% CI，–10%–8%; P = NS）。尽管治疗成功率相似，但两组之间疾病进展的原因不同。 莫西沙星治疗的患者有更大的微生物并发症，包括持续或突破性耐药，而给予环丙沙星联合治疗的患者大多有药物不耐受或导致疾病进展的不良事件。与环丙沙星联合治疗相比，使用诺氟沙星治疗的患者发生严重不良事件（6% vs. 8%; P = 0.23）或任何不良事件（44% vs. 52%; P = 0.13）的比例相似。莫西沙星具有较长的半衰期，允许每日给药一次。与环丙沙星相比，它对革兰氏阴性菌的活性更高，但对铜绿假单胞菌的活性有限。因此，这两种治疗方法都推荐给发热性中性粒细胞减少症感染风险较低的患者，但治疗方案的选择可能会受到局部耐药性和感染模式的影响。

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Two other fluoroquinolones, levofloxacin and ofloxacin, have been tested for the treatment of patients at low risk with febrile neutropenia. Levofloxacin is a category 2A recommendation following studies demonstrating safety and efficacy[79,80] (see *Antibacterial Prophylaxis*). Data from a 2008 self-administered survey indicated that 50% of oncologists were using levofloxacin as empiric therapy for patients at low risk with febrile neutropenia.[398] Ofloxacin was safe in patients at low risk with neutropenic fever in a randomized trial, though an early death in a non-hospitalized patient in this trial underscores the need for close monitoring.[377] Ofloxacin is not currently recommended.

NCCN Recommendations for Outpatient Therapy

The Panel feels that outpatient therapy with a fluoroquinolone should be based on reliable gram-negative bacillary activity of the antibiotic that includes *P aeruginosa* and local antibacterial susceptibilities. Ciprofloxacin plus amoxicillin/clavulanate (or ciprofloxacin plus clindamycin in patients allergic to penicillin) is the standard oral outpatient antibiotic regimen for patients at low risk with neutropenic fever. There is also evidence supporting quinolone monotherapy in this setting.

Moxifloxacin (category 1) and levofloxacin (category 2A) are recommended quinolone monotherapies. These recommendations for quinolone-based outpatient regimens for neutropenic fever only apply to patients at low risk who have not received a quinolone as prophylaxis. Additionally, in order for these patients to receive oral antibiotics, the patient should not present with nausea or vomiting, and must be able to tolerate oral medications (see *Outpatient Therapy for Patients at Low Risk* in the algorithm). IV therapy may also be used for outpatient treatment of patients at low risk with fever and neutropenia when treatment is given either in the home or day clinic setting (see *Outpatient Therapy for Patients at Low Risk* in the algorithm). Several IV outpatient regimens for patients at low risk have been studied in nonrandomized or small open trials, including IV ceftazidime, imipenem/cilastatin, and aztreonam plus clindamycin.[3,371,378,379,399-401]

Once-daily ceftriaxone has been used for empiric antibiotic therapy in a few noncomparative studies in centers where *Pseudomonas* is not a common pathogen.[402] However, most *P aeruginosa* isolates are resistant to ceftriaxone. Although ceftriaxone combined with a once-daily aminoglycoside is a convenient regimen for outpatient IV administration, an aminoglycoside without an antipseudomonal beta-lactam may not be effective against *P aeruginosa*, which remains an infrequent but potentially lethal pathogen. Therefore, the Panel cannot recommend ceftriaxone (with or without an aminoglycoside) as empiric therapy for neutropenic fever. If this regimen is used, it should be restricted to patients of low risk at centers where *P aeruginosa* infection is uncommon. In addition to the antimicrobial spectrum, other factors to consider in the choice of an outpatient regimen include stability of the reconstituted drugs, ability to manage IV infusions, and VADs.

Follow-Up of Outpatients with Fever and Neutropenia

Follow-up management can be performed at the patient's home or in the physician's office or clinic. The Panel recommends that patients be assessed daily while febrile, although some experts feel that less frequent follow-up may be appropriate after fever defervescence (see *Outpatient Therapy for Patients at Low Risk* in the algorithm). For the first 72 hours after initiation of empiric therapy, the patient should be assessed daily at home or at the clinic for treatment response, signs of toxicity, and treatment adherence. If the disease is responding to the treatment regimen, then daily follow-up by telephone is sufficient. A return to the clinic is recommended for any positive culture from blood or other sterile source, for persistent or recurrent fever at 3 to 5 days, if serious subsequent infections or adverse events develop, if the patient is unable

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另外两种氟喹诺酮类药物，左氧氟沙星和氧氟沙星，已被测试用于治疗低风险的发热性中性粒细胞减少症患者。左氧氟沙星是一个 2A 类的建议后，研究证明安全性和有效性（见抗菌预防）。2008 年的一项自我调查数据显示，50%的肿瘤学家使用左氧氟沙星作为经验性治疗低风险的发热性中性粒细胞减少症患者。在一项随机试验中，氧氟沙星在低风险的血小板减少性发热患者中是安全的，尽管在这项试验中，一名非住院患者的早期死亡强调了密切监测的必要性。目前不推荐使用氧氟沙星。

NCCN 门诊治疗建议

The Panel feels that outpatient therapy with a fluoroquinolone should be based on reliable gram–negative bacillary activity of the antibiotic that includes P aeruginosa and local antibacterial susceptibilities. Ciprofloxacin plus amoxicillin/clavulanate (or ciprofloxacin plus clindamycin in patients allergic to penicillin) is the standard oral outpatient antibiotic regimen for patients at low risk with neutropenic fever. There is also evidence supporting quinolone monotherapy in this setting.

重试 错误原因

Moxifloxacin (category 1) and levofloxacin (category 2A) are recommended quinolone monotherapies. These recommendations for quinolone–based outpatient regimens for neutropenic fever only apply to patients at low risk who have not received a quinolone as prophylaxis. Additionally, in order for these patients to receive oral antibiotics, the patient should not present with nausea or vomiting, and must be able to

重试 错误原因

tolerate oral medications (see Outpatient Therapy for Patients at Low Risk

重试 错误原因

在算法中）。当在家庭或日间诊所进行治疗时，IV 治疗也可用于发热和中性粒细胞减少症低风险患者的门诊治疗（参见算法中低风险患者的门诊治疗）。在非随机或小型开放性研究中，

trials, including IV ceftazidime, imipenem/cilastatin, and aztreonam plus clindamycin. [3,371,378,379,399–401]

重试 错误原因

每日一次头孢曲松已被用于经验性抗生素治疗的一些非比较性研究中心，假单胞菌不是一种常见的病原体。虽然头孢曲松联合每日一次氨基糖苷类药物是一种方便的门诊静脉给药方案，但不含抗假单胞菌β–内酰胺的氨基糖苷类药物可能对铜绿假单胞菌无效，铜绿假单胞菌仍然是一种罕见但可能致命的病原体。因此，专家小组不能推荐头孢曲松（含或不含氨基糖苷类）作为经验性治疗血小板减少性发热。如果使用该方案，应仅限于低风险患者，

铜绿假单胞菌感染不常见的中心。除了抗菌谱，在选择门诊治疗方案时要考虑的其他因素包括复溶药物的稳定性、管理 IV 输注的能力和 VAD。

发热和中性粒细胞增多症门诊患者的随访

随访管理可以在患者家中或在医生办公室或诊所进行。专家组建议患者在发热时每天进行评估，尽管一些专家认为在退烧后减少随访频率可能是合适的（参见算法中低风险患者的门诊治疗）。在开始经验性治疗后的前 72 小时内，应每天在家中或诊所评估患者的治疗反应、毒性体征和治疗依从性。如果疾病对治疗方案有反应，那么每天通过电话进行随访就足够了。对于血液或其他无菌来源的任何阳性培养物，如果患者无法在 3 至 5 天内持续或反复发热，如果发生严重的后续感染或不良事件，

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to continue the prescribed antibiotic regimen (eg, intolerance to the oral regimen), or for infusion of IV antibiotics.

Initial Empiric Antibiotic Therapy

The foundation of infection management is to administer empiric antibiotics in patients with fever and neutropenia. This approach is necessary, because currently available diagnostic tests are not sufficiently rapid, sensitive, or specific to identify or exclude microbial causes of fever from other noninfectious causes. All patients with neutropenia should be treated empirically with broad-spectrum antibiotics promptly at the first sign of infection (ie, fever). This is done to avoid the mortality associated with a delay in treatment in patients with a serious infection.[1,403] Many highly effective antibiotic regimens are available and are recommended based on data from randomized clinical trials.

Selection of initial therapy should consider the following:

- The infection risk assessment of patients;
- The antimicrobial susceptibilities of pathogens isolated locally;
- The most common potentially infecting organisms, including antibiotic-resistant pathogens, such as extended spectrum beta-lactamase–producing gram-negative rods, vancomycin-resistant enterococcus (VRE), and colonization with or previous infection with MRSA;
- The potential sites of infection;
- The importance of a broad-spectrum bactericidal antibiotic regimen that includes antipseudomonal coverage;
- Clinical instability (eg, hypotension, organ dysfunction);
- Drug allergy;
- Recent antibiotic use (including prophylaxis); and
- Bactericidal nature of the antibiotic.

Recommended Approaches

The Panel recommends the following approaches to initial empiric management of febrile neutropenia to be appropriate based on the results of large, randomized, controlled clinical trials (see *Initial Inpatient Empiric Therapy for Uncomplicated Fever and Neutropenia* in the algorithm).[1,2,403]

For select patients at low risk of infection with fever and neutropenia, one approach is IV antibiotic monotherapy (all category 1 except where noted) with imipenem/cilastatin, meropenem, piperacillin/tazobactam, or an extended-spectrum antipseudomonal cephalosporin (cefepime [category 1] or ceftazidime [category 2B]).[404-408] Local institutional bacterial susceptibilities should be considered when selecting empiric antibiotic therapy. In hospitals where infections caused by antibiotic-resistant bacteria (eg, MRSA or drug-resistant gram-negative rods) are commonly observed, policies on initial empiric therapy of neutropenic fever may need to be tailored accordingly.

Meta-analyses of randomized trials have reported that cefepime was associated with increased all-cause mortality when used as empiric therapy for neutropenic fever, although no increase in infection-related mortality was noted.[399,409,410] However, a meta-analysis by the FDA, using additional data, did not find a statistically significant increase in mortality for patients treated with cefepime compared with controls. Thus, the FDA concluded that cefepime remains an appropriate therapy for its approved indications. A randomized, dual-center study of 105 patients treated with piperacillin/tazobactam or imipenem/cilastatin as empiric therapy for febrile neutropenia reported imipenem/cilastatin to have superior efficacy, although this area of research requires further investigation.[411]

Another approach for initial empiric therapy for patients at low risk for infections with fever and neutropenia is oral antibiotic therapy (see *Initial Inpatient Empiric Therapy for Uncomplicated Fever and Neutropenia* in the

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继续处方的抗生素方案（例如，对口服方案不耐受），或静脉输注抗生素。

初始经验性抗生素治疗

感染管理的基础是在发热和中性粒细胞减少的患者中使用经验性抗生素。这种方法是必要的，因为目前可用的诊断测试不够快速，灵敏度或特异性，以确定或排除其他非感染性原因的发热微生物原因。所有中性粒细胞减少的患者在出现感染的第一个迹象（即发热）时应立即使用广谱抗生素进行经验性治疗。这样做是为了避免严重感染患者因治疗延迟而导致死亡。许多高效的抗生素方案是基于随机临床试验的数据推荐的。

初始治疗的选择应考虑以下因素：

- 患者感染风险评估;
- 当地分离的病原体的抗菌药物敏感性;
- 最常见的潜在感染微生物，包括耐药病原体，如产超广谱β–内酰胺酶的革兰氏阴性杆菌、耐万古霉素肠球菌（VRE）和 MRSA 定植或既往感染;

- 潜在感染部位;
- 包括抗假单胞菌在内的广谱杀菌抗生素治疗方案的重要性;

- 临床不稳定（例如，低血压、器官功能障碍）;
- 药物过敏;
- 近期使用抗生素（包括预防）;以及
- 抗生素的杀菌特性。

的建议方法的

根据大型、随机、对照临床试验的结果，专家组建议采用以下方法对发热性中性粒细胞减少症进行初始经验性管理（参见算法中的单纯性发热和中性粒细胞减少症的初始住院经验性治疗）。

对于有发热和中性粒细胞减少症的低感染风险的患者，一种方法是静脉抗生素单药治疗（所有 1 类，除非另有说明），包括亚胺培南/西司他丁、美罗培南、哌拉西林/他唑巴坦或广谱抗假单胞菌头孢菌素（头孢吡肟[1 类]或头孢他啶[2B 类]）。在常见耐药菌（如 MRSA 或耐药革兰氏阴性杆菌）引起的感染的医院，可能需要相应地调整贫血热初始经验性治疗的政策。

随机试验的荟萃分析报告，头孢吡肟与增加全因死亡率时，作为经验性治疗的血小板减少性发热，虽然没有增加感染相关的死亡率 noted.However，由 FDA 的荟萃分析，使用额外的数据，没有发现一个统计学显着增加死亡率与头孢吡肟治疗的患者相比，控制。因此，FDA 得出结论，头孢吡肟仍然是其批准适应症的适当治疗方法。一项对 105 例接受哌拉西林/他唑巴坦或亚胺培南/西司他丁作为经验性治疗发热性中性粒细胞减少症的患者进行的随机、双中心研究报告，亚胺培南/西司他丁具有上级疗效，尽管该研究领域需要进一步研究。

对于发热和中性粒细胞减少症感染低风险患者的初始经验性治疗的另一种方法是口服抗生素治疗（参见 Initial Inpatient Empiric Therapy for Uncomplicated Fever and Neutrophil in the

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algorithm). Ciprofloxacin plus amoxicillin/clavulanate (category 1) is an option for oral antibiotic therapy, with the alternative of ciprofloxacin plus clindamycin for patients allergic to penicillin. Moxifloxacin (category 1) or levofloxacin (category 2A) are other recommended options for this approach. Fluoroquinolone regimens should not be administered in patients receiving antimicrobial prophylaxis with a fluoroquinolone. Additionally, while data support the use of fluoroquinolones for prophylaxis, the risks and benefits should be evaluated for empiric therapy or other clinical scenarios. In particular, the side effects of fluoroquinolones should be taken into consideration. In 2016, the FDA issued a warning that fluoroquinolones are associated with disabling side effects involving tendons, muscles, joints, nerves, and the CNS.[412]

IV antibiotic monotherapy is the preferred treatment option for patients at intermediate or high risk with fever and neutropenia. However, IV antibiotic combination therapy, though not routinely recommended, may be considered in higher risk or antimicrobial resistant cases. In such situations, an aminoglycoside combined with an antipseudomonal agent can be considered.[413-415] Aminoglycoside use carries the inherent risk of renal and otic toxicity. Avoiding these toxicities requires careful monitoring and necessitates frequent reassessment, but once-daily aminoglycoside dosing is associated with less renal toxicity than shorter interval dosing.[416] Once-daily aminoglycoside dosing should probably not be used for treating meningitis or endocarditis based on inadequate clinical data. The use of vancomycin, linezolid, daptomycin, or tedizolid is not routinely recommended. Although published studies exist regarding the use of some of these agents in neutropenia, the Panel strongly recommends that these agents not be used routinely as initial empiric therapy because of concerns for resistance and breakthrough infections.

For patients at high risk for *Pseudomonas* infections (eg, history of previous *Pseudomonas* infections, presence of ecthyma gangrenosum), initial combination therapy with the most active antipseudomonal agents available in the local setting should be considered.

For specific indications, the addition of IV vancomycin either to IV monotherapy or to combination therapy (see *Empiric Addition of Vancomycin*) may be considered. Support for the judicious use of vancomycin has developed because of the increased frequency of beta-lactam–resistant gram-positive infections caused by MRSA, most coagulase-negative staphylococci, penicillin-resistant viridans group streptococci and enterococci, and *Corynebacterium jeikeium.* Vancomycin should be reserved for specific indications and should not be considered as a routine component of initial therapy for fever and neutropenia.

Empiric Addition of Vancomycin

Considerable debate has occurred about the use of empiric vancomycin in patients with fever and neutropenia, as the uncontrolled use of vancomycin has facilitated the dissemination of vancomycin-resistant organisms, especially enterococci.[417,418] The clinical concern is that a portion of infections caused by gram-positive pathogens can be fulminant and lead to rapid death in patients who are not treated promptly with appropriate antibiotics. However, a large, prospective, randomized trial from the European Organization for Research and Treatment of Cancer (EORTC) did not show true clinical advantages for empiric vancomycin in adults.[419] This study reported that empiric vancomycin decreased the number of days the patients had fever but did not improve survival. The study also showed that empiric vancomycin was associated with an increased incidence of nephrotoxicity and hepatotoxicity.[419] A prospective randomized trial of fever and neutropenia in children has reported benefit for empiric vancomycin;[420] however, another randomized study in children did not show a benefit for the addition of vancomycin.[421]

In addition to the occurrence of VRE, there are other vancomycin-resistant pathogens of note. Reports of vancomycin-resistant and vancomycin-

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算法）。环丙沙星加阿莫西林/克拉维酸（第 1 类）是口服抗生素治疗的一种选择，对于青霉素过敏的患者，可以选择环丙沙星加克林霉素。莫西沙星（1 类）或左氧氟沙星（2A 类）是其他推荐的选择，

approach.接受氟喹诺酮类药物预防性治疗的患者不应接受氟喹诺酮类药物治疗。此外，虽然数据支持使用氟喹诺酮类药物进行预防，但应评估经验性治疗或其他临床情况的风险和获益。特别是，氟喹诺酮类药物的副作用应予以考虑。2016 年，FDA 发布警告称，氟喹诺酮类药物与涉及肌腱、肌肉、关节、神经和 CNS 的致残性副作用有关。

IV 抗生素单药治疗是发热和中性粒细胞减少症中高危患者的首选治疗选择。然而，静脉抗生素联合治疗，虽然不是常规推荐，可考虑在高风险或耐药的情况下。在这种情况下，可以考虑氨基糖苷类药物联合抗假单胞菌药物。氨基糖苷类药物的使用具有肾和耳毒性的固有风险。避免这些毒性需要仔细监测，并需要经常重新评估，但每日一次氨基糖苷类药物给药与较短的间隔给药相比，肾毒性较低。每日一次氨基糖苷类给药可能不应用于

根据不充分的临床数据治疗脑膜炎或心内膜炎。不推荐常规使用万古霉素、利奈唑胺、达托霉素或泰地唑胺。尽管已发表的研究表明，这些药物中的一些用于治疗中性粒细胞减少症，但专家小组强烈建议，由于对耐药性和突破性感染的担忧，这些药物不应常规用作初始经验性治疗。

对于假单胞菌感染高风险的患者（例如，既往假单胞菌感染史，存在坏疽性脓疮），

应考虑与当地最有效的抗假单胞菌药物进行初始联合治疗。

对于特定适应症，可以考虑在静脉单药治疗或联合治疗中添加静脉万古霉素（见万古霉素经验性添加）。由于 MRSA、大多数凝固酶阴性葡萄球菌、青霉素耐药草绿色链球菌和肠球菌以及杰氏棒状杆菌引起的β–内酰胺耐药革兰氏阳性菌感染的频率增加，因此支持谨慎使用万古霉素。万古霉素应保留用于特定适应症，不应作为发热和中性粒细胞减少症初始治疗的常规成分。

万古霉素的经验性添加

关于经验性万古霉素在发热和中性粒细胞减少患者中的使用存在相当大的争议，因为万古霉素的不加控制的使用促进了万古霉素耐药微生物的传播，特别是肠球菌，临床关注的是，一部分革兰氏阳性病原体引起的感染可能是暴发性的，并导致患者迅速死亡，如果不及时使用适当的抗生素治疗。然而，欧洲癌症研究和治疗组织（EORTC）的一项大型前瞻性随机试验并未显示经验性万古霉素在成人中的真正临床优势，该研究报告经验性万古霉素可减少患者发热天数，但不能提高生存率。 该研究还显示经验性万古霉素与肾毒性和肝毒性发生率增加相关。一项关于儿童发热和中性粒细胞减少的前瞻性随机试验报告了经验性万古霉素的益处；然而，另一项儿童随机研究未显示添加万古霉素的益处。

除了 VRE 的发生，还有其他值得注意的万古霉素耐药病原体。万古霉素耐药和万古霉素–

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intermediate sensitive *S aureus* are currently rare but are of key concern, and they underscore the need for judicious vancomycin use.[422,423] The increase in vancomycin resistance has been associated with use of vancomycin among the patients who are hospitalized. The NCCN Guidelines Panel advises practitioners to adopt the recommendation of the Hospital Infection Control Practices Advisory Committee (HICPAC) of the CDC for preventing the spread of vancomycin resistance.[424,425] Because of the increased risk for vancomycin-resistant organisms, empiric vancomycin use should be considered only in patients at high risk for serious gram-positive infection, and should not be considered as a routine component of initial therapy for fever and neutropenia. Vancomycin should be considered in the following clinical situations:

- Clinically apparent, serious IV catheter-related infection (to cover coagulase-negative staphylococcal isolates, which are usually beta-lactam antibiotic-resistant and MRSA);[426,427]
- Blood cultures positive for gram-positive bacteria before final identification and susceptibility testing;
- Known colonization with penicillin/cephalosporin–resistant pneumococci or MRSA;
- Clinical instability (eg, hypotension or shock), pending the results of cultures;[428,429] and
- Soft tissue infection (particularly in regions where MRSA infection is common).[430]

If empiric vancomycin (or other agents for gram-positive resistant infection) is initiated in any of these situations, its use should be reassessed within 2 to 3 days of initiation. If a resistant gram-positive pathogen (eg, MRSA) is not identified, the Panel recommends discontinuing the agent. Authoritative guidelines have been published on the dosing and therapeutic monitoring of vancomycin.[431] For management of complicated cases of *C difficile* infections, oral vancomycin can be considered (see *Site-Specific Evaluation and Treatment of Infections: Abdominal, Rectal, and Urinary Tract Infections: Clostridium difficile Colitis*).

In patients with acute leukemia receiving mucotoxic regimens, prophylaxis with ciprofloxacin and TMP/SMX have been associated with an increased risk of viridans group streptococcal infections.[432-434] The broad-spectrum, gram-negative bacillary coverage and limited gram-positive pathogen activity of these drugs likely predispose patients to GI colonization and subsequent infection with such organisms.[392,435] One study has reported an increased risk of breakthrough viridans group streptococcal infection following prophylaxis with levofloxacin,[436] which has increased activity against gram-positive bacteria compared to ciprofloxacin; however, this is a single report and more data will be necessary to fully evaluate the use of newer-generation fluoroquinolones.

Although bloodstream infections by viridans group streptococci resistant to all beta-lactams are observed in patients with cancer, cefepime, imipenem/cilastatin, meropenem, and piperacillin-tazobactam have more reliable activity than ceftazidime against viridans group streptococci.[437] The addition of vancomycin provided no benefit compared to placebo with regard to defervescence, episodes of gram-positive bacteremia, or use of empiric antifungal therapy in patients with hematologic malignancies with neutropenic fever of unknown etiology that persisted for 48 to 60 hours after initial empiric piperacillin-tazobactam.[438,439] In patients with neutropenic fever and severe mucositis who are receiving imipenem/cilastatin, meropenem, or piperacillin/tazobactam (ie, antibiotics with activity against oral flora), it does not appear that the addition of vancomycin is advantageous. Thus, the NCCN Guidelines Panel strongly recommends that vancomycin should not be routinely added to an empiric regimen solely based on persistent neutropenic fever of unknown etiology.

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中度敏感的金黄色葡萄球菌目前很少见，但却是一个关键问题，它们强调了明智使用万古霉素的必要性。万古霉素耐药性的增加与住院患者中万古霉素的使用有关。NCCN 指南专家组建议从业者采用 CDC 医院感染控制实践咨询委员会（CAPPAC）的建议，以防止万古霉素耐药性的传播。由于万古霉素耐药微生物的风险增加，仅应在严重革兰氏阳性菌感染的高风险患者中考虑经验性万古霉素使用，不应被视为发热和中性粒细胞减少症初始治疗的常规组成部分。在以下临床情况下应考虑使用万古霉素：

- 临床上明显的严重 IV 导管相关感染（涵盖凝固酶阴性葡萄球菌分离株，通常为β–内酰胺类抗生素耐药和 MRSA）；
- 最终鉴定和药敏试验前革兰氏阳性菌血培养阳性;
- 已知青霉素/头孢菌素耐药肺炎球菌或 MRSA 定植;
- 临床不稳定（例如，低血压或休克），等待培养结果;以及
- 软组织感染（特别是在 MRSA 感染常见的地区）。

如果在这些情况下开始使用经验性万古霉素（或其他革兰氏阳性耐药感染药物），应在开始后 2 – 3 天内重新评估其使用情况。如果未发现耐药革兰氏阳性病原体（如 MRSA），专家组建议停用该药物。关于万古霉素的剂量和治疗监测，已经发表了权威指南。对于艰难梭菌感染的复杂病例的管理，可以考虑口服万古霉素（见感染的部位特异性评价和治疗：

腹部、直肠和尿路感染：艰难梭菌结肠炎）。

在接受粘膜毒性治疗方案的急性白血病患者中，环丙沙星和 TMP/SMX 预防与草绿色链球菌感染风险增加相关。革兰氏阴性杆菌覆盖率和有限的革兰氏–这些药物的阳性病原体活性可能使患者易于胃肠道定植并随后感染这些微生物。左氧氟沙星预防后感染，与环丙沙星相比，左氧氟沙星对革兰氏阳性菌的活性增加;然而，这是一份单一报告，需要更多数据来全面评价新一代氟喹诺酮类药物的使用。

尽管在癌症患者中观察到对所有β–内酰胺类抗生素耐药的草绿色组链球菌的血流感染，但头孢吡肟、亚胺培南/西司他丁、美罗培南和哌拉西林–他唑巴坦对草绿色组链球菌的活性比头孢他啶更可靠。在初始经验性哌拉西林–他唑巴坦治疗后持续 48 – 60 小时的不明原因的血小板减少性发热的血液恶性肿瘤患者中，与安慰剂相比，添加万古霉素在退热、革兰氏阳性菌血症发作或使用经验性抗真菌治疗方面没有提供益处。或哌拉西林/他唑巴坦（即，对口腔植物群具有活性的抗生素），加入万古霉素似乎不是有利的。 因此，NCCN 指南专家组强烈建议，万古霉素不应常规添加到经验性方案仅基于持续的不明病因的血小板减少性发热。

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Agents with Broad-Spectrum Activity Against Gram-Positive Pathogens

Decreased susceptibility to vancomycin is an increasing concern. If decreased susceptibility is found on minimum inhibitory concentration (MIC) assessment, other treatment options for resistant gram-positive infections should be considered. Linezolid, daptomycin, and tedizolid are active against the majority of gram-positive organisms, including beta-lactam–resistant and vancomycin-resistant pathogens. [440-446] Resistance of gram-positive organisms to linezolid is infrequent, but this agent should be administered with caution in patients with compromised bone marrow function because of the marrow toxicity associated with its long-term use. Thrombocytopenia is most common (0.3%–10%) and increases with the duration of linezolid treatment, typically with duration of treatment >2 weeks. In neutropenic patients with cancer, myeloid recovery does not seem to be delayed with short courses of linezolid;[447,448] however, experience with long durations of therapy (eg, >14 days) is limited in patients with cancer.

Vancomycin or linezolid should be used for the treatment of MRSA pneumonia in patients who are ventilated.[449-452] The FDA issued an alert about linezolid indicating that it is not approved for treatment of catheter-related infections, catheter-site infections, or gram-negative infections. In an open-label randomized study, patients treated with linezolid had a higher chance of death compared with those receiving vancomycin, oxacillin, or dicloxacillin for intravascular catheter-related infections with: 1) gram-negative agents alone; 2) both gram-positive and gram-negative organisms; or 3) no infection. No mortality difference by treatment was found among those who had gram-positive infections alone.

Daptomycin is effective against most gram-positive pathogens, but it should not be used for the treatment of pneumonia, because it is inactivated by pulmonary surfactant.[453,454] Daptomycin is indicated for the treatment of complicated skin and skin structure infections caused by susceptible strains of certain gram-positive microorganisms.[455,456] A pharmacokinetic study of daptomycin in patients with cancer with febrile neutropenia showed that this agent was active and well tolerated in this population (N = 29) with a median time to defervescence of 3 days following the start of treatment.[457] A randomized study showed similar efficacy of daptomycin compared with vancomycin or anti-staphylococcal beta-lactams as therapy for *S aureus* bacteremia and endocarditis.[458] In a prospective study in patients with cancer who were treated with daptomycin for gram-positive catheter-related bloodstream infections (N = 40), the rates of symptom resolution at 48 hours (76% vs. 53%) and microbial eradication at 48 hours (78% vs. 34%) were higher with daptomycin compared with historical vancomycin treatment in matched controls.[459] In addition, the overall response rate was higher with daptomycin (68% vs. 32%), and the incidence of nephrotoxicity was lower. The treatment groups were comparable with regard to the rate of neutropenia, complications, adverse events, length of hospital stay, and deaths.[459]

Optimal therapy for VRE infections is not well-defined. Linezolid, tedizolid, and daptomycin have been used with variable success in the treatment of patients with VRE bloodstream infections.[440,443,448,460,461] Removal of an infected catheter should always be strongly considered. In the absence of more definitive data, therapy with one of these agents is advised for VRE bacteremia.

Telavancin, ceftaroline, oritavancin, and dalbavancin have been approved for the treatment of complicated skin and skin structure infections caused by gram-positive pathogens, including MRSA.[462,463] Ceftaroline is also indicated for the treatment of community-acquired bacterial pneumonia caused by susceptible gram-negative and gram-positive (except for MRSA) pathogens; this agent is not active against *Enterococcus faecalis*. There are no directive data on the use of these agents in the oncologic

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对革兰氏阳性病原体具有广谱活性的药物

对万古霉素敏感性降低是一个日益关注的问题。如果在最低抑菌浓度（MIC）评估中发现敏感性降低，则应考虑耐药革兰氏阳性菌感染的其他治疗选择。利奈唑胺、达托霉素和替地唑胺对大多数革兰氏阳性微生物具有活性，包括β–内酰胺耐药和万古霉素耐药病原体。 革兰氏阳性微生物对利奈唑胺的耐药性并不常见，但在骨髓功能受损的患者中应谨慎使用该药物，因为其长期使用具有骨髓毒性。血小板减少症最常见（0.3%–10%），并随利奈唑胺治疗持续时间增加，通常治疗持续时间>2 周。 在患有癌症的血小板减少症患者中，短期利奈唑胺治疗似乎不会延迟骨髓恢复;然而，癌症患者的长期治疗经验（例如，>14 天）有限。

一项达托霉素在伴有发热性中性粒细胞减少症的癌症患者中的药代动力学研究表明，该药物在该人群（N = 29）中具有活性且耐受性良好，治疗开始后中位退热时间为 3 天。内酰胺类药物治疗金黄色葡萄球菌菌血症和心内膜炎在一项对接受达托霉素治疗革兰氏阳性导管相关血流感染的癌症患者（N = 40）的前瞻性研究中，48 小时症状缓解率（76% vs. 53%）和 48 小时时的微生物根除（78%对 34%）与匹配对照中的历史万古霉素治疗相比，使用达托霉素的患者更高。达托霉素的总反应率较高（68%对 32%），肾毒性的发生率较低。 治疗组在中性粒细胞减少症、并发症、不良事件、住院时间和死亡率方面具有可比性。

万古霉素或利奈唑胺应用于治疗机械通气患者的 MRSA 肺炎。FDA 发布了关于利奈唑胺的警告，指出其未被批准用于治疗导管相关感染、导管部位感染或革兰氏阴性菌感染。在一项开放标签随机化研究中，与接受万古霉素、苯唑西林或双氯西林治疗血管内导管相关感染的患者相比，接受利奈唑胺治疗的患者死亡的可能性更高：1）单独使用革兰氏阴性菌; 2）同时使用革兰氏阳性菌和革兰氏阴性菌;或 3）无感染。在单独感染革兰氏阳性菌的患者中，没有发现治疗的死亡率差异。

VRE 感染的最佳治疗尚未明确。利奈唑胺、替地唑胺和达托霉素已用于治疗 VRE 血流感染患者，取得了不同程度的成功。

应始终强烈考虑取出受感染的导管。在缺乏更确切数据的情况下，建议使用这些药物之一治疗 VRE 菌血症。

达托霉素对大多数革兰氏阳性病原体有效，但不应用于治疗肺炎，因为它被肺表面活性物质灭活。达托霉素适用于治疗由以下原因引起的复杂皮肤和皮肤结构感染：

特拉万星、头孢洛林、奥利万星和达巴万星已被批准用于治疗由革兰氏阳性病原体（包括 MRSA）引起的复杂皮肤和皮肤结构感染。头孢洛林还适用于治疗由敏感的革兰氏阴性和革兰氏阳性（MRSA 除外）病原体引起的社区获得性细菌性肺炎;该药物对粪肠球菌无活性。没有关于在肿瘤治疗中使用这些药物的指导性数据。

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setting. Therefore, these agents are not currently recommended as first-line therapy.

The Panel recommends that the use of linezolid, daptomycin, and tedizolid be limited to specific situations involving infections caused by documented vancomycin-resistant organisms, or for patients in whom vancomycin is not an option. Although studies have been published in patients with neutropenia, the NCCN Guidelines Panel strongly recommends that these agents not be used as routine empiric therapy for neutropenic fever because of concerns about the emergence of resistance and toxicity.

Initial Empiric Therapy for Patients Who Are Clinically Unstable

Sepsis is suggested by signs of clinical instability including hypotension, tachypnea, new or worsening tachycardia, mental status changes, decreased urine output, and organ dysfunction. Initial therapy for sepsis should broadly cover pathogens that are likely to cause sepsis while minimizing the potential for inadequate treatment.[428] Unlike the patients who are stable with neutropenic fever, modifying antibiotics based on culture data may not be possible for the patient with sepsis if the initial regimen does not provide adequate coverage. The antibiotic regimen should be modified, if necessary, after culture results and susceptibility are known.

The initial empiric regimen for neutropenia with clinical instability may include a broad-spectrum beta-lactam (eg, imipenem/cilastatin, meropenem, piperacillin-tazobactam) plus an aminoglycoside and vancomycin. Addition of fluconazole or an echinocandin should be strongly considered in patients not receiving antifungal prophylaxis. Local susceptibility patterns and recent antibiotic use should be taken into account when devising the antibiotic regimen.[428] In hospitals where infections by antibiotic-resistant bacteria (eg, MRSA or drug-resistant gram-negative rods) are commonly observed, policies on initial empiric therapy of neutropenic fever may need to be tailored accordingly. Some experts also suggest that patients who have a history of *P aeruginosa* colonization or of invasive disease should receive combination therapy with an antipseudomonal beta-lactam plus an aminoglycoside or ciprofloxacin.

For cases of septic shock, rapid interventions are needed. Fluid resuscitation, oxygen, invasive hemodynamic monitoring, and vasopressor agents may be required. Stress doses of hydrocortisone (IV 50 mg every 6 hours with or without fludrocortisone oral 50 mcg daily) have been associated with decreased mortality in patients with septic shock and with insufficient adrenal reserve.[464-468] Stress-dose corticosteroids are recommended for patients with septic shock who require vasopressor support.[428,469,470] High-dose corticosteroids have not shown any benefit in the setting of septic shock or severe sepsis, and may be associated with increased risks for secondary infections.[471-474]

Empiric Antifungal Therapy in Persistent Neutropenic Fever

Empiric antifungal therapy for persistent febrile neutropenia unresponsive to broad-spectrum antibacterial agents is initiated in neutropenia known to be at risk for invasive fungal infections, but who do not have early detection of those infections following clinical examination and collection of cultures.[6,475-478] Traditionally, empiric antifungal therapy is initiated after 4 or more days of empiric antibiotic therapy for fever and neutropenia, in patients who have remained febrile or who have recrudescent fever (see *Results of Daily Monitoring* in the algorithm). The timing to add empiric antifungal therapy varies with the risk of invasive mold infections, but generally ranges between 7 to 10 days of neutropenic fever despite empiric antibiotic therapy. In patients at high risk for mold infections (eg, neutropenia lasting >10 days, allogeneic HCT recipients, treatment with high-dose corticosteroids), the NCCN Guidelines Panel recommends adding empiric antifungal agents after 4 days unless the patient is receiving prophylaxis with mold-active agents. The concept of using

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设置.因此，目前不推荐这些药物作为一线治疗。

专家小组建议，利奈唑胺、达托霉素和替地唑胺的使用仅限于涉及由记录的万古霉素耐药微生物引起的感染的特定情况，或不选择万古霉素的患者。尽管已经发表了中性粒细胞减少症患者的研究，但 NCCN 指南小组强烈建议，由于担心出现耐药性和毒性，这些药物不应作为血小板减少性发热的常规经验性治疗。

临床不稳定患者的初始经验性治疗

败血症的临床不稳定体征包括低血压、呼吸急促、新发或恶化的心动过速、精神状态改变、尿量减少和器官功能障碍。脓毒症的初始治疗应广泛覆盖可能导致脓毒症的病原体，同时尽量减少治疗不充分的可能性。与稳定的脓毒症发热患者不同，如果初始治疗方案不能提供足够的覆盖率，则可能无法根据培养数据对脓毒症患者进行抗生素调整。如有必要，在了解培养结果和敏感性后，应修改抗生素治疗方案。临床不稳定性中性粒细胞减少症的初始经验性治疗方案可能包括广谱β–内酰胺（例如，亚胺培南/西司他丁、美罗培南、哌拉西林–他唑巴坦）加氨基糖苷类和万古霉素。对于未接受抗真菌预防治疗的患者，应强烈考虑加用氟康唑或棘白菌素。在设计抗生素治疗方案时，应考虑当地的敏感性模式和近期的抗生素使用情况。在常见耐药菌（如 MRSA 或耐药革兰氏阴性杆菌）感染的医院，可能需要相应地调整对血小板减少性发热的初始经验性治疗政策。一些

专家还建议有绿脓杆菌定植或侵袭性疾病史的患者应接受抗假单胞菌β–内酰胺加氨基糖苷类或环丙沙星的联合治疗。

对于感染性休克病例，需要快速干预。可能需要液体复苏、氧气、侵入性血流动力学监测和血管加压剂。氢化可的松的应激剂量（静脉注射 50 mg，每 6 小时一次，每日口服 50 mcg 或不口服 50 mcg 氟氢可的松）与脓毒性休克和肾上腺储备不足患者的死亡率降低有关。对于需要血管加压剂支持的脓毒性休克患者，推荐使用应激剂量的皮质类固醇。大剂量皮质类固醇在脓毒性休克或严重脓毒症的情况下未显示出任何益处，并且可能与继发感染的风险增加有关。

持续性中性粒细胞减少性发热的经验性抗真菌治疗

对广谱抗菌药物无反应的持续性发热性中性粒细胞减少症的经验性抗真菌治疗开始于已知有侵袭性真菌感染风险的中性粒细胞减少症，但在临床检查和收集培养物后没有早期发现这些感染。传统上，经验性抗真菌治疗在发热和中性粒细胞减少症的经验性抗生素治疗 4 天或更长时间后开始，持续发热或复发发热的患者（参见算法中的每日监测结果）。增加经验性抗真菌治疗的时机因侵袭性霉菌感染的风险而异，但尽管经验性抗生素治疗，但通常在 7 至 10 天的贫血性发热范围内。 对于霉菌感染高风险患者（例如，中性粒细胞减少持续>10 天，异基因 HCT 接受者，高剂量皮质类固醇治疗），NCCN 指南小组建议在 4 天后添加经验性抗真菌药物，除非患者正在接受霉菌活性药物预防。使用的概念

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empiric antifungal therapy was established in the 1970s and 1980s when about 20% of patients being treated for acute leukemia or undergoing HCT would develop an invasive fungal infection due to *Candida* or *Aspergillus* species by day 20 of neutropenia.[479] The toxicity of amphotericin B limited its use as routine prophylaxis, which would entail exposing more patients to a toxic drug over a prolonged period compared with empiric therapy. With the widespread use of fluconazole prophylaxis in the 1990s among patients at high risk of infection with acute leukemia and in HCT recipients, the incidence of invasive candidiasis in these patients decreased substantially, although breakthrough candidemia by fluconazole-resistant strains occurred.[69,182] Empiric antifungal therapy for neutropenic fever principally involved switching from fluconazole to amphotericin B to broaden the antifungal spectrum to include molds such as *Aspergillus*. Subsequently, L-AmB proved to be safer than and as effective as conventional amphotericin B for empiric antifungal therapy.[480]

Amphotericin B products are considered a category 2B recommendation for prophylaxis and empiric antifungal therapy for persistent or recurrent neutropenic fever of unknown etiology based on their toxicity and the availability of safer and equally effective alternative agents. In cases where there is a stronger clinical suspicion of mold infection than neutropenic fever alone (eg, a new pulmonary nodule in a patient with fever and prolonged neutropenia), use of an amphotericin B formulation (or a mold-active azole or an echinocandin) should be considered pending additional diagnostic evaluation. In general, lipid formulations of amphotericin B are preferred over the conventional formulation because they are less toxic.[481] This recommendation is stronger in patients with risk factors for acute renal failure, such as pre-existing renal disease, HCT recipients, and coadministration of nephrotoxic agents.[164,165,482]

Fluconazole has been used successfully as empiric therapy for neutropenic fever in patients not receiving prophylaxis but is limited by lack of activity against molds.[483,484] IV itraconazole followed by oral solution was as effective as, but less toxic than, conventional amphotericin B when used as empiric therapy in an open, randomized study.[485] These results led to FDA approval of oral itraconazole solution for this indication. IV itraconazole is no longer available in the United States. Itraconazole in the capsule formulation has erratic oral bioavailability and is therefore not suitable as empiric antifungal therapy. Additionally, the capsule formulation should be used with caution when concurrent with histamine H2-receptor antagonists and PPIs as these medications can reduce absorption of the itraconazole capsule. Itraconazole has negative inotropic effects and is contraindicated in patients with evidence of ventricular dysfunction or a history of congestive heart failure.

Voriconazole was compared with L-AmB in an open, randomized study of empiric antifungal therapy (N = 837 patients, 72% with hematologic malignancies).[486] The overall success rates for preventing invasive fungal infections were 26% with voriconazole and 31% with L-AmB. Empiric voriconazole was associated with fewer breakthrough fungal infections (1.9% vs. 5.0%; $P = .02$), with the greatest protective benefit occurring in patients who are prespecified as high risk (relapsed acute leukemia and allogeneic HCT). Because the noninferiority of voriconazole versus L-AmB was not demonstrated in this study based on prespecified criteria, voriconazole did not receive FDA approval for use as empiric therapy.[476,487] Voriconazole is an option (category 2B) for empiric therapy in patients at high risk for invasive mold infection.

Echinocandins are active against *Candida* and *Aspergillus* species but have unreliable activity against most other opportunistic fungi. Caspofungin was compared with L-AmB as empiric therapy for fungal infections in a randomized double-blind study in patients with persistent fever and neutropenia (N = 1095).[488] The overall success rates were 34% in both caspofungin and L-AmB recipients. The proportion of patients who

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经验性抗真菌治疗建立于 20 世纪 70 年代和 80 年代，当时约 20%的接受急性白血病治疗或进行 HCT 的患者在中性粒细胞减少症的第 20 天将发展为由于念珠菌或曲霉属物种的侵袭性真菌感染。白曲霉素 B 的毒性限制了其作为常规预防的用途，与经验性治疗相比，这将需要使更多的患者在延长的时间内暴露于有毒药物。随着 20 世纪 90 年代在急性白血病感染高危患者和 HCT 接受者中广泛使用氟康唑预防，这些患者中侵袭性念珠菌病的发病率大幅下降，虽然氟康唑引起的突破性念珠菌血症，经验性抗真菌治疗主要包括从氟康唑转换为阿替霉素 B，以扩大抗真菌谱，包括霉菌例如曲霉属。 随后，L–AmB 被证明是安全的，比传统的阿替霉素 B 有效的经验性抗真菌治疗。

两性霉素 B 产品被认为是第 2 类 B 产品，根据其毒性以及更安全和同等有效的替代药物的可用性，推荐用于病因不明的持续性或复发性血小板减少性发热的预防和经验性抗真菌治疗。如果临床上怀疑霉菌感染的可能性大于单纯的血小板减少性发热（例如，发热和持续性中性粒细胞减少的患者出现新的肺结节），则应考虑使用阿替霉素 B 制剂（或霉菌活性唑或棘白菌素），以等待额外的诊断评估。一般来说，阿替霉素 B 的脂质制剂优于常规制剂，因为它们毒性较低。对于有急性肾功能衰竭危险因素的患者，如既存肾脏疾病、HCT 接受者和同时使用肾毒性药物的患者，这一建议更为强烈。

氟康唑已成功地用于未接受预防性治疗的患者的血小板减少性发热的经验性治疗，但受到以下因素的限制：

缺乏抗霉菌的活性。在一项开放随机研究中，当用作经验性治疗时，IV 伊曲康唑随后口服溶液与传统的阿替霉素 B 一样有效，但毒性较低。这些结果导致 FDA 批准口服伊曲康唑溶液用于该适应症。IV 伊曲康唑在美国已不可用。胶囊制剂中的伊曲康唑口服生物利用度不稳定，因此不适合作为经验性抗真菌治疗。此外，胶囊当与组胺 H2 受体拮抗剂和 PPI 同时使用时，应谨慎使用该制剂，因为这些药物可减少伊曲康唑胶囊的吸收。伊曲康唑具有负性肌力作用，禁忌用于有心室功能障碍证据或充血性心力衰竭病史的患者。

在一项经验性抗真菌治疗的开放、随机研究中，比较了伏立康唑与 L–AmB（N = 837 例患者，72%为血液系统恶性肿瘤），伏立康唑预防侵袭性真菌感染的总成功率为 26%，L–AmB 为 31%。经验性伏立康唑与较少的突破性真菌感染相关（1.9% vs. 5.0%; P = 0.02），最大的保护性获益发生在预先指定为高风险（复发性急性白血病和同种异体 HCT）的患者中。因为根据预先规定的标准，伏立康唑与 L–AmB 的非劣效性在本研究中没有得到证明，伏立康唑没有获得 FDA 批准用作经验性治疗。伏立康唑是侵袭性霉菌感染高危患者经验性治疗的一种选择（2B 类）。

棘白菌素对念珠菌属和曲霉属有活性，但对大多数其他机会性真菌的活性不可靠。在一项随机双盲研究中，在持续发热和中性粒细胞减少的患者（N = 1095）中比较了卡泊芬净和 L–AmB 作为真菌感染的经验性治疗。卡泊芬净和 L–AmB 接受者的总成功率均为 34%。的患者比例

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survived at least 7 days after therapy was greater in the caspofungin group (92.6% vs. 89.2%; *P* = .05). The rates of breakthrough fungal infections and resolution of fever during neutropenia were similar between the two groups. Among patients with a baseline invasive fungal infection, the success rate was higher with caspofungin versus L-AmB (52% vs. 26%; *P* = .04) and the mortality rate was lower with caspofungin (11% vs. 44% with L-AmB).[488] Drug-related toxicities and premature withdrawals because of drug-related adverse events were significantly lower in caspofungin recipients. This study supports caspofungin as an option for empiric antifungal therapy. Caspofungin is approved for use as empiric treatment of presumed fungal infection in patients with fever and neutropenia. Micafungin was compared to voriconazole in a randomized, cooperative group, open-label trial as empiric antifungal therapy in patients with hematologic malignancy and febrile neutropenia. This study found no significant differences in clinical efficacy between the two therapies, although discontinuation due to drug-related adverse effects occurred less frequently in patients treated with micafungin.[489] Another echinocandin, anidulafungin, was shown in a 2021 retrospective study to have similar efficacy as caspofungin and micafungin for empiric antifungal therapy.[490] Moreover, a 2022 retrospective case-control study found that patients who were critically ill and treated with anidulafungin as empiric therapy had less incidence of invasive candidiasis (5.2% vs. 29.2%; *P* = .001), and lower 30-day all-cause mortality rate (10.4% vs. 19.4%; *P* = .04) compared to the control group that did not receive empiric antifungal therapy.[491] These studies thus support the use of micafungin and anidulafungin as additional options for empiric antifungal therapy.

Posaconazole and isavuconazonium sulfate can be considered for patients who have invasive, refractory infections or who have intolerance to amphotericin B formulations, although neither is FDA-approved for refractory infections. Both agents are, however, approved by the FDA as primary therapy for invasive fungal infections.

It is unclear whether patients who are already receiving mold-active prophylaxis should subsequently receive empiric antifungal therapy with an additional or different antifungal solely based on persistent neutropenic fever.[492] One approach has been to evaluate such patients with a high-resolution CT scan of the chest, in search of lesions suspicious for invasive fungal disease. CT scanning in this setting has not been validated but it is a reasonable approach, in concert with careful physical examination and blood cultures, in an effort to identify a source of persistent unexplained fever in patients with neutropenia. Laboratory markers (such as serum galactomannan and beta-glucan) have important limitations, including false-negative results in some patients already receiving prophylactic or empiric antifungals.[493,494] A meta-analysis showed the sensitivity of the galactomannan test for proven aspergillosis to be only 70% among patients with hematologic malignancies and 82% among HCT recipients.[495] However, these antigen-based assays have a high negative predictive value in the absence of mold-active antifungal therapy.

In patients undergoing chemotherapy for acute leukemias and receiving only yeast-active prophylaxis with fluconazole, 3% to 4% developed invasive fungal infections despite prophylaxis.[496,497] Empiric antifungal therapy with anti-mold activity would be expected to benefit these few patients without incurring a greater risk of toxicity.

Preemptive antifungal therapy uses characteristic changes in chest or sinus CT scans, laboratory markers, or both to trigger modification of the antifungal regimen, rather than providing empiric antifungals to all patients who are persistently febrile neutropenic. Maertens and colleagues[498] evaluated a preemptive strategy of incorporating L-AmB in patients at high risk with neutropenia (who received fluconazole prophylaxis) based on such prespecified triggers, including serially positive serum galactomannan tests, a bronchoalveolar lavage (BAL) showing mold,

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卡泊芬净组治疗后至少 7 天的存活率更高（92.6% vs. 89.2%; P = 0.05）。中性粒细胞减少期间，两组间的突破性真菌感染率和发热消退率相似。在基线侵袭性真菌感染的患者中，卡泊芬净的成功率高于 L–AmB（52% vs.

26%;卡泊芬净组的死亡率较低（11%对 44%）。卡泊芬净组的药物相关毒性和因药物相关不良事件而提前停药的发生率显著降低。本研究支持卡泊芬净作为经验性抗真菌治疗的一种选择。卡泊芬净被批准用于经验性治疗发热和中性粒细胞减少患者的假定真菌感染。在一项随机、合作组、开放标签试验中，将米卡芬净与伏立康唑作为血液系统恶性肿瘤伴发热性中性粒细胞减少患者的经验性抗真菌治疗进行了比较。本研究发现两种疗法的临床疗效无显著差异，尽管米卡芬净治疗患者因药物相关不良反应而停药的发生率较低，另一种棘白菌素阿尼芬净在 2021 年的回顾性研究中显示，经验性抗真菌治疗的疗效与卡泊芬净和米卡芬净相似。 而且，一项 2022 年的回顾性病例对照研究发现，重症患者使用阿尼芬净作为经验性治疗，侵袭性念珠菌病的发生率更低（5.2% vs. 29.2%; P = .001），30 天全因死亡率更低（10.4% vs. 19.4%; P = 0.04）。因此，这些研究支持使用米卡芬净和阿尼芬净作为经验性抗真菌治疗的额外选择。

泊沙康唑和硫酸艾沙康唑可用于侵袭性、难治性感染或对阿替霉素 B 制剂不耐受的患者，尽管这两种药物均未被 FDA 批准用于难治性感染。然而，这两种药物都被 FDA 批准作为侵袭性真菌感染的主要治疗方法。

目前尚不清楚是否已经接受霉菌活性预防的患者应随后接受经验性抗真菌治疗，根据持续性真菌减少性 fever.One 方法是评估这些患者的胸部高分辨率 CT 扫描，寻找可疑的病变侵袭性真菌病的基础上额外或不同的抗真菌治疗。CT 扫描在这种情况下尚未得到验证，但它是一种合理的方法，与仔细的体格检查和血培养，努力确定中性粒细胞减少症患者持续不明原因发热的来源。 实验室标志物（如血清半乳甘露聚糖和β–葡聚糖）具有重要的局限性，包括在一些已经接受预防性或经验性抗真菌治疗的患者中出现假阴性结果。一项荟萃分析显示，半乳甘露聚糖检测确诊曲霉病的敏感性在恶性血液病患者中仅为 70%，在 HCT 接受者中为 82%。这些基于抗原的测定在没有霉菌活性抗真菌治疗的情况下具有高的阴性预测值。

在急性白血病化疗患者中，仅接受氟康唑酵母活性预防治疗的患者中，尽管有预防治疗，仍有 3%~ 4%的患者发生侵袭性真菌感染，具有抗霉菌活性的经验性抗真菌治疗有望使这少数患者受益，而不会产生更大的毒性风险。

先发制人的抗真菌治疗利用胸部或鼻窦 CT 扫描、实验室标志物或两者的特征性变化来触发抗真菌治疗方案的修改，而不是向所有持续发热性血小板减少的患者提供经验性抗真菌药物。Maertens 及其同事基于预先指定的触发因素，包括血清半乳甘露聚糖试验连续阳性，支气管肺泡灌洗（BAL）显示霉菌，

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and/or suggestive chest CT in patients with persistent fever or with signs of invasive fungal infection. A total of 136 treatment episodes (among 88 patients) were evaluated. Among these, neutropenic fever developed in 117 cases, of which 35% would have met the existing criteria for empiric antifungal therapy. Using the preemptive strategy, antifungal therapy was given to 7.7% (9 of 117 episodes of neutropenic fever) of patients rather than up to one third of patients who might have received it on the basis of fever alone. In addition, seropositivity for galactomannan led to early initiation of antifungal therapy in 10 non-febrile episodes. This approach detected all cases of invasive aspergillosis but missed 1 case of invasive fungal infection that involved disseminated mucormycosis (previously referred to as "zygomycosis") resulting in death. Two cases of breakthrough candidemia were detected by conventional culture methods and successfully treated.[498] In another study by Maertens and colleagues comparing preemptive versus empiric antifungal therapy in patients with neutropenia at high risk and on fluconazole prophylaxis, preemptive antifungal therapy was found to be non-inferior to empiric therapy in terms of overall survival, halving the number of patients receiving antifungals without excess mortality or invasive fungal diseases.[499] In a randomized trial of patients with neutropenic fever, a preemptive strategy was associated with an increased incidence of probable or proven invasive fungal infections (9% vs. 3% in empirically treated group; $P < .05$), although without an increase in overall mortality and ultimately with a decreased cost of antifungal drugs compared to empiric therapy.[500] A 2022 systematic study of 7 randomized controlled trials comparing preemptive antifungal therapy with empiric antifungal therapy found that in patients with cancer at high risk of febrile neutropenia, preemptive antifungal therapy may reduce the duration and rate of use of antifungal agents compared to empiric therapy, without increasing overall and invasive fungal disease-related mortality; but the evidence regarding invasive fungal infection detection and adverse events was inconsistent and uncertain.[501] Taken together, the Panel considers the evidence supporting preemptive antifungal therapy to be insufficient to support its routine use.

Follow-up of Patients with Neutropenic Fever

Daily evaluation by a health care professional who is experienced in treating patients with fever and neutropenia is essential. The daily examination should focus on a site-specific assessment, and an infectious disease consultation should be considered for all complicated cases or progressive infections. Daily follow-up should include an evaluation of response to empiric antimicrobial therapy, both in terms of fever trends and changes in signs and/or symptoms of infections. Time to defervescence ranges from 2 to 7 days (median, 5 days) for febrile patients with cancer with neutropenia who receive appropriate initial antibiotic therapy.[502] This rate of fever response should be considered when assessing the need to adjust initial antibiotics; random additions or changes for persistent fever are discouraged in the absence of clinical or microbiologic evidence. The expected slow defervescence of fever also complicates decisions regarding the need for repeat blood cultures. Although some experts recommend daily blood cultures until the patient becomes afebrile, increasing evidence suggests that daily blood cultures are unnecessary in stable neutropenia with persistent fever of unknown etiology.[503] As part of follow-up, patients should also be evaluated for potential drug toxicities by liver and kidney function tests (generally conducted at least twice weekly).

Current bacterial blood culture systems (such as the BACTEC continuous-monitoring culture system) can detect 90% to 100% of bacterial bloodstream pathogens within 48 hours of culture. For this reason, routine ordering of additional cultures before obtaining the results from the initial series is discouraged. Daily review of previously obtained cultures is critical, and the Panel recommends documenting the clearance of bloodstream bacterial or fungal infections with repeat blood cultures.

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和/或提示性胸部 CT 检查。共评价了 136 次治疗事件（88 例患者）。在这些病例中，117 例发生了血小板减少性发热，其中 35%符合经验性抗真菌治疗的现有标准。使用先发制人的策略，抗真菌治疗给予 7.7%的患者（9/117 次血小板减少性发热），而不是高达三分之一的患者可能接受它的基础上发热。此外，半乳甘露聚糖的血清阳性导致早期

initiation of antifungal therapy in 10 non–febrile episodes. This approach detected all cases of invasive aspergillosis but missed 1 case of invasive fungal infection that involved disseminated mucormycosis (previously referred to as “zygomycosis”) resulting in death. Two cases of breakthrough candidemia were detected by conventional culture methods and successfully treated.In another study by Maertens and colleagues comparing preemptive versus empiric antifungal therapy in patients with neutropenia at high risk and on fluconazole prophylaxis, preemptive antifungal therapy was found to be non–inferior to empiric therapy in terms of overall survival, halving the number of patients receiving antifungals without excess mortality or invasive fungal diseases.In a randomized trial of patients with neutropenic fever, a preemptive strategy was associated with an increased incidence of probable or proven invasive fungal infections (9% vs. 3% in empirically treated group; P < .05), although without an increase in overall mortality and ultimately with a decreased cost of antifungal drugs compared to empiric therapy.A 2022 systematic study of 7 randomized controlled trials comparing preemptive antifungal therapy with empiric antifungal therapy found that in patients with cancer at high risk of febrile neutropenia, preemptive antifungal therapy may reduce the duration and rate of use of antifungal agents compared to empiric therapy, without increasing overall and invasive fungal disease–related mortality; but the evidence regarding invasive fungal infection detection and adverse events was inconsistent and

重试 错误原因

uncertain.Taken together, the Panel considers the evidence supporting preemptive antifungal therapy to be insufficient to support its routine use.

重试 错误原因

Follow–up of Patients with Neutropenic Fever

重试 错误原因

由具有治疗发热和中性粒细胞减少患者经验的医疗保健专业人员进行每日评估至关重要。日常检查应侧重于特定部位的评估，对于所有复杂病例或进行性感染，应考虑进行传染病咨询。每日随访应包括评价经验性抗菌治疗的反应，包括发热趋势和感染体征和/或症状的变化。对于接受适当初始抗生素治疗的发热伴中性粒细胞减少的癌症患者，退热时间范围为 2 – 7 天（中位数为 5 天）。在评估是否需要调整初始抗生素时，应考虑此发热反应率;在缺乏临床或微生物学证据的情况下，不鼓励随机增加或改变持续发热。预期的发热缓慢退热也使关于是否需要重复血培养的决定复杂化。 虽然有些专家建议每天进行血培养，直到患者不再发热，但越来越多的证据表明，对于稳定的中性粒细胞减少伴不明原因持续发热的患者，每天进行血培养是不必要的。作为随访的一部分，患者还应通过肝肾功能检查（通常每周至少进行两次）评估潜在的药物毒性。

目前的细菌血液培养系统（如 BACTEC 连续监测培养系统）可以在培养 48 小时内检测 90%至 100%的细菌血流病原体。因此，不鼓励在获得初始系列的结果之前常规订购额外的培养物。每日检查先前获得的培养物至关重要，专家组建议通过重复血培养记录血流细菌或真菌感染的清除情况。

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The overall response to initial empiric antimicrobial therapy should be evaluated 3 to 5 days from initiation of empiric therapy.

Follow-up Therapy in Responding, Clinically Stable Patients

Patients who have infections that respond to empiric therapy should exhibit decreasing fever trends, show stable or improving signs and symptoms of infection, and be hemodynamically stable. For these patients, no change is needed to the initial empiric regimen, and if patients were started appropriately on an agent for Gram-positive resistant infections, they should continue with the course of therapy. If patients received an agent for Gram-positive–resistant infections as part of their initial empiric therapy, but they do not have a pathogen recovered or a site of infection identified justifying such treatment, then treatment should be discontinued. Similarly, the appropriateness of empiric Gram-negative therapy should be reassessed. It is generally recommended that antibiotics be continued until the ANC is ≥500 cells/mcL, and is increasing (see *Results of Daily Monitoring* in the algorithm).

Patients with fever of unknown origin who become afebrile soon after starting empiric therapy may have empiric antibiotics discontinued with ANC recovery (ANC ≥500 neutrophils/mcL) as long as the neutrophil count is likely to continue to increase (patients are often receiving a growth factor). This recommendation assumes that the patient is clinically well and afebrile for at least 24 hours before antibiotic discontinuation. Patients who become afebrile but remain persistently neutropenic (ANC <500 neutrophils/mcL) should receive a more prolonged course of antibiotic therapy until the neutropenia resolves, although de-escalation to prophylactic antibiotics should be considered[504] (see *Results of Daily Monitoring* in the algorithm). In patients who defervesce for at least 48 hours, it may be appropriate in some cases to de-escalate to fluoroquinolone. Patients at lower risk can also be switched to oral antibiotics until their neutropenia resolves (eg, 500 mg ciprofloxacin every 8 hours plus 500 mg of amoxicillin/potassium clavulanate every 8 hours).

Follow-up Therapy in Persistently Febrile but Otherwise Hemodynamically Stable Patients

Patients with recurrent fever should be reassessed promptly to determine the need for either a change in their antibiotic regimen or for the addition of antifungal therapy. A hemodynamically stable patient with persistent fever of unknown etiology may be safely watched without altering the initial antimicrobial therapy. Modifications of initial empiric antibiotic therapy should be based on specific new clinical findings and/or new microbiologic results; fever alone should not prompt changes in antimicrobial therapy. The exception is consideration of empiric antifungal therapy in patients who have persistent or recurrent fever after 4 to 7 days of empiric antibacterial therapy and who are not receiving mold-active prophylaxis (see *Results of Daily Monitoring* in the algorithm). In patients at high risk for mold infection (ie, neutropenia >10 days, allogeneic HCT recipients, high-dose corticosteroids), the Panel recommends adding empiric antifungal therapy after the fourth day unless the patient is receiving prophylaxis directed against molds. Documented infections are usually treated according to the infection site and pathogen, and at least until ANC recovery (see *Site-Specific Evaluation and Treatment of Infections*).

Follow-up Therapy in Non-responding, Clinically Unstable Patients

Although fever resolution may be slow during neutropenia, persistent fever may result from a noninfectious etiology, such as drug-induced fever. Persistent fever may also represent an inadequately treated infectious process, such as a nonbacterial infection (fungal or viral), a bacterial infection that is resistant to empiric antibiotics, a venous access or closed space infection, or inadequate antimicrobial serum levels. It is important to recognize that documented deep tissue infections may take longer than fever of unknown etiology to respond to antimicrobial therapy. In these cases, daily assessment of clinical improvement or failure depends on

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应在开始经验性抗菌治疗后 3 – 5 天评价对初始经验性抗菌治疗的总体反应。

缓解、临床稳定患者的随访治疗

对经验性治疗有反应的感染患者应表现出发热下降趋势，表现出稳定或改善的感染体征和症状，并且血流动力学稳定。对于这些患者，不需要改变初始经验方案，如果患者开始适当使用革兰氏阳性耐药感染的药物，则应继续治疗。如果患者接受了革兰氏阳性耐药感染的药物作为其初始经验性治疗的一部分，但他们没有恢复病原体或确定感染部位以证明这种治疗是合理的，则应停止治疗。同样，经验性革兰氏阴性菌治疗的适当性也应重新评估。通常建议继续使用抗生素，直至 ANC ⩾500 个细胞/mcL，并且正在增加

(see 算法中的每日监测结果)。

对于开始经验性治疗后不久即无发热的不明原因发热患者，只要中性粒细胞计数可能继续升高（患者通常接受生长因子治疗），则可在 ANC 恢复（ANC ⩾500 中性粒细胞/mcL）后停用经验性抗生素。该建议假定患者在停用抗生素前至少 24 小时临床状况良好且无发热。无发热但持续性中性粒细胞减少（ANC <500 中性粒细胞/mcL）的患者应接受更长时间的抗生素治疗，直至中性粒细胞减少消退，但应考虑降级至预防性抗生素（见算法中的每日监测结果)。在退热至少 48 小时的患者中，在某些情况下可能适合降级为氟喹诺酮。风险较低的患者也可以改用口服抗生素直至中性粒细胞减少症消退（例如，每 8 小时 500 mg 环丙沙星加每 8 小时 500 mg 阿莫西林/克拉维酸钾)。

持续发热但血流动力学稳定患者的随访治疗

对于反复发热的患者，应及时重新评估，以确定是否需要改变抗生素治疗方案或增加抗真菌治疗。一个血流动力学稳定的病人持续发热的原因不明，可以安全地观察，而不改变最初的抗菌治疗。初始经验性抗生素治疗的调整应基于特定的新的临床发现和/或新的微生物结果;发热不应单独促使抗生素治疗的变化。例外情况是经验性抗真菌治疗 4 – 7 天后持续发热或反复发热且未接受霉菌活性预防的患者需要考虑经验性抗真菌治疗（见算法中的每日监测结果)。 对于霉菌感染高风险患者（即中性粒细胞减少>10 天，同种异体 HCT 接受者，高剂量皮质类固醇)，专家组建议在第 4 天后增加经验性抗真菌治疗，除非患者正在接受针对霉菌的预防治疗。记录的感染通常根据感染部位和病原体进行治疗，至少直到 ANC 恢复（见感染的部位特异性评估和治疗)。

无应答、临床不稳定患者的随访治疗

尽管中性粒细胞减少时发热消退缓慢，但持续发热可能由非感染性病因引起，如药物引起的发热。持续性发热也可能代表治疗不充分的感染过程，如非细菌感染（真菌或病毒)、对经验性抗生素耐药的细菌感染、静脉通路或封闭空间感染或抗微生物血清水平不足。重要的是要认识到，记录的深部组织感染可能需要更长的时间比不明原因的发热抗菌治疗作出反应。在这些情况下，临床改善或失败的日常评估取决于

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radiographic, culture, and clinical examination data, and on the fever trends. Unusual infections (eg, toxoplasmosis) may complicate neutropenia, particularly if immunosuppressive agents (eg, high-dose corticosteroids) are also used. The Panel strongly recommends an infectious disease consultation for these patients.

Patients who remain persistently or intermittently febrile, show no improvement in signs/symptoms of infections, have persistent positive blood cultures, and/or may be hemodynamically unstable should be considered non-responsive to initial empiric antimicrobial therapy. These patients pose a serious management challenge and are at increased risk of infection-associated morbidity and mortality. For such patients, antimicrobial coverage should be broadened to include anaerobes, resistant gram-negative rods, and resistant gram-positive organisms, as clinically indicated. Antifungal therapy with activity against molds may be considered for patients with fever continuing for 4 or more days following initiation of empiric antibiotic therapy (see *Results of Daily Monitoring* in the algorithm). The lack of response may suggest an infection with a pathogen resistant to the antimicrobial therapy being used, inadequate serum or tissue levels of the antibiotic(s), infection at a vascular site (ie, catheter or "closed space" infection), or emergence of a second infection. Some documented infections do not respond to appropriate therapy because of associated profound neutropenia. If possible, treatment should be optimized using broad-spectrum antibiotic combinations that minimize other organ toxicity.

Both NCCN and ASCO[505] have guidelines for the use of prophylactic colony-stimulating factors (CSFs) in neutropenia (see NCCN Guidelines for Hematopoietic Growth Factors). It is not clear whether these agents are useful as adjunctive therapy for established infectious events. Although the data supporting their use are limited, adjunctive therapy with G-CSF or granulocyte-macrophage CSF (GM-CSF) should be considered (category 2B) in patients with neutropenia with serious infectious complications such as pneumonia, invasive fungal infections, or any type of progressive infection.

De-escalation and Duration of Therapy for Patients with Documented Infections

Targeted treatment of documented infections should be continued for patients whose infections are responding to therapy. The need to continue empiric Gram-negative therapy may be reassessed in these patients, discontinuing Gram-negative therapy if appropriate.[506] The duration and de-escalation of antimicrobial therapy is dictated by the: 1) underlying site of infection; 2) causative organism(s); and 3) patient's clinical condition, response to treatment, and neutrophil recovery (see *Follow-up Therapy for Responding Disease* in the algorithm). For example, most skin and soft tissue infections can be treated with 5 to 14 days of therapy. For most bacterial bloodstream infections, 7 to 14 days of therapy is usually adequate, with longer durations (10–14 days) recommended for more complicated bacteremias. For all *S aureus* bloodstream infections, treatment should be continued for at least 4 weeks after documentation of a first negative blood culture, although some institutions may use a shorter duration based on infectious disease consultation. In cases of endovascular involvement, treatment may need to be prolonged. Treatment for bloodstream infections caused by yeast should be continued for at least 2 weeks after the first negative blood culture is obtained. Catheter removal is recommended for septic phlebitis, tunnel infection, or port pocket infection and if bloodstream infection is caused by *Candida, S aureus, P aeruginosa, Corynebacterium jeikeium, Acinetobacter, Bacillus* organisms, atypical mycobacteria, yeasts, molds, VRE, *Stenotrophomonas maltophilia*, and other MDROs. A duration of treatment lasting 7 to 14 days is usually indicated for infections of the lungs (eg, bacterial pneumonia) or sinuses.[507] Complex intra-abdominal infections, such as typhlitis, should be treated until all evidence of infection has

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放射学、培养和临床检查数据，以及发热趋势。异常感染（如弓形虫病）可能使中性粒细胞减少症复杂化，特别是如果同时使用免疫抑制剂（如大剂量皮质类固醇）。专家小组强烈建议对这些病人进行传染病咨询。

持续或间歇性发热、感染体征/症状没有改善、血培养持续阳性和/或血流动力学不稳定的患者应被视为对初始经验性抗菌治疗无反应。这些患者构成了严重的管理挑战，感染相关的发病率和死亡率的风险增加。对于此类患者，应根据临床指征扩大抗菌药物覆盖范围，包括厌氧菌、耐药革兰氏阴性杆菌和耐药革兰氏阳性微生物。对于开始经验性抗生素治疗后发热持续 4 天或以上的患者，可考虑使用具有抗霉菌活性的抗真菌治疗（见算法中的每日监测结果）。缺乏反应可能表明感染了

病原体对正在使用的抗菌治疗耐药、血清或组织中抗生素水平不足、血管部位感染（即导管或“封闭空间”感染）或出现二次感染。一些记录的感染对适当的治疗没有反应，因为相关的严重中性粒细胞减少。如果可能，应使用广谱抗生素组合优化治疗，以最大限度地减少其他器官毒性。

NCCN 和 ASCO 都有使用预防性集落刺激因子（CSF）治疗中性粒细胞减少症的指南（见 NCCN 造血生长因子指南）。目前尚不清楚这些药物是否可用于已确定的感染事件的连续治疗。虽然支持其应用的数据有限，但应考虑使用 G–CSF 或粒细胞–巨噬细胞 CSF（GM–CSF）进行连续治疗

（2B 类）中性粒细胞减少伴严重感染并发症（如肺炎、侵袭性真菌感染或任何类型的进行性感染）的患者。

记录感染患者的治疗降级和持续时间

对于感染对治疗有反应的患者，应继续对记录的感染进行靶向治疗。这些患者可能需要重新评估是否需要继续经验性革兰氏阴性菌治疗，如果合适，则停止革兰氏阴性菌治疗。抗菌药物治疗的持续时间和减量取决于：1）潜在感染部位; 2）病原微生物; 3）患者的临床状况，治疗反应和中性粒细胞恢复（参见算法中的疾病反应随访治疗）。例如，大多数皮肤和软组织感染可以用 5 至 14 天的治疗来治疗。对于大多数细菌性血流感染，7 至 14 天的治疗通常是足够的，对于更复杂的细菌性疾病，建议使用更长的持续时间（10–14 天）。 对于所有金黄色葡萄球菌血流感染，治疗应在首次血培养阴性后至少持续 4 周，尽管有些机构可能会根据传染病咨询使用较短的持续时间。在血管内受累的情况下，可能需要延长治疗时间。由酵母菌引起的血流感染的治疗应在获得首次阴性血培养后至少持续 2 周。对于脓毒性静脉炎、隧道感染或输液港袋感染，以及由念珠菌、金黄色葡萄球菌、铜绿假单胞菌、杰氏棒状杆菌、不动杆菌、芽孢杆菌、非典型分枝杆菌、酵母菌、霉菌、VRE、嗜麦芽窄食单胞菌和其他 MDRO 引起的血流感染，建议拔除导管。 肺部感染（如细菌性肺炎）或鼻窦感染通常需要持续 7 – 14 天的治疗。复杂的腹腔内感染，如盲肠炎，应治疗至所有感染迹象消失。

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resolved and the patient has recovered from neutropenia. For fungal infections with *Candida*, treatment should be continued for at least 2 weeks after documentation of a first negative blood culture. Invasive mold infections (eg, aspergillosis) generally require treatment for a minimum of 12 weeks.

The duration of treatment for HSV (uncomplicated, localized disease to the skin) and VZV (uncomplicated, localized disease to a single dermatome) infections is typically 7 to 10 days (category 1). Life-threatening infections, such as invasive fungi or CMV, require individualized courses of therapy that are often prolonged. The duration of anti-infective therapy may need to be extended if further chemotherapy is required while treating a significant infection. This may occur with infections that complicate leukemia or lymphoma treatments in which multiple cycles of intensive chemotherapy are required.

In patients with influenza, oseltamivir is approved for a minimum duration of 5 days in ambulatory patients who are otherwise healthy individuals with intact immune systems. A longer course of treatment (eg, at least 10 days) or higher dose (eg, 150 mg) that continues until resolution of symptoms should be considered in the highly immunocompromised.

Patients with documented infections who become afebrile after the initiation of the empiric antibiotic regimen and who are at low risk for complications associated with infection may be candidates for outpatient antibiotic therapy. The regimen, whether oral or IV, should be appropriate for neutropenic fever and have activity against the specific infection.

Development of Clinical Instability While Receiving Antibacterial Therapy

It is essential to recognize the early signs of breakthrough infections after the initiation of antibacterial therapy. Although persistent neutropenic fever alone is not an indication to modify the antibacterial regimen, signs of breakthrough infection should prompt additional evaluation and consideration of therapy modification.

New findings suggestive of sepsis (eg, hypotension, tachycardia, mental status changes, organ dysfunction) require the following: 1) repeat physical examination to identify the source of infection; 2) repeat blood cultures; 3) consideration of radiologic studies; and 4) empiric modification of antimicrobial therapy pending culture results.[428] Information about previous use of antibiotics and local sensitivity patterns of gram-negative pathogens should guide empiric changes. Empiric addition of vancomycin is warranted in patients who are unstable. In patients receiving ceftazidime, the possibility of breakthrough infections (either from extended spectrum beta-lactamase–producing or from cephalosporinase-producing gram-negative rods) should be considered and switching to imipenem/cilastatin or meropenem is appropriate pending culture results. *Stenotrophomonas maltophilia* or carbapenem-resistant *P aeruginosa* may cause breakthrough sepsis in patients receiving imipenem/cilastatin or meropenem; consider empiric modification to a regimen containing piperacillin-tazobactam, an aminoglycoside, and TMP/SMX. In patients not receiving a systemic antifungal agent, addition of fluconazole or an echinocandin should be strongly considered for possible candidemia. The antibiotic regimen should then be tailored based on culture and radiologic results.

Site-Specific Evaluation and Treatment of Infections

The NCCN Guidelines provide recommendations for site-specific evaluation and therapy for infections of the mouth and esophagus, sinuses, liver, abdomen, rectum, vascular access sites, lungs, skin/soft tissue, urinary tract, and CNS. This section is tailored to patients with neutropenia or those who are otherwise significantly immunocompromised (eg, HCT recipients).

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已消退，患者已从中性粒细胞减少症中恢复。对于念珠菌属真菌感染，在首次血培养阴性后应继续治疗至少 2 周。侵袭性霉菌感染（如曲霉病）通常需要至少 12 周的治疗。

HSV（单纯性局部皮肤病）和 VZV（单纯性局部皮肤病）感染的治疗持续时间通常为 7 至 10 天（第 1 类）。危及生命的感染，如侵入性真菌或 CMV，需要个性化的治疗过程，往往是延长。如果在治疗严重感染时需要进一步化疗，则抗感染治疗的持续时间可能需要延长。这种情况可能发生在白血病或淋巴瘤治疗复杂化的感染中，需要多个周期的强化化疗。

在流感患者中，奥司他韦被批准用于非卧床患者，至少持续 5 天，这些患者是免疫系统完整的健康个体。在高度免疫功能低下的患者中，应考虑使用更长的疗程（例如，至少 10 天）或更高的剂量（例如，150 mg），直至症状消退。

有记录的感染患者在开始经验性抗生素治疗方案后不再发热，并且感染相关并发症的风险较低，可能适合门诊抗生素治疗。该方案，无论是口服或静脉注射，应适用于贫血热，并对特定的感染有活性。

抗菌药物治疗中临床不稳定性的发展

在开始抗菌治疗后，识别突破性感染的早期体征至关重要。尽管持续性血小板减少性发热本身并不是修改抗菌治疗方案的指征，但突破性感染应促使额外的评估和治疗调整的考虑。

提示败血症的新发现（如低血压、心动过速、精神状态改变、器官功能障碍）需要：1）重复体格检查以确定感染源; 2）重复血培养; 3）考虑放射学检查; 4）在细菌培养结果出来之前，根据经验调整抗菌药物治疗。阴性病原体应指导经验性变化。对于病情不稳定的患者，经验性加用万古霉素是必要的。在接受头孢他啶治疗的患者中，应考虑突破性感染的可能性（来自产超广谱β–内酰胺酶或产头孢菌素酶的革兰氏阴性杆菌），在等待培养结果期间，改用亚胺培南/西司他丁或美罗培南是适当的。 嗜麦芽窄食单胞菌或碳青霉烯类耐药铜绿假单胞菌可能导致接受亚胺培南/西司他丁或美罗培南治疗的患者发生突破性脓毒症;考虑经验性修改含哌拉西林–他唑巴坦、氨基糖苷类和 TMP/SMX 的方案。对于未接受全身性抗真菌药物治疗的患者，应强烈考虑加用氟康唑或棘白菌素治疗可能的念珠菌血症。抗生素方案应根据培养和放射学结果进行调整。

感染的部位特异性评价和治疗

NCCN 指南提供了针对口腔和食管、鼻窦、肝脏、腹部、直肠、血管穿刺部位、肺部、皮肤/软组织、尿路和 CNS 感染的部位特异性评价和治疗建议。本节适用于中性粒细胞减少症患者或其他显著免疫功能低下的患者（例如，HCT 接受者）。

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Mouth and Esophageal Infections

The mouth and esophagus are common sites of infection in patients with fever and neutropenia. This site predilection occurs because of the propensity of the mouth and alimentary tract mucosa to be disrupted by cytotoxic therapy, which can cause mucositis. Unfortunately, the characteristics of this disruption are not etiology specific, and important viral and fungal pathogens are often only distinguished by microbiologic culture. Empiric antibiotic therapy must consider the endogenous anaerobic flora and the shift in oral flora, which occur with serious illness or antibiotic use. The increased frequency of HSV reactivation and severity of these infections in patients with cancer are well known and preventable. The incidence of HSV reactivation in patients who are immunocompromised may approach 50% to 75%, but it is nearly zero in those who receive prophylaxis with appropriate antiviral agents.[508] HSV infections are associated with more extensive mucosal damage, increased secondary infections, and significantly prolonged healing time. Baglin et al[509] reported that patients with fever and neutropenia who experienced concomitant HSV reactivation and were treated with appropriate antiviral therapy had a significant decrease in the number of days with fever.[509] Ulcerations of the oral mucosa may be due to HSV infections or fungal sources. A culture should be obtained to determine the pathogenic organism, and addition of antiviral or systemic antifungal therapy should be considered, pending results. Vesicular lesions are most often caused by herpes virus infections and should be treated with antivirals pending culture (or other diagnostic assay) results (see *Initial Clinical Presentation: Mouth/Mucosal Membrane* in the algorithm).

Systemic or topical antifungal agents can be used to treat thrush. Because of the risk of candidemia, systemic antifungal therapy is advised in neutropenia. Fluconazole is recommended as first-line therapy for thrush (see *Initial Clinical Presentation: Mouth/Mucosal Membrane* in the algorithm). If the infection does not respond, the dose of fluconazole can be increased up to 800 mg daily (in adults with normal renal function).[510] Although cross-resistance among azoles may occur, oral voriconazole or posaconazole are reasonable oral options for thrush that is refractory to fluconazole. Echinocandins can be used for patients with azole-refractory mucosal candidiasis. Though amphotericin B formulations are also effective, they are not recommended because of toxicity.

Thrush along with retrosternal burning, chronic nausea, or odynophagia should raise suspicion for *Candida* esophagitis. However, *Candida* esophagitis may occur in the absence of oral thrush, especially in patients receiving oral topical antifungal agents. Definitive diagnosis of esophageal candidiasis is made by endoscopy. Empiric systemic antifungal therapy is often used to treat presumed *Candida* esophagitis.

The presence of thrush favors esophageal candidiasis in patients with symptoms compatible with esophagitis, although the symptoms of HSV and *Candida* esophagitis are similar. Other causes of esophagitis (eg, radiation esophagitis, GVHD of the esophagus or stomach) also produce similar symptoms. A trial of fluconazole and/or acyclovir should be considered in patients with neutropenia and in other persons who are highly immunocompromised with symptoms that suggest esophagitis. CMV esophagitis is a rare complication of chemotherapy-induced neutropenia and is most commonly observed in allogeneic HCT recipients with GVHD. Negative CMV surveillance results from PCR studies would make CMV disease very unlikely. If CMV esophagitis is diagnosed, treatment with valganciclovir or ganciclovir should be initiated. Foscarnet or cidofovir should be reserved for ganciclovir-resistant CMV or for patients who cannot tolerate ganciclovir. Empiric treatment may be considered in patients at high risk for CMV disease with symptoms suggestive of esophagitis.

For patients with esophagitis that does not respond to empiric therapy with these agents, careful upper endoscopy with platelet support (if required)

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口腔和食管感染

口腔和食管是发热和中性粒细胞减少患者的常见感染部位。由于口腔和消化道粘膜易于被细胞毒性治疗破坏，这可能导致粘膜炎，因此发生该部位偏好。不幸的是，这种破坏的特征不是病原学特异性的，并且重要的病毒和真菌病原体通常仅通过微生物培养来区分。经验性抗生素治疗必须考虑内源性厌氧植物群和口腔内植物群的变化，这在严重疾病或抗生素使用时发生。癌症患者中 HSV 再激活的频率和这些感染的严重程度增加是众所周知的，并且是可以预防的。 免疫功能低下的患者中 HSV 再激活的发生率可能接近 50%至 75%，但在接受适当抗病毒药物预防的患者中几乎为零。HSV 感染与更广泛的粘膜损伤、继发感染增加和愈合时间显著延长相关。Baglin et al.报告称，伴有发热和中性粒细胞减少的患者，如果同时发生 HSV 再激活，并接受适当的抗病毒治疗，发热天数显著减少。口腔粘膜溃疡可能是由于 HSV 感染或真菌来源。应进行培养以确定致病微生物，并应考虑增加抗病毒或全身抗真菌治疗，等待结果。 水疱性病变最常由疱疹病毒感染引起，在等待培养（或其他诊断检测）结果期间应使用抗病毒药物治疗（参见算法中的初始临床表现：口腔/粘膜）。

全身或局部抗真菌剂可用于治疗鹅口疮。由于念珠菌血症的风险，建议对中性粒细胞减少症患者进行系统性抗真菌治疗。推荐氟康唑作为鹅口疮的一线治疗

(see 初始临床表现：口腔/粘膜
算法)。如果感染没有反应，氟康唑的剂量可以每日增加至 800 mg（肾功能正常的成人）。虽然唑类药物之间可能发生交叉耐药，但口服伏立康唑或泊沙康唑是治疗氟康唑难治性鹅口疮的合理口服选择。棘白菌素可用于唑类难治性粘膜念珠菌病患者。虽然阿替霉素 B 制剂也有效，但由于毒性，不推荐使用。

鹅口疮沿着胸骨后烧灼感、慢性恶心或吞咽痛应怀疑念珠菌性食管炎。然而，念珠菌性食管炎可能发生在没有口腔鹅口疮的情况下，特别是在接受口服局部抗真菌药物的患者中。食管念珠菌病的确诊是通过内窥镜检查。经验性全身抗真菌治疗常用于治疗假丝酵母菌性食管炎。

鹅口疮的存在有利于食管念珠菌病患者的症状符合食管炎，虽然 HSV 和念珠菌性食管炎的症状是相似的。食管炎的其他原因（例如，放射性食管炎，食管或胃的 GVHD）也会产生类似的症状。在中性粒细胞减少症患者和其他高度免疫功能低下伴食管炎症状的患者中，应考虑进行氟康唑和/或阿昔洛韦试验。CMV 食管炎是化疗诱导的中性粒细胞减少症的罕见并发症，最常见于患有 GVHD 的同种异体 HCT 受者。PCR 研究的阴性 CMV 监测结果将使 CMV 疾病非常不可能。如果诊断为 CMV 食管炎，应开始使用缬更昔洛韦或更昔洛韦治疗。膦甲酸或西多福韦应保留用于更昔洛韦耐药的 CMV 或不能耐受更昔洛韦的患者。 经验性治疗可能被认为是在高风险的 CMV 疾病的患者症状

可能是食管炎

对于对这些药物经验性治疗无效的食管炎患者，应在血小板支持下仔细进行胃镜检查（如需要）

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may be considered to obtain cultures. Tissue biopsies are the gold standard for the diagnosis of invasive esophageal infections. However, endoscopy and biopsy may be associated with complications in patients who are profoundly neutropenic and/or thrombocytopenic; therefore, the procedure should be performed with caution. Radiographic procedures, such as barium studies, lack sensitivity and add little clinically significant information; therefore, they are not recommended.

Sinus or Nasal Infections

The sinuses are a common site of bacterial infection. Patients with severe and prolonged neutropenia (eg, >10 days) and allogeneic HCT recipients with GVHD are particularly susceptible to invasive mold infections. Cytotoxic therapy disrupts the natural cleansing mechanisms in the nasal passages and increases colonization. A preceding chronic infection may also become active in the setting of neutropenia. Sinusitis during the early neutropenic period (<7 days) is principally caused by respiratory and gram-negative bacterial pathogens. In patients with longer-duration neutropenia or in patients receiving concomitant high-dose corticosteroid therapy, invasive mold infections are an important concern.

Initial symptoms of sinusitis may be mild. A high-resolution CT scan of the sinuses is the radiographic procedure of choice to evaluate patients with pain or tenderness of the sinuses, nasal erosions, unilateral facial swelling, unilateral eye tearing, or epistaxis. An MRI that includes evaluation of the orbital and cavernous sinuses is useful to evaluate proptosis of the eye or cranial nerve abnormalities (see *Initial Clinical Presentation: Sinus/Nasal* in the algorithm). Bony erosion on CT scan suggests invasive fungal disease. Ear, nose, and throat (ENT) and ophthalmologic examinations should be performed for symptomatic patients with abnormalities on CT scans, with biopsy and culture of any abnormal tissues. Broad-spectrum coverage for aerobes and anaerobes is appropriate for patients who are neutropenic and otherwise highly immunocompromised with sinus infections. Vancomycin (or another gram-positive active agent) should be added for periorbital cellulitis, which is frequently caused by *S aureus*.

Sinus endoscopy with biopsy and culture are often required to definitively establish the diagnosis and should be pursued aggressively in patients at high risk for mold infection. Invasive fungal sinusitis in patients with hematologic malignancies and with prolonged neutropenia is principally caused by *Aspergillus* species (*A flavus* and *A fumigatus*) and Mucorales (previously referred to as Zygomycetes). In a case-control study of invasive aspergillosis and mucormycosis in patients with either acute leukemia or who were allogeneic HCT recipients, the risk factors that favored the diagnosis of mucormycosis included fungal sinusitis and use of voriconazole.[511] A lipid formulation of amphotericin B should be used for suspected or confirmed invasive sinus mold infection, pending definitive histology and culture results. Isavuconazonium sulfate or posaconazole can be considered for treatment of refractory infection or if there is intolerance to amphotericin B formulations; isavuconazonium sulfate has been approved by the FDA for invasive aspergillosis and mucormycosis; and posaconazole has been approved for invasive aspergillosis. Urgent debridement of necrotic tissue should be performed, when feasible.[512]

Abdominal, Rectal, and Urinary Tract Infections

Most infections in the abdomen, rectum, or liver are discovered because of a combination of clinical signs and symptoms (eg, abdominal pain, perirectal pain, diarrhea) and of biochemical abnormalities (eg, abnormal liver function tests). These infections are usually diagnosed and managed based on the radiologic, GI, and surgical expertise of the treating oncology center. Improved imaging techniques (including ultrasonography, CT scans, MRI, and radionuclide and endoscopic procedures) have decreased the need for surgical intervention. The choice of diagnostic studies should be based on the clinical presentation and relative clinical

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可以考虑培养。组织活检是诊断侵袭性食管感染的金标准。然而，内窥镜检查和活检可能与严重血小板减少和/或血小板减少患者的并发症有关;因此，应谨慎进行该操作。放射学检查，如钡剂检查，缺乏灵敏度，几乎没有增加临床重要信息;因此，不推荐使用。

鼻窦或鼻腔感染

鼻窦是细菌感染的常见部位。患有严重和长期中性粒细胞减少症（例如，>10 天）的患者和患有 GVHD 的同种异体 HCT 接受者特别容易受到侵袭性霉菌感染。细胞毒性治疗破坏了鼻道中的自然清洁机制并增加了定植。先前的慢性感染也可能在中性粒细胞减少症的情况下变得活跃。鼻窦炎早期（<7 天）主要由呼吸道和革兰氏阴性细菌病原体引起。在持续时间较长的中性粒细胞减少症患者或同时接受高剂量皮质类固醇治疗的患者中，侵袭性霉菌感染是一个重要的问题。

鼻窦炎的早期症状可能是轻微的。鼻窦高分辨率 CT 扫描是评价鼻窦疼痛或压痛、鼻糜烂、单侧面部肿胀、单侧眼撕裂或鼻出血患者的首选影像学检查。包括眼眶和海绵窦评价的 MRI 可用于评价眼球突出或颅神经异常（参见算法中的初始临床表现：鼻窦/鼻腔）。CT 扫描显示骨质侵蚀说明是侵袭性真菌病。对于 CT 扫描有异常的有症状的患者，应进行耳、鼻、喉（ENT）和眼科检查，并对任何异常组织进行活检和培养。需氧菌和厌氧菌的广谱覆盖范围适用于贫血和其他高度

免疫力低下的鼻窦感染对于经常由金黄色葡萄球菌引起的眶周蜂窝织炎，应加用万古霉素（或另一种革兰氏阳性活性剂）。

通常需要鼻窦内窥镜活检和培养来明确诊断，并应在霉菌感染高危患者中积极进行。侵袭性真菌性鼻窦炎

血液恶性肿瘤和长期中性粒细胞减少症主要由曲霉属（黄曲霉和烟曲霉）和毛霉目（以前称为接合菌）引起。在急性白血病患者或异基因 HCT 接受者中进行的侵袭性曲霉病和毛霉菌病的病例对照研究中，有利于毛霉菌病诊断的风险因素包括真菌性鼻窦炎和伏立康唑的使用。在等待明确的组织学和培养结果的情况下，对于疑似或确诊的侵袭性鼻窦霉菌感染，应使用阿替霉素 B 的脂质制剂。硫酸异沙康唑或泊沙康唑可考虑用于治疗难治性感染，

对阿替霉素 B 制剂的不耐受;硫酸艾沙康唑已被 FDA 批准用于侵袭性曲霉病和毛霉菌病;泊沙康唑已被批准用于侵袭性曲霉病。如果可行，应对坏死组织进行紧急清创。

腹部、直肠和尿路感染

大多数腹部、直肠或肝脏感染是由于临床体征和症状（例如腹痛、直肠周围疼痛、腹泻）以及生化异常（例如肝功能检查异常）的组合而发现的。这些感染通常是诊断和管理的基础上，放射学，胃肠道和外科专业知识的治疗肿瘤中心。改进的成像技术（包括超声检查、CT 扫描、MRI、放射性核素和内窥镜检查）减少了手术干预的需要。诊断性研究的选择应基于临床表现和相关临床表现。

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benefit. Antimicrobial therapy for GI infections must take into account the high likelihood of polymicrobial pathogens and the presence of the endogenous anaerobic GI flora. Acceptable therapeutic options in this setting include monotherapy with a carbapenem (imipenem/cilastatin, meropenem, doripenem, or ertapenem), piperacillin/tazobactam, or pairing ceftriaxone with metronidazole. In patients with neutropenia, the antibiotic regimen should have antipseudomonal activity. Percutaneous aspiration and drainage should be performed, if feasible, for suspicious infected collections. Cholangitis may complicate obstructive tumors or previous hepatobiliary surgery. If cholangitis is suspected (ie, patients have fever with or without abdominal tenderness and liver enzyme abnormalities compatible with obstruction), a CT scan should be performed to detect biliary tract dilatation and abscess or infected collections. An endoscopic cholangiogram is useful to document the level of obstruction; if present, endoscopic stent placement may resolve the obstruction, which is a key component in managing cholangitis.

The GI tract and central venous catheters are the principal portals of entry of systemic candidiasis. *Candida* species are frequently components of the colonic flora in normal adults. Patients are susceptible to candidal bloodstream infection because of the mucosal damage induced with cytotoxic therapy and neutropenia. Breaches in the GI tract after anastomotic leaks also predispose patients to candidal peritonitis and bloodstream infections,[513] and antifungal prophylaxis (eg, fluconazole) should be considered.

Clostridium Difficile Colitis

Clostridium difficile colitis is principally a complication of antibiotic therapy and hospitalization, but it is also a complication of neutropenia, occurring in about 7% of patients.[514] Diarrhea should be evaluated with at least 2 stool *C difficile* toxin screens. Additionally, depending on clinical circumstances, a GI multiplex panel may be considered for identification of other pathogens, including adenovirus, rotavirus, and norovirus. The rate and severity of *C difficile* colitis in the United States may be increasing, partly because of the emergence of a more virulent strain of *C difficile*. Multi-institutional outbreaks of *C difficile* colitis have been reported that were associated with high morbidity and mortality; these outbreaks were caused by a distinct strain with variations in toxin genes and with resistance to fluoroquinolones.[90,91] Early reports suggested that metronidazole cured >90% of cases of *C difficile* colitis, and the rate of recurrence was low.[515,516] However, Musher et al[517] reported that among patients (N = 207) treated with metronidazole for *C difficile* colitis, only 50% were cured and had no recurrence of disease.

A multicenter, double-blind, randomized trial was conducted to evaluate the efficacy and safety of oral fidaxomicin versus oral vancomycin in patients with *C difficile* infection (N = 629).[518] The primary endpoint of this study was clinical cure, defined as the resolution of diarrhea and no further therapy necessary following completion of study treatment. The clinical cure rate with fidaxomicin was noninferior to vancomycin (88.2% vs. 85.8%) in the modified intent-to-treat analysis.[518] The frequency and severity of adverse events were similar between treatment arms. In addition, fidaxomicin was associated with a significantly decreased recurrence rate compared with vancomycin (15.4% vs. 25.3%; $P = .005$) and a significantly higher rate of resolution of diarrhea without recurrence (74.6% vs. 64.1%; $P = .006$).[518] A decrease in recurrence of *C difficile* diarrhea was not observed in the treatment of the current epidemic strain, NAP1/BI/027. The investigators postulate that the improved duration of infection resolution with fidaxomicin may be due to its preservation of normal intestinal anaerobic flora, which may help to prevent the reemergence of *C difficile*.[518]

Another multicenter, double-blind, randomized trial evaluated the efficacy and safety of oral fidaxomicin versus oral vancomycin in adults with acute

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效益 GI 感染的抗菌治疗必须考虑到多种微生物病原体的高可能性和内源性厌氧 GI 植物群的存在。在这种情况下，可接受的治疗选择包括碳青霉烯（亚胺培南/西司他丁、美罗培南、多利培南或厄他培南）、哌拉西林/他唑巴坦单药治疗或头孢曲松与甲硝唑配对治疗。中性粒细胞减少症患者的抗生素治疗方案应具有抗假单胞菌活性。如果可行，对于可疑感染的收集物，应进行经皮抽吸和引流。胆管炎可能使阻塞性肿瘤或既往肝胆外科手术并发症。如果怀疑胆管炎（即患者发热伴或不伴腹部压痛和肝酶异常，与梗阻相符），应进行 CT 扫描以检测胆道扩张和脓肿或感染的集合。 内窥镜胆管造影有助于记录梗阻的程度;如果存在，内窥镜支架放置可以解决梗阻，这是治疗胆管炎的关键组成部分。

胃肠道和中心静脉导管是系统性念珠菌感染的主要入口。念珠菌属是正常成人结肠植物群的常见组成部分。由于细胞毒性治疗和中性粒细胞减少引起的粘膜损伤，患者易受念珠菌血流感染。吻合口漏后胃肠道破裂也使患者易患念珠菌性腹膜炎和血流感染，应考虑预防性抗真菌治疗（如氟康唑）。

艰难梭菌结肠炎

艰难梭菌结肠炎主要是抗生素治疗和住院治疗的并发症，但它也是中性粒细胞减少症的并发症，发生在约 7%的患者中。腹泻应至少进行 2 次粪便艰难梭菌毒素筛查。此外，根据临床情况，可以考虑使用 GI 多重检测板来识别其他病原体，包括腺病毒、轮状病毒和诺如病毒。在美国，艰难梭菌结肠炎的发病率和严重程度可能正在增加，部分原因是出现了更具毒性的艰难梭菌菌株。据报道，艰难梭菌结肠炎的多机构爆发与高发病率和死亡率相关;这些爆发是由毒素基因变异和对氟喹诺酮耐药的独特菌株引起的。早期报告表明，甲硝唑治愈了 90%以上的艰难梭菌结肠炎病例，复发率低。然而，Musher 等报道，在使用甲硝唑治疗艰难梭菌结肠炎的患者（N = 207）中，只有 50%的患者治愈，且无疾病复发。

本研究采用多中心、双盲、随机对照研究方法，对 629 例艰难梭菌感染患者口服非达霉素和万古霉素的疗效和安全性进行了比较，主要终点为临床治愈，即腹泻消退且完成研究治疗后无需进一步治疗。在改良的意向治疗分析中，非达霉素的临床治愈率不劣于万古霉素（88.2% vs. 85.8%）。

治疗组之间不良事件的严重程度相似。此外，与万古霉素相比，非达霉素可显著降低复发率（15.4% vs. 25.3%; P = 0.005），腹泻缓解率显著更高，无复发（74.6%对 64.1%;在当前流行菌株 NAP 1/BI/027 的治疗中未观察到艰难梭菌腹泻复发的减少。研究者推测，非达霉素改善感染消退的持续时间可能是由于其保留了正常的肠道厌氧植物群，这可能有助于防止艰难梭菌的再次出现。

另一项多中心、双盲、随机试验评价了口服非达霉素与口服万古霉素治疗成人急性

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C difficile infection (N = 535; n = 509 evaluable).[519] The primary endpoint of this study was clinical cure; fidaxomicin was noninferior to vancomycin (87.7% vs. 86.8%) in the modified intent-to-treat analysis. Interestingly, among the subgroup of patients receiving concomitant antibiotics for other infections (n = 96), treatment with fidaxomicin resulted in a higher cure rate compared with vancomycin (90.2% vs. 73.3%; $P = .031$).[519] The incidence of treatment-emergent adverse events was similar between treatment arms.

Both of these large randomized controlled studies showed that treatment of *C difficile* infection with fidaxomicin was noninferior to vancomycin. A subgroup analysis combining data from the two randomized studies was conducted to evaluate the efficacy of these agents in patients with a cancer diagnosis who had *C difficile* infection.[520] Overall, the cure rate was significantly lower among patients with cancer (n = 183) compared with patients without cancer in these trials (n = 922; 79.2% vs. 88.6%; $P < .001$). In addition, the median time to resolution of diarrhea was delayed among patients with cancer (100 hours vs. 55 hours; $P < .001$). An analysis by treatment regimen showed that among the subgroup of patients with cancer, those treated with fidaxomicin had a more rapid median time to resolution of diarrhea compared with patients treated with vancomycin (74 hours vs. 123 hours; $P = .045$).[520] Subtotal colectomy, diverting ileostomy, or colostomy may be required in cases involving toxic dilatation or perforation of the colon.

Multiple recurrences of *C difficile* are a challenge in the patient with cancer and may respond to a prolonged, tapered treatment with oral vancomycin dose over several weeks.[521] The use of oral vancomycin followed by duodenal infusion of donor feces (fecal microbiota transplant, FMT) may also be an effective strategy for patients with recurrent *C difficile* infection, although there is a lack of data on the safety and efficacy of FMT in patients with cancer. In one randomized study, patients with recurrent *C difficile* infection were assigned to receive treatment with a short course of initial oral vancomycin (500 mg orally 4 times daily for 4 days) followed by bowel lavage and infusion of donor feces (n = 16) or standard oral vancomycin (500 mg orally 4 times daily for 14 days) alone (n = 13) or standard oral vancomycin with bowel lavage (n = 13). Resolution was achieved in 81% of patients in the FMT group compared with 31% in the vancomycin alone group and 23% in the group treated with vancomycin plus bowel lavage ($P < .001$ for both comparisons with the infusion group).[522] Another randomized study assigned patients with recurrent *C difficile* infection to receive an initial short course of vancomycin (125 mg orally 4 times daily for 3 days) followed by FMT via colonoscopy (n = 20) or vancomycin (125 mg orally 4 times daily for 10 days, followed by 125-500 mg/day every 2–3 days for at least 3 weeks) alone (n = 19). In this study, resolution was achieved in 90% of patients in the FMT group compared with 26% of patients in the vancomycin group ($P < .0001$).[523] While these studies should be interpreted with caution as they excluded patients who had neutropenia or recent chemotherapy, some institutions consider FMT for treatment of refractory *C difficile* infection in select cases.

Another consideration for recurrent *C difficile* is bezlotoxumab, a human monoclonal antibody against *C difficile* toxin B, used in conjunction with antibiotic treatment. Bezlotoxumab was approved by the FDA in 2016 to reduce recurrence for *C difficile* for patients receiving antibacterial treatment and who are at high risk for *C difficile* recurrence. Two double-blind, randomized, placebo-controlled, phase 3 trials of 2655 patients receiving oral antibiotics for *C difficile* infection studied the efficacy and safety of bezlotoxumab. Both trials showed that the rate of recurrent *C difficile* infection was significantly lower in patients given bezlotoxumab than in those given placebo (MODIFY I: 17% vs. 28%; $P < .001$; MODIFY II: 16% vs. 26%; $P < .001$). Rates of adverse events were similar between bezlotoxumab and placebo.[524]

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艰难梭菌感染（N = 535; n = 509 可评估）。本研究的主要终点是临床治愈;在改良的意向治疗分析中，非达霉素不劣于万古霉素（87.7% vs. 86.8%）。有趣的是，在接受合并抗生素治疗其他感染的患者亚组（n = 96）中，与万古霉素相比，非达霉素治疗的治愈率更高（90.2% vs. 73.3%; P = 0.031）。治疗后出现的不良事件的发生率在治疗组之间相似。

Both of these large randomized controlled studies showed that treatment of C difficile infection with fidaxomicin was noninferior to vancomycin. A subgroup analysis combining data from the two randomized studies was conducted to evaluate the efficacy of these agents in patients with a cancer diagnosis who had C difficile infection.Overall, the cure rate was significantly lower among patients with cancer (n = 183) compared with patients without cancer in these trials (n = 922; 79.2% vs. 88.6%; P < .001). In addition, the median time to resolution of diarrhea was delayed among patients with cancer (100 hours vs. 55 hours; P < .001). An analysis by treatment regimen showed that among the subgroup of patients with cancer, those treated with fidaxomicin had a more rapid median time to resolution of diarrhea compared with patients treated with vancomycin (74 hours vs. 123 hours; P = .045).Subtotal colectomy, diverting ileostomy, or colostomy may be required in cases involving toxic dilatation or perforation of the colon.

重试 错误原因

艰难梭菌的多次复发是癌症患者面临的挑战，并且可能对口服万古霉素剂量持续数周的长期逐渐减少的治疗有反应。（粪便微生物群移植，FMT）也可能是复发性艰难梭菌感染患者的有效策略，尽管缺乏关于 FMT 在癌症患者中的安全性和有效性的数据。在一项随机研究中，复发性 C 艰难梭菌感染的患者被分配接受短期初始口服万古霉素（500 mg 口服，每日 4 次，持续 4 天），随后洗肠并输注供体粪便（n = 16）或单独标准口服万古霉素（500 mg 口服，每日 4 次，持续 14 天）（n = 13）或标准口服万古霉素并洗肠（n = 13）。FMT 组 81%的患者缓解，而万古霉素单用组为 31%，万古霉素加肠冲洗组为 23 另一项随机研究将复发性艰难梭菌感染的患者分配为接受初始短期万古霉素治疗，（125 mg 口服，每日 4 次，持续 3 天），随后通过结肠镜检查进行 FMT（n = 20）或单独使用万古霉素（125 mg 口服，每日 4 次，持续 10 天，随后 125500 mg/天，每 2–3 天，持续至少 3 周）（n = 19）。在这项研究中，FMT 组 90%的患者获得了缓解而万古霉素组为 26%（P <0.0001）。虽然这些研究应谨慎解释，因为它们排除了中性粒细胞减少症或近期化疗的患者，但一些机构认为 FMT 可用于治疗选定病例中的难治性艰难梭菌感染。

复发性艰难梭菌的另一个考虑因素是 bezlotoxumab，一种针对艰难梭菌毒素 B 的人单克隆抗体，与抗生素治疗联合使用。Bezlotoxumab 于 2016 年获得 FDA 批准，用于减少接受抗菌治疗和艰难梭菌复发高风险患者的艰难梭菌复发。两项双盲、随机、安慰剂对照、3 期试验研究了 2655 名接受口服抗生素治疗艰难梭菌感染的患者对 Bezlotoxumab 的疗效和安全性。两项试验均显示，与安慰剂组相比，接受 Bezlotoxumab 治疗的患者的艰难梭菌感染复发率显著降低（MODIFY I：17% vs. 28%; P <0.001; MODIFY II：16% vs. 26%; P <0.001）。贝洛妥尤单抗和安慰剂之间的不良事件发生率相似。

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The NCCN Panel recommends vancomycin (preferred in adults), or metronidazole for the treatment of suspected *C difficile* colitis. Oral vancomycin has a similar efficacy rate compared to oral metronidazole and can be considered an option for initial therapy for *C difficile* colitis despite the risk of selection for VRE and the substantial expense. Oral vancomycin should be considered over metronidazole for more complicated cases, such as those associated with severe diarrhea, dehydration, clinical instability, significant comorbidities, or recurrent or refractory *C difficile* colitis. The NCCN Panel recommends maintaining good antibiotic stewardship and reducing unnecessary antibiotic therapy at each step of the treatment process. For confirmed *C difficile* infection, the Panel recommends oral vancomycin or fidaxomicin for 10 days. In case of ongoing or worsening infection, switching to fidaxomicin can be considered if the initial treatment was with vancomycin. FMT may be considered, but should be avoided in patients with neutropenia. For relapse/recurrent *C difficile* infection, fidaxomicin is recommended if not given previously. Alternatively, a vancomycin taper, FMT (with appropriate consultation and to be avoided in patients with neutropenia) or bezlotoxumab may be considered. In case of fulminant disease, vancomycin is recommended either orally or via a nasogastric (NG) tube in combination with IV metronidazole. If ileus is present, vancomycin via rectal instillation may be considered. Efforts should be made to deliver vancomycin by the NG route in patients with severe *C difficile* colitis.[525,526] Limited data suggest that IV metronidazole may be useful in this setting, and it is best used as an adjunct to oral vancomycin.[527,528] IV vancomycin is not recommended in this setting because of inadequate luminal levels. IV metronidazole should be used in patients who cannot be treated with oral agents. Upon resolution of *C difficile* infection, secondary prophylaxis may be considered if continuing antibiotic therapy.

Enterocolitis

Neutropenic enterocolitis is a serious, potentially life-threatening disease characterized by fever, diarrhea, and abdominal pain.[529,530] When it occurs in the cecum, it is commonly referred to as typhlitis. The cecum is more vulnerable because of its size and shape, but any portion or the entire colon may be involved. This illness has frequently been associated with acute leukemia, neutropenia, and intensive cytotoxic therapy. CT scanning is the preferred diagnostic test and usually identifies any thickening of the bowel wall. The differential diagnosis for this syndrome includes *C difficile* colitis, CMV enteritis (most common in allogeneic HCT recipients), and GI tract GVHD. Bloodstream infections and sepsis (frequently polymicrobial), bowel perforation, and hemorrhage may occur. The natural history of typhlitis is quite variable, but all patients should be assessed for *C difficile* infection and should be treated with bowel rest and broad-spectrum antibiotics, including coverage for *C difficile*, aerobic pathogens, and anaerobic pathogens. Parenteral nutrition should be considered if clinical signs and symptoms do not resolve promptly. Approximately 5% of patients with typhlitis develop complications requiring surgical intervention (eg, perforation, uncontrolled sepsis, rectal bleeding).[531] Consequently, the Panel recommends that surgical and other subspecialty consultations be obtained early in the course of treatment.

The NCCN Panel recommends an abdominal exam, abdominal CT scan (preferred), or ultrasound for a confirmatory diagnosis of infection in patients experiencing abdominal pain. Additional recommendations include liver function tests (eg, alkaline phosphatase, transaminases, bilirubin, amylase, lipase). The Panel also recommends consideration of early surgical and other subspecialty (eg, gastroenterology, interventional radiology) consultations based on the clinical diagnosis. In patients with perirectal pain, the Panel recommends perirectal inspection and consideration of imaging such as abdominal/pelvic CT or MRI. Addition of antimicrobial therapy with adequate anaerobic activity is recommended in

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NCCN 专家组推荐万古霉素（成人首选）或甲硝唑治疗疑似艰难梭菌结肠炎。与口服甲硝唑相比，口服万古霉素的有效率相似，尽管选择 VRE 存在风险且费用高昂，但仍可被视为艰难梭菌结肠炎初始治疗的一种选择。对于更复杂的病例，如与严重腹泻、脱水、临床不稳定、显著合并症或复发性或难治性艰难梭菌结肠炎相关的病例，应考虑口服万古霉素而不是甲硝唑。NCCN 专家组建议在治疗过程的每一步保持良好的抗生素管理并减少不必要的抗生素治疗。对于确诊的艰难梭菌感染，专家组建议口服万古霉素或非达霉素 10 天。在感染持续或恶化的情况下，如果初始治疗是万古霉素，则可以考虑改用非达霉素。 可以考虑 FMT，但在中性粒细胞减少症患者中应避免。对于复发性/复发性艰难梭菌感染，如果先前未给予，则建议使用非达霉素。或者，可以考虑万古霉素减量、FMT（适当咨询，中性粒细胞减少患者应避免）或 Bezlotoxumab。在暴发性疾病的情况下，建议口服万古霉素或通过鼻胃管（NG）与 IV 甲硝唑联合使用。如果出现肠梗阻，可考虑经直肠滴注万古霉素。在重度艰难梭菌性结肠炎患者中，应努力通过 NG 途径给药万古霉素。有限的数据表明静脉注射甲硝唑在这种情况下可能是有用的，最好是作为口服万古霉素的辅助治疗。静脉注射万古霉素在这种情况下是不推荐的，因为腔内水平不足。静脉注射甲硝唑应用于不能用口服药物治疗的患者。 在艰难梭菌感染消退后，如果继续抗生素治疗，可以考虑二级预防。

小肠结肠

中性粒细胞减少性小肠结肠炎是一种严重的，可能危及生命的疾病，其特征是发热，腹泻和腹痛。当它发生在盲肠时，通常被称为盲肠炎。盲肠由于其大小和形状而更容易受到攻击，但任何部分或整个结肠都可能受到影响。这种疾病经常与急性白血病，中性粒细胞减少症和强化细胞毒治疗有关。CT 扫描是首选的诊断测试，通常可以识别肠壁的任何增厚。该综合征的鉴别诊断包括艰难梭菌结肠炎、CMV 肠炎（最常见于同种异体 HCT 接受者）和胃肠道 GVHD。可能发生血流感染和败血症（通常为多种微生物）、肠穿孔和出血。的自然史

盲肠炎是相当可变的，但所有的患者都应评估艰难梭菌感染，并应与肠道休息和广谱抗生素，包括覆盖艰难梭菌，需氧病原体和厌氧病原体治疗。如果临床体征和症状不能迅速缓解，应考虑肠外营养。大约 5%的盲肠炎患者出现需要手术干预的并发症（如穿孔、无法控制的脓毒症、直肠出血）。因此，专家组建议在治疗过程的早期进行手术和其他专科会诊。

The NCCN Panel recommends an abdominal exam, abdominal CT scan (preferred), or ultrasound for a confirmatory diagnosis of infection in patients experiencing abdominal pain. Additional recommendations include liver function tests (eg, alkaline phosphatase, transaminases, bilirubin, amylase, lipase). The Panel also recommends consideration of early surgical and other subspecialty (eg, gastroenterology, interventional radiology) consultations based on the clinical diagnosis. In patients with perirectal pain, the Panel recommends perirectal inspection and consideration of imaging such as abdominal/pelvic CT or MRI. Addition of antimicrobial therapy with adequate anaerobic activity is recommended in

重试 错误原因

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both patient populations. In patients experiencing diarrhea who are not on laxatives, stool softeners, or tube feeds, *C. difficile* testing is strongly recommended. Other medical causes of diarrhea should be considered, including neutropenic enterocolitis, GI GVHD, and mucositis. Colonoscopy may be required to evaluate for colitis and determine etiology of diarrhea. Moreover, based on clinical circumstances, diagnostic testing for viral, bacterial, and/or parasitic pathogens may be considered. In patients with symptoms of urinary tract infection, a urine culture is recommended with no additional therapy until specific pathogen identified.

Lung Infections

Pulmonary infiltrates pose a difficult diagnostic challenge in patients with cancer. Noninfectious causes of pulmonary infiltrates include congestive heart failure, pulmonary edema, hemorrhage, infarction, drug-induced pneumonitis, radiation injury, tumor, bronchiolitis obliterans, and acute respiratory distress syndrome. Common processes can have atypical radiographic appearances, and two or more pulmonary processes can exist simultaneously. A careful history should include the time course of respiratory symptoms, sick contacts (eg, community respiratory viral infections, tuberculosis), recent hospitalization, travel, exposure to animals, and exposure to droplets from water distribution systems (*Legionella*). Community outbreaks of specific pathogens (eg, influenza, pertussis) should be considered in the differential diagnosis and should guide initial therapy.

Community-Acquired Pneumonia in the Absence of Neutropenia and Immunosuppressive Therapy

The diagnostic evaluation and initial therapy for community-acquired pneumonia must consider host factors and previous use of antibiotics. The IDSA has published guidelines on community-acquired pneumonia.[316] If feasible, sputum and blood cultures should be collected before starting therapy. In patients who are not neutropenic, receiving IST, or requiring hospital admission (based on a validated pneumonia severity index), therapy includes either 1) a respiratory fluoroquinolone (levofloxacin 750 mg/day, moxifloxacin); or 2) a beta-lactam (eg, high-dose amoxicillin or amoxicillin-clavulanate) plus a macrolide (eg, azithromycin).[316] These regimens will treat most of the common community-acquired pathogens, including "atypical" pneumonia (*Chlamydia, Mycoplasma,* and *Legionella* species). Although daptomycin is effective against most gram-positive pathogens, it should not be used for the treatment of pneumonia, because it is inactivated by pulmonary surfactant.[453,454]

In patients requiring hospital admission, monotherapy with a respiratory fluoroquinolone or combination therapy with a macrolide plus either ceftriaxone, cefotaxime, or ertapenem is recommended. Ertapenem has gram-positive, gram-negative (excluding *P aeruginosa* and *Acinetobacter* species), and anaerobic activity useful for suspected aspiration or postobstructive pneumonia. In patients with severe community-acquired pneumonia (eg, who require admission to an intensive care unit), the Panel advises broad-spectrum coverage with an antipseudomonal beta-lactam plus either a respiratory fluoroquinolone or azithromycin. In patients with previous MRSA infection or known colonization with MRSA, addition of vancomycin or linezolid should be considered for pneumonia requiring hospitalization (see *Treatment Modifications* in the algorithm).[316] A nasopharyngeal wash for respiratory viruses and initiation of empiric antiviral therapy should be considered during peak influenza season in the local area. Note that rapid immunofluorescent viral antigen tests may result in a false negative for H1N1 (swine flu). A parapneumonic effusion should be aspirated and submitted for Gram stain, bacterial culture, protein, lactate dehydrogenase, and pH.

Community respiratory viral infections (such as influenza, RSV, adenovirus, rhinoviruses, and metapneumoviruses) have a seasonal pattern (generally November through April); however, parainfluenza viral

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两个患者群体。在腹泻的病人中，如果没有服用泻药、大便软化剂或管饲，C。强烈建议进行艰难测试。腹泻的其他医学原因也应该考虑，包括贫血性小肠结肠炎、GI GVHD 和粘膜炎。可能需要结肠镜检查来评估结肠炎并确定腹泻的病因。此外，根据临床情况，可以考虑对病毒、细菌和/或寄生虫病原体进行诊断检测。在有尿路感染症状的患者中，建议进行尿培养，在确定特定病原体之前无需额外治疗。

肺部感染

肺浸润对癌症患者的诊断提出了困难的挑战。肺浸润的非感染性原因包括充血性心力衰竭、肺水肿、出血、梗死、药物性肺炎、放射性损伤、肿瘤、闭塞性细支气管炎和急性呼吸窘迫综合征。共同的过程可以有不典型的影像学表现，两个或两个以上的肺过程可以同时存在。详细的病史应包括呼吸道症状的时间进程、患病接触者（如社区呼吸道病毒感染、结核病）、近期住院、旅行、暴露于动物，以及接触配水系统的飞沫（军团菌）。在鉴别诊断中应考虑特定病原体（例如流感、百日咳）的社区爆发，并应指导初步治疗。

无中性粒细胞和免疫抑制治疗的社区获得性肺炎

社区获得性肺炎的诊断评估和初始治疗必须考虑宿主因素和既往抗生素使用情况。IDSA 已经发布了社区获得性肺炎的指南。如果可行，在开始治疗前应收集痰和血培养。在非血小板减少、接受 IST 或需要入院（基于经验证的肺炎严重程度指数），治疗包括 1）呼吸性氟喹诺酮类药物（左氧氟沙星 750 mg/天，丙沙星）;或 2）β–内酰胺（例如，大剂量阿莫西林或阿莫西林–克拉维酸）加大环内酯类这些疗法将治疗大多数常见的社区获得性病原体，包括“非典型”肺炎（衣原体、支原体和军团菌属）。 虽然达托霉素对大多数革兰氏阳性病原体有效，但不应用于治疗肺炎，因为它会被肺表面活性物质灭活。对于需要住院治疗的患者，建议使用呼吸系统氟喹诺酮类药物单药治疗或大环内酯类药物加头孢曲松、头孢噻肟或厄他培南的联合治疗。厄他培南具有革兰氏阳性、革兰氏阴性（不包括铜绿假单胞菌和不动杆菌属）和厌氧活性，可用于疑似吸入性或阻塞后肺炎。对于重度社区获得性肺炎患者（例如，需要入住重症监护室的患者），专家组建议使用抗假单胞菌β–内酰胺类药物加呼吸道氟喹诺酮类药物或阿奇霉素进行广谱覆盖。在既往 MRSA 感染或已知 MRSA 定植的患者中，对于需要住院治疗的肺炎，应考虑添加万古霉素或利奈唑胺（见算法中的治疗调整）。在当地流感高峰期，应考虑鼻咽冲洗呼吸道病毒并开始经验性抗病毒治疗。请注意，快速免疫荧光病毒抗原测试可能会导致 H1N1（猪流感）假阴性。应吸出旁胸腔积液并进行革兰氏染色、细菌培养、蛋白质、乳酸脱氢酶和 pH 值检测。

社区呼吸道病毒感染（如流感病毒、RSV、腺病毒、鼻病毒和偏肺病毒）具有季节性模式（通常为 11 月至 4 月）;然而，副流感病毒感染（如流感病毒、RSV、腺病毒、鼻病毒和偏肺病毒）具有季节性模式（通常为 11 月至 4 月）。

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infections can occur throughout the year. During the influenza season, consider empiric antiviral therapy for patients within 48 hours after symptoms develop that are suggestive of influenza (eg, high fever, coryza, myalgia, dry cough), especially during community outbreaks. Both the IDSA (2018) and CDC guidelines (2023) recommend antiviral treatment with the neuraminidase inhibitors oseltamivir, zanamivir, or peramivir, which are active against both influenza A and B viruses.[532,533] Both agents are approved by the FDA for the treatment of influenza within 48 hours of symptomatic onset; the indicated duration of treatment is 5 days. However, longer courses of treatment (eg, 10 days) and treatment until resolution of symptoms should be considered in patients who are immunocompromised. Some centers have used higher doses (eg, 150 mg BID) of oseltamivir in these patients with mixed results (see *Suggested Minimum Duration of Therapy for Documented Infection* in the algorithm). Pandemic influenza does not have a predictable seasonal pattern, and may spread in the community concurrently with a seasonal influenza strain. Antiviral susceptibility of influenza strains is variable and cannot be predicted based on previous influenza outbreaks. In cases of seasonal influenza and pandemic strains, it is necessary to be familiar with susceptibility patterns and guidelines on appropriate antiviral treatment.[317] Peramivir has been shown to have similar clinical outcomes as oral oseltamivir[318] and can be considered for patients who cannot have oral oseltamivir or inhaled zanamivir, though it is available only as an IV injection.

Hospital-Acquired Pneumonia

Guidelines on the management of adults with hospital-acquired pneumonia from the American Thoracic Society (ATS) emphasize that the time of onset is an important risk factor for specific pathogens that may be resistant to antibiotics.[534] Early-onset hospital-acquired pneumonia (occurring within the first 4 days of hospitalization) is likely to be caused by antibiotic-sensitive bacteria and usually carries a better prognosis. However, patients with cancer may be at risk for acquisition of antibiotic-resistant bacteria based on prior hospitalizations, prior antibiotic use, and impaired immune status regardless of when pneumonia begins in the course of the current hospitalization. The ATS guidelines define the following as risk factors for multidrug-resistant pathogens in patients with health care-associated pneumonia: 1) received antibiotics in the preceding 90 days; 2) hospitalization for ≥2 days in the preceding 90 days; 3) resident in nursing home or extended care facility; 4) chronic dialysis within 30 days; 5) home wound care; and 6) family member with a multidrug-resistant pathogen.[534] Late-onset hospital-acquired pneumonia (occurring after ≥5 days of hospitalization) is more likely to be caused by multidrug-resistant pathogens, and is associated with greater morbidity and mortality.

The population of multidrug-resistant bacteria (notably, MRSA and antibiotic-resistant gram-negative pathogens) varies among different hospitals and geographic distributions. Therefore, the selection of initial therapy for hospital-acquired pneumonia requires knowledge of the local patterns of antibiotic susceptibility. For example, at some centers, a high frequency of extended-spectrum beta lactamase-producing gram-negative bacterial infections may make a carbapenem the drug of choice as initial therapy for pneumonia. At other centers, carbapenem-resistant gram-negative infections are an increasing problem, and an alternative class of antibiotics may be preferred based on prior local susceptibility results.[535]

In patients with late-onset hospital-associated pneumonia or risk factors for multidrug-resistant pathogens regardless of when pneumonia developed in relation to hospitalization, a broad-spectrum antibiotic regimen is recommended. An antipseudomonal beta-lactam (eg, ceftazidime, cefepime, imipenem/cilastatin, meropenem, piperacillin/tazobactam) plus an antipseudomonal fluoroquinolone (eg,

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全年都可能发生感染。在流感季节，在出现提示流感的症状（如高热、鼻炎、肌痛、干咳）后 48 小时内，尤其是在社区暴发期间，应考虑对患者进行经验性抗病毒治疗。IDSA（2018）和 CDC 指南（2023）均推荐使用神经氨酸酶抑制剂奥司他韦、扎那米韦或帕拉米韦进行抗病毒治疗，这些药物对甲型和 B 型流感病毒都有活性。FDA 批准这两种药物在症状发作后 48 小时内治疗流感;指示的治疗持续时间为 5 天。然而，对于免疫功能低下的患者，应考虑更长的疗程（如 10 天）和治疗直至症状消退。一些中心在这些患者中使用了更高剂量（例如，150 mg BID）的奥司他韦，结果好坏参半（参见算法中记录的感染的建议最短治疗持续时间）。 大流行性流感没有可预测的季节性模式，可能与季节性流感毒株同时在社区中传播。流感病毒株的抗病毒敏感性是可变的，不能根据以往的流感疫情进行预测。在季节性流感和大流行性流感病毒株的情况下，有必要熟悉易感性模式和适当抗病毒治疗的指南。帕拉米韦已被证明具有与口服奥司他韦相似的临床结果，并且可以考虑用于不能口服奥司他韦或吸入扎那米韦的患者，尽管它只能作为 IV 注射剂使用。

医院获得性肺炎

美国胸科学会（ATS）的成人医院获得性肺炎管理指南强调，发病时间是可能对抗生素产生耐药性的特定病原体的重要风险因素。

（发生在住院的前 4 天内）很可能是由抗生素敏感的细菌引起的，通常预后较好。

然而，癌症患者可能存在获得耐药性细菌的风险，这是基于既往住院治疗、既往抗生素使用和免疫状态受损，无论当前住院治疗过程中肺炎何时开始。ATS 指南将以下定义为医疗相关性肺炎患者中多重耐药病原体的风险因素：1）在过去 90 天内接受抗生素治疗; 2）在过去 90 天内住院≥2 天; 3）居住在疗养院或延长护理机构; 4）30 天内长期透析; 5）家庭伤口护理;（6）家族成员携带多重耐药病原体，迟发性医院获得性肺炎（住院≥5 天后发生）更易由多重耐药病原体引起，且发病率和死亡率较高。

多重耐药细菌（特别是 MRSA 和耐药性革兰氏阴性病原体）的数量在不同的医院和地理分布中有所不同。因此，医院获得性肺炎初始治疗的选择需要了解当地的抗生素敏感性模式。例如，在一些中心，产超广谱β内酰胺酶的革兰氏阴性细菌感染的频率很高，这可能使碳青霉烯成为肺炎初始治疗的首选药物。在其他中心，碳青霉烯类耐药

革兰氏阴性菌感染是一个日益严重的问题，基于先前的局部敏感性结果，可优选另一类抗生素。

对于迟发性医院相关性肺炎或多重耐药病原体风险因素的患者，无论何时发生与住院相关的肺炎，均建议使用广谱抗生素方案。抗假单胞菌β–内酰胺（如头孢他啶、头孢吡肟、亚胺培南/西司他丁、美罗培南、哌拉西林/他唑巴坦）加抗假单胞菌氟喹诺酮（如，

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ciprofloxacin or levofloxacin) or aminoglycoside, plus either linezolid or vancomycin (to cover MRSA) is a reasonable initial regimen (aim for vancomycin trough level of 15–20 mcg/mL).[534] If *Legionella* is suspected, a quinolone (ciprofloxacin, levofloxacin, or moxifloxacin) should be used instead of an aminoglycoside. The antibiotic regimen should be subsequently tailored based on culture results.

Pulmonary Infiltrates in Patients with Neutropenia

In patients with neutropenia for <7 days, pulmonary infections are likely to be caused by Enterobacteriaceae (eg, *E coli, Klebsiella* species), *P aeruginosa*, *S aureus*, and pathogens encountered in non-immunocompromised persons (as previously described). Because of the neutropenia, consolidation and sputum production may be absent.[536] Blood cultures, a chest radiograph, and, if possible, a sputum sample for Gram stain and culture should be obtained. In suspected acute bacterial pneumonia, appropriate empiric antibiotic therapy must be initiated promptly and the response must be closely monitored in an inpatient setting. The therapeutic regimen depends on several variables, including recent use of antibiotics, community or nosocomial (hospital-acquired) pneumonia, and the local antibiotic sensitivity data.

If community-acquired pneumonia is suspected (ie, pneumonia is present before admission or develops within 3 to 4 days of hospitalization), addition of a macrolide or fluoroquinolone to an antipseudomonal beta-lactam is warranted to treat atypical pathogens. For nosocomial pneumonia, therapy with an antipseudomonal beta-lactam is advised, and addition of an aminoglycoside or fluoroquinolone should be considered. For cases of nosocomial pneumonia in which hospital-acquired legionellosis is suspected, empiric addition of a macrolide or fluoroquinolone is also warranted. Vancomycin or linezolid should be added for pneumonia in patients colonized with MRSA and for nosocomial pneumonia at centers in which MRSA is common. Community respiratory viruses should also be considered, especially during winter months. RSV, parainfluenza, and influenza are significant pathogens during neutropenia in patients receiving chemotherapy for acute leukemia and in HCT recipients.

If clinical improvement occurs within 48 to 72 hours of therapy, no further diagnostic measures are necessary; antibiotic therapy should be continued until neutropenia resolves and for at least 7 to 14 days thereafter. Once neutropenia resolves, an appropriate oral antibiotic regimen can be administered for the remainder of the course.

In cases of refractory pneumonia, bacterial infection resistant to the initial antibiotic regimen and nonbacterial pathogens should be considered, particularly filamentous fungi.[536] A CT scan of the chest is useful in defining the location and morphology of the lesions, and in guiding diagnostic procedures. A “halo sign” in a persistently febrile neutropenic patient is highly suggestive of invasive aspergillosis;[537] however, angioinvasive infections including other filamentous fungi and *P aeruginosa* may produce similar findings.

A new or progressive infiltrate developing in patients with prolonged neutropenia (eg, >10 days) receiving broad-spectrum antibacterial agents is concerning for the possibility of invasive aspergillosis or infection with other molds.[536] Consider adding an extended-spectrum azole or a lipid formulation of amphotericin B while waiting for diagnostic results. Empiric modification of the antibacterial regimen based on the predominant local hospital pathogens (eg, MRSA, antibiotic-resistant gram-negative bacteria) is also warranted in patients with rapidly progressive pneumonia.

Pulmonary Infiltrates in Patients with Impaired Cellular Immunity

Patients with impaired cellular immunity are at increased risk for common bacterial infections and opportunistic infections, including fungi (eg, *Aspergillus* and other filamentous fungi, *Cryptococcus neoformans*,

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环丙沙星或左氧氟沙星）或氨基糖苷类，加上利奈唑胺或万古霉素（以覆盖 MRSA）是合理的初始方案（目标万古霉素谷浓度为 15–20 mcg/mL）。如果怀疑是军团菌，则应使用喹诺酮类（环丙沙星，左氧氟沙星或阿托沙星）而不是氨基糖苷类。抗生素方案应随后根据培养结果进行调整。

中性粒细胞增多症患者的肺损伤

在中性粒细胞减少<7 天的患者中，肺部感染可能由肠杆菌科（例如，大肠杆菌、克雷伯氏菌属）、铜绿假单胞菌、金黄色葡萄球菌和非免疫功能低下者中遇到的病原体引起（如前所述）。由于中性粒细胞减少，可能没有实变和痰液产生。血培养，胸片，如果可能的话，痰标本进行革兰氏染色和培养。在疑似急性细菌性肺炎时，必须立即开始适当的经验性抗生素治疗，并在住院环境中密切监测反应。治疗方案取决于几个变量，包括最近使用抗生素，社区或医院（医院获得性）肺炎，以及当地抗生素敏感性数据。

如果怀疑社区获得性肺炎（即，肺炎在入院前存在或在住院后 3 – 4 天内发生），则需要在抗假单胞菌β–内酰胺类药物的基础上加用大环内酯类或氟喹诺酮类药物治疗非典型病原体。对于医院获得性肺炎，建议使用抗假单胞菌β–内酰胺治疗，并应考虑添加氨基糖苷类或氟喹诺酮类药物。对于怀疑医院获得性军团菌病的医院获得性肺炎病例，也需要经验性地加用大环内酯类或氟喹诺酮类药物。对于 MRSA 定植患者的肺炎和 MRSA 常见中心的院内肺炎，应加用万古霉素或利奈唑胺。社区呼吸

还应考虑病毒，尤其是在冬季。RSV、副流感病毒和流感病毒是急性白血病化疗患者和 HCT 患者中性粒细胞减少症的重要病原体

受惠人士

如果在治疗后 48 至 72 小时内出现临床改善，则无需进一步的诊断措施;应继续抗生素治疗，直至中性粒细胞减少症消退，此后至少持续 7 至 14 天。一旦中性粒细胞减少症消退，可以在剩余的疗程中给予适当的口服抗生素方案。

对于难治性肺炎，应考虑对初始抗生素方案耐药的细菌感染和非细菌病原体，特别是丝状真菌。胸部 CT 扫描有助于确定病变的位置和形态，并指导诊断程序。持续性发热性贫血患者的“晕轮征”高度提示侵袭性曲霉病;然而，血管侵袭性感染，包括其他丝状真菌和铜绿假单胞菌可能产生类似的结果。

长期使用广谱抗菌药物的中性粒细胞减少症患者（如>10 天）出现新的或进行性浸润，应考虑侵袭性曲霉菌病或其他霉菌感染的可能性。在等待诊断结果时，应考虑加用广谱唑类或脂质制剂阿替霉素 B。在快速进展性肺炎患者中，也需要根据当地医院的主要病原体（如 MRSA、耐药革兰氏阴性菌）对抗菌方案进行经验性调整。

细胞免疫功能低下患者的肺功能损害

细胞免疫受损的患者发生常见细菌感染和机会性感染的风险增加，包括真菌（如，

曲霉和其他丝状真菌，新型隐球菌，

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dimorphic fungi), *Legionella*, *Pneumocystis jirovecii, M tuberculosis*, nontuberculous mycobacteria, *Nocardia* species, and viral pathogens.

In patients with clinical and radiographic findings suggestive of acute bacterial pneumonia (eg, acute onset fever, respiratory symptoms, focal infiltrate), the diagnosis and management are similar to the treatment of patients with neutropenia. An antipseudomonal beta-lactam plus either a respiratory fluoroquinolone or azithromycin is a reasonable initial regimen in patients with pneumonia requiring hospitalization. In allogeneic HCT recipients with GVHD not receiving mold-active prophylaxis, addition of a mold-active drug (eg, voriconazole) should be considered. Particularly among the most highly immunocompromised (eg, chronic GVHD), the differential diagnosis is very broad, and an initial empiric regimen cannot have activity against all possible pathogens. It is critical to establish a definitive diagnosis in patients with negative diagnostic results who are deteriorating clinically after a 2- to 3-day trial of broad-spectrum antibiotics.

Diffuse infiltrates have a broad differential diagnosis,[536] including PJP, viral infections, hemorrhage, and drug-induced pneumonitis. A diagnosis of PJP should be considered in patients with significantly impaired cellular immunity not receiving PJP prophylaxis who present with diffuse pulmonary infiltrates. BAL is the standard approach for diagnosing PJP. In patients with substantial respiratory disease (eg, labored breathing, requiring supplemental oxygen), empiric therapy should be initiated before BAL. Pending BAL results, an initial regimen can include a respiratory fluoroquinolone against community-acquired pathogens and TMP/SMX against possible PJP. TMP/SMX desensitization or atovaquone, dapsone, or pentamidine (aerosolized or IV) can be considered when PJP prophylaxis is required in patients who are TMP/SMX intolerant. For patients receiving dapsone, consider assessing G6PD levels prior to initiating therapy.

Patients at the highest risk for CMV pneumonia include allogeneic HCT recipients in the post-engraftment setting (particularly if receiving IST for GVHD) and patients receiving treatment with alemtuzumab. Negative results from CMV surveillance testing (peripheral blood PCR) make CMV pneumonia very unlikely. CMV pneumonia is uncommon in patients who have not undergone transplant but are receiving immunosuppressive chemotherapy for leukemia.[538] Community respiratory viruses can cause severe pulmonary infection in patients with neutropenia and in those who are non-neutropenic but with impaired cellular immunity. Noninfectious etiologies must also be considered, as previously stated. BAL is sensitive in diagnosing bacterial and viral pneumonia and PJP, and is often used as the initial invasive diagnostic procedure (see *Invasive Diagnostic Procedures for Pulmonary Infiltrates*).

Non-Invasive Diagnosis of Pneumonia

In patients with suspected pneumonia, routine sputum and blood cultures should be obtained, ideally before antibiotics are initiated or modified. Sputum cultures for *Legionella* species are sensitive if obtained before initiating antibiotics; however, specific culture conditions are required. Legionellosis can also be diagnosed based on urine antigen testing, which only detects *Legionella pneumophila* type I, the cause of most (but not all) cases of *Legionella* pneumonia.[316] A nasopharyngeal wash is useful to diagnose community respiratory viral infections. The rapid test for influenza A and B may be performed using a throat or nasopharyngeal swab. Rapid antigen detection methods can provide a diagnosis within hours; however, if results are negative, a shell vial culture will take about 5 days.

Fungal pneumonia is suggested by the following: host factors predisposing the patient to invasive aspergillosis; appropriate symptoms or signs of infection; a compatible pulmonary lesion; and a positive serum galactomannan or beta-glucan assay. Host factors indicative of high risk

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二型真菌），军团菌，耶氏肺孢子虫，结核分枝杆菌，
非结核分枝杆菌、诺卡氏菌属和病毒病原体。

In patients with clinical and radiographic findings suggestive of acute bacterial pneumonia (eg, acute onset fever, respiratory symptoms, focal infiltrate), the diagnosis and management are similar to the treatment of patients with neutropenia. An antipseudomonal beta–lactam plus either a respiratory fluoroquinolone or azithromycin is a reasonable initial regimen in patients with pneumonia requiring hospitalization. In allogeneic HCT recipients with GVHD not receiving mold–active prophylaxis, addition of a mold–active drug (eg, voriconazole) should be considered. Particularly among the most highly immunocompromised (eg, chronic GVHD), the differential diagnosis is very broad, and an initial empiric regimen cannot have activity against all possible pathogens. It is critical to establish a definitive diagnosis in patients with negative diagnostic results who are deteriorating clinically after a 2– to 3–day trial of broad–spectrum antibiotics.

重试 错误原因

弥漫性浸润性肺炎有广泛的鉴别诊断，包括 PJP、病毒感染、出血和药物引起的肺炎。在出现弥漫性肺浸润且细胞免疫功能显著受损而未接受 PJP 预防治疗的患者中，应考虑 PJP 的诊断。BAL 是诊断 PJP 的标准方法。对于患有严重呼吸系统疾病（例如，呼吸困难，需要补充氧气）的患者，应在 BAL 前开始经验性治疗。在等待 BAL 结果时，初始方案可包括针对社区获得性病原体的呼吸道氟喹诺酮和针对可能的 PJP 的 TMP/SMX。当 TMP/SMX 不耐受的患者需要 PJP 预防时，可以考虑 TMP/SMX 脱敏或阿托伐醌、氨苯砜或喷他脒（雾化或 IV）。对于接受氨苯砜治疗的患者，考虑在开始治疗前评估 G6 PD 水平。

CMV 肺炎风险最高的患者包括植入后环境中的同种异体 HCT 接受者（特别是如果接受 IST 治疗 GVHD）和接受 Alemtuzumab 治疗的患者。CMV 监测检测（外周血 PCR）的阴性结果使 CMV 肺炎非常不可能。巨细胞病毒肺炎在未接受移植但正在接受免疫抑制化疗的白血病患者中并不常见。社区呼吸道病毒可导致中性粒细胞减少患者和非血小板减少但细胞免疫受损的患者严重肺部感染。如前所述，还必须考虑非感染性病因。BAL 在诊断细菌性和病毒性肺炎和 PJP 方面很敏感，通常用作初始侵入性诊断程序（见肺损伤的侵入性诊断程序）。

肺炎的非侵入性诊断

In patients with suspected pneumonia, routine sputum and blood cultures should be obtained, ideally before antibiotics are initiated or modified. Sputum cultures for Legionella species are sensitive if obtained before initiating antibiotics; however, specific culture conditions are required. Legionellosis can also be diagnosed based on urine antigen testing, which only detects Legionella pneumophila type I, the cause of most (but not all) cases of Legionella pneumonia.A nasopharyngeal wash is useful to diagnose community respiratory viral infections. The rapid test for influenza A and B may be performed using a throat or nasopharyngeal swab. Rapid antigen detection methods can provide a diagnosis within hours; however, if results are negative, a shell vial culture will take about 5 days.

重试 错误原因

真菌性肺炎的诊断依据如下：宿主因素使患者易患侵袭性曲霉菌病;感染的适当症状或体征;相容的肺部病变;血清半乳甘露聚糖或β–葡聚糖检测阳性。表明高风险的宿主因素

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for invasive aspergillosis include neutropenia for >10 days, receipt of an allogeneic HCT, prolonged use of high-dose systemic corticosteroids, or treatment with T-cell suppressants. The galactomannan assay is specific for invasive aspergillosis,[493,539] whereas the beta-glucan assay detects aspergillosis and other invasive fungal infections (including invasive candidiasis, *Pneumocystis jirovecii*, and fusariosis).[540-542] Mucormycosis yields negative serum galactomannan and beta-glucan test results.

Antigen-based detection systems have advantages and limitations. A meta-analysis showed that the galactomannan assay had a sensitivity of 70% and specificity of 89% for proven invasive aspergillosis, though the accuracy of the test varied.[495] The lack of consistent results likely relates to different cutoff values for a positive result, differences in patient populations, and possibly the use of mold-active prophylaxis. Several variables can affect the performance of the galactomannan assay,[543,544] which may account for the different results. The sensitivity of the assay is significantly reduced by concomitant mold-active antifungal agents.[494,545] False-positive results may be more common in children and allogeneic HCT recipients.[546] Historically, concomitant piperacillin/tazobactam has caused false-positive galactomannan results;[547,548] however, current formulations available in the United States rarely cause false positives.[549] False-positive beta-glucan results have also been reported in patients with surgical packing who are receiving immunoglobulin therapy and in patients receiving IV amoxicillin-clavulanate.[550,551] Despite these limitations, a patient at high risk for invasive aspergillosis (eg, prolonged neutropenia or allogeneic HCT recipient) with clinical and radiologic findings (eg, a new pulmonary nodule ≥1 cm, infiltrate) compatible with invasive aspergillosis and with a positive serum galactomannan is likely to have invasive aspergillosis, and therefore a mold-active agent (voriconazole is preferred) should be added.

Additional assays can detect histoplasmosis, coccidioidomycosis, and *Pneumocystis jirovecii* as part of the noninvasive diagnosis of pneumonia. The assay for serum or urine Histoplasma antigen is a sensitive and specific test in patients with disseminated histoplasmosis (histoplasmosis is endemic to the Central United States). Coccidioidomycosis is endemic to the Southwestern United States. Disseminated coccidioidomycosis can be diagnosed based on appropriate symptoms and signs of infection and on positive serum titers. As previously discussed, BAL is the diagnostic gold standard for PJP. However, in a small study, sputum induction with hypertonic saline was diagnostic of PJP in patients who are not HIV-infected in about 60% of cases.[552] A BAL should be performed if sputum induction is attempted, and the results are negative.

Invasive Diagnostic Procedures for Pulmonary Infiltrates

Invasive diagnostic procedures may be required in the following situations: 1) the clinical course does not suggest an acute bacterial process; 2) the infection has not responded to initial antibiotic therapy and/or; 3) noninvasive testing yields negative results. BAL has a high diagnostic yield in alveolar infiltrates, such as pneumonia caused by *Pneumocystis jirovecii*, *M tuberculosis,* and respiratory viruses. The sensitivity of BAL for focal lesions (such as nodules) is variable. In lesions >2 cm, the sensitivity of BAL ranges from 50% to 80%; however, in smaller lesions, the diagnostic yield is usually about 15%.[553] Quantitative cultures from either BAL or a protected brush catheter may increase the specificity in the diagnosis of bacterial pneumonia as distinguished from upper airway colonization in ventilated patients. It is recommended to use galactomannan and special stains or molecular techniques with BAL to aid in the identification of additional viral, protozoal, fungal, or bacterial pathogens, particularly if there is no response to the initial therapy or if diffuse infiltrates are present.

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侵袭性曲霉病的治疗包括中性粒细胞减少>10 天、接受同种异体 HCT、长期使用高剂量全身性皮质类固醇或用 T 细胞抑制剂治疗。半乳甘露聚糖检测对侵袭性曲霉病具有特异性，而β–葡聚糖检测可检测曲霉病和其他侵袭性真菌感染（包括侵袭性念珠菌病、耶氏肺孢子虫病和镰刀菌病）。毛霉菌病的血清半乳甘露聚糖和β–葡聚糖检测结果为阴性。

Antigen–based detection systems have advantages and limitations. A meta–analysis showed that the galactomannan assay had a sensitivity of 70% and specificity of 89% for proven invasive aspergillosis, though the accuracy of the test varied.The lack of consistent results likely relates to different cutoff values for a positive result, differences in patient populations, and possibly the use of mold–active prophylaxis. Several variables can affect the performance of the galactomannan assay, which may account for the different results. The sensitivity of the assay is significantly reduced by concomitant mold–active antifungal agents. False–positive results may be more common in children and allogeneic HCT recipients.Historically, concomitant piperacillin/tazobactam has caused false–positive galactomannan results;however, current formulations available in the United States rarely cause false positives. False–positive beta–glucan results have also been reported in patients with surgical packing who are receiving immunoglobulin therapy and in patients receiving IV amoxicillin–clavulanate.Despite these limitations, a patient at high risk for invasive aspergillosis (eg, prolonged neutropenia or allogeneic HCT recipient) with clinical and radiologic findings (eg, a new pulmonary nodule ⩾1 cm, infiltrate) compatible with invasive aspergillosis and with a positive serum galactomannan is likely to have invasive aspergillosis, and therefore a mold–active agent (voriconazole is preferred) should be added.

重试 错误原因

其他检测方法可以检测组织胞浆菌病、球孢子菌病和耶氏肺孢子虫病，作为肺炎非侵入性诊断的一部分。血清或尿液组织胞浆菌抗原检测是播散性组织胞浆菌病（组织胞浆菌病是美国中部的地方病）患者的敏感和特异性检测。球孢子菌病是美国西南部的地方病。播散性球孢子菌病可以根据感染的症状和体征以及阳性血清滴度进行诊断。如前所述，BAL 是 PJP 的诊断金标准。然而，在一项小型研究中，60%的非 HIV 感染者用高渗盐水诱导痰可诊断为 PJP，如果尝试诱导痰，结果为阴性，则应进行 BAL。

肺损伤的侵入性诊断程序

在以下情况下可能需要侵入性诊断程序：1）临床过程并不表明急性细菌过程; 2）感染对初始抗生素治疗没有反应和/或; 3）非侵入性测试产生阴性结果。支气管肺泡灌洗对肺泡浸润有很高的诊断率，如肺孢子虫、结核分枝杆菌和呼吸道病毒引起的肺炎。BAL 对局灶性病变（如结节）的敏感性是可变的。对于直径>2 cm 的病变，BAL 的敏感性为 50%~ 80%;而对于较小的病变，诊断率通常为 15%左右。从 BAL 或带保护刷的导管中定量培养可提高诊断细菌性肺炎的特异性，以区别于通气患者的上气道定植。 建议对 BAL 使用半乳甘露聚糖和特殊染色或分子技术，以帮助识别其他病毒、原虫、真菌或细菌病原体，特别是如果对初始治疗无反应或存在弥漫性浸润。

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BAL cultures only detect about 50% of pulmonary infection cases; therefore, it is relatively insensitive for diagnosing aspergillosis.[554] Galactomannan detection in BAL fluid appears to be more sensitive than serum detection[555,556] and can be used to support a diagnosis of probable aspergillosis.[557] In patients with focal peripheral lesions, percutaneous biopsy may increase the diagnostic yield; however, in patients with thrombocytopenia, the risk of bleeding may be unacceptably high. The microbiologic evaluation should take into account the clinical manifestations and nature of immunosuppression. In patients who are highly immunocompromised (eg, those receiving chemotherapy for acute leukemia, HCT recipients), the following studies on BAL and lung biopsies should be considered: culture and stains for bacteria, fungi, *Legionella*, mycobacteria, *Nocardia*, HSV, CMV, and community respiratory viruses (both rapid antigen and shell vial culture), and cytology or immunofluorescent studies for *Pneumocystis jirovecii*. In a patient with compatible host factors and radiologic findings, a positive galactomannan result from BAL is also indicative of probable invasive aspergillosis.[557]

For nondiagnostic BAL or percutaneous lung biopsy results, a thoracoscopic lung biopsy should be considered if an adequate platelet count is achievable. The thoracoscopic approach has less morbidity than an open lung biopsy and generally provides adequate tissue samples for the diagnosis of most infectious and noninfectious etiologies. This invasive procedure may identify the causative pathogen or the presence of a noninfectious etiology (eg, treatment-associated lung toxicity, hemorrhage, bronchiolitis obliterans organizing pneumonia [BOOP]), which may allow for the elimination of potentially toxic or unnecessary antimicrobial therapies. Thoracoscopic and open lung biopsies sometimes do not provide a definitive diagnosis, either due to sampling error or nonspecific pathologic findings.

Skin and Soft Tissue Infections

When evaluating the potential for a skin/soft tissue infection, careful examination of all line sites and perineal areas is essential. Antimicrobial therapy should be tailored to the probable organism(s): staphylococci and streptococci for catheter-associated processes, and gram-negative and anaerobic organisms for perineal processes. Vancomycin may be considered for cellulitis, disseminated papules/lesions, and infections associated with VAD (see *Treatment Modifications* in the algorithm and *Vascular Access Device Infections* in the discussion). Acyclovir, famciclovir, or valacyclovir should be considered for vesicular lesions after appropriate diagnostic tests (ie, scraping base of vesicle for HSV or VZV, direct fluorescent antibody tests, herpes virus culture) have been performed.

Skin lesions can be manifestations of systemic infection. Ecthyma gangrenosum is the most characteristic skin lesion associated with systemic *P aeruginosa* infection.[558] Similar lesions can be caused by *S aureus*, enteric gram-negative bacilli infection, and filamentous fungi (including *Aspergillus*, Mucorales, and *Fusarium* species). A rapidly progressive deep soft tissue infection with gas formation suggests clostridial myonecrosis (or polymicrobial necrotizing fascitis).[559] Broad-spectrum antibiotics and surgical debridement may be lifesaving if initiated early. Hematogenously disseminated candidiasis with skin involvement manifests as fever and erythematous cutaneous papules; blood cultures are expected to be positive for *Candida* species.

In patients with cancer who are highly immunocompromised, the differential diagnosis of skin lesions is often broad and includes noninfectious etiologies such as drug reactions, Sweet syndrome, erythema multiforme leukemia cutis, and (in the case of allogeneic HCT recipients) GVHD. Biopsy of skin lesions for histology and culture is recommended. In allogeneic HCT recipients, the differential diagnosis of

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支气管肺泡灌洗液培养仅能检出约 50%的肺部感染病例，因此，支气管肺泡灌洗液培养对曲霉病的诊断相对不敏感。BALF 液中半乳甘露聚糖的检测似乎比血清检测更敏感，可用于支持可能的曲霉病的诊断，在局灶性外周病变患者中，经皮活检可提高诊断率;然而，在血小板减少症患者中，出血风险可能高得不可接受。微生物学评价应考虑免疫抑制的临床表现和性质。 在高度免疫功能低下的患者（例如，接受化疗治疗急性白血病的患者、HCT 接受者）中，应考虑对 BAL 和肺活检进行以下研究：细菌、真菌、军团菌、分枝杆菌、诺卡氏菌、HSV、CMV 和社区呼吸道病毒的培养和染色（快速抗原和壳瓶培养），以及耶氏肺孢子虫的细胞学或免疫荧光研究。在具有相容的宿主因素和放射学结果的患者中，BAL 的半乳甘露聚糖阳性结果也提示可能的侵袭性曲霉病。

对于非诊断性 BAL 或经皮肺活检结果，如果可以获得足够的血小板计数，则应考虑进行胸腔镜肺活检。胸腔镜方法的发病率低于开胸肺活检，通常可提供足够的组织样本用于大多数感染性和非感染性病因的诊断。这种侵入性程序可以识别致病病原体或非感染性病因（例如，治疗相关的肺毒性、出血、闭塞性细支气管炎机化性肺炎[BOOP]）的存在，从而可以消除潜在毒性或不必要的抗菌治疗。胸腔镜和开放式肺活检有时不能提供明确的诊断，无论是由于采样错误或非特异性病理结果。

皮肤和软组织感染

在评估皮肤/软组织感染的可能性时，仔细检查所有线部位和会阴区域至关重要。抗菌治疗应针对可能的微生物：葡萄球菌和链球菌用于导管相关过程，革兰氏阴性和厌氧微生物用于会阴过程。对于蜂窝织炎、播散性丘疹/病变和 VAD 相关感染，可考虑使用万古霉素（参见算法中的治疗修改和讨论中的血管通路器械感染）。在进行了适当的诊断试验（即，刮除囊泡基底部的 HSV 或 VZV、直接荧光抗体试验、疱疹病毒培养）后，应考虑使用阿昔洛韦、泛昔洛韦或伐昔洛韦治疗水疱性病变。

皮肤损害可表现为全身感染。坏疽性脓疱是全身性铜绿假单胞菌感染最具特征性的皮肤病变，金黄色葡萄球菌、肠道革兰阴性杆菌感染和丝状真菌（包括曲霉菌、毛霉菌和镰刀菌属）也可引起类似病变。快速进行性深部软组织感染伴气体形成提示梭菌性肌坏死（或多微生物坏死性筋膜炎）。如果及早开始，广谱抗生素和手术清创可能会挽救生命。累及皮肤的血源性播散性念珠菌病表现为发热和皮疹性皮肤丘疹;血培养预期念珠菌属阳性。

在高度免疫功能低下的癌症患者中，皮肤病变的鉴别诊断通常很广泛，包括非感染性病因，如药物反应、Sweet 综合征、皮肤多形性红斑白血病和（在同种异体 HCT 接受者的情况下）GVHD。建议对皮肤病变进行组织学和培养活检。在异基因 HCT 受者中，

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infectious etiologies is particularly broad, and cultures from skin biopsies for bacteria, fungi, viruses, and mycobacteria should be considered when infection is suspected.

Vascular Access Device Infections

VAD infections are common due to the ubiquity of VADs in patients undergoing intensive or cyclic chemotherapy. The risk of infection varies with the device used (long-term implanted catheters versus short-term central catheters), duration of placement, and extent of the patient's immunosuppression. Short-term central catheters coated with the antimicrobial agent chlorhexidine-silver sulfadiazine (CHSS) have been shown to significantly decrease the incidence of both catheter colonization and catheter-related bloodstream infections compared with standard (non-coated) catheters.[560-562] However, this benefit with CHSS coating was not observed in the setting of patients with hematologic malignancies requiring longer use of central catheters (eg, duration of catheterization 20 days).[563] In subsequent studies that evaluated the use of CHSS-coated short-term catheters compared with controls, CHSS-coated catheters significantly decreased the incidence of colonization but showed no difference in terms of incidence of catheter-related bloodstream infections.[564-566] The use of short-term catheters coated with minocycline and rifampin has been shown to significantly decrease the risks for catheter colonization and bloodstream infections compared with either controls or CHSS-coated catheters.[567,568] However, conflicting results were reported by another study in which minocycline- and rifampin-coated catheters reduced the risk for coagulase-negative staphylococci colonization, but they increased the risk for colonization with *Candida* spp. Moreover, no significant difference was noted in the incidence of catheter-related bloodstream infections compared with controls.[569] Only limited data are available on the use of long-term catheters coated with minocycline and rifampin. In a prospective, randomized, double-blind study in patients with cancer requiring long-term catheterization (mean duration of catheterization, 63–66 days), a significant risk reduction in catheter-related bloodstream infections was observed with the coated catheter (1.6% vs. 8%; RR for uncoated vs. coated, 1.8; 95% CI, 1.4–2.3; $P = .003$).[570] Published guidelines for the prevention of catheter-related infections (based on an interdisciplinary working group involving the IDSA and CDC) recommend the use of catheters impregnated with CHSS or minocycline/rifampin in patients requiring catheterization for >5 days, if the rate of catheter-related bloodstream infections does not decrease despite implementation of comprehensive prevention measures at the local institution.[571] A meta-analysis of prospective, randomized studies showed that use of a vancomycin lock solution in patients being treated with long-term central VADs reduced the risk of bloodstream infection.[572] The Panel does not currently endorse this practice due to concerns over the emergence of bacterial resistance if this approach were widely employed. The IDSA has published guidelines on the diagnosis and management of intravascular catheter-related infections.[427]

VAD infections are categorized as entry or exit site inflammation versus tunnel infection, port pocket infection, or septic phlebitis (see *Initial Clinical Presentation* in the algorithm). The majority of these infections are caused by gram-positive pathogens, with coagulase-negative staphylococci recovered most frequently.[427] Accordingly, IV vancomycin is recommended for those infections that are serious and clinically obvious.

Most VAD exit-site infections can be treated effectively with appropriate antimicrobial therapy without the need for catheter removal. If clinical signs of catheter infection are present, a skin swab for culture from the exit site, blood culture from each port of VAD, and a peripheral culture should be obtained. In a patient with neutropenic fever and clinical signs of a VAD-associated infection, an appropriate initial regimen would consist of an agent recommended for neutropenic fever and vancomycin (see *Initial Inpatient Empiric Therapy for Uncomplicated Fever and Neutropenia* and

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感染的病因特别广泛，当怀疑感染时，应考虑皮肤活检的细菌、真菌、病毒和分枝杆菌培养。

血管通路器械感染

由于 VAD 在接受强化或周期化疗的患者中普遍存在，因此 VAD 感染很常见。感染的风险因所用器械（长期植入导管与短期中心导管）、放置时间和患者免疫抑制程度而异。 与标准的中心静脉导管相比，涂有抗菌剂氯己定–磺胺嘧啶银（CHSS）的短期中心静脉导管可显著降低导管定植和导管相关血流感染的发生率。然而，在需要更长时间使用中心导管的恶性血液病患者中，没有观察到 CHSS 涂层的这种益处（例如，导管插入持续时间 20 天）。在随后的研究中，评估了 CHSS 涂层短期导管与对照组相比的使用，CHSS 涂层导管显著降低了定植的发生率，但在导管相关血流感染的发生率方面没有显示出差异。与对照或 CHSS–涂覆的导管相比，用米诺环素和利福平涂覆的长期导管已显示出显著降低导管定植和血流感染的风险。然而，另一项研究报告了矛盾的结果，其中米诺环素和利福平涂覆的导管降低了凝固酶阴性葡萄球菌定植的风险，但它们增加了念珠菌定植的风险。 此外，与对照组相比，导管相关性血流感染的发生率无显著差异，关于长期使用米诺环素和利福平涂层导管的数据有限。在一项对需要长期导管插入术的癌症患者进行的前瞻性、随机、双盲研究中（平均导管插入持续时间，63–66 天），使用涂层导管观察到导管相关血流感染的风险显著降低（1.6% vs. 8%;未涂层 vs.涂层的 RR，1.8; 95% CI，1.4–2.3; P = .003）。已发表的导管相关感染预防指南（基于涉及 IDSA 和 CDC 的跨学科工作组）建议在需要导管插入术>5 天的患者中使用 CHSS 或米诺环素/利福平浸渍的导管，如果尽管在当地机构实施了综合预防措施，随机研究表明，在接受长期中心 VAD 治疗的患者中使用万古霉素封管液可降低血流感染的风险。专家组目前不支持这种做法，因为担心如果广泛使用这种方法会出现细菌耐药性。 IDSA 已经发布了血管内导管相关感染的诊断和管理指南。

VAD 感染被分类为入口或出口部位炎症与隧道感染、输液港袋感染或脓毒性静脉炎（参见算法中的初始临床表现）。这些感染大多数由革兰氏阳性病原体引起，其中凝固酶阴性葡萄球菌最常见。因此，对于严重和临床明显的感染，建议静脉注射万古霉素。

大多数 VAD 出口部位感染可以通过适当的抗菌治疗有效治疗，而无需移除导管。如果存在导管感染的临床体征，则应获得出口部位的皮肤拭子培养、VAD 每个端口的血液培养和外周培养。在具有血小板减少性发热和 VAD 相关感染的临床体征的患者中，适当的初始方案将包括推荐用于血小板减少性发热的药物和万古霉素（参见用于无并发症发热和中性粒细胞增多症的初始住院经验性治疗，

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Treatment Modifications in the algorithm). Linezolid is not advised as routine therapy for catheter-related infections nor is it FDA-approved for this indication. For a clinically apparent, serious, catheter-related infection (such as a tunnel or port pocket infection, or septic phlebitis), catheter removal should be performed immediately.

Determining the role of the catheter in bloodstream infections is frequently difficult if local catheter inflammation is not evident. A useful diagnostic tool for detecting VAD infections is the DTP. Early positivity of central venous blood cultures predicts catheter-related bacteremia and may be used to avoid unnecessary catheter removal in critically ill patients. It was shown that a DTP of ≥120 minutes (between centrally and peripherally drawn blood cultures) is highly sensitive and specific for diagnosing catheter-related bacteremia.[573-577] However, these studies were only performed in patients with removable catheters, not implanted catheters (eg, Hickman or Mediport) that are frequently used in patients undergoing cancer treatment.

Most catheter-associated bloodstream infections respond to antimicrobial therapy alone without catheter removal, but immediate catheter removal is favored for patients with bloodstream infections caused by fungi (yeasts or molds) or nontuberculous mycobacteria (eg, *Mycobacterium chelonae*, *Mycobacterium abscessus*, *Mycobacterium fortuitum*).[427] Bloodstream infections caused by *Bacillus* organisms, *Candida, S aureus, Acinetobacter*, *C jeikeium*, *P aeruginosa*, *S maltophilia,* and VRE may be difficult to eradicate with antimicrobial therapy alone; therefore, catheter removal should be considered as part of initial therapy. In patients with mucositis, the bowel is likely to be the portal of entry for bloodstream infection by GI flora such as *Candida* spp. and enterococci. DTP may be useful to distinguish whether bloodstream infection by these organisms is catheter-related and to guide whether catheter removal should be performed. If not removed initially, catheter removal is advised for known or suspected VAD-associated bloodstream infections if the organism is recovered from blood obtained 48 hours after initiation of appropriate antibiotic therapy. In patients with VAD infection and clinical instability, removal of the infected catheter should be performed immediately.

The Panel recognizes that certain conditions may preclude the ability to immediately replace IV catheters, such as limited options for IV access and thrombocytopenia refractory to platelet products. Administering antibiotics through each lumen of the involved catheter has been suggested to avoid disease progression caused by microbial sequestration. Some experts believe supplemental urokinase infusions can be helpful in patients with catheter-related infections.[578] However, the Panel believes data are insufficient to recommend either of these approaches.

Central Nervous System Infections

CNS infections in patients with cancer can be divided into surgical and nonsurgical complications. The IDSA has published guidelines on the management of bacterial meningitis.[579] The most common organisms infecting intraventricular devices are coagulase-negative staphylococci, *S aureus,* and *Propionibacterium acnes.* Enterobacteriaceae and *P aeruginosa* account for only 10% of these infections. Coagulase-negative staphylococci and *Propionibacterium acnes* usually cause indolent late postoperative infections. Therapy with systemic antibiotics and removal of the entire device are the most effective approaches to eradicate infection. Use of parenteral and intraventricular instillation of antibiotics without removal of the device may not be effective, and recrudescence of infection is common. Antibiotic therapy should be tailored to the specific pathogen isolated from cerebrospinal fluid. In an acutely ill patient with suspected meningitis related to previous neurosurgery, empiric therapy can include parenteral vancomycin (which has activity against *Staphylococcus*, *Streptococcus*, and *Propionibacterium* species) in combination with

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算法中的治疗修改）。利奈唑胺不建议作为导管相关感染的常规治疗，也未获得 FDA 批准用于该适应症。对于临床上明显的严重导管相关感染（如隧道或输液港袋感染或脓毒性静脉炎），应立即取出导管。

如果局部导管炎症不明显，则通常难以确定导管在血流感染中的作用。检测 VAD 感染的一个有用的诊断工具是 DTP。中心静脉血培养早期阳性可预测导管相关菌血症，可用于避免危重患者不必要的导管拔除。研究表明，DTP ≥120 分钟（中心和外周抽取的血培养之间）对于诊断导管相关菌血症具有高度敏感性和特异性。然而，这些研究仅在使用可移除导管的患者中进行，而不是在接受癌症治疗的患者中经常使用的植入导管（例如，Hickman 或 Mediport）。

大多数导管相关性血流感染对单独的抗菌治疗有反应而不拔除导管，但对于真菌引起的血流感染患者，立即拔除导管是有利的（酵母菌或霉菌）或非结核分枝杆菌（例如，龟分枝杆菌（Mycobacterium chelonae）、龟分枝杆菌（Mycobacterium dumessus）、偶发分枝杆菌（Mycobacterium fortuturonae））。由芽孢杆菌属生物体、念珠菌属、金黄色葡萄球菌、不动杆菌属、杰氏梭菌、铜绿假单胞菌、嗜麦芽窄食单胞菌、VRE 可能难以单独用抗菌药物治疗根除;因此，导管移除应被视为初始治疗的一部分。在粘膜炎患者中，肠道可能是胃肠道植物群（如念珠菌属）血流感染的入口。和肠球菌。DTP 可能有助于区分这些微生物的血流感染是否与导管相关，并指导是否应进行导管移除。如果最初未取出，建议取出导管，或疑似 VAD 相关的血流感染，如果在开始适当的抗生素治疗后 48 小时从血液中回收到微生物。在 VAD 感染和临床不稳定的患者中，应立即取出受感染的导管。

专家小组认识到，某些情况可能会妨碍立即更换静脉导管的能力，例如静脉通路的选择有限和血小板产品难治性血小板减少症。已建议通过相关导管的每个管腔施用抗生素，以避免由微生物隔离引起的疾病进展。一些专家认为，补充尿激酶输注对导管相关感染的患者有帮助。然而，专家组认为，数据不足以推荐这两种方法。

中枢神经系统感染

癌症患者的 CNS 感染可分为手术和非手术并发症。IDSA 已经发布了细菌性脑膜炎的治疗指南。感染脑室内装置的最常见微生物是凝固酶阴性葡萄球菌、金黄色葡萄球菌和痤疮丙酸杆菌。肠杆菌科和 P

铜绿假单胞菌仅占这些感染的 10%。凝固酶阴性葡萄球菌和痤疮丙酸杆菌常引起无痛性晚期术后感染。全身抗生素治疗和移除整个装置是根除感染的最有效方法。在不取出装置的情况下使用肠外和脑室内滴注抗生素可能无效，并且感染复发是常见的。抗生素治疗应针对从脑脊液中分离的特定病原体。在疑似与既往神经外科手术相关的脑膜炎的急性病患者中，经验性治疗可包括肠外万古霉素（其对葡萄球菌、链球菌和丙酸杆菌属具有活性）联合

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ceftazidime, cefepime, or meropenem (which have activity against Enterobacteriaceae and *P aeruginosa*).[579]

CNS infections unrelated to neurosurgery are relatively uncommon in patients with cancer. Initial evaluation generally involves a head MRI (preferred) to rule out intracranial bleeding and/or a CT scan in addition to a lumbar puncture (assuming there are no contraindications). Cerebrospinal fluid studies should be tailored to specific host factors, epidemiologic exposures (eg, travel history), and clinical presentation. At a minimum, cell counts with differential, glucose and protein levels, Gram stain and bacterial culture, cryptococcal antigen, and fungal culture on cerebrospinal fluid should be obtained. Noninfectious causes of meningitis include nonsteroidal anti-inflammatory agents, TMP/SMX, carcinomatous meningitis, and serum sickness (eg, associated with anti-lymphocyte immunoglobulin preparations).

For suspected CNS infections, infectious disease and neurology consultation is strongly recommended, and empiric therapy should be initiated pending infectious disease consult. Empiric therapy for presumed meningitis should include an anti-pseudomonal beta-lactam agent that readily enters the CSF (eg, cefepime, ceftazidime, meropenem) plus vancomycin plus ampicillin (to cover listeriosis) (see *Treatment Modifications* in the algorithm). If meropenem is used, addition of ampicillin is unnecessary because meropenem is active against *Listeria*. This regimen has activity against the common causes of bacterial meningitis, including penicillin-resistant pneumococci and listeriosis. In patients at risk for *P aeruginosa* meningitis (eg, neutropenia, neurosurgery within the past 2 months, allogeneic HCT, history of *P aeruginosa* infection), use of cefepime or meropenem instead of ceftriaxone in the initial empiric regimen is advised. The antibiotic regimen should be tailored based on culture results.

The IDSA guidelines (2004) for the management of bacterial meningitis support the incorporation of adjuvant dexamethasone in pediatric patients with *H influenzae* type B meningitis and in adult patients with pneumococcal meningitis.[580] In patients with suspected encephalitis (ie, fever, mental status changes, CSF pleocytosis), high-dose IV acyclovir with sufficient hydration should be considered as empiric therapy for HSV in addition to an appropriate antibacterial regimen.[581] An MRI and the following CSF studies should be performed: 1) cell count with differential; 2) glucose and protein levels; 3) Gram stain and culture for bacteria; 4) cryptococcal antigen and fungal culture; and 5) PCR for HSV. PCR for West Nile virus and other arboviruses should be considered in patients with exposure to endemic areas. Culture and PCR for tuberculosis should be considered in patients with known or suspected exposure to tuberculosis (eg, residence in an endemic area, shelter, or prison; previous positive PPD [purified protein derivative]). In patients with severe impairment of cellular immunity (eg, allogeneic HCT recipients, advanced AIDS), additional CSF studies should be considered (such as PCR for CMV, VZV, human herpesvirus–6 type B [HHV-6B], and toxoplasmosis). For cases of HHV-6B–associated encephalitis in patients who are severely immunocompromised, such as those who have received an allogeneic transplant, treatment is recommended; however, the optimal therapy is not known (with either foscarnet or ganciclovir).[581] Cytology to evaluate for CNS malignancy as a cause of meningitis or encephalitis should also be considered.

Brain abscesses usually manifest with headache, focal neurologic findings, or seizures. An MRI typically shows single or multiple lesions with edema and ring enhancement.[582] Bacterial abscesses in patients who are non-immunocompromised are typically caused by dental flora. In patients with prolonged neutropenia and in allogeneic HCT recipients, CNS aspergillosis must be considered. A chest CT showing a new nodule or infiltrate and a positive serum galactomannan result in this setting is highly

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头孢他啶、头孢吡肟或美罗培南（对肠杆菌科和铜绿假单胞菌具有活性）。

与神经外科手术无关的 CNS 感染在癌症患者中相对少见。初始评估通常包括头部 MRI（首选）以排除颅内出血和/或 CT 扫描以及腰椎穿刺（假设无禁忌症）。脑脊液研究应根据特定的宿主因素、流行病学暴露（如旅行史）和临床表现进行调整。至少应获得脑脊液细胞分类计数、葡萄糖和蛋白水平、革兰氏染色和细菌培养、隐球菌抗原和真菌培养。脑膜炎的非感染性病因包括非甾体类抗炎药、TMP/SMX、癌性脑膜炎和血清病（例如，与抗淋巴细胞免疫球蛋白制剂相关）。

对于疑似 CNS 感染，强烈建议进行感染性疾病和神经病学会诊，并应在感染性疾病会诊前开始经验性治疗。对假定脑膜炎的经验性治疗应包括易进入 CSF 的抗假单胞菌β–内酰胺类药物（如头孢吡肟、头孢他啶、美罗培南）加万古霉素加氨苄青霉素（以覆盖脑膜炎）（见算法中的治疗修改）。如果使用美罗培南，则不需要添加氨苄青霉素，因为美罗培南对李斯特菌有活性。该方案对细菌性脑膜炎的常见原因，包括青霉素耐药肺炎球菌和脑膜炎有活性。在有铜绿假单胞菌脑膜炎风险的患者中（例如，中性粒细胞减少症、过去 2 个月内接受过神经外科手术、同种异体 HCT、铜绿假单胞菌感染史），

建议在初始经验方案中使用头孢吡肟或美罗培南代替头孢曲松。抗生素治疗方案应根据培养结果进行调整。

IDSA 关于细菌性脑膜炎管理的指南（2004 年）支持在患有 B 型流感嗜血杆菌脑膜炎的儿科患者和患有肺炎球菌脑膜炎的成人患者中加入辅助性地塞米松。（即发热、精神状态改变、CSF 细胞增多），表示“高”之义除了适当的抗菌治疗方案外，静脉注射阿昔洛韦并充分水化应被视为 HSV 的经验性治疗。MRI 和随后的 CSF 检查应执行：1）细胞计数和分类; 2）葡萄糖和蛋白质水平; 3）革兰氏染色和细菌培养; 4）隐球菌抗原和真菌培养;和 5）HSV 的 PCR。应考虑对暴露于流行区的患者进行西尼罗河病毒和其他虫媒病毒的 PCR 检测。 对于已知或疑似接触结核病的患者（例如，居住在流行区、收容所或监狱;既往 PPD [纯化蛋白衍生物]阳性），应考虑进行结核病培养和 PCR。在细胞免疫严重受损的患者中（例如，同种异体 HCT 接受者、晚期 AIDS），应考虑进行额外的 CSF 研究（例如 CMV、VZV、人疱疹病毒 6 型 B [HHV–6 B]和弓形虫病的 PCR）。对于 HHV–6 B 相关性脑炎的严重免疫功能低下的患者，如接受过同种异体移植的患者，建议进行治疗;然而，最佳治疗尚不清楚（膦甲酸或更昔洛韦）。还应考虑进行细胞学检查，以评估 CNS 恶性肿瘤是否是脑膜炎或脑炎的原因。

脑水肿通常表现为头痛、局灶性神经学检查结果或癫痫发作。MRI 通常显示单个或多个病灶伴水肿和环形强化。非免疫功能低下患者的细菌性脓肿通常由牙齿植物群引起。在长期中性粒细胞减少的患者和同种异体 HCT 接受者中，必须考虑 CNS 曲霉菌病。在这种情况下，胸部 CT 显示新的结节或浸润以及血清半乳甘露聚糖阳性，这是非常重要的

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suggestive of pulmonary aspergillosis with CNS dissemination. In patients with impaired cellular immunity, other causes of CNS abscesses include toxoplasmosis, nocardiosis, cryptococcosis, and mycobacterial infections. Noninfectious etiologies in patients with impaired cellular immunity include CNS malignancies (such as secondary lymphomas) and Epstein-Barr virus (EBV)–associated post-transplantation lymphoproliferative disorder (PTLD). Given the broad differential diagnosis of new CNS lesions in highly immunocompromised patients, a brain biopsy is strongly recommended (if feasible) with material submitted for histology and culture. Cultures and stains should include bacteria, fungi, mycobacteria, and *Nocardia* species.

In non-immunocompromised patients with a bacterial brain abscess, initial therapy with ceftriaxone plus metronidazole is advised.[17,582,583] In patients with prolonged neutropenia without corticosteroids or lymphocyte-depleting agents, a reasonable initial regimen consists of combination cefepime, metronidazole, and voriconazole; however, IV voriconazole (but not the oral formulation) may worsen renal disease in patients with significant pre-existing renal impairment. Voriconazole (as well as itraconazole and posaconazole) has important drug-drug interactions with certain antiseizure agents (eg, phenytoin); therefore, the voriconazole package insert should be reviewed to guide dosing of these agents. In allogeneic HCT recipients and other patients with severe T-cell impairment, addition of high-dose TMP/SMX should be considered to cover toxoplasmosis and nocardiosis, pending a definitive diagnosis. An infectious disease consultation is advised in all cases of suspected or documented CNS infection.

Therapy for Invasive Fungal Infections

Invasive Candidiasis

Candida species are the fourth most common cause of nosocomial bloodstream infections in the United States.[584,585] The crude mortality of candidemia ranges from 20% to 40%.[585,586] This variable mortality rate reflects the presence of serious comorbidities (such as malignancy and neutropenia), patient population (adult versus pediatric), and illness requiring prolonged periods in the intensive care unit. *Candida albicans* is the most common *Candida* species isolated from the blood.[585] The proportion of non-albicans *Candida* species varies among different centers, but accounts for approximately 50% of blood stream isolates.

A randomized study comparing IV fluconazole (400 mg daily) with amphotericin B as therapy for candidemia in patients who are non-neutropenic found both regimens equally effective, but fluconazole had less toxicity.[587] In a subsequent study of non-neutropenic cases with candidemia, combination therapy with a higher dose of fluconazole (800 mg daily) and amphotericin B led to improved clearance of candidemia compared with fluconazole alone, but the combination regimen was associated with significantly more nephrotoxicity and with no survival benefit.[510] Voriconazole was as equally effective as, but less nephrotoxic than a strategy of amphotericin B followed by fluconazole in non-neutropenia with invasive candidiasis.[588] In trials of “invasive candidiasis,” most patients had candidemia, but those with deep organ involvement (eg, peritoneal, hepatic, or renal candidiasis) without positive blood cultures were also eligible for enrollment.

Four phase III randomized trials have been performed evaluating echinocandins as initial therapy for invasive candidiasis.[178,589-591] When caspofungin was compared with conventional amphotericin B, there was a trend for a higher favorable response (defined as resolution of clinical symptoms and culture-confirmed eradication) rate in the caspofungin arm (73% vs. 62%) in the modified intent-to-treat analysis.[590] Among patients who met prespecified criteria for evaluation (those who met eligibility criteria and received at least 5 days of the study drug), caspofungin resulted in a significantly higher success rate compared with amphotericin

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提示肺部曲霉菌病伴中枢神经系统播散在细胞免疫受损的患者中，CNS 疾病的其他原因包括弓形虫病、诺卡氏菌病、隐球菌病和分枝杆菌感染。细胞免疫受损患者的非感染性病因包括 CNS 恶性肿瘤（如继发性淋巴瘤）和 EB 病毒（EBV）相关的移植后淋巴组织增生性疾病（PTLD）。鉴于高度免疫功能低下患者新发 CNS 病变的广泛鉴别诊断，强烈建议（如果可行）进行脑活检，并提交材料进行组织学和培养。培养物和染色剂应包括细菌、真菌、分枝杆菌和诺卡氏菌。

对于非免疫功能低下的细菌性脑脓肿患者，建议采用头孢曲松+甲硝唑作为初始治疗方案。对于长期中性粒细胞减少且无皮质类固醇或淋巴细胞清除剂的患者，合理的初始治疗方案包括头孢吡肟、甲硝唑和伏立康唑联合治疗;然而，静脉注射伏立康唑（而非口服制剂）可能会加重既往存在严重肾损害的患者的肾脏疾病。伏立康唑（与伊曲康唑和泊沙康唑一样）是重要的药物，

与某些抗癫痫药物（如苯妥英）的相互作用;因此，应审查伏立康唑药品说明书，以指导这些药物的给药。在异基因 HCT 接受者和其他严重 T 细胞损伤的患者中，应考虑添加高剂量 TMP/SMX 以覆盖弓形虫病和诺卡氏菌病，等待明确诊断。对于所有疑似或记录的中枢神经系统感染病例，建议进行传染病咨询。

侵袭性真菌感染的治疗

侵袭性念珠菌病

念珠菌属是美国医院内血流感染的第四大常见原因。该可变死亡率反映了严重合并症（如恶性肿瘤和中性粒细胞减少症）、患者人群（成人与儿童）和需要在重症监护室长期治疗的疾病的存在。白色念珠菌是最常见的念珠菌属，非白色念珠菌的比例在不同的中心不同，但约占 50%的血流分离株。

A randomized study comparing IV fluconazole (400 mg daily) with amphotericin B as therapy for candidemia in patients who are non–neutropenic found both regimens equally effective, but fluconazole had less toxicity.In a subsequent study of non–neutropenic cases with candidemia, combination therapy with a higher dose of fluconazole (800 mg daily) and amphotericin B led to improved clearance of candidemia compared with fluconazole alone, but the combination regimen was associated with significantly more nephrotoxicity and with no survival benefit.Voriconazole was as equally effective as, but less nephrotoxic than a strategy of amphotericin B followed by fluconazole in non–neutropenia with invasive candidiasis.In trials of “invasive candidiasis,” most patients had candidemia, but those with deep organ involvement (eg, peritoneal, hepatic, or renal candidiasis) without positive blood cultures were also eligible for enrollment.

重试 错误原因

已经进行了四项 III 期随机试验，评估棘白菌素作为侵袭性念珠菌病的初始治疗。当卡泊芬净与常规阿替霉素 B 相比时，有一种倾向，卡泊芬净组的根除率（定义为临床症状缓解和培养证实的根除）在符合预先规定的评估标准的患者中，（符合合格标准并接受至少 5 天研究药物治疗的患者），卡泊芬净的成功率显著高于阿替沙星

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B (81% vs. 65%; 95.6% CI, 1.1–29.7; *P* = .03). Caspofungin was less toxic than amphotericin B. Similarly, micafungin was shown to be as effective as L-AmB for invasive candidiasis, with fewer treatment-related adverse events (including those that led to treatment discontinuation).[589] Anidulafungin was not inferior to fluconazole as therapy for invasive candidiasis and was possibly more efficacious.[178] At the end of IV therapy, successful outcomes (based on both clinical and microbiologic responses; primary endpoint) were achieved in a higher proportion of patients treated with anidulafungin compared with fluconazole (76% vs. 60%; 95% CI, 3.9–27.0; *P* = .01), though a center effect was observed in this study. Finally, caspofungin and micafungin were shown to be equally safe and efficacious as treatment for invasive candidiasis.[591]

The IDSA has published detailed updated guidelines for the management of candidiasis recommending fluconazole or an echinocandin as initial therapy for most non-neutropenic adult patients; an echinocandin is preferred in critically ill patients.[592] Transition from an echinocandin to fluconazole is recommended for patients who have isolates that are likely to be susceptible to fluconazole (eg, *Candida albicans*), who are clinically stable, and who have not had recent azole exposure.[592] Fluconazole-resistant *Candida* isolates are frequently cross-resistant to other azoles;[593] therefore, if candidemia occurs in a patient with recent azole exposure, a switch in class (eg, to an echinocandin) is recommended. *Candida krusei* is generally resistant to fluconazole. An echinocandin is the preferred therapy for *Candida glabrata* stains due to their variable sensitivity to azoles;[592] however, transition to fluconazole or voriconazole can be considered if azole susceptibility is documented. *Candida auris* may be resistant to fluconazole or echinocandins. Echinocandins have reduced sensitivity to *Candida parapsilosis* compared to other candidal strains; fluconazole is recommended in this setting.[592]

The IDSA recommends an echinocandin as initial therapy for candidemia in most neutropenic patients.[592] The NCCN Guidelines Panel agrees with this recommendation (category 1), but notes that because studies evaluating echinocandins have included very small numbers of neutropenic patients, the optimal therapy for invasive candidiasis in this population is not definitive. Given the availability of safer alternatives, the Panel does not recommend amphotericin B products routinely for candidemia, although such agents may be considered in unusual or complicated cases, such as instances of meningitis and endocarditis.

Invasive Aspergillosis

Voriconazole has been evaluated as primary therapy for invasive aspergillosis. In an open-label, multicenter, randomized trial, voriconazole resulted in a significantly higher success rate (including complete and partial responses) compared with amphotericin B (53% vs. 32%; 95% CI, 10.4–32.9) and was associated with an improved survival rate at 12 weeks (71% vs. 58%; HR, 0.59; 95% CI, 0.40–0.88) in this patient population.[594] Success rates were similar for the two treatment arms in the subgroup of patients with neutropenia (51% with voriconazole vs. 32% with amphotericin B). In a retrospective analysis of 86 patients with CNS aspergillosis treated with voriconazole either as primary or subsequent-line therapy, 35% had a complete or partial response.[595] This success rate compares favorably to a previous series in which the frequency of successful responses to amphotericin B in CNS aspergillosis was almost nil.[596] Considerable inter-individual variability in voriconazole exposure can occur, and the utility of monitoring drug levels is controversial.[597,598] Studies with a few patients have noted a relationship between low plasma voriconazole levels and disease progression,[147] and between high voriconazole levels and toxicity.[145,599] Voriconazole blood levels that are at least 1 to 2 mcg/mL are thought to be required for efficacy. One week after initiating treatment with voriconazole, it is recommended that trough levels by TDM be obtained to ensure adequate

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B（81% vs. 65%; 95.6% CI，1.1–29.7; P = .03）。卡泊芬净的毒性低于阿替霉素 B。同样，米卡芬净被证明与 L–AmB 一样有效治疗侵袭性念珠菌病，治疗相关不良事件（包括导致治疗中止的不良事件）较少。阿尼芬净治疗侵袭性念珠菌病的疗效不劣于氟康唑，可能更有效。（基于临床和微生物学缓解;主要终点），阿尼芬净治疗的患者比例高于氟康唑（76% vs. 60%; 95% CI，3.9– 27.0; P = 0.01），尽管在本研究中观察到中心效应。最后，卡泊芬净和米卡芬净作为侵袭性念珠菌病的治疗同样安全有效。

IDSA 已经发布了详细的更新的念珠菌病管理指南，推荐氟康唑或棘白菌素作为大多数非血小板减少症成人患者的初始治疗;重症患者首选棘白菌素。对于分离株可能对氟康唑敏感的患者，建议从棘白菌素过渡到氟康唑（例如，白色念珠菌），临床稳定，近期未暴露于唑类药物。

氟康唑耐药念珠菌分离株经常对其他唑类药物交叉耐药;因此，如果最近接触唑类药物的患者发生念珠菌血症，建议转换类别（例如，使用棘白菌素）。克柔念珠菌对氟康唑普遍耐药。由于光滑念珠菌对唑类药物的敏感性不同，棘白菌素是其首选治疗方法;然而，如果记录到唑类药物敏感性，则可以考虑过渡到氟康唑或伏立康唑。

耳念珠菌可能对氟康唑或棘白菌素耐药。与其他念珠菌菌株相比，棘白菌素对近平滑念珠菌的敏感性降低;在这种情况下推荐使用氟康唑。

IDSA 推荐棘白菌素作为大多数血小板减少症患者念珠菌血症的初始治疗。NCCN 指南小组同意这一建议（第 1 类），但指出，由于评估棘白菌素的研究包括了非常少的血小板减少症患者，因此对这一人群侵袭性念珠菌病的最佳治疗方法并不确定。考虑到更安全的替代品的可用性，专家小组不建议将阿替霉素 B 产品常规用于念珠菌血症，尽管在不寻常或复杂的情况下可能会考虑使用此类药物，例如脑膜炎和心内膜炎。

侵袭性曲霉病

伏立康唑已被评价为侵袭性曲霉病的主要治疗方法。在一项开放标签、多中心、随机试验中，伏立康唑的成功率显著较高（包括完全缓解和部分缓解）与阿替西霉素 B 相比（53%与 32%; 95%CI，10.4–32.9），并与该患者人群 12 周生存率的提高相关（71%与 58%; HR，0.59; 95%CI，0.40–0.88）。在中性粒细胞减少患者亚组中，两个治疗组的成功率相似（伏立康唑组为 51%，阿替霉素 B 组为 32%）。 在一项对 86 例接受伏立康唑作为主要或辅助治疗的 CNS 曲霉病患者进行的回顾性分析中，35%的患者完全或部分缓解。这一成功率与之前的一系列研究相比是有利的，在之前的一系列研究中，CNS 曲霉病中对阿替霉素 B 的成功缓解率几乎为零。伏立康唑暴露可能存在相当大的个体间差异，监测药物水平的效用是有争议的。对少数患者的研究已经注意到低血浆伏立康唑水平与疾病进展之间的关系，以及高伏立康唑水平与毒性之间的关系。伏立康唑血液浓度至少为 1 至 2 微克/毫升被认为是有效的。开始伏立康唑治疗后一周，建议通过 TDM 获得谷浓度，以确保足够的

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plasma concentration of the drug. Obtaining a serum voriconazole level should be considered in cases of breakthrough or refractory fungal disease or drug toxicity.

It is not clear what the optimal therapy is for breakthrough invasive aspergillosis in patients receiving mold-active prophylaxis. Breakthrough invasive aspergillosis in a patient receiving oral posaconazole prophylaxis may be caused by inadequate oral bioavailability due to mucositis or poor oral intake, or possibly resistance. Some experts would advise changing to a different class of antifungals (such as a lipid formulation of amphotericin B, with or without an echinocandin). Others would use IV voriconazole with or without an echinocandin.

Lipid formulations of amphotericin B have at least comparable efficacy and reduced renal toxicity compared to conventional AmB-D. Some investigators have persuasively argued that lipid formulations should be considered suitable replacements for amphotericin B for primary therapy for many invasive fungal infections.[481] ABCD was equally as effective as, but less nephrotoxic than amphotericin B as primary therapy for invasive aspergillosis.[600] ABLC was shown to be as safe and efficacious as therapy for invasive aspergillosis based on an analysis of a registry database.[601]

A randomized study compared L-AmB at either 3 or 10 mg/kg/day for 14 days, followed by 3 mg/kg/day as therapy for invasive mold infections.[602] Response rates (both complete and partial responses) after completion of treatment with the 3 mg/kg/day and 10 mg/kg/day dose groups were similar (50% vs. 46%); the 12-week survival rates were 72% and 59%, respectively (95% CI, -0.2–26%). The high-dose group was associated with significantly higher incidences of nephrotoxicity and hypokalemia, which suggested that the 3 mg/kg/day dosing was more optimal in this patient population.[602] Because 97% of enrolled patients had invasive aspergillosis, this study does not permit conclusions about optimal L-AmB dosing in patients with other mold infections (such as mucormycosis).

Echinocandins have not been evaluated as initial monotherapy for invasive aspergillosis in clinical trials. Caspofungin for treatment of refractory infections in patients with invasive aspergillosis led to a favorable response in 37 (45%) of 83 patients.[170] It might be possible to use combination antifungal therapy pairing an echinocandin with either an amphotericin B preparation or an azole with activity against *Aspergillus* species. The rationale is that echinocandins target a unique site (the beta-glucan constituent of the fungal cell wall), which is distinct from the polyenes and azoles that target the fungal cell membrane. The combination of an echinocandin with an azole or amphotericin B has shown neutral to synergistic activity in vitro. Enhanced efficacy of combination regimens pairing an echinocandin with either an azole or an amphotericin B formulation was observed in some animal models of invasive aspergillosis[603-606] but not in others.[607-609] In two small retrospective series, the combination of caspofungin and L-AmB for infections refractory to first-line therapy led to a favorable outcome in approximately 40% to 60% of patients with invasive aspergillosis, although these series included cases of “possible” or “probable” aspergillosis.[610,611] Marr et al reported a significant improvement in the 3-month survival rate with voriconazole plus caspofungin compared with voriconazole alone in a small retrospective analysis (N = 47) of invasive aspergillosis refractory to first-line thearpy.[612] This database study, although encouraging, involved small numbers of patients and the two groups of patients evaluated were non-contemporaneous; therefore, other host and infection-related factors may have influenced the outcome. A noncomparative study of caspofungin combined with other mold-active drugs as subsequent-line therapy for invasive aspergillosis reported a success rate of 49% (25/51) at 12 weeks after initiation of combination therapy,[613] which was similar to caspofungin monotherapy.[170] In an open-label study of invasive aspergillosis, micafungin combined with other antifungals led to a successful response in 29% (5/17) of patients treated as primary therapy and 35% (60/174) of patients with infection refractory to first-line therapy.[614] These results did

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药物的血浆浓度。在出现突破性或难治性真菌病或药物毒性的情况下，应考虑获得血清伏立康唑水平。

目前尚不清楚接受霉菌活性预防的患者中突破性侵袭性曲霉病的最佳治疗方法是什么。口服泊沙康唑预防性治疗患者的突破性侵袭性曲霉病可能是由于粘膜炎或口服摄入量不足导致的口服生物利用度不足，或可能是耐药性。一些专家会建议换用不同种类的抗真菌药（如阿替霉素 B 的脂质制剂，含或不含棘白菌素）。其他人将使用静脉伏立康唑与或不与棘白菌素。

与常规 AmB-D 相比，阿替霉素 B 的脂质制剂具有至少相当的功效和降低的肾毒性。一些研究者令人信服地认为脂质制剂应该被认为是阿替霉素 B 的合适替代品，用于许多侵袭性真菌感染的主要治疗。作为侵袭性曲霉病的主要治疗，ABCD 与阿替霉素 B 同样有效，但肾毒性较低。基于对登记数据库的分析，ABLC 被证明与侵袭性曲霉病的治疗一样安全有效。

一项随机研究比较了 L-AmB 以 3 或 10 mg/kg/天持续 14 天，随后以 3 mg/kg/天作为侵入性霉菌感染的治疗。3 mg/kg/天和 10 mg/kg/天剂量组完成给药后的缓解率（完全和部分缓解）为

相似（50% vs. 46%）；12 周生存率分别为 72%和 59%（95% CI，-0.2-26%）。高剂量组与肾毒性和低钾血症的发生率显著较高相关，这表明 3 mg/kg/天的剂量在该患者人群中更优，因为 97%的入组患者患有侵袭性曲霉病，本研究无法得出关于其他霉菌感染（如毛霉菌病）患者的最佳 L-AmB 剂量的结论。

Echinocandins have not been evaluated as initial monotherapy for invasive aspergillosis in clinical trials. Caspofungin for treatment of refractory infections in patients with invasive aspergillosis led to a favorable response in 37 (45%) of 83 patients.It might be possible to use combination antifungal therapy pairing an echinocandin with either an amphotericin B preparation or an azole with activity against Aspergillus species. The rationale is that echinocandins target a unique site (the beta-glucan constituent of the fungal cell wall), which is distinct from the polyenes and azoles that target the fungal cell membrane. The combination of an echinocandin with an azole or amphotericin B has shown neutral to synergistic activity in vitro. Enhanced efficacy of combination regimens pairing an echinocandin with either an azole or an amphotericin B formulation was observed in some animal models of invasive aspergillosisbut not in others.In two small retrospective series, the combination of caspofungin and L-AmB for infections refractory to first-line therapy led to a favorable outcome in approximately 40% to 60% of patients with invasive aspergillosis, although these series included cases of “possible” or “probable” aspergillosis. Marr et al reported a significant improvement in the 3-month survival rate with voriconazole plus caspofungin compared with voriconazole alone in a small retrospective analysis (N = 47) of invasive aspergillosis refractory to first-line thearpy.This database study, although encouraging, involved small numbers of patients and the two groups of patients evaluated were non-contemporaneous; therefore, other host and infection-related factors may have influenced the outcome. A noncomparative study of caspofungin combined with other mold-active drugs as subsequent-line therapy for invasive aspergillosis reported a success rate of 49% (25/51) at 12 weeks after initiation of combination therapy,which was similar to caspofungin monotherapy.In an open-label study of invasive aspergillosis, micafungin combined with other antifungals led to a successful response in 29% (5/17) of patients treated as primary therapy and 35% (60/174) of patients with infection refractory to first-line therapy.These results did

重试 错误原因

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not appear favorable to response rates observed with micafungin alone (50% and 41% in primary and refractory treatment groups, respectively); however, the patient numbers in the micafungin monotherapy arms were too small to permit comparisons. In addition, the initial micafungin dose (75 mg/day) used in this study was low by current standards. More recently, data from a randomized, prospective clinical trial comparing voriconazole versus voriconazole plus anidulafungin as primary therapy for invasive aspergillosis evaluated response based on 6-week mortality (N = 454 patients with hematologic malignancies or HCT).[615] The combination therapy had a trend towards reduced mortality compared to voriconazole alone (19.3% vs. 27.5%, respectively; 95% CI, -19.0–1.5; *P* = .087).[615]

Isavuconazonium sulfate was approved in 2015 for the treatment of invasive aspergillosis and invasive mucormycosis. Unlike other azoles, isavuconazonium sulfate is dosed as a prodrug that is broken down to the active component, isavuconazole, upon infusion.[616] A phase III, randomized trial comparing isavuconazonium sulfate to voriconazole for the primary treatment of invasive aspergillosis and other filamentous fungi showed the non-inferiority of isavuconazonium sulfate compared with voriconazole (all-cause mortality of 19% vs. 20%; adjusted difference -1.0%; 95% CI, -7.8–5.7%).[617] Treatment-emergent adverse events were similar between isavuconazonium sulfate and voriconazole (96% vs. 98%; *P* = .122) with GI disorders and infections or infestations as the most common (see *Toxicities and Drug-Drug Interactions of Azoles* for more information on safety). Isavuconazonium sulfate demonstrated a lower incidence of hepatobiliary disorders, eye disorders, and skin or subcutaneous tissue disorders. Drug-related adverse events were also lower for isavuconazonium sulfate compared to voriconazole (42% vs. 60%; *P* < .001). Based on these data, isavuconazonium sulfate can be considered for patients who have invasive, refractory infections or who have intolerance to amphotericin B formulations. A 2016 update to the IDSA Practice Guidelines lists isavuconazonium sulfate as an alternative therapy option for primary treatment of invasive aspergillosis.[618]

Posaconazole has shown activity as a second-line agent against a broad spectrum of invasive fungal infections.[619-622] In an open-label study in patients with invasive aspergillosis refractory to or who had intolerance to standard antifungal therapy (N = 107), 42% had a complete or partial response with posaconazole.[623] Posaconazole is approved in the European Union for the treatment of invasive aspergillosis and certain other invasive fungal infections refractory to standard antifungal agents. In the United States, posaconazole is approved by the FDA for prophylaxis of invasive *Aspergillus* and *Candida* infections, and for treatment of oropharyngeal candidiasis (including cases refractory to fluconazole or itraconazole), but is not indicated as primary or subsequent-line therapy for invasive fungal disease.

The NCCN Guidelines Panel recommends voriconazole monotherapy (category 1) as primary therapy for invasive aspergillosis (see *Antifungal Agents: Azoles* in the algorithm). Although combination antifungal therapy is used as treatment for invasive aspergillosis in some centers, the clinical evidence is inadequate to make conclusions about whether any combination regimen is more effective than voriconazole alone, the current gold standard.

For patients receiving treatment with an echinocandin, the Panel recommends TDM following initiation of treatment to ensure adequate plasma concentrations of the drug. Ongoing TDM is generally warranted.

Mucormycosis and Other Invasive Mold Infections

A higher frequency of mucormycosis has emerged at some institutions with the increased use of voriconazole.[511,624,625] In a case-control study of invasive aspergillosis and mucormycosis in patients with acute leukemia and allogeneic HCT recipients, use of voriconazole and presence of fungal

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与米卡芬净单药治疗组观察到的缓解率相比（主要治疗组和难治性治疗组分别为 50%和 41%），似乎并不有利;然而，米卡芬净单药治疗组的患者数量太少，无法进行比较。此外，本研究中使用的初始米卡芬净剂量（75 mg/天）低于现行标准。最近，一项比较伏立康唑与伏立康唑加阿尼芬净作为侵袭性曲霉菌病主要治疗的随机前瞻性临床试验的数据基于 6 周死亡率评估了反应（N = 454 例血液恶性肿瘤或 HCT 患者）。与伏立康唑单药治疗相比，联合治疗有降低死亡率的趋势（分别为 19.3%和 27.5%; 95% CI，−19.0–1.5; P = .087）。

硫酸异沙唑氮铵于 2015 年获批用于治疗侵袭性曲霉病和侵袭性毛霉菌病。与其他唑类药物不同，硫酸沙武康唑是一种前药，在输注时分解为活性成分艾沙康唑。一项 III 期随机试验比较了硫酸沙武康唑与伏立康唑在侵袭性曲霉病和其他丝状真菌的主要治疗中的作用，结果显示硫酸沙武康唑与伏立康唑相比具有非劣效性（全因死亡率为 19% vs. 20%;校正差异为−1.0%，95%CI 为−7.8–5.7%）。硫酸艾沙康唑和伏立康唑治疗后出现的不良事件相似（96% vs. 98%; P = 0.122），胃肠道疾病和感染或侵染是最常见的（有关安全性的更多信息，请参见唑类药物的毒性和药物相互作用）。硫酸异沙唑氮铵显示肝胆疾病、眼部疾病和皮肤或皮下组织疾病的发生率较低。 与伏立康唑相比，硫酸艾沙康唑的药物相关不良事件也较低（42% vs. 60%; P <0.001）。基于这些数据，硫酸沙武康唑铵可用于侵袭性、难治性感染或对阿替霉素 B 制剂不耐受的患者

2016 年更新的 IDSA 实践指南将硫酸艾沙康氮铵列为侵袭性曲霉病主要治疗的替代疗法。

Posaconazole has shown activity as a second–line agent against a broad spectrum of invasive fungal infections.In an open–label study in patients with invasive aspergillosis refractory to or who had intolerance to standard antifungal therapy (N = 107), 42% had a complete or partial response with posaconazole.Posaconazole is approved in the European Union for the treatment of invasive aspergillosis and certain other invasive fungal infections refractory to standard antifungal agents. In the United States, posaconazole is approved by the FDA for prophylaxis of invasive Aspergillus and Candida infections, and for treatment of oropharyngeal candidiasis (including cases refractory to fluconazole or itraconazole), but is not indicated as primary or subsequent–line therapy for invasive fungal disease.

重试 错误原因

NCCN 指南专家组建议伏立康唑单药治疗（第 1 类）作为侵袭性曲霉病的主要治疗（参见算法中的抗真菌药：唑类）。尽管联合抗真菌治疗在一些中心被用作侵袭性曲霉病的治疗，但临床证据不足以得出结论，即任何联合方案是否比伏立康唑单药更有效，伏立康唑是目前的金标准。

对于接受棘白菌素治疗的患者，专家组建议在开始治疗后进行 TDM，以确保药物的足够血药浓度。通常需要持续进行 TDM。

毛霉菌病和其他侵袭性霉菌感染

随着伏立康唑使用的增加，一些机构出现了更高频率的毛霉菌病。在急性白血病患者和异基因 HCT 接受者中进行的侵袭性曲霉病和毛霉菌病的病例对照研究中，使用伏立康唑和存在真菌

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sinusitis each favored a diagnosis of mucormycosis.[511] However, some transplant centers reported an increased frequency of mucormycosis that pre-dated the availability of voriconazole,[626,627] a finding that likely reflects a greater proportion of patients with severe host defense impairment. Mucormycosis typically manifests as rhinocerebral or pulmonary disease. Histopathology showing broad aseptate or hyposeptate hyphae with 90-degree branching is suggestive of mucormycosis, although culture is required for confirmation.

To date, there have been no positive results from randomized studies for treatment of mucormycosis and other uncommon invasive mold infections. Therefore, recommendations for therapy are based on a limited number of patients from retrospective analyses, data registries, and open-label trials for refractory infections. Treatment of mucormycosis involves amphotericin B (a lipid formulation is advised over AmB-D to reduce the chance of nephrotoxicity) plus early and aggressive surgical debridement, when feasible. A gap in knowledge exists regarding optimal dosing of amphotericin B lipid formulations for invasive non-*Aspergillus* mold infections; an initial dose of 5 mg/kg/day is commonly used. Isavuconazonium sulfate and posaconazole have shown promising results as therapy in mucormycosis refractory to or intolerant of amphotericin B formulations and may be considered for these patients.[619,628,629] Data from an open-label, single-arm, case-control study showed that isavuconazonium sulfate had activity against rare fungi, including mucormycosis, compared to matched controls treated with amphotericin B-based treatment (crude all-cause mortality of 33% vs. 39%; 95% CIs, 14.6–57.0% and 22.9–57.9%, respectively).[629] Ninety-five percent of patients treated with isavuconazonium sulfate had one or more adverse events during treatment, most commonly GI disorders (see *Toxicities and Drug-Drug Interactions of Azoles* for more information on safety).

While isavuconazonium sulfate is approved for the treatment of invasive mucormycosis, posaconazole has not been FDA approved for this indication. Isavuconazonium sulfate and posaconazole can also be considered as maintenance therapy for mucormycosis following control of infection with an amphotericin B formulation and/or surgical debridement.

Fusarium species[630-632] and *Scedosporium* species have emerged as important causes of invasive fungal infection-related mortality in leukemia and in allogeneic HCT recipients at some centers.[627,633,634] The likelihood of infection by a *Fusarium* species is substantially increased by the presence of disseminated cutaneous lesions and isolation of a mold from blood culture.[630] Therapy for invasive fusariosis generally involves voriconazole,[635] posaconazole,[622] or a lipid formulation of amphotericin B.[636] *Scedosporium* species are resistant to amphotericin B; therapy generally involves itraconazole, voriconazole, or posaconazole.[637,638] An infectious disease consultation is advised in all cases of invasive mold infections, particularly for cases involving uncommon and resistant molds.

Early Diagnosis of Invasive Mold Infections

The frequency and diversity of invasive fungal pathogens have increased, and effectively treating these pathogens remains a major challenge. CT scanning of the chest may facilitate early detection of aspergillosis and other filamentous fungi.[639,640] A CT scan may show peripheral or subpleural nodules that are not apparent on plain chest radiographs. The “halo sign” is a characteristic, but not pathognomonic, early chest CT feature of angioinvasive organisms.[537] The hazy alveolar infiltrates surrounding the central nodule or region of consolidation appear to correspond to regions of hemorrhage and are highly suggestive of invasive mold disease, aspergillosis being the most common. The Panel recommends a chest CT scan in patients with 10 to 14 days of neutropenia and with persistent or recurrent fever of unknown origin that is unresponsive to empiric antibacterial agents. A chest CT scan may be

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然而，一些移植中心报告在伏立康唑可用之前毛霉菌病的频率增加，这一发现可能反映了更大比例的患者患有严重的宿主防御功能障碍。

毛霉菌病通常表现为鼻脑或肺部疾病。组织病理学显示宽的无隔或下隔菌丝，90 度分支提示毛霉菌病，尽管需要培养来证实。

到目前为止，还没有积极的结果，从随机研究治疗毛霉菌病和其他罕见的侵袭性霉菌感染。因此，治疗建议是基于回顾性分析、数据登记和难治性感染开放标签试验中有限数量的患者。毛霉菌病的治疗包括阿替霉素 B（一种脂质制剂，建议优于 AmB-D，以减少肾毒性的机会）加上早期和积极的外科清创术（如可行）。对于侵袭性非曲霉菌属霉菌感染的最佳剂量，存在关于阿替霉素 B 脂质制剂的知识缺口;通常使用 5 mg/kg/天的初始剂量。 硫酸异沙康唑和泊沙康唑已显示出治疗难治性或对阿替霉素 B 制剂不耐受的毛霉菌病的良好效果，可考虑用于这些患者。一项开放标签、单组、病例对照研究的数据显示，硫酸异沙康唑对罕见真菌，包括毛霉菌病，与接受基于阿替霉素 B 的治疗的匹配对照相比（粗全因死亡率分别为 33%和 39%; 95%CI 分别为 14.6–57.0%和 22.9– 57.9%）。95%的硫酸沙武康氮铵治疗患者在治疗期间发生了一次或多次不良事件，最常见的是胃肠道疾病（有关安全性的更多信息，请参见唑类药物的毒性和药物相互作用）。

虽然硫酸艾沙康唑铵被批准用于治疗侵袭性毛霉菌病，但泊沙康唑尚未被 FDA 批准用于该适应症。硫酸异沙康唑铵和泊沙康唑也可以被认为是毛霉菌病的维持治疗后，控制感染与阿替霉素 B 制剂和/或手术清创。

镰刀菌属和 Scedosporium 属已成为白血病和同种异基因 HCT 接受者中侵袭性真菌感染相关死亡的重要原因。存在播散性皮肤病变和从血培养中分离出霉菌可显著增加镰刀菌属感染的可能性。侵袭性镰刀菌病的治疗通常包括伏立康唑、泊沙康唑、或脂质制剂的阿替霉素 B。Scedosporium 物种是耐阿替霉素 B;治疗通常涉及伊曲康唑，伏立康唑，或泊沙康唑。传染病咨询建议在所有情况下的侵袭性霉菌感染，特别是对于涉及罕见和耐药霉菌的情况下。

侵袭性霉菌感染的早期诊断

侵袭性真菌病原体的频率和多样性增加，有效治疗这些病原体仍然是一个重大挑战。胸部 CT 扫描有助于早期发现曲霉菌病和其他丝状真菌，CT 扫描可以显示胸部平片上不明显的周围或胸膜下结节。"晕征"是一种特征性的，但不是典型的，早期胸部 CT 表现的血管侵袭性生物。围绕中央结节或实变区的模糊肺泡浸润似乎与出血区域相对应，高度提示侵袭性霉菌病，曲霉病是最常见的。专家组建议对中性粒细胞减少 10 至 14 天，以及对经验性抗菌药物无反应的持续性或复发性发热（原因不明）的患者进行胸部 CT 扫描。胸部 CT 扫描可能是

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considered earlier in patients with multiple prior cycles of potently cytotoxic chemotherapy and in patients receiving systemic corticosteroid therapy.

Studies differ regarding whether serum galactomannan is a useful surveillance tool in asymptomatic patients at high risk for mold infections and in patients with persistent neutropenic fever of unknown etiology. In one study, prospective serial monitoring of galactomannan antigenemia in allogeneic HCT recipients yielded positive and negative predictive values of 94.4% and 98.8%, respectively, and antigenemia preceded radiographic findings by >1 week in 80% of cases of invasive aspergillosis.[641] In another study, the sensitivity was only 64.5% in cases of definite invasive aspergillosis.[546] The PPV was poor when serum galactomannan was used as a surveillance tool in patients with persistent neutropenic fever (PPV = 7.1%) and in HCT (mostly autologous) recipients (PPV = 10%); the negative predictive value was 100% in both groups.[546]

Odabasi et al[540] evaluated the beta-glucan assay as an early diagnostic marker for invasive fungal infections in patients with acute leukemia or MDS receiving antifungal prophylaxis.[540] At least one serum sample was positive at a median of 10 days before the clinical diagnosis in all patients with a proven or probable invasive fungal infection, including candidiasis, fusariosis, trichosporonosis, and aspergillosis. The negative predictive value was 100%, and the specificity of the test was 90% for a single positive test result and at least 96% for two or more sequential positive results.[540] The experience of the beta-glucan assay in HCT recipients is limited and requires additional study.

Although valuable as diagnostic adjuncts to support a diagnosis of probable invasive aspergillosis in patients with compatible host factors, clinical findings, and radiologic findings[642] (see *Initial Clinical Presentation for Lung Infiltrates: Evaluation* in the algorithm), the value of these laboratory markers as surveillance tools for invasive fungal infections is controversial. Use of surveillance markers as a trigger for additional diagnostic evaluation or to modify antifungal therapy is at an exploratory level,[498] and more research is required. Currently, the evidence is inadequate to recommend any of these methods as a surveillance tool in patients who are asymptomatic immunocompromised or in patients with neutropenic fever alone.

COVID-19 Infection in Patients with Cancer

Coronavirus disease 2019 (COVID-19), caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) can cause increased disease severity and mortality in patients with active cancer compared to healthy individuals. Patients with cancer who are immunocompromised, undergoing chemotherapy, older in age, or having other medical conditions are at an increased risk for severe COVID-19. Moreover, patients with hematologic malignancies may be at higher risk of prolonged infection and death from COVID-19 than patients with solid tumors. The CDC provides a comprehensive list of recommendations on testing for SARS-CoV-2 and managing COVID-19 and cancer-directed therapies in people with concurrent cancer and COVID-19. For a summary of NCCN recommendations, see *Management of Concurrent COVID-19 and Cancer in Patients* and *General Recommendations for Vaccination in Patients with Cancer* in the algorithm.

Summary

Substantial progress has been made in the prevention and treatment of infectious complications associated with neutropenia and IST in patients with cancer. Certain populations of patients are at increased risk for developing infectious complications during the course of their disease and cancer treatment. Infectious complications remain an important cause of morbidity and mortality in patients undergoing anti-tumor therapy. The extent of infectious risk is highly dependent on an individual patient's underlying malignancy, degree of neutropenia, past history of infections and exposure to pathogens, treatment with myelosuppressive regimens,

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在既往接受过多个周期的强效细胞毒性化疗的患者和接受全身性皮质类固醇治疗的患者中，更早考虑。

关于血清半乳甘露聚糖是否是霉菌感染高风险无症状患者和病因不明的持续性血小板减少性发热患者的有用监测工具，研究存在差异。在一项研究中，对同种异体 HCT 受者半乳甘露聚糖抗原血症的前瞻性系列监测分别产生了 94.4%和 98.8%的阳性和阴性预测值，并且在 80%的侵袭性曲霉病病例中，抗原血症先于放射学发现>1 周。在确诊为侵袭性曲霉病的病例中，敏感性仅为 64.5%，在持续性贫血发热患者中，血清半乳甘露聚糖作为监测工具，PPV 值较差（PPV = 7.1%）和 HCT（主要是自体）受体（PPV = 10%）;两组的阴性预测值均为 100%。

Odabasi 等评价了β–葡聚糖检测作为接受抗真菌预防治疗的急性白血病或 MDS 患者侵袭性真菌感染的早期诊断标志物，在临床诊断前 10 天的中位数，所有确诊或可能的侵袭性真菌感染（包括念珠菌病、镰刀菌病、毛孢子虫病和曲霉病）患者至少有一份血清样本呈阳性。阴性预测值为 100%，并且该测试的特异性对于单个阳性测试结果为 90%，对于两个或更多个连续阳性结果为至少 96%。

虽然作为诊断标志物对于支持具有相容的宿主因素、临床表现和放射学表现的患者诊断可能的侵袭性曲霉菌病是有价值的（参见肺损伤的初始临床表现：算法中的评价），但这些实验室标志物作为侵袭性真菌感染的监测工具的价值是有争议的。使用监视标志作为触发器，

诊断评估或修改抗真菌治疗尚处于探索阶段，需要更多的研究。目前，没有足够的证据推荐任何这些方法作为无症状免疫功能低下患者或单纯血小板减少性发热患者的监测工具。

癌症患者的 COVID–19 感染

2019 冠状病毒病（COVID–19）由严重急性呼吸道综合征冠状病毒 2（SARS–CoV–2）引起，与健康个体相比，活动性癌症患者的疾病严重程度和死亡率会增加。免疫功能低下、正在接受化疗、年龄较大或患有其他疾病的癌症患者患严重 COVID–19 的风险增加。此外，恶性血液病患者因 COVID–19 长期感染和死亡的风险可能高于实体瘤患者。疾病预防控制中心提供了一份关于检测 SARS–CoV–2 和管理 COVID–19 以及并发癌症和 COVID–19 患者的癌症导向疗法的全面建议清单。关于 NCCN 的摘要

建议，参见算法中的并发 COVID–19 和癌症患者的管理和癌症患者疫苗接种的一般建议。

总结

在预防和治疗与癌症患者中性粒细胞减少和 IST 相关的感染性并发症方面取得了实质性进展。某些患者群体在其疾病和癌症治疗过程中发生感染性并发症的风险增加。感染并发症仍然是接受抗肿瘤治疗的患者发病率和死亡率的重要原因。感染风险的程度高度依赖于个体患者的潜在恶性肿瘤、中性粒细胞减少症的程度、既往感染史和病原体暴露史、骨髓抑制方案治疗，

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and the overall status of immune function in the patient. It is therefore imperative that patients be evaluated individually for risk of infection in order to minimize the occurrence of infection-related complications. Preventive measures for infection management in patients with cancer include routine surveillance to monitor for early laboratory indications of infection (especially in the context of viral reactivations) and the appropriate use of prophylaxis and/or preemptive therapy with antimicrobial agents in patient groups who are at high risk. It is important to note that upfront prophylaxis is not necessary in all patients with cancer; prophylactic measures should only be used in patients at high risk for specific pathogens during the high-risk period in order to avoid the emergence of resistant pathogens.

The development of antipseudomonal beta-lactam agents and the routine use of empiric antimicrobial therapy at the onset of neutropenic fever have contributed to reductions in mortality from bacterial infections. With more patients undergoing treatment with potent cytotoxic regimens and receiving allogeneic HCT, opportunistic viral and fungal infections have become important causes of mortality in these patients. In addition, the increasing prevalence of antibiotic-resistant pathogens is a challenge. Infection control should not only rely on anti-infective prophylaxis but should continue to incorporate standard infection control measures (eg, careful hand-washing by health care professionals). When selecting antimicrobial agents for prophylaxis and/or preemptive therapy, consideration should be given to the local susceptibility and resistance patterns of pathogens.

In summary, prevention and treatment of infections in patients with cancer is a complex and continuously evolving field. However, these advances in treatment have only further emphasized the need for multidisciplinary care. The NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections aim to provide an overview of the risk categorization and recommended strategies for prevention of infections in patient populations at high risk, and recommendations for empiric therapy, evaluation, follow-up, and monitoring in patients with signs and/or symptoms of infections. Individualized risk evaluation for infections, incorporation of preventive measures, and prompt identification and treatment of active infections are essential components of the overall spectrum of care in cancer management, and can contribute to optimizing treatment outcomes in patients with cancer.

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以及患者免疫功能的总体状态。因此，必须对患者进行单独的感染风险评估，以尽量减少感染相关并发症的发生。癌症患者感染管理的预防措施包括常规监测，以监测感染的早期实验室指征（特别是在病毒再激活的情况下），以及在高危患者群体中适当使用预防和/或抗菌药物先发制人治疗。值得注意的是，并非所有癌症患者都需要提前预防;预防措施仅应用于高危期特定病原体高风险患者，以避免耐药病原体的出现。

抗假单胞菌β–内酰胺类药物的开发和在血小板减少性发热发作时常规使用经验性抗菌治疗有助于降低细菌感染的死亡率。随着越来越多的患者接受强效细胞毒性方案治疗并接受同种异体 HCT，机会性病毒和真菌感染已成为这些患者死亡的重要原因。此外，抗药性病原体的日益流行也是一个挑战。感染控制不应仅仅依靠抗感染预防，

应继续采用标准的感染控制措施（例如，由卫生保健专业人员仔细洗手）。在选择预防和/或预防性治疗的抗菌药物时，应考虑病原体的局部敏感性和耐药模式。

总之，预防和治疗癌症患者的感染是一个复杂且不断发展的领域。然而，这些治疗方面的进展只是进一步强调了多学科护理的必要性。NCCN 癌症相关感染预防和治疗指南旨在概述分类和推荐的预防高危患者人群感染的策略，以及对有感染体征和/或症状的患者的经验性治疗、评估、随访和监测的建议。感染的个体化风险评估、预防措施的结合以及活动性感染的及时识别和治疗是癌症管理中整体护理的重要组成部分，有助于优化癌症患者的治疗结果。

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